Metrical grammar: Variation, gradience, and cumulativity at the syntax–phonology interface

A dissertation presented

by

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to

The Department of Linguistics

in partial fulfillment of the requirements

for the degree of

Doctor of Philosophy

in the subject of

Linguistics

Harvard University

Cambridge, Massachusetts

August 2026

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Abstract

Poetic syntax is less rigid than ordinary language's: Poets avail themselves of otherwise impermissible syntactic configurations to make verse lines more phonologically well-formed. However, the system is by no means "anything goes," since not all deviations from ordinary-language syntax are permitted in meter. This dissertation examines phonologically conditioned syntactic variation in the Bosnian/Croatian/Montenegrin/Serbian (BCMS) oral folk tradition. BCMS folk meter's highly variable constituent order in line-final noun and verb phrases is largely predictable from several phonological factors, which compete against, and sometimes cooperate with, meter's baseline linearization preferences. The results bear on two extensively debated topics in linguistic theory: the choice of framework for modeling variation and gradience in phonology and at its interfaces, and the nature of the syntax-phonology interaction in poetic meter.

To model the patterned word-order variation in the BCMS folk tradition, I adopt a probabilistic interleaving model, in which phonological and linearization constraints interact freely in a single parallel constraint grammar, extending Youmans's (1983, 1989) and Fitzgerald's (2007) proposals into the probabilistic Maximum Entropy (MaxEnt) grammar. MaxEnt is compared against two rival stochastic frameworks, Stochastic Optimality Theory (StOT) and Noisy Harmonic Grammar (NHG), using two diagnostics: ganging-up cumulativity and simple harmonic bounding. Even when augmented

with conjoined constraints, StOT suppresses interaction effects between independent factors. Meanwhile, NHG underperforms MaxEnt on harmonic bounding: Both frameworks produce relative harmonic bounding, but NHG is shown to be overly restrictive. Only MaxEnt captures all qualitative generalizations in the data.

The free, direct interaction between phonological and linearization constraints in meter lends itself to two interpretations. On one, syntactic and phonological information is processed in parallel, allowing direct, bidirectional flow of information between the two modules. I show that a parallel interactionist architecture is required only insofar as linearization takes place in narrow syntax. The other option is that linear order is assigned post-syntactically and surface word-order variation has no bearing on the character of the syntax-phonology interaction. On this account, the probabilistic interleaving model is, in effect, a model of PF grammar, which includes both phonological and linearization constraints.

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Acknowledgments

Much like oral poetry, this dissertation is in many ways the result of a collective effort of my entire family. I say this because every step that has led me to my doctoral studies and this magnum opus as their conclusion was effectively a collective enterprise. My mother Snežana, to whom this dissertation is dedicated, has always been the greatest source of unwavering and unconditional support in my life. When, entering the final semester of my undergraduate studies at the very beginning of the COVID-19 pandemic, I first communicated my plan to pursue a doctorate in the US, she simply asked: "okay, but what do you need to get there?" And as it turns out, I needed a lot. My preparation expenses, application fees, visa application, and plane tickets exceeded the yearly income of an average Serbian household. The bulk of these expenses were covered by my mother. To say that I would not have been here without my mother's support is therefore true in the most literal sense. More importantly, she was never overly happy about my decision to continue my education in a land far away, which was, at the time, making headlines for all the wrong reasons, but she never once questioned my decision. Just like four years before that, she never questioned my choice of major, or really any choice I have made, however questionable. I had the privilege of being introduced to the object of this study, the South Slavic oral epic tradition, at a very early age by my late grandfather Dragoslav (Dragan) Milenković. He knew thousands of verse lines of this epic tradition, which he would fondly quote on various occasions. My aunt Jasmina, uncle Dejan, and late grandmother Elena have also been a great support and immensely important figures in my life.

My journey in linguistics started at the University of Belgrade. I am very grateful to all my professors and advisors across several departments (Serbian, General Linguistics, and Classics) who showed me the study of language as one of the most fascinating fields

of inquiry, took great care of my early research, and, in effect, taught me how to be a linguist. Professor Ana Batas was my first phonetics and phonology teacher. Her courses in the phonology of Serbian and intonation were a real game changer for me, as they dragged me into synchronic linguistics at a time when I was primarily interested in diachrony. Professor Batas also advised my MA project, on the acoustic correlates of stress in dialectal Serbian. It was during this time that the ideas for what are now my first journal articles were born. I owe the key ingredients of my research program to my time as Professor Batas's student: an interest in (various aspects of) the sound patterns of the South Slavic languages and dialects, an inductive, data-driven approach, and a quantitative grounding. Professor Orsat Ligorio taught me historical linguistics and more ancient languages than I could list here, and was, unofficially, my undergraduate research advisor. It was under his supervision that I conducted my research projects in etymology and comparative Indo-European linguistics. In fact, my first introduction to the linguistic study of poetic meter was in Professor Ligorio's Old Avestan course. Professor Đorđe Božović introduced me to linguistic theory. His introductory courses and his course in the history of linguistics remain, to this day, among the best general linguistics courses I have had the fortune to attend.

I am indebted to my advisor, Kevin Ryan, and my committee members, Jeremy Rau and Aleksei Nazarov. Many thanks go to Kevin. I was spoiled by the time and attention my work received from Kevin. I have learned a great deal from him not only about prosody and linguistics, but about what kind of scholar, teacher, advisor, and collaborator I want to be. Arriving at Harvard, I never imagined I would work on meter, let alone write a dissertation on it, but by the end of my third year, Kevin's work had already convinced me that poetic meter could tell us a great deal about the inner workings of language. This project could hardly have had a better supervisor. Jeremy always pushed me to frame my work in ways accessible to a broader interdisciplinary

audience, which is an underrated skill in our field, and one I hope this dissertation reflects (at least to some extent). He also taught me a great deal about linguistic pedagogy and took a great interest in my professionalization and job applications. Aleksei's advising has been instrumental in shaping my approach to linguistic issues from a computational perspective, and my embracing the chaos of real-life linguistic data, particularly in learning to live with, and eventually come to like, exceptions and noise.

I would most certainly not have applied to Harvard had it not been for Marek Majer, an alumnus of the program. Professor Majer also gave me valuable first-hand information that helped me tremendously with my graduate applications. Many thanks to Jonathan Bobaljik, Kate Davidson, Jim Huang, Yağmur Sağ, and Susi Wurmbrand for showing me the thrilling world of linguistics outside phonology, and to Jay Jasanoff for inspiring lectures in historical Indo-European linguistics and casual, first-hand recollections of the history of phonology.

The idea for this dissertation was born and developed in two graduate phonology seminars: the joint Harvard–MIT Metrics seminar, co-taught by Donca Steriade and Kevin Ryan, and a Morphophonology seminar at UMass (which I was visiting at the time), co-taught by Joe Pater and Michael Becker. Various stages of this project, and of earlier projects that eventually led to it, were presented at the MIT Phonology Circle, UMass Sound Workshop, Harvard PhonLab meetings, AMP 2023, AMP 2025, FASL 32, and CLS 59; I thank the audiences at all of these venues for fruitful discussion and constructive feedback. I am also grateful to Canaan Breiss, Edward Flemming, Miloje Despić, and Jonathan Bobaljik for talking through certain aspects of this project with me at various stages. It goes without saying that all errors in this dissertation, which are unavoidable in stochastic systems, are exclusively mine.

At Harvard Linguistics, I am indebted to everyone: my professors, peers, and staff.

It was a great privilege to be part of this community. Huge thanks to the people who made this such a wonderful experience for me, especially my colleagues, Daria Bikina, Joost Botman, Jian Cui, Wei-Fang Hsieh, Ian Kirby, Hayley Ross, Ankana Saha, Hande Sevgi, Natasha Thalluri, Melody Wang, and Yvette Wu, and my roommates, Eli and Briana Spradling, and Mark Durham. Dissertation year was a particularly challenging time for me, and in a way, a number of factors, academic and nonacademic, ganged up (in the sense of Chapter 3) against this project. I would certainly not have kept my focus (and sanity) during this time without my only cohort-mate, Natasha Thalluri, Mark Durham, Sunyoung Park, and Jian-Leat Siah.

Mojoj majci.

Introduction

0.1 Phonological effects on word order

Phonological factors can modulate various aspects of sentence structure, including word order (Hetzron, 1972; Zec and Inkelas, 1990; Harford and Demuth, 1999; Wasow, 2002; Rosenbach, 2014; Shih et al., 2015; Shih, 2017b; Shih and Zuraw, 2018; Anttila, 2016; Richards, 2016; Szmrecsanyi et al., 2017; Ryan, 2019b; Breiss and Hayes, 2020). Coordinate constructions (or binomials) are a prime environment for phonologically conditioned word-order preferences. When semantic and other non-phonological factors are controlled for, binomial order is robustly affected by the conjuncts' phonological properties, such as syllable count (Wright et al., 2002, 2005; Benor and Levy, 2006; Morgan and Levy, 2016; Ryan, 2019a). The syllable count effect is illustrated in (1): The binomial order in (1a), with the longer, trisyllabic *sciences* following monosyllabic *arts*, is preferred (per Google Books Ngram data) to its transposition in (1b).[1]

(1) Syllable count effect in coordinate constructions: (1a) > (1b)

   a. arts and sciences

   b. sciences and arts

Phonological markedness constraints may trigger deviation from the canonical order

[1] Available here.

in some syntactic constructions. In Tagalog (Shih and Zuraw, 2017), the default order of adjectival modifiers is prenominal, as evidenced by the fact that about 90% of NP tokens in Shih and Zuraw's corpus show that order (and the strong negative intercept in the regression model; *op. cit.*, e334). As example (2) shows, Tagalog speakers preferentially use the inverted, adjective-final order (2b) over the default, adjective-first order (2a) when the default order brings about a nasal cluster (underlined). The canonical order is therefore more phonologically marked, since it violates OCP-NASAL, which penalizes adjacent nasal segments (Goldsmith, 1976; McCarthy, 1986). (The linker particle is realized as *-ng* [ŋ] if the first NP member ends in a vowel, and as *na* postconsonantally.)

(2) OCP-NASAL triggers adjective inversion in Tagalog: (2b) > (2a)

a. Canonical order (Adj–N)

itim na pelúka
black LINK wig

b. Inversion (N–Adj)

pelúka -ng itim
wig -LINK black

'black wig' [Shih and Zuraw, 2017: e325, ex. (6)]

Similarly, the German intensifier *ganz* 'quite' usually follows the article *ein*, but may also precede it. There are subtle, but statistically robust, rhythmic effects on intensifier order (Kentner, 2018). For instance, the likelihood of the intensifier-first order (*ganz ein*) is higher when the article-first alternative (*ein ganz*) results in a clash, that is, adjacent primary stressed syllables (Prince, 1983; Hayes, 1995). In (3), the canonical order (3a) produces a stress clash. Inversion (3b) separates the primary stressed syllables (underlined) by moving the article *ein* between them. In (4), the canonical order of the disyllabic intensifier (4a) does not result in a stress clash, which

renders inversion (4b) rhythmically unmotivated and therefore strongly dispreferred. Moreover, the non-canonical order of (4b) brings about a lapse and is in fact more rhythmically marked than (4a).

(3) Inversion frequent when canonical order results in stress clash

a. Canonical order

ein ganz junger Mann
a quite young man

b. Inversion

ganz ein junger Mann
quite a young man

'a quite young man' [Kentner, 2018: 126; ex. (3b)]

(4) Inversion strongly dispreferred when canonical order does not result in stress clash

a. Canonical order

ein gänzlich junger Kerl
a quite young lad

b. Inversion

*gänzlich ein junger Kerl
quite a young lad

'a quite young lad' [Kentner, 2018: 126; ex. (5a)]

Similar cases of phonologically conditioned syntactic variation include variable adjective–noun order in French (Abeillé and Godard, 1999; Thuilier, 2012), English genitive (Shih et al., 2015; Szmrecsanyi et al., 2017) and dative alternations (Wasow, 2002; Anttila et al., 2010), *to*-optionality in English infinitives (Wasow et al., 2015), and many others.

Poetic meter offers a unique window into phonological effects on word order. Meter tends to maximize phonological well-formedness. This allows phonological constraints to override syntax under certain conditions, on which I elaborate in §0.2 and Chapter 5. Thus, some ill-formed syntactic configurations become available in meter by way of reducing phonological markedness (Kiparsky, 1975; Youmans, 1983, 1989; Fitzgerald, 2003, 2007; Gunkel and Ryan, 2011; Hayes et al., 2012). Consider the Shakespearian line in (5b). The canonical order (*should be blunter*; 5a) maps an initially stressed, trochaic word (*blunter*) onto an iambic template, resulting in the so-called inverted foot (Halle and Keyser, 1971; Kiparsky, 1975; Youmans, 1983; boxed below). This mismatch is readily avoided by syntactic inversion (5b).

(5) Syntactic inversion makes the verse line scannable

a. Canonical order: *should be blunter*

*	W	S	W	S	W	S	W	S	W	S	[unmetrical]
	Thy	édge	should	bé	blún-	ter	than	áp-	pe-	-títe	

b. Inversion: *should blunter be* (Youmans, 1983, 67)

W	S	W	S	W	S	W	S	W	S	[Son 56.2]
Thy	édge	should	blún-	ter	bé	than	áp-	pe-	-títe	

Cases like (5) show that phonological requirements can override syntax in meter in ways that are rarely, if ever, permitted in ordinary language. Oral folk poetry offers a particularly apt testing ground for probing the syntax–phonology relationship in metered verse. Folk poetry is a collective enterprise, composed and transmitted across generations of performers in the community. This presupposes a shared knowledge of the fine-grained regularities that shape folk verse. Folk meter therefore shows a higher

degree of uniformity across poets than artistic verse, the product of individual poets' exceptional literary gifts (Jakobson, 1960; Foley, 1988; Hayes and MacEachern, 1998).

0.2 Aims and scope

This dissertation examines variable conjunct, adjective–noun, noun–possessive, and verb–object order in the Bosnian/Croatian/Montenegrin/Serbian (BCMS) oral folk tradition. I limit the discussion to the epic decasyllable, the most extensively studied verse form in this oral tradition (Karadžić, 1824; Maretić, 1901, 1907; Vaillant, 1932; Jakobson, 1933, 1966; Lord, 1960; Ružić, 1975; Batinić, 1975; Foley, 1993; Zec, 2008, among others). The key demonstration is that this oral tradition's highly variable word order is fully predictable from a handful of phonological factors such as relative word size, syllable weight, and stress. I develop models of poetic word order that predict the surface order of constituents in verse lines based on their phonological properties. Although the empirical domain of this dissertation is poetic meter, its theoretical stakes are more general. The findings have implications for the choice of grammatical framework for modeling variation and gradience, an actively debated question in phonological theory, and for the nature of the syntax–phonology interface (see §0.3).

It is customarily assumed that the syntax of poetic language, including its word order, is less rigid than the syntax of natural language (Rice, 1997; Golston, 1998; Golston and Riad, 2000; Fitzgerald, 2003, 2007; Kentner and Franz, 2019). This flexibility manifests primarily in the fact that many syntactic configurations that are downright impermissible in natural language become available in poetic meter. These deviations from the syntactic rules of natural language are by no means random: Poets avail themselves of noncanonical word orders to make verse lines more phonologically well-

formed.

In natural language, syntax usually takes precedence over phonology (Golston, 1995; Tranel, 1998; Anttila, 2016). Phonological factors arguably never get to tamper with syntactic operations and preferences, which is customarily referred to as the principle of phonology-free syntax (Zwicky and Pullum, 1986a,b; Miller et al., 1997).[2] In meter, however, sensitivity to phonological factors is amplified and phonology gets to override syntax, which makes deviations from the canonical order possible.

Thus, poetic word order differs from prose word order because poetic language prioritizes phonology over syntax. This raises the question of just how far poets will go to satisfy phonological pressures. In fact, not all kinds of syntactic inversions are permitted in meter: There are limits as to how much the ordinary-language syntax may be distorted in poetic language (Youmans, 1983; Fitzgerald, 2007). This dissertation aims to discover where those limits are, or at least contribute to our understanding of those limits, using the South Slavic oral folk tradition as a case study.

In this spirit, Fitzgerald (2007) proposes a constraint-based model of poetic grammar, couched in Optimality Theory (OT; Prince and Smolensky, 2004), in which phonological and linearization constraints are *interleaved*, i.e. freely rankable. This dissertation develops a specific implementation of Fitzgerald's interleaving model, which I call the PROBABILISTIC INTERLEAVING MODEL. Unlike Fitzgerald, who adopts deterministic OT as a model of poetic grammar, I adopt Maximum Entropy (MaxEnt; Smolensky, 1986; Goldwater and Johnson, 2003; Hayes and Wilson, 2008), a probabilistic variant of Harmonic Grammar (HG; Legendre et al., 1990). MaxEnt accommodates the fundamentally probabilistic nature of grammatical knowledge that folk poets possess and that governs their patterned manipulation of word order in verse. In so doing,

[2]Though see Harford and Demuth (1999); Teeple (2008); Shih and Zuraw (2017, 2018); Breiss and Hayes (2020) for possible exceptions.

MaxEnt outperforms two prominent rival frameworks: Stochastic OT (StOT; Boersma and Hayes, 2001) and Noisy HG (NHG; Boersma and Pater, 2016).

The present study differs from earlier work in its empirical scope and quantitative approach. Previous studies in the generative tradition (e.g., Youmans, 1983; Fitzgerald, 2007) analyze samples of syntactic inversions and demonstrate that most instances of noncanonical word orders are phonologically motivated. Their focus is therefore on a particular aspect of poetic word order, namely departures from the canonical word order. In contrast to examining syntactic inversions in isolation, the present study analyzes all attested instances of multiple construction types in a large metrical corpus (see §1.2.3), including both canonical and noncanonical orders. The more comprehensive empirical coverage allows us to model poetic grammar as a system of competing phonological and syntactic pressures. The analysis identifies the conditions under which phonological pressures override baseline linearization preferences and estimates with quantitative rigor the relative contributions of phonological and syntactic factors to poetic word order.

Another departure from previous work is this study's focus on folk poetry, whereas the previous studies cited above examine word order in artistic verse. Unlike artistic poetry, which is individualistic and idiosyncratic, folk epic poetry is highly FORMULAIC: Poets repeatedly deploy conventionalized, petrified word collocations, larger phrasal units, and at times entire verse lines. This formulaic, densely repetitive character of oral poetry enables poets to memorize the thousands of verse lines that are part of their repertoire. At first glance, then, the word order of folk verse may simply reflect formula, that is, the petrified order of frequently collocated lexical items, memorized in a particular order rather than actively manipulated by the poet.

The results of this study suggest otherwise: Oral folk poetry, for all its repetitiveness and uniformity, is grammar-governed. Folk meter's word order is largely rule-governed,

subject to phonological and other grammatical constraints. Poets possess internalized probabilistic knowledge that lets them gauge the plausibility of word-order alternatives given the phonological makeup of phrasal members. A productive mental device, the METRICAL (or POETIC) GRAMMAR, actively governs poets' deployment of linguistic means (Lord, 1960). Poetic word order is thus not learned through memorization; folk poets internalize the fine-grained statistical generalizations by which they order words in folk verse.

0.3 Contributions beyond metrics

The results of this dissertation extend beyond poetic meter, bearing on several topics of more general interest to linguists with no stake in verse. First, the study has framework-level implications for phonological theory. In Chapter 2, I provide further evidence that, despite its ability to model ganging-up interactions between independent constraints, HG still necessitates conjoined constraints to accommodate a limited subset of interaction effects in phonology (Green and Davis, 2014; Shih, 2017a; Milenković, 2026).

Additionally, Chapter 2 makes several methodological contributions. I provide a blueprint for comparing stochastic variants of OT and HG in a maximally controlled fashion. I also show that the learning algorithms used in stochastic simulations skew parameter estimates and model performance in a way that confounds model comparison. Finally, I identify limitations in the goodness-of-fit metrics commonly used for model comparison in computational linguistics.

In Chapters 3–4, I develop a two-pronged argument for MaxEnt as a theory of variation and gradience. In Chapter 3, I use the phenomenon of ganging-up cumulativity, or interaction effects between independent factors in the grammar, to show that

the stochastic variants of HG, MaxEnt and Noisy Harmonic Grammar, are superior to StOT. This result is not surprising, given that the same conclusion has been reached in several recent studies (Zuraw and Hayes, 2017; Smith and Pater, 2020; Flemming and Magri, 2026). However, Chapter 3 goes one step further: I show that even when StOT is augmented with additional, non-default machinery which should in theory help it capture ganging-up cumulativity, it still fails to do so. By extension, StOT is not a plausible theory of variation, given that ganging-up interactions are ubiquitous in variable linguistic data (Jäger and Rosenbach, 2006; McPherson, 2016; Breiss and Hayes, 2020; Breiss and Albright, 2022; Kawahara and Breiss, 2021; Hayes, 2022, 2023). Arguably more interesting is Chapter 4, in which I directly compare MaxEnt and NHG using harmonic bounding as a test case. The results favor MaxEnt.

In Chapter 5, I discuss models of the syntax–phonology interface in poetic meter. I identify two models that allow phonology to override syntax and directly shape poetic word order, in ways consistent with the BCMS metrical data. This aspect of the study is more narrowly meter-specific, but it still bears on linguistic theory more generally, since it shows what kind of empirical data can be brought to bear on major theoretical questions about the syntax–phonology interface in natural language.

This dissertation thus contributes to two largely independent lines of inquiry in linguistic theory: the choice of framework for modeling variation and gradience in phonology and at its interfaces (Zuraw and Hayes, 2017; Anttila and Magri, 2017; Smith and Pater, 2020; Flemming, 2021; Hayes, 2022; Magri and Anttila, 2026), and the nature of the syntax–phonology interaction in poetic meter (Youmans, 1983; Golston, 1998; Golston and Riad, 2000; Fitzgerald, 2007).

0.4 Dissertation roadmap

The dissertation proceeds as follows. Chapter 1 provides background on BCMS word order, segmental phonology, word prosody, and folk meter. I introduce the phonological constraints that affect word order in the BCMS oral tradition and adduce statistical evidence for them.

Chapter 2 has several goals. First, I outline the main properties and predictions of the three stochastic constraint-based frameworks compared in this dissertation: StOT, NHG, and MaxEnt HG. I address the challenges and confounds that complicate the meaningful comparison of these stochastic constraint-based models. Second, I carry out a model-selection procedure to identify the best-fitting constraint set. I present overall fit metrics for the stochastic models before zooming in on their handling of phenomena with major theoretical import in Chapters 3–4.

The focus of Chapter 3 is ganging-up cumulativity. There are several patterns of constraint interaction in meter. Multiple syntactic constraints can gang up against a linearization constraint, jointly driving inversion. Phonological constraints may also join forces with syntactic linearization constraints and jointly figure as inversion inhibitors. I explore the range of interaction effects that stochastic frameworks can generate, given the way they make the grammar probabilistic. Although StOT, unlike its deterministic counterpart, produces a limited range of gang effects, it is an overly restrictive theory of ganging-up cumulativity, failing to accommodate the full range of ganging-up interactions in the meter. In addition, I show that even when supplemented with conjoined constraints, the standard remedy for strict-ranking OT's inability to model gang effects (Mai and Bakovic, 2020; Tebay and Zimmermann, 2020; Milenković, 2026), StOT still fails to capture the full range of ganging-up patterns in the data.

Chapter 4 turns to simple harmonic bounding, that is, phonologically unmotivated syntactic inversions, which are abundantly attested in meter. Having ruled out StOT in Chapter 3 as insufficiently powerful (on the grounds of its failure to accommodate the pervasive gang effects in the data), I use harmonic bounding to compare the predictions of the stochastic variants of HG: MaxEnt and NHG. While ample work has shown stochastic HGs (MaxEnt and NHG) to be superior to StOT, comparison between MaxEnt and NHG has turned out to be quite challenging (Alderete and Finley, 2023, 590), Flemming (2021) being a rare case study that compares the two frameworks. I first show that while both MaxEnt and NHG (as implemented here) maintain relative harmonic bounding, they make different testable predictions about the range of probability mass that harmonically bounded candidates can receive. The main finding is that NHG bounds more aggressively than MaxEnt, consistently assigning lower probability to harmonically bounded candidates. The BCMS metrical data offer the exact kind of case study that can be used to probe these newly identified testable predictions of MaxEnt and NHG.

Chapter 5 takes up the syntax–phonology interface in poetic meter, from the perspective of phonologically conditioned word-order variation in the BCMS oral tradition. I compare several models of the syntax–phonology interface and argue that the data generally support an interactionist model, whereby phonological information is directly accessible to the syntactic component. I then show, however, that this conclusion holds only insofar as linearization is part of narrow syntax: If linear order is established post-syntactically, a non-interactionist architecture with strict separation between the two modules is still tenable.

Chapter 6 concludes.

Chapter 1

The grammar of meter

1.1 Chapter overview

This chapter lays the descriptive and theoretical groundwork for the simulations and formal analysis in Chapters 2–5. I introduce the BCMS oral folk tradition and present the phonological factors that modulate poetic word order. The chapter is organized as follows. §1.2 provides background on BCMS phonology, word order, and folk meter and introduces the metrical corpus used in this dissertation. §1.3 outlines the phonological factors responsible for syntactic variation in meter. §1.4 reports two regression models of the corpus data, which evaluate the effects of phonological factors on word-order variation. §1.5 concludes.

1.2 Background

1.2.1 Segmental phonology and word prosody

This section provides the essential background on BCMS phonology: the language's segment inventory (Miller-Ockhuizen and Zec, 2003; Morén, 2006; Petrović and Gudurić,

2010) and its word prosody (Ivić, 1958; Lehiste and Ivić, 1986; Inkelas and Zec, 1988). Throughout, I adopt an IPA-based rendition of BCMS; the relationship between the IPA transcription and the standard orthography is given below.

BCMS has a symmetric, typologically unremarkable five-vowel system, shown in Figure 1.1 (Morén, 2006; Petrović and Gudurić, 2010). The coronal trill r is syllabic word-initially before a consonant, as in *rvati* [ˈr̩.ʋa.ti] 'wrestle.INF,' and word-internally between two consonants, as in *prvi* [ˈpr̩ː.ʋiː] 'first.NOM.SG.M.'[1] All six syllabic segments can be short ([a]) or long ([aː]). In Neoštokavian dialects, whose prosodic system constitutes the basis for standard BCMS prosody, vowel length is contrastive in accented and post-tonic syllables. Pre-tonic syllables permit only short vowels (Lehiste and Ivić, 1986; Petrović and Gudurić, 2010), though distributional constraints on vowel length vary considerably across BCMS dialects (Ivić, 1958, 1985; Ligorio, 2016).

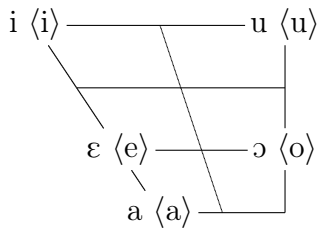


Figure 1.1: BCMS vowel chart

The consonant inventory is shown in Table 1.1.

Turning to word prosody, the Neoštokavian dialects of BCMS are traditionally described as pitch-accent varieties (Daničić, 1925; Ivić, 1958; Nikolić, 1970; Lehiste and Ivić, 1986; Ligorio, 2016). The classic Neoštokavian system has two contrastive pitch-accent types: falling and rising. Both accent types can be located on a short or a long vowel, which results in four accent categories in total (Table 1.2).

[1]The other two alveolar sonorants, l and n, can also be syllabic in recent loanwords.

	labial	dental	alveolar	postalveolar	palatal	velar
stop	p ⟨p⟩ b ⟨b⟩	t̪ ⟨t⟩ d̪ ⟨d⟩				k ⟨k⟩ g ⟨g⟩
fricative	f ⟨f⟩		s ⟨s⟩ z ⟨z⟩	ʃ ⟨š⟩ ʒ ⟨ž⟩		x ⟨h⟩
affricate		ʦ ⟨c⟩		ʧ ⟨č⟩ ʤ ⟨dž⟩ ʨ ⟨ć⟩ ʥ ⟨dj⟩		
nasal	m ⟨m⟩		n ⟨n⟩		ɲ ⟨nj⟩	
lateral			l ⟨l⟩		ʎ ⟨lj⟩	
trill			r ⟨r⟩			
glide	ʋ ⟨v⟩				j ⟨j⟩	

Table 1.1: BCMS consonants: IPA and ⟨orthographic renditions⟩.

accent type	example	gloss
long falling	mêso	'meat'
short falling	krȁva	'cow'
long rising	rúka	'arm'
short rising	sèstra	'sister'

Table 1.2: BCMS pitch accents and their orthographic rendering.

Accent position is subject to distributional restrictions, summarized in Table 1.3. Monosyllables must be falling-accented, and in longer words, falling accents are restricted to the initial syllable. Rising accents can fall on any non-final syllable. The initial syllable of polysyllabic forms is the only position in which accent type is contrastive; everywhere else, accent type is fully predictable.

	ONLY	initial	medial	final
falling	✓	✓	✗	✗
rising	✗	✓	✓	✗

Table 1.3: Distributional constraints on Neoštokavian pitch accents.

BCMS accents are an amalgam of two prosodic ingredients: TONE, cued by an F0 peak, and STRESS, cued by vowel duration and intensity (Inkelas and Zec, 1988; Zec, 1999; Zec and Zsiga, 2010; Zsiga and Zec, 2013). Accent type is cued primarily by the

timing of the F0 peak relative to the stressed syllable. In falling accents, the pitch peak is located on the stressed vowel; in rising accents, the stressed syllable is characterized by a pitch plateau, with the F0 peak on the following, post-tonic vowel (Lehiste and Ivić, 1986; Zsiga and Zec, 2013).

I follow Inkelas and Zec (1988); Zec (1999); Zec and Zsiga (2010) in analyzing NS as a lexical tone system with tone-driven stress. Underlyingly, moras can be high-toned or lack tone altogether.[2] Tone observes strict culminativity (in the sense of Hyman, 2006), in that prosodic words can bear at most one, singly-linked high tone. The locus of stress is entirely predictable from the position of the lexical high. Stress regularly falls on the syllable immediately preceding the lexical high (6a); otherwise on the initial syllable (6b). Initial stress is observed in two classes of forms: forms with a lexical high on the word-initial mora and forms that lack a high-toned mora underlyingly. The latter class of forms receives a high on the initial mora by default (Zec and Zsiga, 2010).

(6) NS stress pattern

a. Stress the syllable preceding the syllable that contains the word's only high-toned mora:

/u.sta.nɔ.ʊí.ti/	[u.sta.ˈnɔ.ʊí.ti]	'establish.INF'
/in.sti.túut/	[in.ˈsti.túut]	'institute.NOM.SG'

b. Else stress the initial syllable:

/prá.ʋi.ti/	[ˈprá.ʋi.ti]	'make.INF'
/prɔ.pa.dɛ/	[ˈprɔ́.pa.dɛ]	'fail.AOR.2/3SG'

Batinić (1975) and Milenković and Ryan (to appear) report that tone plays a role in meter. Falling-accented syllables are more robustly avoided in even, weak positions

[2]See Zsiga and Zec (2013) for evidence for the mora as the tone-bearing unit in Neoštokavian.

than their rising counterparts. This study does not examine the role of tone in poetic word order, since tonally annotating the large metrical corpus proved infeasible. However, the models that predict poetic word order based on other prosodic properties (syllable count, vowel length, and stress position) already achieve excellent predictive performance, suggesting that the role of tone, if any, is marginal at best.

1.2.2 Word order in BCMS

The default, unmarked word order in BCMS is SVO (Progovac, 2005; Urošević et al., 1986), as in (7a). At the same time, the order of clausal constituents is relatively free in the language (modulo clitics; Schütze, 1994; Bošković, 2001), in that permutations of the default SVO order are possible, as in (7b), though these are associated with particular interpretations (e.g. focus and topic).

(7) Default head-initiality in BCMS VPs: *Oni pišu knjigu.* 'They are writing the book.'

a. Default order (VO)

ɔ.ni piː.ʃuː kɲi.gu
3PL.NOM write.PRS.3PL book.ACC.SG

'They are writing the book.'

b. Nondefault order (OV)

ɔ.ni kɲi.gu piː.ʃuː
3PL.NOM book.ACC.SG write.PRS.3PL

'They are WRITING the book.'

The order of adjectives in NPs is somewhat less flexible. Adjectival modifiers are prenominal by default, as (8a) shows. The noncanonical order (noun–modifier), shown in (8b), has a high degree of expressivity, being effectively unavailable in most registers. Exceptions include poetry, persuasive speech, and ecclesiastical texts, which

occasionally allow the noun–modifier order. Effectively, postnominal adjectives serve as an immediate cue that we are in a poetic register.

(8) Adjectival modifiers are prenominal: *ljuta guja* 'angry snake'

a. Unmarked order (modifier–noun)

ʎuː.ta guː.ja
angry.NOM.SG.F snake.NOM.SG

b. Marked order (noun–modifier)

? guː.ja ʎuː.ta
snake.NOM.SG angry.NOM.SG.F

'angry snake'

In nonpoetic registers, possessive pronouns such as *moj-* [mɔj-] 'my' behave exactly like regular adjectives: being invariably prenominal (9a). Postnominal possessives are highly expressive and as such available in poetic language (9b). As we will see towards the end of this chapter, folk poets tend overwhelmingly to use possessive pronouns postnominally, even when phonological factors do not motivate departure from the prose-canonical order. Thus, meter's default order of possessives appears to be postnominal, deviating from the prose default.

(9) Possessive pronouns are prenominal: *moja knjiga* 'my book'

a. Default order (possessive–noun)

mɔ.ja kɲi.ga
my.NOM.SG.F book.NOM.SG

b. Nondefault order (noun–possessive)

? kɲi.ga mɔ.ja
book.NOM.SG my.NOM.SG.F

'my book'

1.2.3 BCMS folk meter

BCMS has a rich oral folk tradition that can be traced back to Common Slavic and, more distantly, Indo-European antiquity (Loma, 2002). The tradition has been abundantly documented between the late 18th and mid 20th centuries. In this dissertation, I focus on the epic decasyllable, the most extensively studied verse form of BCMS oral folk tradition (particularly in Western scholarship). Poems in this oral tradition were performed live by folk poets known as guslars, named after the instrument that accompanied their performance, the gusle. The epic decasyllable is primarily associated with heroic poems (or "masculine;" BCMS *junačke/muške p(j)esme*), but a nonnegligible number of lyric poems (or "feminine;" BCMS *ženske p(j)esme*) are also composed in this meter.

The epic decasyllable is a trochaic meter in which each line consists of ten syllables. The trochaicity of this verse form was first observed by the Serbian philologist Vuk Stefanović Karadžić (hereafter Vuk, per BCMS tradition) in the foreword to the Leipzig edition of his *Serbian Folk Poems* (1824). Drawing a distinction between recitation (*kazivanje*, "saying") and musical performance with the gusle (*pjevanje*, "singing"), Vuk observes that recitation follows the prosody of ordinary language, whereas sung performance adheres to a strict metrical template, with ictus, i.e., prominence on odd-numbered syllables. Furthermore, Vuk discusses the correspondence between ordinary-language prosody (i.e. rhythm) and the metrical template, discussing the metrical patterning of long and short syllables. However, it is clear from his examples (*ibid.*) that his "long" and "short" refer to stressed and unstressed, respectively. In this regard, Vuk's terminology is influenced by Classical metrics, where long vs. short is the operative distinction in terms of prominence (see West, 1987, among many others).

The epic decasyllable is a trochaic pentameter: In scansion, verse lines are organized into five trochaic feet, such that odd positions are strong and even positions are

weak, as Figure 1.2 depicts. A caesura, or break, divides the line into two hemistichs. The first hemistich comprises the initial two feet, while the final three feet form the second hemistich. Hemistich edges are invariably crisp: Prosodic words may not span hemistich boundaries (Karadžić, 1824; Maretić, 1907; Lord, 1960; Jakobson, 1966; Batinić, 1975; Zec, 2008). Although a strain of work has challenged the periodic rhythmic organization of the epic decasyllable, growing evidence supports the trochaic pentameter analysis of the epic decasyllable (see Zec, 2008; Milenković and Ryan for recent discussion).

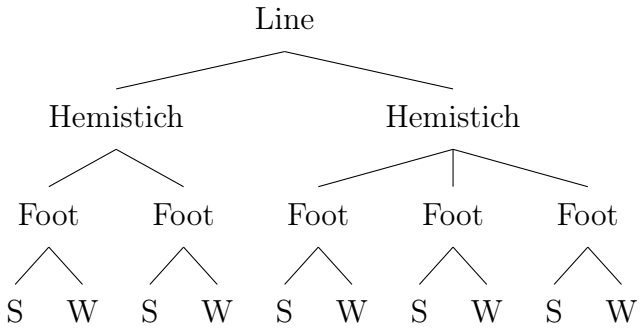


Figure 1.2: Metrical constituency of the line: Epic decasyllable as trochaic pentameter.

A property of the epic decasyllable that directly bears on the study design is its strict regulation of monosyllables. Monosyllabic prosodic words are generally prohibited in weak (even-numbered) positions (Zec, 2008). Given that monosyllables have a life of their own in this verse form and do not appear to be subject to the same phonological constraints as longer words, I exclude them from the present survey. The corpus study examines the pattering of disyllabic and longer words.

Further, phonologically conditioned word-order variation is primarily associated with line-final phrases (Maretić, 1907). Accordingly, the analysis is restricted to word-order variation in the second hemistich (final six positions).

1.2.4 Metrical corpus

1.2.4.1 The corpus

The data for this study come from the second edition of Vuk's collection of BCMS folk poetry (Karadžić, 1841–1862). The corpus includes all poems from this collection that are composed in the epic decasyllable, that is, the entirety of books 2–4, and all poems of book 1 composed in this verse form (book 1 also contains folk poems in other meters). The resulting corpus totals approximately 60,000 verse lines and 230,000 word tokens.

1.2.4.2 Corpus construction and annotation

The annotated corpus was built from plain-text source files. Before annotation, the files were pre-processed. Punctuation was removed, and hash symbols (#) were inserted to mark the boundaries of verse lines. The cleaned files were annotated using Sketch Engine (Kilgarriff et al., 2004), which automated tokenization, lemmatization, and morphological tagging. Figure 1.3 shows the annotated corpus as displayed in the Sketch Engine interface.

Morphological annotation was carried out with a pre-trained part-of-speech tagger, which assigned to each token a part-of-speech label and a bundle of contextual morphosyntactic features: case, number, and (where applicable) gender for nominals, and tense and agreement features for verbs.[3] The tagger was trained on contemporary standard Serbian. This creates a mismatch with the folk epic corpus, which contains archaic and dialectal forms that the tagger is unlikely to have encountered in training. For that reason, tagging accuracy on this corpus was a major concern at the outset.

[3]Agreement features in verbs include person, number, and gender (the latter only in participial forms).

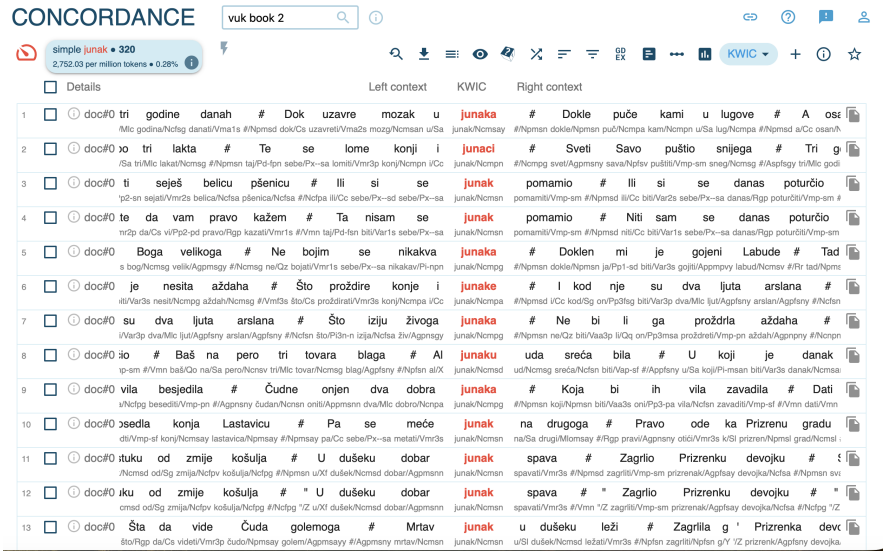


Figure 1.3: A Sketch Engine concordance view of the lemma *junak* 'hero' in the annotated corpus (the second book of Vuk's collection). The figure shows the left and right context of the target word and part-of-speech tags and lemmas beneath each token. Line ends are marked by #.

In practice, the tagger performed reasonably well despite this mismatch. More importantly, the target data for this study were subsequently curated by hand, so that false hits and other annotation errors were excluded before analysis. An example of a false hit for the noun–adjective construction is provided in (10). The tagger misidentifies the dialectal GEN.PL form of a noun as an adjective.[4] The results of the corpus study are therefore not affected by annotation errors.

(10) ˈkɔ.lɔ ʤɛ.ˈʊɔ.jaː.kaːx
coven girl.GEN.PL
'a coven of girls'

To give another example of an annotation error, consider (11). In spelling, the

[4]The GEN.PL form ʤɛ.ˈʊɔ.jaː.kaːx contains the dialectal case ending -aːx, found only in the Zeta-Sjenica dialect (Ivić, 1958, 1985, 2009). The corresponding ending in standard BCMS is -aː. This ending is largely restricted to poems in the fourth book of Vuk's collection, which contains poems from the parts of Montenegro where this dialect is spoken. This is therefore a patterned, rather than random, annotation error.

form *dvori* can represent two homographs: the NOM.PL form of the noun *dvor* 'castle': [ˈdʋɔː.ri],[5] and the PRS.3SG form of the verb *dvoriti* 'host': [ˈdʋɔ.riː]. In (11), *dvori* is a noun modified by a possessive genitive ('my Peter's castle'). The tagger erroneously identifies *moga Petra dvori* as an inverted VP: '(S)he is hosting my Peter.'

(11) ˈpɔ.ʎɛp.ʃiː =su ‖ ˈmɔː.ga ˈpɛ.tra ˈdʋɔː.ri
beautiful.COMP.NOM.PL.M are my.GEN.SG.M Petar.GEN castle.NOM.PL
'My Peter's castle is more beautiful.' (I, 4:4)

1.2.4.3 Data availability

User-created corpora in Sketch Engine are private and cannot be made publicly available via the platform. The annotated corpus is, however, publicly available on Harvard dataverse (https://doi.org/10.7910/DVN/JQPVRU),[6], as a collection of tab-delimited plain-text files downloaded from Sketch Engine, whose format is shown in Figure 1.4. These files can be downloaded and searched in Python or R.

doc_id	verse_line	token_pos	word	tag	lempos	lemma
file36696431	1	1	Na	Sa	na-s	na
file36696431	1	2	daleko	Rgp	daleko-r	daleko
file36696431	1	3	lijepu	Agpfsay	lep-a	lijepa
file36696431	1	4	đevojku	Ncfsa	đevojka-n	đevojka
file36696431	2	1	Baš	Qo	baš-q	baš
file36696431	2	2	đevojku	Ncfsa	đevojka-n	đevojka
file36696431	2	3	prelijepu	Agpnsdn	prelep-a	preprelijep
file36696431	2	4	Anu	Npfsa	ana-n	ana

Figure 1.4: An excerpt from the tab-delimited plain-text format of the annotated corpus (the first book of Vuk's collection), available on Harvard Dataverse. The figure shows one token per row, with columns for document ID, verse-line number, token position in the line, word form, part-of-speech tag, part of speech, and lemma.

1.2.4.4 Data extraction and filtering

Once the corpus was built, I retrieved all instances of line-final NPs, VPs, and coordinate constructions using regular expressions. From this initial dataset, I excluded

[5]Though note that the translation in (11) is singular.

[6]This repository also contains all supplementary materials and replication data for this dissertation.

false hits, which fall into two categories: tagging errors (discussed above) and noun–adjective sequences involving secondary predication. Out of context, every adjective in a noun-adjective sequence like [ˈdʊɔː.ri bi.ˈjɛ.li] ‘palaces white’ is ambiguous between an inverted adjectival modifier and a secondary predicate. Given that the annotation encodes only morphological information, not syntactic constituency, secondary-predicate configurations could not be filtered out automatically at the query stage. Disambiguation had to rely on the surrounding textual context and was carried out manually.

The final dataset used for modeling in Chapters 2–4 comprises 15,397 line-final NPs, VPs, and coordinate constructions.

1.2.4.5 Prosodic annotation

The target data were prosodically annotated. This process was only partially automated: Syllable count was added for each token using an R script (R Core Team, 2023). Syllabic *r* required special attention, since syllabic and non-syllabic *r* are not distinguished orthographically. I annotated syllabic *r* with the symbol ∗, which the syllable-counting script treats as a vocalic segment. Other prosodic features (vowel length and stress position) were coded manually by the author, following Vuk’s *Serbian Dictionary* (1852). Vuk’s dictionary was chosen over contemporary BCMS dictionaries because it is diachronically closer to the BCMS dialects spoken by the folk poets whose work Vuk documented, and because it includes many epic-specific lemmata which are absent from contemporary dictionaries. Whenever a word’s contemporary stress pattern differs from the one given in Vuk’s dictionary, I adhered to Vuk’s stress pattern, as in *sokola* ‘falcon.GEN/ACC.SG’: contemporary [ˈsɔ.kɔ.la], Vuk [sɔ.ˈkɔ.la].

Another annotation decision that potentially impacts prosodic analysis concerns the coding of definiteness in adjectival forms. As in other Slavic languages, BCMS distinguishes “short,” indefinite adjective forms from “long,” definite forms. In some

case forms, the two forms are morphologically distinct, as in (12).

(12) a. Short, indefinite form

bi.ˈjɛ.l-a ˈdaː.na

‘of a bright day’

b. Long, definite form

ˈbi.jɛ.l-ɔː.ga ˈdaː.na

‘of the bright day’

However, in forms like the feminine NOM.SG (13), the indefinite and definite forms are segmentally identical and therefore indistinguishable in writing. In (13), the only difference resides in the length of the final vowel, which is long in the definite form. When context and syntactic environment did not disambiguate a given adjectival form, I coded it as indefinite by default.

(13) a. Short, indefinite form

ˈʎu.t-a ˈguː.ja

‘an angry snake’

b. Long, definite form

ˈʎu.t-aː ˈguː.ja

‘the angry snake’

1.3 Variable word order in BCMS folk meter

This section examines variable word order in BCMS folk meter. NPs and VPs vacillate between canonical and noncanonical orders, as illustrated in (14–15). The syntactic variants in (14–15) do not convey different meanings in the folk epic.

(14) Variable adjective–noun order: *careve dvore* [ˈtsa.rɛ.ʊɛ ˈdʊɔː.rɛ] ‘tsar’s palace’

a. Canonical order: Adj N

ɔd.nɛː.t͡ɕu =ga ‖ u= ˈtsa.rɛ.ʋɛ ˈdʋɔː.rɛ
carry.FUT.1SG 3SG.M.ACC in tsar's palace.ACC.PL

'I will carry him to tsar's palace.' (II; 29:118)

b. Noncanonical order: N Adj

ɔd.nɛ.sɛ =ga ‖ u= ˈdʋɔː.rɛ ˈtsa.rɛ.ʋɛ
carry.AOR.3SG 3SG.M.ACC in palace.ACC.PL tsar's

'He carried him to tsar's palace.' (II; 29:163)

(15) Variable verb–object order: *izgubiti glavu* [iz.ˈgu.bi.ti ˈglaː.ʋu] 'lose (one's) head'

a. Canonical order: VO

jaː =t͡ɕu sʋɔ.ju ‖ iz.ˈgu.bi.ti ˈglaː.ʋu
1SG.NOM will.1SG self's lose.INF head.ACC

'I will lose my head (i.e. die).' (IV, 57: 559)

b. Noncanonical order: OV

da= t͡ɕɛ= a.gi ‖ ˈglaː.ʋu iz.ˈgu.bi.ti
that will.3PL agha.DAT head.ACC lose.INF

'That they will lose the agha's head (i.e. kill him).' (IV, 57: 74)

Since Vuk collected folk poetry from different regional varieties of BCMS and from many folk poets, one might wonder whether the observed word-order variation is a by-product of aggregating data across different near-categorical (regional or individual) poetic grammars. Data like (14–15) argue against the view that individual grammars are invariably deterministic. Both line pairs display syntactic variation in the same poem and, moreover, with the same lexical items, suggesting that variation is speaker-internal and the metrical grammar is fundamentally stochastic.

The data for this study include 15,397 NPs, VPs, and binomials from the corpus described in §1.2.4. NPs constitute the largest portion of the data. Adjective–noun and

possessive–noun bigrams are analyzed separately, given that the two constructions show different word-order preferences in meter despite their identical patterning in modern BCMS. Figure 1.5 plots aggregate rates of word-order alternatives by construction type. The binomial bar is hatched, since binomials do not have a syntactically determined canonical order (though phonology and semantics may skew binomial order; see §1.3.2.2).

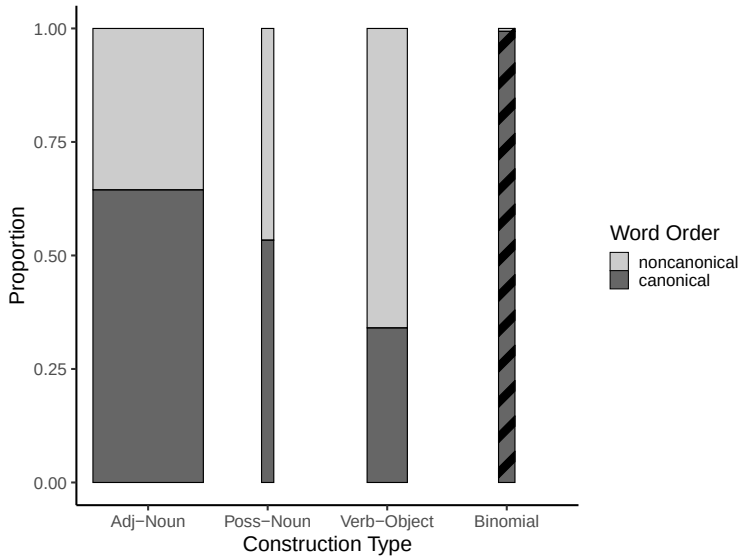


Figure 1.5: Word-order distribution by construction type. Bar thickness is proportional to sample size.

The discussion in this section proceeds as follows. In §1.3.1, I introduce the phonological constraints that modulate poetic word order using adjective–noun data, in which, thanks to the large sample size, all phonological environments are populated with ample observations. In §1.3.2, I demonstrate that syntactic variation in other construction types follows the same phonological regularities as in NPs with adjectival modifiers. In §1.4, I analyze the corpus data using mixed effects logistic regression.

1.3.1 Adjective–noun order

Adjectival modifiers are prenominal in BCMS, as *ljuta guja* 'angry snake' in (8a) shows. In folk meter, NPs may undergo inversion, placing the modifier after the head noun: *guja ljuta* 'snake angry.' Deviations from canonical order are not random. First, the noncanonical order is positionally restricted, being available only line-finally. Inversion rarely, if ever, targets line-internal NPs. Second, folk poets employ the syntactically marked order (noun–adjective) to satisfy phonological markedness constraints. Inversion is therefore phonologically motivated.

When phonological factors are held constant, the canonical order is strongly preferred to noun–adjective. I formalize the syntactic pressure to preserve the canonical order as a violable constraint, which I define in (16).

(16) AdjectiveFirst

Adjectival modifiers are prenominal. Assign a violation for every postnominal adjectival modifier.

The probability that the modifier follows the head noun depends on the prosodic properties of the two constituents, such as their relative size, prosodic weight,[7] and stress position. All relevant prosodic factors have precedents in the literature. In particular, the effect of weight on word order in the epic decasyllable was first observed by Maretić (1907) and has since been discussed extensively (Jakobson, 1966; Batinić, 1975; Ružić, 1975; Foley, 1993; Zec, 2008). The role of word size has likewise received some attention (e.g., Foley, 1993, 99).

Figure 1.6 visualizes inversion propensity, that is, the proportion of NPs with a postnominal modifier (as in *guja ljuta* 'snake angry'), as a function of their size (syllable count) and weight profile. For the latter effect, the meter tracks only the weight

[7]Codas do not contribute weight in BCMS; only CVV syllables are heavy (Zec, 2000).

of penultimate syllables (Maretić, 1907; Zec, 2008). For reasons discussed below, the meter prefers to place words with a heavy penult in line-final position. Inversion is therefore most likely when the modifier's penult is heavy and the head's is light (top-right panel), since this satisfies the weight-mapping constraint. The reverse configuration (bottom-left) resists inversion more than any other weight condition. The weight factor is moot in the top-left and bottom-right panels, since the modifier and the head have the same penult weight.

The second hemistich consists of six metrical positions. The head noun and the modifier may therefore occupy six positions at most in aggregate. Empty cells in Figure 1.6 represent metrically impossible word combinations, which exceed the six-syllable limit. Monosyllables are excluded, so each NP member can be minimally disyllabic and maximally quadrisyllabic. Disyllables freely combine with all other admissible word sizes, trisyllables pair with disyllables and trisyllables, and quadrisyllables only pair with disyllables.

1.3.1.1 End-Weight

The effect of word size is consistent across all weight profiles. The likelihood of the noncanonical, modifier-final order is inversely correlated with the head noun size, and directly correlated with the modifier size. Within each weight profile in Figure 1.6, inversion rates are highest in the upper-left region, where the modifier is syllabically longer than the head ($4\sigma \times 2\sigma$),[8] as in *bijeloga grada* 'white city' (18b). As head size increases (moving rightward along the horizontal axis), the rates of modifier-finality gradually decrease, reaching their lowest points in the lower-right region, where the head is longer than the modifier ($2\sigma \times 4\sigma$), as in *dragi gospodaru* 'dear master' (18c).

[8]Hereon, in $x\sigma \times y\sigma$, $x\sigma$ is the syllable count of the dependent, and $y\sigma$ is the syllable count of the head.

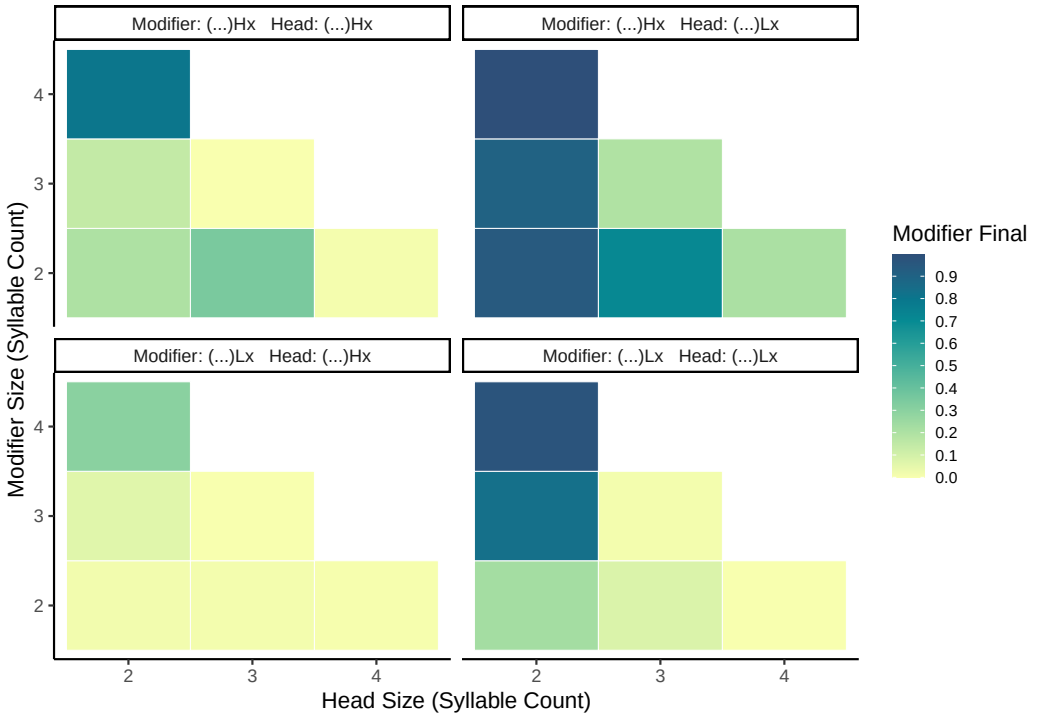


Figure 1.6: Proportion of NPs that exhibit noncanonical (noun–modifier) order by head size (horizontal axis), modifier size (vertical axis), and weight profile (facets; H = heavy penult, L = light penult, x = final syllable (any weight)). Darker colors indicate higher inversion rates. Empty cells correspond to metrically impossible word size combinations.

Along the diagonal of each panel, the relative size effect is moot, since both NP members are of equal size. I address the consistently lower inversion rates in $3\sigma \times 3\sigma$ relative to the corresponding $2\sigma \times 2\sigma$ conditions in §1.3.1.4.

The word-size effect on adjective–noun order is an instantiation of PROSODIC END-WEIGHT, a preference for longer words to occupy final positions in larger phrasal domains (here, verse lines) (Behaghel, 1909; Malkiel, 1959; Bever, 1970; Quirk, 1972; Cooper and Ross, 1975; Pinker and Birdsong, 1979; Wasow, 1997a,b, 2002; Eitelmann, 2016; Ryan, 2019a).[9] Cross-linguistically, end-weight effects show sensitivity to a range

[9]Known in the literature by a number of alternative names, such as the LAW OF GROWING MEMBERS (Behaghel, 1909), the LONG-LAST PRINCIPLE (Hayes and MacEachern, 1998), or the SHORT-BEFORE-LONG PREFERENCE (Benor and Levy, 2006).

of factors, such as syllable count, vowel length and sonority, and margin complexity. In BCMS folk meter, the only factor contributing to end-weight propensity is a word's syllable count.[10] I formalize the end-weight propensity as a violable constraint in (17); this definition follows Benor and Levy (2006, 242).

(17) End-Weight

Assign a violation for every line whose final prosodic word is preceded by a syllabically longer prosodic word within the second hemistich.

Effectively, End-Weight tracks the syllable count of the line-final prosodic word and compares it to the syllable counts of other prosodic words in the second hemistich (in practice, one or two preceding prosodic words). The constraint penalizes a line-final word that is shorter than a preceding word, but does not require it to be longer than the preceding word(s). End-Weight is defined here as a categorical constraint (in the sense of McCarthy, 2003b), primarily for simplicity. The gradient nature of end-weight effects can be attributed, at least in part, to an independent alignment constraint (§1.3.1.3).

Much of the variation observed in Figure 1.6 can be attributed to prosodic end-weight, whose role in the meter is immense (the regression and MaxEnt models further bear this out). The data in tableau (18a) provide a baseline for word-order distribution in meter, isolating the effect of syntax (AdjFirst). With syllable count and penultimate weight both controlled for, the meter strongly prefers the canonical order (adjective–noun) to inversion. When the adjective is longer than the head noun, as in (18b), inversion is preferred, that is, AdjFirst gives way to End-Weight. Conversely, when the head noun is longer than the modifier, as in (18c), AdjFirst and

[10]For the role of prosodic end-weight in other metrical traditions, see Hayes and MacEachern (1998); Hayes and Minkova (2023); Ryan (2019b). For the opposite, "short-last" structures in meter, see Blumenfeld (2016).

End-Weight jointly favor the canonical order, which, being violation-free, approaches the ceiling. The observed frequencies reported in each tableau are not specific to the bigram shown; the bigram is an illustration of the relevant prosodic configuration. Frequencies represent proportions computed from aggregated counts across all bigrams in the corpus with the requisite prosodic properties.

(18) a. *ljuta guja* [ˈʎuː.ta ˈguː.ja] 'angry snake'

	End-W	AdjFirst	p
Adj N ˈʎuː.ta ˈguː.ja			0.80
N Adj ˈguː.ja ˈʎuː.ta		−1	0.20

b. *bijeloga grada* [ˈbi.jɛ.lɔː.ga ˈgraː.da] 'white city.GEN'

	End-W	AdjFirst	p
Adj N ˈbi.jɛ.lɔː.ga ˈgraː.da	−1		0.20
N Adj ˈgraː.da ˈbi.jɛ.lɔː.ga		−1	0.80

c. *dragi gospodaru* [ˈdraː.giː ˈgɔ.spɔ.daː.ru] 'dear master.VOC'

	End-W	AdjFirst	p
Adj N ˈdraː.giː ˈgɔ.spɔ.daː.ru			0.98
N Adj ˈgɔ.spɔ.daː.ru ˈdraː.giː	−1	−1	0.02

1.3.1.2 Weight mapping in the final foot

Another factor at play in Figure 1.6 is prosodic weight. As originally discovered by Tomo Maretić (1901; 1907), the ninth position of the line (the final strong position; see Figure 1.2) is preferably assigned a heavy syllable. Accordingly, the meter favors word orders that place a heavy syllable in the ninth position. Because monosyllables are banned from the tenth position (Zec, 2008), the ninth position is always occupied by the penultimate syllable of the line-final word. This explains why the weight mapping

effect tracks only the weight of penults.

The attraction of heavy syllables to the final strong position is driven by the constraint in (19).

(19) *Light/FinalStrong (*L/FinS)

Assess a violation for every light syllable assigned to the final strong position.

*L/FinS is a variant of the weight-mapping constraint *Light/Strong, known from Classical quantitative meters, which penalizes light syllables in strong positions (Hanson and Kiparsky, 1996; Ryan, 2017b; Kiparsky, 2020). In BCMS folk meter, weight alone (i.e., with no recourse to stress) is strictly regulated only in the ninth position (Jakobson, 1966; Zec, 2008), as Figure 1.7 further substantiates.

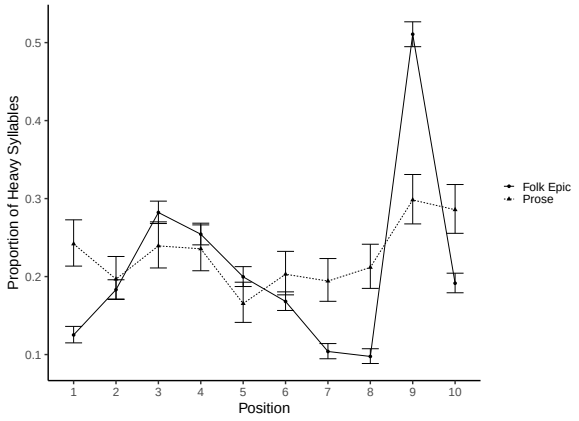


Figure 1.7: BCMS folk meter regulates weight in the ninth position, but not elsewhere in the line. The plot shows the proportion of heavy syllables (vertical axis) by position (horizontal axis) in folk meter (3,771 randomly selected lines; solid line) against Milenković and Ryan's (2026) prose sample (dashed line); error bars represent Wilson confidence intervals. In quantitative meters, one expects a zigzag pattern throughout the line, with peaks in strong and dips in weak positions.

The tableaux in (20) show *Light/FinalStrong at work (unconfounded by end-weight). Tableau (20a), repeated from (18a), is the baseline: Both NP members have heavy penults, such that *Light/FinalStrong is uniformly satisfied. In (20b),

*Light/FinalStrong favors inversion over canonical order, which results in a high inversion rate.[11] Tableau (20c) shows the opposite of (20b): Placing the modifier after the head is less metrical than the canonical order. Syntax and phonology therefore jointly militate against inversion, which is almost categorically inhibited.

(20) a. *ljuta guja* [ˈʎuː.ta ˈguː.ja] 'angry snake'

	*L/FinS	AdjFirst	p
Adj N ˈʎuː.ta ˈguː.ja			0.80
N Adj ˈguː.ja ˈʎuː.ta		−1	0.20

b. *ravno polje* [ˈraːʋ.nɔ ˈpɔ.ʎɛ] 'flat field'

	*L/FinS	AdjFirst	p
Adj N ˈraːʋ.nɔ ˈpɔ.ʎɛ	−1		0.06
N Adj ˈpɔ.ʎɛ ˈraːʋ.nɔ		−1	0.94

c. *mila majka* [ˈmi.la ˈmaːj.ka] 'beloved mother'

	*L/FinS	AdjFirst	p
Adj N ˈmi.la ˈmaːj.ka			0.98
N Adj ˈmaːj.ka ˈmi.la	−1	−1	0.02

1.3.1.3 Alignment

The patterning of trisyllables is somewhat exceptional, calling for two additional metrical constraints: Align(ω_{max}, R, Ft, R) and TrisyllableInertia. I start by introducing the former.

Recall from Figure 1.6 that modifier size and inversion likelihood are positively correlated: Controlling for the size of the head noun, the longer the modifier, the

[11] *L/FinS is not the sole driving force behind inversion in (20b); another contributing factor is avoidance of stressed lights in position 9 (§1.3.1.5).

higher the probability of modifier-finality. As a result, we expect a consistent gradient along the left vertical: Modifier-finality rate should increase steadily going from the origin ($2\sigma \times 2\sigma$) to the upper-left region ($2\sigma \times 4\sigma$).

However, the top two panels in Figure 1.6 depart from the expected gradient. In these weight profiles, the $3\sigma \times 2\sigma$ conditions show lower modifier-finality rates than the corresponding $2\sigma \times 2\sigma$ conditions. This is completely unexpected from End-Weight's perspective, given that inversion is slightly more frequent in $2\sigma \times 2\sigma$, where End-Weight is moot, than in $3\sigma \times 2\sigma$, where End-Weight directly benefits from it. I illustrate this puzzling exception in (21–22).

(21) *ljuta guja* [ʎuː.ta guː.ja] 'angry snake' ($2\sigma \times 2\sigma$)

a. Adj N: ˈʎuː.ta ˈguː.ja 0.80

b. N Adj: ˈguː.ja ˈʎuː.ta 0.20

(22) *jarkoga sunca* [ˈjaːr.kɔː.ga ˈsuːn.tsa] 'bright sun.GEN' ($3\sigma \times 2\sigma$)

a. Adj N: ˈjaːr.kɔː.ga ˈsuːn.tsa 0.85

b. N Adj: ˈsuːn.tsa ˈjaːr.kɔː.ga 0.15

In the same vein, recall that, holding modifier size constant, the size of the head noun and inversion likelihood are inversely correlated: The longer the head noun, the less likely the noncanonical, modifier-final order. This correlation manifests itself in the bottom horizontal. Moving from the origin ($2\sigma \times 2\sigma$) to the bottom-right region ($2\sigma \times 4\sigma$), one expects the rate of modifier-finality to decline monotonically, as the size of the head noun increases.

The gradient holds in three out of four weight profiles, the sole exception being the top-left panel in Figure 1.6. In this exceptional case, the $2\sigma \times 3\sigma$ condition undergoes inversion more frequently than the corresponding $2\sigma \times 2\sigma$ condition (Fisher's exact test OR = 2.11, $p < 0.0001$). This pattern contravenes End-Weight, since the likelihood

of moving a disyllable over a trisyllable (in $2\sigma \times 3\sigma$) is higher than the likelihood of moving a disyllable over a disyllable (in $2\sigma \times 2\sigma$), as the data in (23–24) indicate.

(23) *ljuta guja* [ˈʎuː.ta ˈguː.ja] 'angry snake' ($2\sigma \times 2\sigma$)

a.	Adj N:	ˈʎuː.ta ˈguː.ja	0.80
b.	N Adj:	ˈguː.ja ˈʎuː.ta	0.20

(24) *mlada slavuja* [ˈmlaː.da sla.ˈʋuː.ja] 'young nightingale.GEN' ($2\sigma \times 3\sigma$)

a.	Adj N:	ˈmlaː.da sla.ˈʋuː.ja	0.65
b.	N Adj:	sla.ˈʋuː.ja ˈmlaː.da	0.35

The unified explanation for both exceptions is that the orders which place a trisyllable in final position are less frequent than END-WEIGHT predicts. Examining the word-size distribution in a sample of poems from Vuk's collection, Ružić (1975, 143–146) reports that lines with a trisyllable in final position are underrepresented in meter, being less frequent than lines ending in a disyllable, and especially than lines ending in a quadrisyllable. This dispreference for final trisyllables explains both departures from the general END-WEIGHT-driven gradient in Figure 1.6.

What, then, drives this trisyllable-final markedness? I analyze final trisyllable avoidance as an alignment effect (McCarthy and Prince, 1993). The alignment constraint in question, defined in (25), dictates that the right edge of a prosodic word must coincide with the right edge of a foot, penalizing strong positions which contain a word-final syllable. Line-final trisyllables are unproblematic themselves, because their right edge mechanically coincides with the end of the line. However, when a trisyllable is line-final, the preceding prosodic word ends in position 7, the strong position of the fourth foot, which results in an edge misalignment.

(25) ALIGN(ω_{max}, R, Ft, R)

In a foot $Ft(S, W)$, S and W must belong to the same maximal prosodic word, ω_{max}.

Somewhat paradoxically, binomials show a marked tendency to place trisyllables in line-final position (Maretić, 1907; Ružić, 1975), as (26) shows.

(26) *zlata i bisera* ['zlaː.ta i='bi.sɛ.ra] 'gold.GEN and pearl.GEN'

a. $2\sigma - 3\sigma$: 'zlaː.ta i='bi.sɛ.ra 0.97

b. $3\sigma - 2\sigma$: 'bi.sɛ.ra i='zlaː.ta 0.03

Unlike NPs, final trisyllables in binomials are not preceded by the right edge of another prosodic word, but by the proclitic conjunction *i* 'and,' which forms a recursive prosodic word (Selkirk, 1996) with the following trisyllable. As a result, the line-final maximal prosodic word is quadrisyllabic. If the disyllabic conjunct is final, as in (26b), it hosts the proclitic and forms with it a trisyllabic prosodic word, the very configuration penalized by ALIGN.[12]

Thus, the two seemingly contradictory patterns, namely the strong preference for final trisyllables in binomials and their avoidance in NPs, fall out from the same alignment constraint. The effect of edge alignment is illustrated in Figure 1.8. In each of the four diagrams, the structure above the melodic tier is the prosodic structure of the linguistic material. The structure below the melodic tier represents the metrical template. Metrical constraints govern the mapping of prosodic structure onto the metrical template. ALIGN governs one aspect of this mapping: the correspondence between the edges of ω_{max} and metrical feet (see Hayes et al., 2012 on the role of edge alignment in meter).

The tableaux in (27) show the full violation profiles of the structures in Figure 1.8.

[12] ALIGN has the added benefit of potentially explaining why monosyllables are avoided in weak positions (Zec, 2008): If a monosyllable occupies a weak position, the preceding prosodic word's right edge occurs foot-internally.

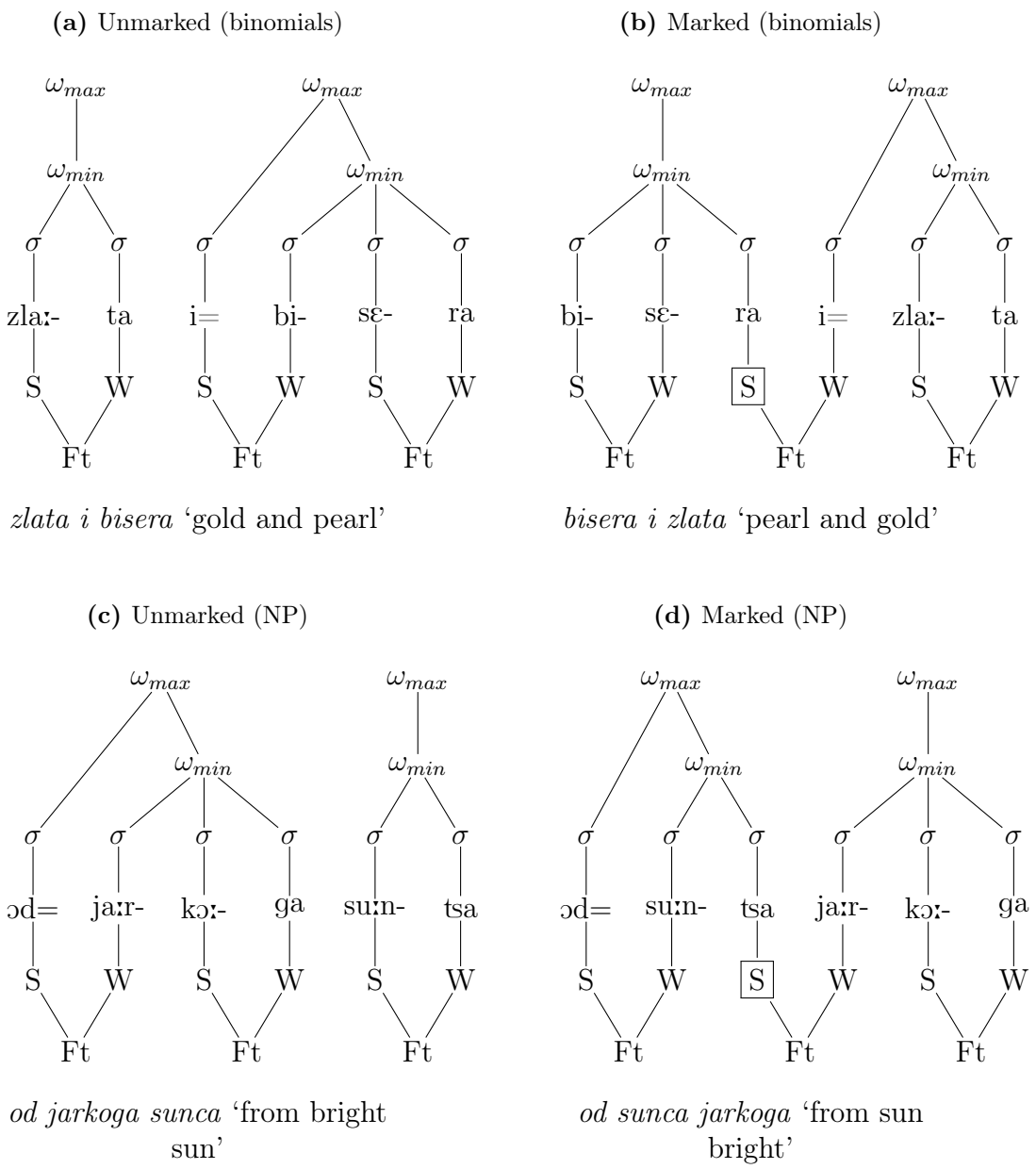


Figure 1.8: Edge alignment markedness in binomials (**a**–**b**) and NPs (**c**–**d**). Strong positions filled by ω_{max}-final syllables incur ALIGN penalties (marked with boxes). Structures **(b)** and **(d)** are more marked than their counterparts in **(a)** and **(c)**, respectively.

(27) Dispreference for prosodic word-foot misalignment

a. *zlata i bisera* ['zlaː.ta i='bi.sɛ.ra] 'gold and pearl'

	End-W	Align	*L/FinS	AdjFirst	p
$2\sigma + 3\sigma$ 'zlaː.ta i='bi.sɛ.ra			-1		0.97
$3\sigma + 2\sigma$ 'bi.sɛ.ra i='zlaː.ta	-1	-1			0.03

b. *od jarkoga sunca* [ɔd='jaːr.kɔː.ga 'suːn.tsa] 'of bright sun'

	End-W	Align	*L/FinS	AdjFirst	p
Adj N ɔd='jaːr.kɔː.ga 'suːn.tsa	-1				0.85
N Adj ɔd='suːn.tsa 'jaːr.kɔː.ga		-1		-1	0.15

In (27a), the preferred binomial order is $2\sigma + 3\sigma$. This candidate violates *L/FinS but is nonetheless chosen near-categorically because the alternative order violates both End-Weight and Align. In (27b), Align and AdjFirst gang up (in the sense of Jäger and Rosenbach, 2006) against End-Weight: The violator of End-Weight (adjective–noun) is strongly preferred to the doubly violating adjective-final alternative.

In purely metrical terms, the alignment effect detected in this section can be understood as a bridge effect (or zeugma). Jakobson (1966) identifies bridge effects in the final foot of each hemistich: positions 3–4 and positions 9–10 must each belong to the same word unit, which explains the prohibition on monosyllables in positions 4 and 10 (see also Zec, 2008). The dispreference for line-final trisyllables can similarly be attributed to a bridge effect in the fourth foot, which operates as a tendency: Positions 7 and 8 preferentially belong to the same word unit. A line-final trisyllable necessarily begins in position 8, placing a word boundary between positions 7 and 8 and thereby

contravening the fourth-foot zeugma.

1.3.1.4 Trisyllable inertia

Consider the difference between (28) and (29). In (28), the modifier and the head are both disyllabic, whereas in (29), both NP members are trisyllabic. END-WEIGHT, which is sensitive to relative, not absolute, word size, fails to discriminate between the two environments. Nevertheless, these environments show different inversion propensities: Trisyllables (29) resist inversion more than disyllables (28). In other words, while the relative size effect is moot, *absolute* word size appears to play a role.

(28) *ljuta guja* [ˈʎuː.ta ˈguː.ja] 'angry snake' ($2\sigma \times 2\sigma$)

a. Adj N: ˈʎuː.ta ˈguː.ja 0.80

b. N Adj: ˈguː.ja ˈʎuː.ta 0.20

(29) *hrišćanska zlamenja* [ˈxri.ʃʧaːn.skaː ˈzla.mɛː.ɲa] 'Christian insignia' ($3\sigma \times 3\sigma$)

a. Adj N: ˈxri.ʃʧaːn.skaː ˈzla.mɛː.ɲa 1.00

b. N Adj: ˈzla.mɛː.ɲa ˈxri.ʃʧaːn.skaː 0.00

No constraint introduced thus far accounts for the different inversion propensities in $2\sigma \times 2\sigma$ (28) vs. $3\sigma \times 3\sigma$ (29). Within each environment, the modifier and the head noun are identical along all relevant prosodic dimensions: syllable count, weight, and stress (all items in 28 and 29 have initial stress). Similarly, ALIGN fails to discriminate between canonical and noncanonical orders, since in $2\sigma \times 2\sigma$, both disyllable orders are ALIGN-compliant, and in $3\sigma \times 3\sigma$, the word boundary falls within a foot irrespective of how the two trisyllables are arranged. The asymmetry appears to be based solely on the absolute word size.

The contrast between (28) and (29) is not an isolated pattern. As Figure 1.9 shows, across all weight profiles and construction types (including now VPs and possessives),

the $3\sigma \times 3\sigma$ environments consistently exhibit lower proportions of noncanonical order than the corresponding $2\sigma \times 2\sigma$ environments (modulo the attenuation effect in the bottom-left panel, where inversion rates are pushed to the floor by the weight mapping constraints). Although the directionality is constant, the strength of this effect is later shown to vary by construction (§1.4.3.3).

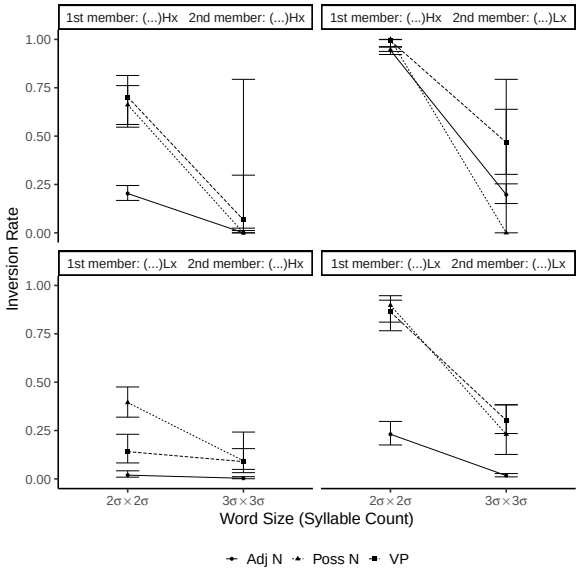


Figure 1.9: Trisyllable inertia: In each weight profile (panels) and construction type (line types), XPs whose both members are trisyllabic are less likely to deviate from canonical order than their disyllabic counterparts.

The constraint responsible for this effect, dubbed TrisyllableInertia (3σInert), is perplexing: Its sole purpose is to prevent trisyllables from inverting in the presence of another trisyllable. Trisyllables may invert when the head is disyllabic (subject to an Align penalty), showing that the constraint is not *MoveTrisyllable. Likewise, disyllables frequently invert over trisyllables (§1.3.1.3), which indicates that the constraint is also not *MoveOverTrisyllable. Puzzling as it is, the effect is statistically robust, as Figure 1.9 indicates and the logistic and MaxEnt models further confirm. The fact that all weight profiles and construction types show the same down-

ward slope is particularly telling, as it suggests the effect is not an artifact of any single prosodic environment. An explanatory account of trisyllable inertia must await future research.[13]

1.3.1.5 Stress-weight interaction

In addition to the general avoidance of light syllables in the ninth position, stressed light syllables are avoided in this position even more strongly. If a stressed syllable is assigned to the final strong position, it is preferably heavy (Jakobson, 1966). The effect is robust: In a random sample of 3,771 lines analyzed by Milenković and Ryan (2026), 84% of stressed syllables occupying position 9 are heavy (the grand mean is 29%). The effect is driven by the constraint in (30).

(30) *StressedLight/FinalStrong (*'L/FinS)

Assess a violation for every stressed light syllable assigned to the final strong position.

Note that *'L/FinS and *L/FinS are in a stringency relation: Every violation of the former is a violation of the latter but not vice versa. Stressed lights are therefore more marked than their unstressed counterparts in the final strong position, as they violate both weight-mapping constraints. Unstressed lights in position 9 are penalized only by *L/FinS. By contrast, heavy syllables do not incur any penalties in the ninth position, regardless of whether they are stressed or not.

*'L/FinS's effect on adjective order is isolated in (31–32). The two environments differ only in the position of stress in the trisyllabic modifier: antepenultimate in (31) and penultimate in (32).

[13] I suspect that XPs with longer constituents (e.g. quadrisyllabic adjective and quadrisyllabic noun) may well exhibit inertia, but this effect cannot be probed in the second hemistich given its limited size.

(31) *u careve dvore* [u=ˈtsa.rɛ.ʊɛ ˈdʊɔː.rɛ] ‘in tsar’s castle’

a. Adj N: u=ˈtsa.rɛ.ʊɛ ˈdʊɔː.rɛ 0.93

b. N Adj: u=ˈdʊɔː.rɛ ˈtsa.rɛ.ʊɛ 0.07

(32) *i bijela dana* [i=bi.ˈjɛ.la ˈdaː.na] ‘and white day.GEN’

a. Adj N: i=bi.ˈjɛ.la ˈdaː.na 1.00

b. N Adj: i=ˈdaː.na bi.ˈjɛ.la 0.00

In both environments, the canonical order is strongly preferred to the modifier-final alternative. The syntactic constraint AdjFirst and several phonological constraints gang up against End-Weight. Inversion satisfies End-Weight, but violates Adj-First along with at least two phonological constraints: Align and *L/FinS. In (32), inversion additionally results in a *stressed* light syllable in the ninth position, incurring a further *ˈL/FinS violation on top of the penalties shared with (31); *ˈL/FinS therefore predicts inversion to be less likely in (32) than in (31). This prediction is borne out: Inversion is significantly more likely in (31) than in (32) (Fisher’s exact test, OR = 10.50, $p = 0.001$). The tableaux in (33) illustrate the additive effect of *ˈL/FinS.

(33) Stress effect in light penults

a. *u careve dvore* [u=ˈtsa.rɛ.ʊɛ ˈdʊɔː.rɛ] ‘in tsar’s castle’

	End-W	Align	*L/FinS	*ˈL/FinS	AdjFirst	p
Adj N u=ˈtsa.rɛ.ʊɛ ˈdʊɔː.rɛ	−1					0.93
N Adj u=ˈdʊɔː.rɛ ˈtsa.rɛ.ʊɛ		−1	−1		−1	0.07

b. *i bijela dana* [i=bi.ˈjɛ.la ˈdaː.na] ‘and white day.GEN’

	End-W	Align	*L/FinS	*'L/FinS	AdjFirst	p
Adj N i=bi.ˈjɛ.la ˈdaː.na	−1					1.00
N Adj i=ˈdaː.na bi.ˈjɛ.la		−1	−1	−1	−1	0.00

The minimally distinct environments which isolate the effect of *'L/FinS (like 33) are rare because polysyllabic forms with stressed light penults are uncommon in BCMS. In our random sample, only 17% of trisyllables have penultimate stress, approximately half of which are light. Besides lexical statistics, Simonović and Kager (2020) document an ongoing process which shifts prominence to stem-initial syllables in some BCMS verbal paradigms.[14] This shift further depletes the already sparse pool of polysyllables with penultimate stress. Despite the sparsity of the relevant items, the logistic and MaxEnt models support *'L/FinS's inclusion in the grammar.

1.3.2 Other construction types

In this section, I analyze the phonological effects on word order in possessive–noun pairs, VPs, and coordinate constructions (or binomials). The crucial takeaway is that no additional phonological constraints are needed to capture syntactic variation in these construction types.

Inversion rates are higher across the board in possessives and VPs than in the corresponding adjective–noun conditions, as I show in §1.3.2.1. I capture the construction-specific skews by introducing two additional linearization constraints, defined in (34).

(34) a. VerbFirst

VPs are head-initial (see 7). Assign a violation for every head-final VP.

[14] E.g. *stojimo* 'stand.prs.1pl;' older [stɔ.ˈjiː.mɔ] → [ˈstɔ.jiː.mɔ].

b. PossessiveFirst (PossFirst)

Possessives are prenominal (see 9). Assign a violation for every postnominal possessive.

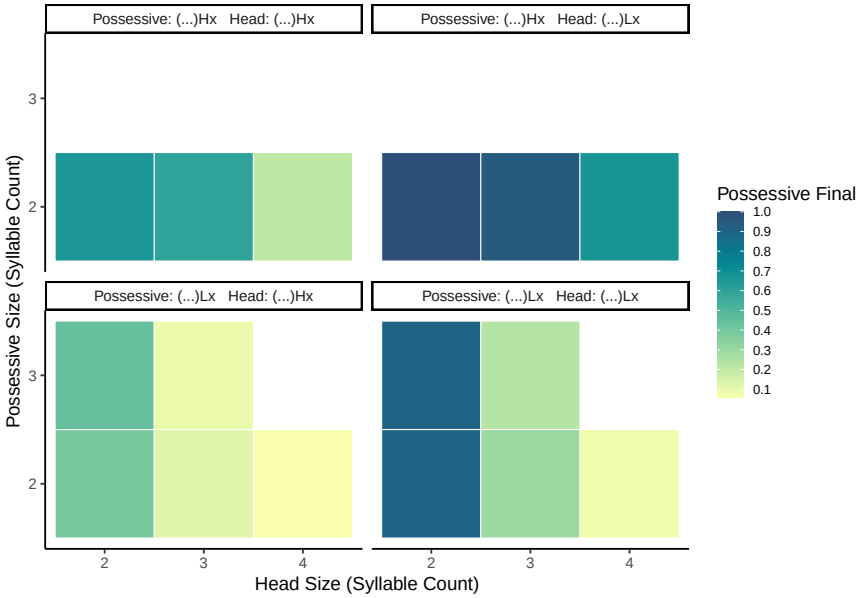
Further, binomial order is shown in §1.3.2.2 to be more strictly regulated in meter than in ordinary language. I attribute this strictness to the meter's amplified sensitivity to phonological factors, which completely override competing semantic predictors of binomial order.

1.3.2.1 Possessives and VPs

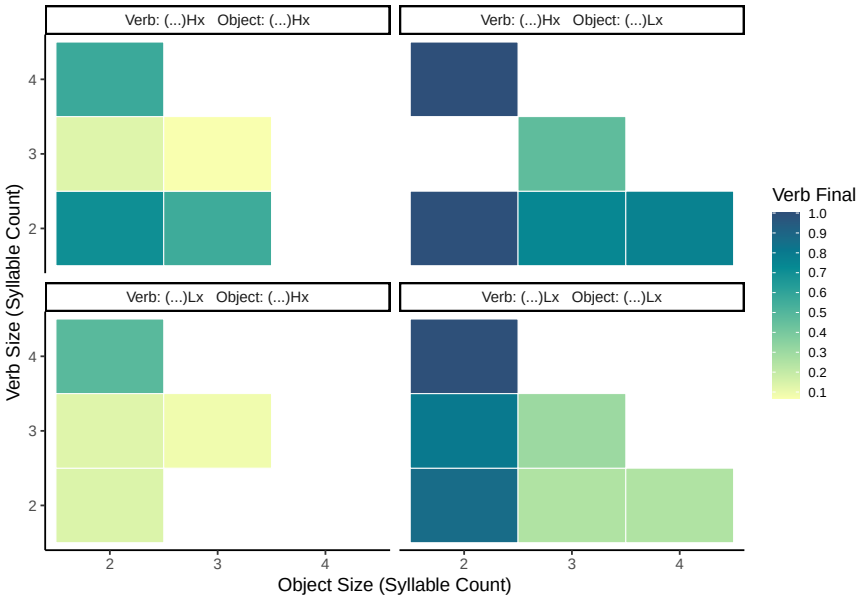
Figure 1.10 shows inversion rates for the possessive–noun construction and VPs. In possessives, the canonical order is possessive–noun, and in VPs, verb–object. The data are sparser than in adjective–noun pairs, as indicated by missing cells in the heatmaps, especially in possessives. The sparsity stems from the fact that possessives are a closed class comprising exclusively monosyllabic (e.g., *moj-* [mɔj-] 'mine') and disyllabic stems (like *njihov-* [ɲi.xɔʊ-] 'theirs'). The VP data are skewed because verb forms tend to be syllabically longer than their complements,[15] though most prosodic environments are populated reasonably well.

The main findings are the following. Comparing inversion rates in the adjective–noun construction with the possessive–noun construction and VPs (Figure 1.10), the latter show higher rates of deviation from canonical order. This is true across the board: Any cell in the possessive or VP plots with ample data shows a higher inversion rate than the corresponding adjective–noun condition in Figure 1.6. The construction types' divergent inversion propensities indicate that each construction has its own syntactic baseline. Further, as mixed effects logistic regression in §1.4 will confirm, the default

[15]Due primarily to the rich verbal morphology in BCMS (including theme vowels, tense and agreement suffixes, and prefixal aspect markers).



(a) Possessive–Noun



(b) verb–object

Figure 1.10: Rates of deviation from BCMS canonical order in **(a)** possessive–noun pairs and **(b)** VPs.

possessive and VP order in meter (when phonological factors are moot) does not match the canonical order in ordinary language.[16]

Beyond the shifted baselines in possessives and VPs, the same phonological factors seem to be at play as in the adjective–noun data (though see §1.4 for the statistical assessment of this claim). In each weight profile, inversion propensity is positively correlated with the size of the first member (vertical axis) and negatively correlated with the size of the second member (horizontal axis), consistent with prosodic end-weight (see §1.3.1). Deviations from the gradient along the leftmost vertical and bottom horizontal (especially in VPs) are attributable to ALIGN (§1.3.1.3). The weight effect (§1.3.1.2) also follows the same general pattern as in the adjective–noun construction: Inversion propensity is highest when the first member has a heavy penult and the second has a light penult (top-right panel), and lowest when the opposite is true (bottom-left panel). Along the diagonal, $2\sigma \times 2\sigma$ conditions consistently display higher inversion rates than the corresponding $3\sigma \times 3\sigma$, indicative of trisyllable inertia (§1.3.1.4). The assessment of the stress effect (§1.3.1.5) in possessives and VPs is hampered by sparse data, but logistic and MaxEnt models provide supporting evidence.

1.3.2.2 Binomials

Binomial order in meter is less variable than in other construction types (Maretić, 1907). Consider the data in (35–36). Both binomial conditions display (near-)categorical distributions.

(35) *smilje i bosilje* [ˈsmiː.ʎɛ i ˈbɔ.siː.ʎɛ] ‘immortelle and basil’

a. $2\sigma - 3\sigma$: ˈsmiː.ʎɛ i ˈbɔ.siː.ʎɛ 1.00

[16] A clarification is in order, to ward off any potential confusion. I draw a distinction between the canonical order that each construction exhibits in ordinary language (“prose-canonical”: adjective–noun, possessive–noun, verb–object) and their default order in meter, i.e. the order poets preferably choose when phonological factors are not at play (see further §1.4).

b. $3\sigma - 2\sigma$: ˈbɔ.siː.ʎɛ i ˈsmiː.ʎɛ 0.00

(36) *zlata i bisera* [ˈzlaː.ta i ˈbi.sɛ.ra] 'gold and pearl'

a. $2\sigma - 3\sigma$: ˈzlaː.ta i ˈbi.sɛ.ra 0.97

b. $3\sigma - 2\sigma$: ˈbi.sɛ.ra i ˈzlaː.ta 0.03

In both examples, the preferred binomial order places the longer, trisyllabic conjunct in final position. As already shown in (27a), this strong preference is driven by two constraints: ALIGN and END-WEIGHT. In (35), no constraint favors (35b) over (35a). Consequently, the marked binomial order (35b) is unattested. The tableau for (36) has already been provided in (27a): The disyllable-final order (36b) is penalized by END-WEIGHT and ALIGN, but favored by *L/FINS, which explains its marginal attestation in the corpus.

Binomial order is strikingly fixed in folk meter, as folk poets almost invariably choose the phonologically unmarked options. Meter thus departs from ordinary language, where binomial order is more flexible. To illustrate, Table 1.4 compares the frequencies of binomial order in four frequently conjoined noun pairs in meter versus ordinary language (srWaC 1.2 corpus; Ljubešić and Klubička, 2014).

A		B		meter % AB	meter total	ord. lang. % AB	ord. lang. total
[ˈɔ.tats]	'father'	[ˈmaːj.ka]	'mother'	100%	17	71%	1,410
[ˈsrɛ.brɔ]	'silver'	[ˈzlaː.to]	'gold'	100%	12	31%	996
[ˈnɔ.ga]	'leg'	[ˈruː.ka]	'arm'	100%	11	19%	1,757
[ˈʊiː.nɔ]	'wine'	[ˈra.ki.ja]	'rakija'	100%	37	71%	658

Table 1.4: Binomial order in folk epic vs. ordinary language. AB order is less phonologically marked than BA.

Binomial order varies considerably in ordinary language and tends to be fixed in meter. This marks a departure from NPs and VPs, whose order is largely fixed in

ordinary language and varies more liberally in meter. The apparent discrepancy lends itself to a straightforward explanation. In ordinary language, binomial order is affected by a series of factors, both phonological and, importantly, non-phonological. The latter include semantic factors such as animacy, gender, specificity and many others (Cooper and Ross, 1975; Benor and Levy, 2006; Lohmann and Takada, 2014; Morgan and Levy, 2016, a.o.). Phonological predictors thus interact with, and compete against, non-phonological factors. These conflicting pressures lead to more variable binomial order.

Meter heightens sensitivity to phonological factors, effectively suppressing semantic ones. A good example is *ruke i noge* 'arms and legs' versus *noge i ruke* 'legs and arms' from Table 1.4. In ordinary language, 'arms and legs' is strongly preferred to its transposition (81% vs. 19%), likely due to semantic factors (arms are closer to the perceptual source than legs). In meter, however, 'legs and arms' is the only attested order because *ruke* [ruː.kɛ] 'arms' has a heavy penult, being attracted to final position by the weight-mapping constraints. With competing syntactic and semantic pressures virtually out of the way, phonological markedness alone governs binomial order in meter.

Further evidence that phonological markedness is the sole determinant of binomial order in meter comes from the fact that the only binomials that show word-order alternation are those whose conjuncts have identical prosodic properties, such that phonology is indifferent, as in (37). Note that the lines with different conjunct orders are attested in the same poem, only a few lines apart.

(37) *i viša i lipša* [i='ʋi.ʃaː i='lip.ʃaː] 'taller and prettier'

a. ɔd= mɛ.nɛ =jɛ ‖ i= 'ʋi.ʃaː i= 'lip.ʃaː
from 1SG.GEN be.PRS.3SG and taller and prettier
'She is taller and prettier than me.' (I, 342:27)

b. nɛk =jɛ maːn.da ‖ i= 'lip.ʃaː i= 'ʋi.ʃaː
though be.PRS.3SG Manda.NOM and prettier and taller

'though Manda is prettier and taller' (I, 342:31)

1.4 Regression analysis

To statistically assess the effects of the phonological factors discussed in §1.3, I fit a Bayesian hierarchical (or mixed-effects) logistic model using the `brms` R package (Bürkner, 2017). Unlike frequentist models, which return single point estimates of parameter values, Bayesian models return probability distributions over plausible parameter values, known as posterior distributions. A posterior distribution reflects, for a given parameter, how probable each candidate value is, given the observed data and the model's prior assumptions. The posterior distribution is reported here by its mean and by a 95% credible interval (CI), the range within which the parameter is estimated to fall with 95% probability. An effect is credible if its 95% CI does not include zero. Posterior distributions are approximated via Markov Chain Monte Carlo (MCMC) sampling, specifically the No-U-Turn Sampler (NUTS; Hoffman and Gelman, 2014) as implemented in Stan and called through `brms`. For accessible introductions to Bayesian data analysis, see Kruschke (2014), and Nicenboim and Vasishth (2016) and Vasishth et al. (2018) for linguistics-oriented introductions.

I employ a hypothesis-driven approach to model design that tends to maximize the random-effects structure, in order to reduce the risk of Type I errors (Schielzeth and Forstmeier, 2009; Barr et al., 2013). The choice of Bayesian over frequentist regression analysis (e.g., Bates et al., 2015) was motivated by the complex random-effects structure of the model. Frequentist models can fail to converge when complex random-effects structures, especially random slopes, are specified, which poses no comparable problem to Bayesian models (Vasishth et al., 2018; Laurenceau et al., 2026).

Since model specification falls out from the research questions of this dissertation,

I begin in §1.4.1 by outlining those questions, which I aim to evaluate statistically. I explain how each research question is reflected in the model design. Along the way, I present the key modeling decisions that bear on the interpretation of the model output. These include the coding of fixed effects, the choice of which factors to treat as fixed versus random effects, the operationalization of subject-level (i.e., poet-level) idiosyncrasy, and the treatment of infrequent group-level categories, which provide little basis for detecting idiosyncratic behavior. I then turn to the model results. Given the complex random-effects structure, discussing the results all at once would be ineffective, so I break them down and report them question by question.

1.4.1 Research questions

The central hypotheses of this dissertation are, first, that folk poets possess internalized knowledge which they productively utilize to establish the linear order of words in verse lines, and second, that this knowledge is essentially stochastic rather than categorical: Informally, this mental device generates probability distributions over possible word orders, not ones and zeros.

The high variability of word order in the metrical corpus need not automatically indicate that the metrical grammar is probabilistic in nature. A strain of work in the generative tradition maintains that variation observed in corpus data is a mirage (prominently Newmeyer, 2003), that is, a consequence of aggregating data across many items and subjects whose individual grammars differ from one another but are, each in isolation, categorical (ones and zeros). On this view, variation arises at the population level, not at the level of the individual speaker, so no model of grammar needs to capture it. Being external to the mental grammar, variation is traditionally relegated to "performance" (in the standard generativist sense; Chomsky, 1965) and is therefore the object of sociolinguistics, not the theory of grammar.

The Bayesian hierarchical model allows us to test this straightforwardly, by estimating how much of the variation in the data is attributable to subject- (poet-) and item- (here, bigram-) level idiosyncrasy. The results indicate: very little. That is, the probabilistic nature of the word-order variation itself is not an artifact of aggregating across idiosyncratic poets or idiosyncratic lexical items; it holds fairly uniformly across both. Put another way, stochasticity is not a byproduct of pooling; it is a property of the shared grammar.

There are five specific questions that the model is designed to address (and, in some cases, control for):

1. Is inversion likelihood really affected by the phonological factors discussed so far? Are the proposed phonological constraints statistically supported, once subject- and item-level variation is controlled for?

2. Does inversion likelihood vary across construction types, that is, is there statistical support for construction-specific linearization constraints?

3. Do the effects of phonological factors on the likelihood of inversion vary across construction types? For example, does prosodic end-weight modulate word order in NPs the same way it does in VPs, or does its effect vary among construction types?

4. Subject-level idiosyncrasy: to what extent do baseline inversion propensities, and the phonological effects on inversion likelihood, vary across individual poets?

5. Item-level idiosyncrasy: to what extent do baseline inversion propensities, and the phonological effects on inversion likelihood, vary across bigrams?

The dependent variable in the model is INVERSION, coded "0" for (prose-)canonical word order and "1" for noncanonical order. Given that binomials do not have a syntac-

tically defined canonical order (both linear orders of conjuncts are equally well-formed syntactically), they cannot undergo inversion and are therefore excluded from the regression model (though they are included in the StOT, NHG, and MaxEnt models). This leaves 14,167 tokens of line-final NPs with adjectival modifiers, possessive–noun constructions, and VPs (5,770 unique bigrams, i.e. unique combinations of two lemmata).

Question (1) (the statistical evidence for, and robustness of, the phonological constraints) is addressed by including five phonological predictors as fixed effects. Each phonological predictor corresponds to one phonological constraint on poetic word order discussed in §1.3. All five predictors are relational, since each references phonological properties of both members of the construction (e.g., both the adjective and the noun, for adjective–noun NPs). The phonological predictors and their coding are given in Table 1.5. Note that, unlike the StOT/NHG/MaxEnt models, phonological predictors here are not coded as constraint violations but in terms of observable phonological properties of the two construction members (e.g., "first member longer than second" rather than as a penalty for an End-Weight violation).

The coding logic for size difference, penult length, and misalignment is as follows. Negative values mark environments in which these factors figure as inversion inhibitors: The canonical order is more phonologically well-formed than its transposition. At positive values, these constraints figure as inversion triggers: The noncanonical order is favored over the canonical order. At zero, the effect is moot; the two orders fare equally on a given phonological constraint.

To illustrate, for penultimate length (which is a proxy for *Light/FinalStrong), the negative value ('−1') marks the environment in which the canonical order satisfies this constraint, but inversion violates it: ['mi.la 'ma:j.ka] 'beloved mother' (bottom-left panel in Figure 1.6). Zero means that the effect is moot, i.e. both phrasal members have

Predictor	Type	Coding	Corresponding constraint
size diff. (Adj − N)	continuous	< 0: Adj shorter than N 0: no size diff. > 0: Adj longer than N	End-Weight
penult length (Adj − N)	continuous	−1: Adj penult lighter 0: no penult weight diff. 1: Adj penult heavier	*Light/FinalStrong
stressed light penult (Adj − N)	continuous	−1: only N penult stressed and light 0: no diff. 1: only Adj penult stressed and light (inversion increases markedness)	*ˈLight/FinalStrong
misalignment	continuous	−1: inversion introduces misaligment 0: alignment moot 1: inversion removes misaligment	Align
both trisyllabic	binary	1: both Adj and N trisyllabic 0 otherwise	3σInertia

Table 1.5: Fixed effect coding.

the same quantity of the penultimate vowel: [ˈʎuː.ta ˈguː.ja] 'angry snake' (top-left and bottom-right panels in Figure 1.6). Finally, the positive value of the penultimate length variable ('1') indicates that the canonical order violates this phonological constraint, while inversion satisfies it: [ˈraːʊ.nɔ ˈpɔ.ʎɛ] 'flat field' (top-right panel in Figure 1.6).

For stressed light penult, the coding logic is the opposite: The factor functions as an inversion trigger at negative values, and as an inversion inhibitor at positive values. When true, the 'both trisyllabic' predictor is by definition an inversion inhibitor.

Questions (2–3) (whether baseline inversion propensities vary across the three construction types, and whether phonological effects on inversion likelihood vary across construction types) could in principle be addressed in two ways. Construction type could be treated as a random effect (random intercepts and slopes), or as a fixed effect, with two-way interaction terms between construction and the phonological predictors as the analogue to random slopes. I treat construction as a fixed effect. Treating a factor as random presumes that its observed levels are a sample from some larger pop-

ulation of possible levels, and that it is the population-level variance, not the identity of individual levels, that is of interest. That assumption does not hold for construction type: adjective–noun, possessive–noun, and verb–object are an exhaustive, theoretically motivated set of categories that are the object of the research questions above, not a sample drawn from a broader population of possible constructions. In practice, construction type has only three levels, which complicates the estimation of a variance component. The thresholds proposed for treating a variable as a random effect vary considerably by field; commonly cited figures range from roughly 5 to 8 levels in ecology (Bolker, 2015; Harrison, 2015; Harrison et al., 2018) to 10–20 in psychology (McNeish and Stapleton, 2016) to 30–50 in sociology (Maas and Hox, 2005); see Oberpriller et al. (2022) for an overview. Looking at the standard literature on mixed-effects models in linguistics (Baayen et al., 2008; Barr et al., 2013), I am aware of no equivalent, field-specific recommendation for linguistics. Gelman and Hill (2007) and Bolker et al. (2009) similarly suggest that, as a rule of thumb, fewer than four or five levels is insufficient for stable estimation of the variance component.

The rationale for addressing questions (4–5) (the degree of subject- and item-level idiosyncrasy in the data) is twofold. Understanding how word-order distributions vary across poets and items is, of course, interesting in and of itself. Crucially, failure to include a maximal random-effects structure motivated by the research questions inflates the rates of Type I error (Barr et al., 2013). Further, including random effects allows us to determine whether the effects of the phonological factors on word order persist once subject- and item-level variation is controlled for. This helps us address the standard objection to probabilistic grammar, namely that variable output is an artifact of aggregating across different subjects and items (Newmeyer, 2003).

I use poem title as a proxy for the folk poet. In Vuk's collection, the correspondence between poems and poets is not strictly one-to-one, since some folk poets authored

multiple pieces. Consequently, multiple poems by the same poet are treated as distinct levels of this grouping factor. I do, however, believe this to be a methodologically defensible, albeit imperfect, choice, given that authorship information is not available in the collection and would have to be retrieved from various sources in the literature. For one, using poem, rather than poet, as the grouping factor keeps the levels of the random effect *more specific* rather than coarser, which would otherwise come at the expense of fixed-effect precision. There are 525 unique levels of this grouping factor (i.e., unique poems).

Bigram is defined as a unique combination of two lemmata. Bigrams are coded as word-1 lemma plus word-2 lemma under the (prose) default order (adjective–noun, possessive–noun, verb–object). For example, the attested NP tokens [ˈbi.jɛ.li.jɛm ˈgraː.dɔm] 'white city.INS.SG' and [ˈgraː.da ˈbi.jɛ.lɔː.ga] 'city white.GEN.SG' have the same bigram label: *bijel*, *grad* 'white, city.' Note that bigram coding is insensitive to surface word order. There are 5,770 unique bigrams in the data (for 14,167 tokens). Most bigrams in the data (3,766, or 65%) are hapaxes, i.e., attested only once in the corpus. In fact, over 9 in 10 bigrams (5,258 bigrams) are attested fewer than five times. These infrequent bigrams account for slightly over half of all tokens in the regression dataset (7,524 of 14,167). The remaining 512 bigrams, attested five times or more, account for the other 6,643 tokens, just under half the corpus.

Bigram-level idiosyncrasy must be controlled for primarily because of the formulaic nature of oral poetry (Lord, 1960; Jakobson, 1960; Foley, 1988, 1993). Frequently used bigrams that are part of a formula may occur in a petrified order that folk poets have memorized, and such bigrams, being frequent, could skew the overall distribution of word-order alternatives. The model lets us see whether such items contribute to the observed word-order probabilities (i.e. the descriptive statistics reported in Figures 1.6–1.10), and if so to what extent.

Since we are interested in bigram-level idiosyncratic behavior, the preponderance of hapax legomena and infrequent bigrams raises a critical modeling challenge. Detecting by-item idiosyncratic behavior requires observing ample instances of that item (here, tokens of a given bigram). Infrequent bigrams, especially hapaxes, provide insufficient information to reliably estimate item-specific deviations. Moreover, a preliminary model which specified an individual random effect for each of the 5,770 unique bigrams exceeded the memory limit of the R session, so that the model output was not saved.

To resolve this computational bottleneck without excluding data, I assign an individual random intercept and random slopes only to bigrams attested at least five times in the corpus, where sufficient empirical evidence exists to estimate idiosyncratic variance. All remaining low-frequency bigrams are pooled into a single consolidated 'other' category. A somewhat comparable practice exists in lexicography, where minimum frequency cutoffs are standardly recommended before a lexicographer can reliably characterize the behavior of a given lexical unit before attempting to characterize the stable behavior of individual lexical units (Sinclair, 2005; Baroni et al., 2009). Importantly, no tokens were excluded from the dataset. This pooling strategy alters only the granularity of the grouping factor level assigned to a token.

1.4.2 Model fitting and convergence

The model was fit using `brms` (Bürkner, 2017), which called Stan's No-U-Turn Sampler over four independent chains, each run for 8,000 iterations with the first 4,000 discarded as warmup, for a total of 16,000 post-warmup draws. A weakly informative prior ($\mu = 0, \sigma = 2.5$) was placed on all population-level (fixed) effects.

The prior acts as a conservative regularization mechanism. The standard deviation of 2.5 assigns a high prior probability to realistic effect sizes and at the same time precludes astronomically large coefficients that lead to complete separation or unstable

estimates (Gelman et al., 2008). With $\sigma = 2.5$, 68% of the prior belief falls within one standard deviation (i.e. between -2.5 and 2.5) on the log-odds scale; 95% of the prior belief falls within approximately two standard deviations (between -5 and 5). A coefficient of ± 5 on the logit scale translates into a near-categorical effect. Effectively, the prior tells the model that fixed effects are expected to be somewhere between zero and very strong, but any astronomically higher magnitude is highly unlikely.

Convergence was assessed using the potential scale reduction factor ($\hat{R}$; Gelman and Rubin, 1992) and effective sample size (ESS). All $\hat{R}$ values were below 1.01, indicative of successful convergence. ESS values were over 1,000 for all parameters, which supports successful mixing across chains.

1.4.3 Results: Fixed effects

The fixed effects and their interactions bear on three research questions defined in §1.4.1 (1–3). In this section, I address each of those questions individually.

1.4.3.1 Question 1: The role of phonological factors

Figure 1.11 plots the posterior distributions of the fixed phonological predictors. The 95% credible intervals (shaded) for all five phonological predictors exclude zero, which indicates that all five predictors have a credible effect on the log-odds of inversion.

The directionality of the effects reflects the coding of the fixed predictors (Table 1.5), not different directionalities of the actual phonological phenomena. Recall from Table 1.5 that size difference, penult length, and misalignment avoidance are inversion-inducing at positive values, whereas stressed light penult avoidance is inversion-inducing at negative values. That the first three predictors have credible positive effects, and the fourth has a credible negative effect, indicates that all four modulate inversion likelihood in the exact same, and fully expected, direction: Each

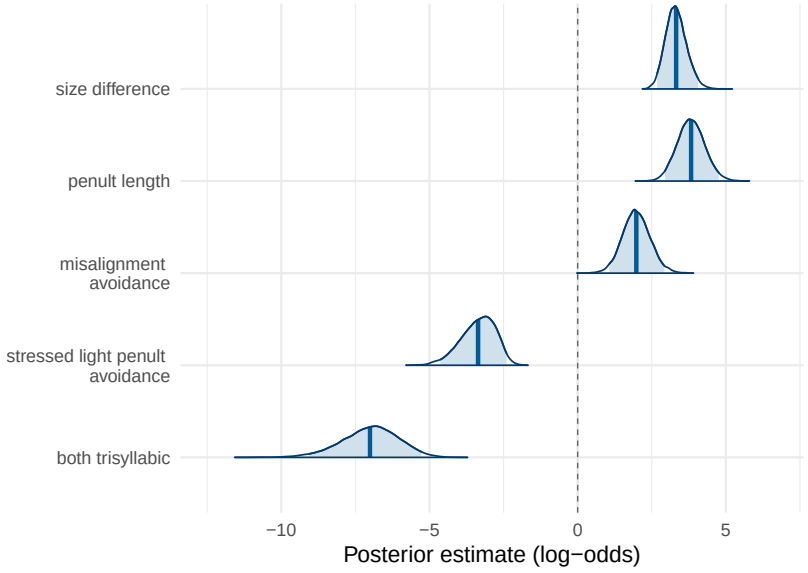


Figure 1.11: Posterior distributions of the main effects of phonological factors. Points indicate posterior means. The shaded areas represent 95% credible intervals. An effect is considered credible if zero (dashed vertical line) is outside the 95% credible interval.

factor increases the probability of inversion when it favors the noncanonical over the canonical order. "Both trisyllabic" is by definition an inversion inhibitor: Inversion is predicted to be dispreferred when both phrasal constituents are trisyllabic. The credible negative effect on this predictor confirms exactly that: Inversion probability drops credibly when both phrasal constituents are trisyllabic ("1") relative to when this is not the case ("0").

In sum, all five phonological factors examined here are found to have a credible effect on inversion probability, even when a maximally complex random-effects structure is specified in the model. This result corroborates the descriptive statistical analysis in §1.3 and previous literature that identified some of these effects in prose, without a quantitative model (Maretić, 1907; Jakobson, 1966; Foley, 1993; Zec, 2008).

1.4.3.2 Question 2: Construction-specific inversion propensities

Whether the three constructions tested here differ in their propensity to undergo inversion is estimated through the fixed effect for construction, which, as a reminder, has three levels: adjective–noun, possessive–noun, and VP. Adjective–noun was set as the reference level, so the model directly estimates the inversion propensity of the other two constructions, possessive–noun and VP, relative to adjective–noun. Figure 1.12 plots the posterior distributions for all three pairwise comparisons: Possessive–noun vs. adjective–noun and VP vs. adjective–noun are estimated directly by the model, while the third, VP vs. possessive–noun, is obtained via the `hypothesis()` function of the `brms` package, which tests linear combinations of posterior draws (here, the difference between the two non-reference-level coefficients) without requiring the model to be refit with a different reference level.

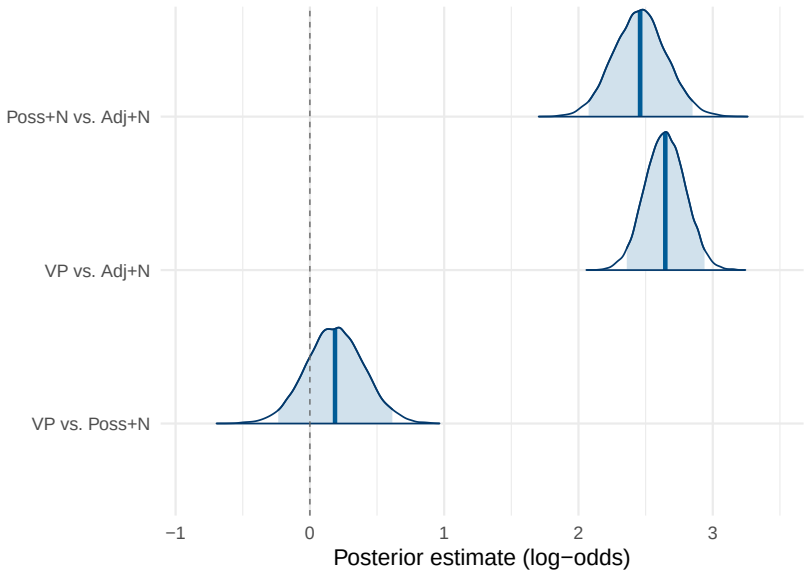


Figure 1.12: Posterior distributions of the pairwise contrasts between construction types. Each row shows one pairwise comparison; the *Poss+N vs. Adj+N* and *VP vs. Adj+N* contrasts are estimated directly as fixed-effect coefficients relative to the adjective–noun reference level, while *VP vs. Poss+N* is obtained *post-hoc*.

The posterior distributions for possessive–noun vs. adjective–noun and VP vs. adjective–noun are both positive and credibly different from zero. This indicates that the possessive construction and VPs are credibly more likely to display noncanonical (inverted) order than adjective–noun NPs. This result is consistent with the descriptive results in §1.3.2 (recall that the possessive and VP heatmaps are considerably darker overall than the adjective–noun heatmap). The posterior distribution of the third comparison, VP vs. possessive–noun, includes zero, suggesting no credible difference in inversion propensity between the two construction types. In other words, while both possessive–noun and VP constructions differ credibly from the adjective–noun construction, they do not credibly differ from each other.

The regression model therefore confirms that linearization pressures apply differently to different construction types, much like the descriptive analysis in §§1.3–1.3.2 suggests. The pressure to preserve the prose-canonical order is credibly stronger in the adjective–noun construction than in the other two construction types. However, the results in Figure 1.12 do not tell us what the meter's default order is for each construction type, that is, which order, prose-canonical or inverted, is more likely when all phonological predictors are at their baseline levels. This information can also be obtained from the model.

NPs with adjectival modifiers credibly favor the prose-canonical order (adjective–noun) at baseline, as indicated by the negative global intercept of the model: -1.18, 95% CI $[-1.67, -0.68]$, which maps onto 0.235 inversion probability by the inverse logit function. For possessive–noun and VP, the baseline likelihood of deviation from the prose-canonical order can be obtained as the linear combination of the intercept and construction coefficient, obtained via `brms`'s `hypothesis()` function. For the possessive–noun construction, these sum to 1.28 (95% CI $[0.69, 1.87]$), which maps onto 0.78 baseline inversion probability. For VPs, the sum is 1.47 (95% CI $[0.95, 2.00]$),

corresponding to a baseline inversion probability of approximately 0.81. For both constructions, deviation from the prose-canonical order is more likely than adherence to it even at baseline.

Summing up, the results suggest that in meter, the possessive–noun and VP constructions are not only more likely to deviate from the prose-canonical order than NPs with adjectival modifiers, they are also more likely to deviate from the prose-canonical order at baseline. Thus, for these two constructions, the meter-canonical order does not match their prose-canonical order. This regression model establishes that the three constructions differ credibly in their propensity to undergo inversion and, for possessive–noun and VP, that the meter-canonical order departs from the prose-canonical one. It does not, however, tell us whether this surface pattern requires a dedicated linearization constraint for each construction in the grammar, as opposed to following from the general phonological constraints established in response to Question 1. I address this question through a MaxEnt model-selection procedure in Chapter 2, which finds no support for a linearization constraint specific to VPs.

1.4.3.3 Question 3: Do phonological effects on inversion vary across construction types?

The preliminary analysis in §§1.3–1.3.2 suggests that the same phonological factors that modulate word-order alternations in the adjective–noun construction are also at play in the possessive–noun and VP constructions. The analysis does not, however, tell us whether the effect of these phonological factors on inversion probability is consistent across the constructions. The hierarchical logistic model allows us to test the consistency of phonological effects on inversion probability across construction types, through the two-way interaction terms between construction and the phonological predictors.

Just as for the main effect for construction, the model estimates the difference in

application of a phonological predictor in possessive–noun and VP relative to adjective–noun. For the third pairwise comparison, VP vs. possessive–noun, the interaction effects are obtained *post-hoc* using the `hypothesis()` function, as described in §1.4.3.2.

The posterior distributions of the interaction effects for each pairwise comparison are reported in Figure 1.13. The interpretation is as follows: If the 95% credible interval for a given parameter includes zero, there is no credible evidence that the phonological constraint in question behaves differently across the two constructions compared.

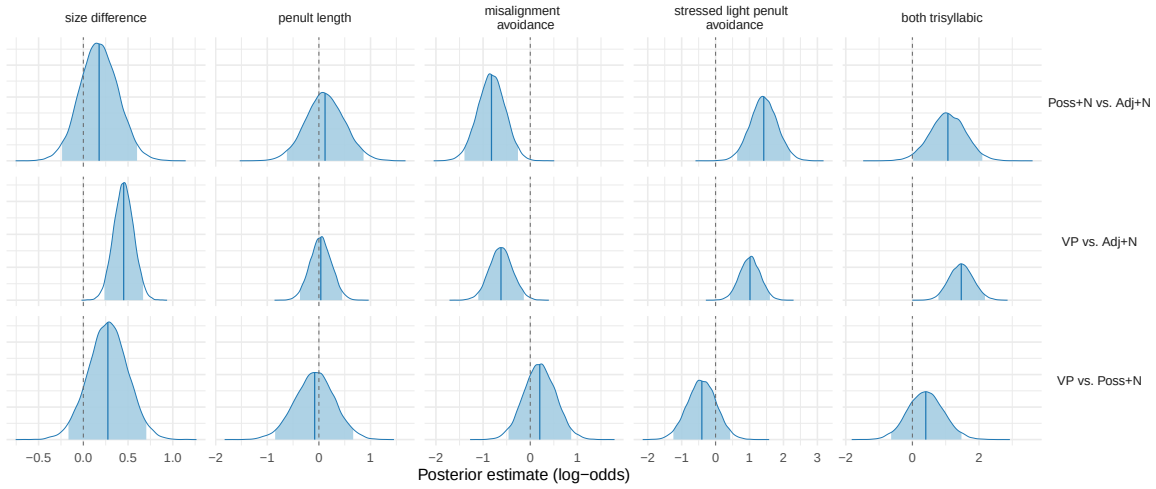


Figure 1.13: Posterior distributions of the construction × phonological-predictor interaction effects, for each phonological predictor (faceting category: columns) and pairwise construction comparison (faceting category: rows).

None of the interaction terms for the VP vs. possessive–noun pair credibly differs from zero, which suggests that the effect of all five phonological predictors is relatively consistent in these two construction types. This finding complements the absence of credible difference in the two constructions' baseline inversion propensity (§1.4.3.2).

Turning to the possessive–noun vs. adjective–noun pair, three factors show credible between-construction differences: misalignment avoidance, stressed light penult avoidance, and both trisyllabic. For misalignment avoidance, the interaction effect is negative, while for the other two phonological factors, the effect is positive. Given the

coding of these phonological predictors (see §1.4.1), the negative interaction effect between misalignment avoidance and possessive–noun (relative to adjective–noun) means that the effect of misalignment avoidance on inversion probability is credibly weaker in the possessive–noun construction than in the adjective–noun construction. Given the opposite coding for stressed light penult avoidance and both trisyllabic, the positive interaction effects indicate the same effect directionality as for misalignment avoidance: The effects of these predictors are likewise attenuated in the possessive–noun construction relative to the adjective–noun construction.

These three effects show the same attenuation pattern in the VP vs. adjective–noun pair. The only difference in this pair is that size difference, a proxy for End-Weight, shows a credibly positive interaction effect, meaning that the effect of this constraint on inversion probability is stronger in VPs than in NPs with adjectival modifiers.

1.4.4 Results: random effects

Turning to random effects, I report the posterior distributions of their standard deviations in Table 1.6.

1.4.4.1 Question 4: Between-poem variation

I focus on two issues here: between-subject differences in baseline inversion propensity (random intercepts) and between-subject differences in the way phonological factors affect inversion probability (random slopes). Starting with the random intercepts, the catepillar plot in Figure 1.14 reports the estimates and 95% credible intervals for each of the 525 poems. If the poem's 95% credible interval includes zero, the model provides no evidence that the poem in question deviates from the population mean inversion behavior.

The vast majority of poems in the corpus (515 out of 525) do not credibly show

	SD estimate	Est. error	95% CI	Prop. credibly different from zero
Poem (n = 525)				
Intercept	0.71	0.08	[0.57, 0.87]	0.02
Size difference	0.49	0.07	[0.36, 0.62]	0.02
Penult length	1.16	0.12	[0.94, 1.40]	0.02
Misalignment	0.97	0.14	[0.71, 1.25]	0.01
Stressed light penult	1.64	0.20	[1.27, 2.04]	0.03
Both trisyllabic	1.58	0.26	[1.09, 2.11]	0.01
Bigram (n = 513)				
Intercept	2.44	0.31	[1.88, 3.09]	0.03
Size difference	1.68	0.38	[0.97, 2.45]	0.00
Penult length	2.53	0.50	[1.56, 3.52]	0.01
Misalignment	2.01	0.82	[0.30, 3.50]	0.00
Stressed light penult	1.08	0.66	[0.07, 2.50]	0.00
Both trisyllabic	1.97	0.74	[0.79, 3.67]	0.00

Table 1.6: Summaries of posterior distributions for the standard deviations of random effects in the hierarchical Bayesian model, by grouping factor (Poem, Bigram) and term (six per grouping factor: intercept plus the five phonological predictors).

any idiosyncratic tendencies in terms of their baseline inversion propensities, since the 95% of the credible intervals of their random intercept posterior distributions include zero. The model found credible evidence for idiosyncratic behavior only for 10 out of 525 poems (2%). These are listed in Table 1.7. Note that idiosyncratic poems do not behave uniformly. Four of these 10 idiosyncratic poems show credibly negative random intercepts, meaning that their overall inversion propensity is lower than the population mean. The remaining six idiosyncratic poems have credibly positive random intercepts, meaning that at baseline, their inversion propensity is higher than the population mean.

A minimal degree of poem-level idiosyncrasy likewise resides in random slopes. The proportion of poems that deviate from the population mean in terms of how individual phonological predictors modulate word order ranges between 1% and 3% (see Table 1.6).

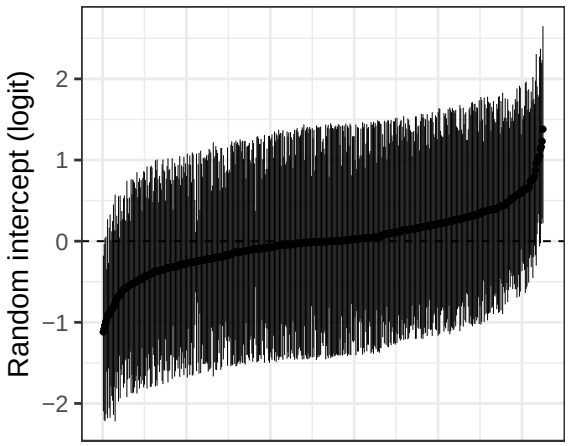


Figure 1.14: By-poem random intercepts: posterior means (points) and 95% credible intervals (error bars). Each point on the x-axis represents one of the 525 unique poems in the corpus (ordered by their posterior means). Poems whose 95% credible intervals exclude zero (dashed horizontal line) credibly differ from the population mean.

These results were largely expected, given the uniformity and formulaic nature of oral poetry (Lord, 1960; Jakobson, 1960; Foley, 1988, 1993; Hayes and MacEachern, 1998). The metrical regularities and word order across folk poets are governed by essentially the same metrical grammar, that is, folk poets share a largely uniform poetic grammar subject to marginal speaker- (i.e. poet-) level idiosyncrasy. This near-uniformity holds not only for the poems' baseline inversion propensities (random intercepts) but also for the degree to which phonological factors condition the choice between orders (random slopes), where only 1–3% of poems credibly deviate from the population-level pattern.

These findings are difficult to reconcile with a population-heterogeneity account of gradience in grammar à la Newmeyer (2003). An important caveat is that the absence of a credible deviation for 515 poems is in some cases a function of statistical power: Shorter poems, which contribute fewer tokens to the corpus, have wider credible intervals. Still, nothing in the data or the model suggests that poem-level idiosyncrasies meaningfully skew the overall distributions of word-order alternatives in

Poem	Book	Estimate	Lower	Upper
Đakon Stefan i dva anđela	2	−1.12	−2.10	−0.18
Jakšići kušaju ljube	2	−1.08	−2.20	−0.04
Boj na Loznici	4	−0.96	−1.61	−0.34
Sestra Leke kapetana	2	−0.91	−1.57	−0.27
Ropstvo i ženidba Jakšića Šćepana	2	0.97	0.09	1.91
Nahod Momir	2	0.99	0.23	1.81
Moračani i Kolašinci	4	1.03	0.33	1.78
Nevjera ljube Gruičine	3	1.16	0.21	2.20
Ženidba bega Ljubovića	3	1.23	0.28	2.23
Ženidba Jakšića Mitra	2	1.38	0.22	2.65

Table 1.7: Poems with the most extreme poem-level random-intercept estimates (log-odds), with 95% credible intervals.

the corpus, and more importantly, that the fine-grained phonological effects largely reduce to population-level differences.

1.4.4.2 Question 5: Between-item variation

Finally, I discuss the degree of bigram-level idiosyncrasy. Figure 1.15 plots the posterior means and 95% credible intervals for the bigram-specific random intercepts, which capture item-specific baseline inversion propensities for the 513 modeled levels of this grouping factor. As explained in §1.4.1, individual levels were specified only for bigrams attested at least five times in the corpus.

Looking at random intercepts, credible deviation from the population mean was found for only 17 out of 513 bigrams (3%). Of these idiosyncratic items, 9 are more inversion-resisting at baseline than the population mean. The remaining 8 bigrams are more inversion-prone than the population mean. Interestingly, two NP bigrams with the same head noun, *vino* [ˈʋiː.nɔ] 'wine' show opposite baseline inversion propensities: *rujan*, *vino* 'claret wine,' as in [ruːj.nɔ ˈʋiː.nɔ], is strongly inversion-resisting, whereas *ladan*, *vino* 'cold wine,' as in [ˈlaːd.nɔ ˈʋiː.nɔ], is inversion-favoring.

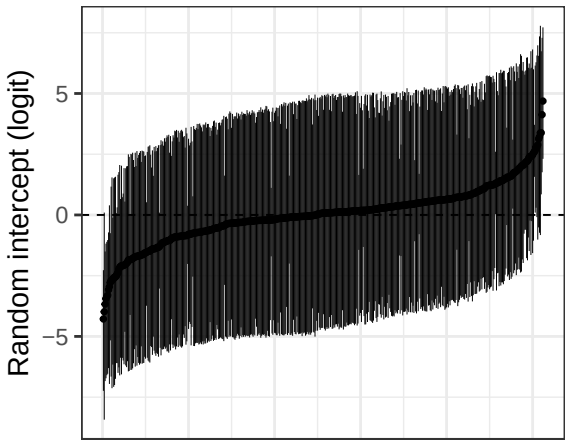


Figure 1.15: By-bigram random intercepts, with 95% credible intervals. A separate level was specified only for bigrams attested at least five times in the corpus.

The proportions of items credibly different from the population mean in terms of by-bigram random slopes are close to zero. This suggests that much of bigram-level idiosyncratic behavior resides in the random intercepts (baseline inversion propensity) rather than the random slopes (how phonological factors modulate inversion likelihood across bigrams).

While the analysis identifies some item-level idiosyncrasy, there is little evidence to suggest that the patterned variation observed in the corpus data largely reduces to lexical propensities. A caveat that must be acknowledged is the pooling of low-frequency bigrams into a single category. This consolidation potentially masks idiosyncratic tendencies (both in terms of random intercepts and random slopes) in less frequent bigrams, which account for slightly over half of the regression dataset.

1.4.5 Section summary

By way of conclusion, it is worth mentioning that the logistic regression model reported in this section and the StOT/NHG/MaxEnt models reported in Chapters 2–4 serve different purposes. The logistic model is a statistical instrument for detecting

Bigram	Gloss	Construction	Estimate	Lower	Upper
star, majka	‘old mother’	Adj–N	−4.28	−7.24	−2.27
vjeran, sluga	‘faithful servant’	Adj–N	−3.67	−6.84	−1.50
rujan, vino	‘claret wine’	Adj–N	−3.44	−6.72	−1.18
vjeran, ljuba	‘faithful wife’	Adj–N	−3.36	−6.60	−1.11
sitan, knjiga	‘tiny book’	Adj–N	−3.30	−6.54	−1.03
vedar, nebo	‘clear skies’	Adj–N	−3.08	−6.30	−0.72
svoj, sin	‘self’s son’	Poss–N	−2.79	−5.16	−0.75
suv, zlato	‘dry (pure) gold’	Adj–N	−2.66	−6.16	−0.11
beo, dvor	‘white palace’	Adj–N	−1.84	−4.12	−0.03
mio, snaha	‘beloved daughter-in-law’	Adj–N	1.31	0.03	2.60
ladan, vino	‘cold wine’	Adj–N	1.67	0.26	3.10
sinj, more	‘gray sea’	Adj–N	2.60	1.57	3.71
dati, vatra	‘give fire’ (‘shoot’)	VP	2.65	0.09	6.15
roniti, suza	‘shed tears’	VP	2.79	0.30	6.28
mlad, bula	‘young Turkish woman’	Adj–N	3.38	0.85	6.74
mlad, kada	‘young spouse’	Adj–N	4.13	1.76	7.29
drag, duša	‘dear soul’ (‘darling’)	Adj–N	4.69	2.31	7.72

Table 1.8: Bigrams with the most extreme bigram-level random-intercept estimates (log-odds), with 95% credible intervals.

regularities in the corpus data; it is not put forth here as a model of the grammar itself. The StOT/NHG/MaxEnt models, by contrast, are proposed as models of poets' actual grammatical knowledge. The logistic model in this chapter therefore plays an evidentiary role. Controlling for subject- (i.e. poem-) and item- (i.e. bigram-) level idiosyncrasy via random effects, the logistic model establishes credible effects for all relevant phonological predictors on the adjective-noun order alternation.

Two differences in scope are worth noting. First, as implemented here, the StOT/NHG/MaxEnt models do not account for lexical propensities, which the logistic model captures via by-bigram random effects. This could in principle be incorporated via lexically indexed constraints (Pater, 2000; Nazarov and Smith, 2022) or lexical scaling (Linzen et al., 2013; Hughto et al., 2019). Second, the logistic model is fit only to the NP and VP data (n = 14,167), for which a clear prose-canonical baseline exists, whereas

the StOT/NHG/MaxEnt models are fit to NP and VP data along with binomials (n = 15,397).

1.5 Conclusion

The highly variable word order of the BCMS folk epic is largely predictable from a handful of phonological factors, most of which have solid precedent in the literature on South Slavic folk meter (Maretić, 1907; Jakobson, 1966; Ružić, 1975; Zec, 2008). These factors manifest as fine-grained statistical tendencies rather than categorical effects.

The regression analysis in §1.4 largely substantiates this view and adds several important nuances to it. All five phonological factors credibly modulate inversion probability (§1.4.3.1). The way these phonological factors affect word order is not uniform across construction types. At baseline, the adjective–noun construction resists inversion more strongly than the possessive–noun and VP constructions. Moreover, the meter's default order for possessives and VPs departs from their prose-canonical order. Crucially, word-order variation is not an artifact of speaker- or item-level idiosyncrasy: Only about 2% of poems and 3% of bigrams credibly deviate from the population-level mean in their baseline inversion propensity. Idiosyncrasy in how phonological factors condition inversion is similarly low for both grouping factors. Much of the word-order variation observed in the corpus is a property of a shared internalized poetic grammar rather than as an artifact of aggregating over heterogeneous individual poets or combinations of lexical items.

Accordingly, the grammar that governs poetic word order must be stochastic. Several stochastic constraint-based frameworks have been proposed, of which I focus on StOT, NHG, and MaxEnt HG. The following three chapters examine the predictions of these stochastic frameworks for BCMS folk meter.

Chapter 2

Variation and stochastic grammars

2.1 Introduction

In generative phonology, variation has traditionally been treated as external to the grammar proper, attributed to performance-related factors (Newmeyer, 2003), and therefore relegated to sociolinguistics. An expansive body of recent work has identified cases of intra-speaker variation driven by purely grammatical factors, which indicates that not all variation is external to the phonological grammar (see Coleman and Pierrehumbert, 1997; Pierrehumbert, 2001; Bresnan et al., 2001; Ernestus and Baayen, 2003; Rosenbach, 2005; Jäger and Rosenbach, 2006; Jäger, 2007; Hayes and Londe, 2006; Hayes and Wilson, 2008; Hayes et al., 2009; Coetzee and Pater, 2011; Alderete and Finley, 2023, among others). This shift in perspective coincides with a more general methodological pivot towards naturalistic data, such as unelicited corpus data and gradient acceptability judgments obtained through controlled experimentation.

This growing body of evidence further indicates that phonological knowledge consists not only of categorical rules, but also of fine-grained statistical generalizations. Real-life phonological data are messier than customarily acknowledged in formal theo-

retical models: They are rife with subtle, gradient effects (Frisch et al., 2004; Coetzee and Pater, 2008; Ryan, 2011) and admit of exceptions (Zuraw, 2000, 2010; Guy, 2007; Nazarov and Smith, 2022). The goal of this dissertation is to establish what formal machinery is needed to capture the fundamentally stochastic nature of the shared grammatical knowledge that folk poets use to determine the order of constituents in verse lines, and how those subtle generalizations are inferred from noisy input data. This chapter introduces three constraint-based frameworks that can accommodate variable output: StOT, NHG, and MaxEnt, as a prelude to a detailed framework comparison in Chapters 3–4.

These three frameworks differ along two dimensions. The first is the mode of evaluation. In StOT, constraints are strictly ranked (but their rankings may change across evaluations). The winner is determined by the highest-ranked constraint that distinguishes between the competing candidates; every lower-ranked constraint is rendered irrelevant to the outcome. NHG and MaxEnt, by contrast, evaluate candidates holistically. Candidates are evaluated based on their performance on all constraints in the grammar. Not all constraints contribute equally to candidates' probability, but they all matter to some extent.

The second dimension is what I call the source of stochasticity, that is, how the grammar becomes probabilistic. StOT and NHG are sampling-based. The parameters of the grammar are perturbed by random noise, using a Monte Carlo procedure, every time a speaker uses their grammar to evaluate a set of competing forms, so that the grammar's output can vary from one evaluation to the next. MaxEnt, by contrast, is inherently probabilistic, in that candidate probabilities are computed analytically, with no appeal to noise or repeated sampling. The output of a MaxEnt grammar is not a single winning form, but a probability distribution over the candidate set.

The chapter is structured as follows. §2.2 introduces the stochastic variants of

HG, MaxEnt and NHG, while §2.3 introduces StOT. §2.4 then reports the model selection procedure and its outcome. With the constraint set in place, §2.5 describes the simulation setup for the MaxEnt, NHG, and StOT models. §2.6 discusses the utility and limitations of goodness-of-fit metrics for framework comparison. §2.7 presents the models' overall fit to the data.

2.2 Stochastic Harmonic Grammars

In HG, constraints are assigned numeric weights commensurate with their relative contributions to the grammar. Following Keller (2000); Potts et al. (2010), constraint weights are nonnegative, which ensures that violations result in penalties rather than rewards. Candidates are assigned Harmony scores as a negative weighted sum of all violations they incur: For each candidate i, the grammar assigns a Harmony score $\mathscr{H}_i$ as the negative weighted sum of all violations incurred by i, by scaling for each constraint k in constraint set C the number of violations i incurs to k (c_{ik}) by k's weight (w_k).

(38) Harmony

$$\mathscr{H}_i = -\sum_{k \in C} w_k c_{ik}$$

HG predicts widespread interactions between multiple independent factors (ganging-up cumulativity) and aggregate effects of multiple instances of individual factors (counting cumulativity); see Jäger and Rosenbach (2006). HG's ability to derive cumulativity effects is a desirable property, since cumulativity is ubiquitous in variable phonological data (Jäger and Rosenbach, 2006; Jäger, 2007; Keller, 2006; Pater, 2009b; Breiss, 2020; Breiss and Albright, 2022; Kawahara and Breiss, 2021).

Deterministic HG selects a single candidate with the highest Harmony score as

winner. Stochastic HGs (MaxEnt and NHG),[1] by contrast, return probability distributions over candidates. The two variants rely on fundamentally different machinery to introduce stochasticity and therefore make disparate predictions about the relationship between Harmony and probability, which I review in the remainder of this section.

2.2.1 MaxEnt

MaxEnt is an intrinsically probabilistic variant of HG. Rather than designating candidates as winners or losers, the grammar assigns a probability distribution over the candidate set. Probabilities are calculated directly from Harmony scores via the softmax function (39): The probability of candidate i, p_i, equals its exponentiated Harmony score, $e^{\mathscr{H}_i}$, divided by the sum of $e^{\mathscr{H}_j}$ of all candidates j in the candidate set K.

(39) Probability mapping in MaxEnt (general case)

$$p_i = \frac{e^{\mathscr{H}_i}}{\sum_{j \in K} e^{\mathscr{H}_j}}$$

where

e = Euler's number, the base of the natural logarithm (≈ 2.72)

The formula in (39) applies to candidate sets irrespective of the number of contenders in a tableau. For the two-candidate case (binomial response), the softmax function reduces to the logistic cumulative distribution function of the Harmony difference between the two candidates, $\Delta\mathscr{H} = \mathscr{H}_a - \mathscr{H}_b$ (Zuraw and Hayes, 2017; Flemming, 2021).

(40) Probability mapping in MaxEnt (binomial response)

$$p_a = \frac{1}{1 + e^{-\Delta\mathscr{H}}}$$

[1] A note on terminology is in order. I use the term "stochastic HG" as a cover term for any variant of HG that generates variable, non-deterministic output, as in, e.g., Flemming (2021). By contrast, some authors reserve the term "stochastic HG" exclusively for NHG (Magri, 2018, 2024), emphasizing its parallelism with StOT.

Figure 2.1 plots, for binomial response, a's fitted probability as a function of $\Delta\mathscr{H}$ between a and its only rival b. The sigmoid curve, MaxEnt's quantitative signature (Zuraw and Hayes, 2017; Kawahara, 2020, 2021; Smith and Pater, 2020; Hayes, 2022; Hayes and Moore-Cantwell, 2025), reflects the monotonic increase in $p(a)$ as $\Delta\mathscr{H}$ approaches positive infinity (and the monotonic decrease towards 0 as $\Delta\mathscr{H}$ approaches negative infinity). A $\Delta\mathscr{H}$ of approximately 5 units in either direction effectively maps to a floor/ceiling distribution, though neither candidate is ever assigned a probability of exactly 0 or 1.

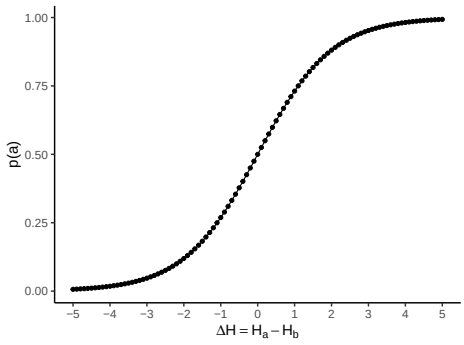


Figure 2.1: MaxEnt's predicted probability of candidate a (vertical axis) as a function of the Harmony difference between a and its only rival b.

MaxEnt probability mapping is not directly sensitive to the source of $\Delta\mathscr{H}$, i.e. which specific penalties the candidates incur. Thus, for instance, violating one constraint with the weight of x has the same effect on probability as violating two constraints with weights of $\frac{1}{2}x$. By way of illustration, the violation profiles of the b-candidates in (41) are different, but their $\Delta\mathscr{H}$s relative to the a-candidates, and the resulting MaxEnt-generated probability distributions, are the same.

(41) No sensitivity to the source of Harmony difference in MaxEnt

a. $p(b) = 0.62$

	$\mathbb{A}$ 2.0	$\mathbb{B}$ 1.5	$\mathscr{H}$	$e^{\mathscr{H}}$	p
a. Candidate 1	−1		−2.0	0.14	0.38
b. Candidate 2		−1	−1.5	0.22	0.62

b. $p(b) = 0.62$

	$\mathbb{A}$ 2.0	$\mathbb{C}$ 0.75	$\mathbb{D}$ 0.75	$\mathscr{H}$	$e^{\mathscr{H}}$	p
a. Candidate 3	−1			−2.0	0.14	0.38
b. Candidate 4		−1	−1	−1.5	0.22	0.62

c. $p(b) = 0.62$

	$\mathbb{A}$ 2.0	$\mathbb{E}$ 0.5	$\mathbb{F}$ 0.5	$\mathbb{G}$ 0.5	$\mathscr{H}$	$e^{\mathscr{H}}$	p
a. Candidate 5	−1				−2.0	0.14	0.38
b. Candidate 6		−1	−1	−1	−1.5	0.22	0.62

For this reason, violations of zero-weighted constraints contribute nothing to the candidate's Harmony, like in Classical HG (and unlike NHG; see §2.2.2). This is illustrated by the difference between (42a) and (42b): All else being equal, the added violation of the zero-weighted constraint $\mathbb{C}$ incurred by Candidate 4 in (42b) does not affect the probability distribution. In other words, zero-weighted constraints are completely switched off in MaxEnt.

(42) Zero-weighted constraints play no role in MaxEnt probability mapping

a. Zero-weighted constraint $\mathbb{C}$ inactive

	$\mathbb{A}$ 1	$\mathbb{B}$ 1	$\mathbb{C}$ 0	$\mathscr{H}$	$e^{\mathscr{H}}$	p
a. Candidate 1	−1			−1.0	0.37	0.50
b. Candidate 2		−1		−1.0	0.37	0.50

b. Zero-weighted constraint $\mathbb{C}$ violated: no effect on probability

	$\mathbb{A}$ 1	$\mathbb{B}$ 1	$\mathbb{C}$ 0	$\mathscr{H}$	$e^{\mathscr{H}}$	p
a. Candidate 3	-1			-1.0	0.37	0.50
b. Candidate 4		-1	-1	-1.0	0.37	0.50

On a final note, MaxEnt employs an analytic (i.e. closed-form) solution for probability estimation: the softmax function (39). The use of an analytic method distinguishes MaxEnt from Monte Carlo, simulation-based approaches to probability mapping (like OT-GLA and HG-GLA), in that closed-form solutions can never return a probability of exactly 0 or 1, since the softmax function of any finite value lies strictly between 0 and 1. The Monte Carlo method is based on a finite number of draws and can return exactly 0 or 1 if one candidate never (or always) wins across the simulated evaluations. This has consequences for replicability and goodness-of-fit comparison (I address the latter in §2.6).

2.2.2 Noisy Harmonic Grammar

Much like in Classical HG, NHG selects the most harmonic candidate as the winner at every evaluation. Stochasticity is introduced by perturbing the base weights, at each evaluation, with independent noise terms drawn from Gaussian distributions with mean $\mu = 0$ and standard deviation σ. The weighting conditions are altered across evaluations, which may lead to different winners.

Depending on the stage of evaluation when random noise is added, NHG comes in several varieties, which make distinct predictions (Hayes, 2017, 2021). In the present implementation, noise is added to base constraint weights before multiplication with the number of violations (for other options, see §2.5.2 and references therein). For every constraint k, I distinguish between its base weight w_k and its evaluation-specific

perturbed weight $\hat{w}_k$ (43).

(43) Perturbed weight

$$\hat{w}_k = w_k + n_k$$

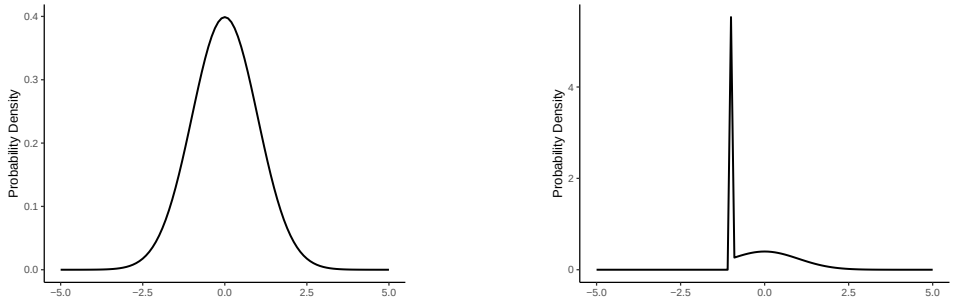
Harmony is calculated as the summation over constraint violations scaled by *perturbed* weights, as shown in (44).

(44) Harmony (NHG update)

$$\mathscr{H}_i = -\sum_{k \in C} \hat{w}_k c_{ik}$$

The presence of random terms n_k makes NHG's probability mapping more complex than MaxEnt's. Noise terms can be positive or negative, pushing the perturbed weight above or below the base value. Given the undesirable effects of negative weights (Keller, 2000; Potts et al., 2010), perturbed weights must be nonnegative. When random noise pushes $\hat{w}_k$ below zero, $\hat{w}_k$ is truncated to zero, which is known as CLIPPING (Magri, 2015; Boersma and Pater, 2016; Hayes and Kaplan, 2025). Variants of NHG that employ clipping are collectively referred to as censored NHG (cNHG). Unlike uncensored NHG, where noise terms are drawn from standard normal distributions (Figure 2.2a), in cNHG each n_k added to w_k is drawn from a normal distribution left-censored at $-w_k$ (Flemming, 2021; Figure 2.2b), which ensures that $\hat{w}_k$ cannot go below zero.

Given that each constraint weight is perturbed by its own independent Gaussian noise, NHG probability mapping is sensitive not only to the Harmony difference between candidates but also to the *source* of the Harmony difference. Thus, in contrast to MaxEnt (41), violating a single constraint with weight x will not have the same effect on probability in NHG as coincident violations of two constraints with weights $\frac{1}{2}x$ each.



(a) **(b)**

Figure 2.2: Probability density functions of **(a)** standard normal distribution ($\mu = 0$, $\sigma = 1$); **(b)** normal distribution ($\mu = 0$, $\sigma = 1$) left-censored at -1.

Example (45), the NHG counterpart of (41), puts this observation on a more concrete footing. In all tableaux in (45), the a- and b-candidates share the same fixed terms: -2 for the a-candidates and -1.5 for the b-candidates. The a-candidates have the same random term ($-n_{\mathbb{A}}$) in all tableaux, whereas the random terms of the b-candidates vary across tableaux: $-n_{\mathbb{B}}$ in (45a), $-(n_{\mathbb{C}}+n_{\mathbb{D}})$ in (45b), and $-(n_{\mathbb{E}}+n_{\mathbb{F}}+n_{\mathbb{G}})$ in (45c). This difference in random terms drives the variation in the probabilities assigned to the b-candidates. The more constraints a b-candidate violates, the lower its probability.[2]

(45) Sensitivity to violation profile in NHG

a. $p(b) = 0.64$

[2]Candidate probabilities are obtained from simulations in R. Random noise perturbation is simulated using the `rnorm()` function, which returns N draws (here, N = 1,000,000) from a Gaussian distribution. Censoring is ensured by feeding `rnorm()` into `pmax()`. Ties are resolved by randomly selecting one of the tied candidates as the winner (Boersma and Pater, 2016).

	$\mathbb{A}$	$\mathbb{B}$		
w_k	2.00	1.50		
n_k	$n_{\mathbb{A}}$	$n_{\mathbb{B}}$	$\mathscr{H}$	p
a. Candidate 1	-1		$-1(2.0 + n_{\mathbb{A}})$	0.36
b. Candidate 2		-1	$-1(1.5 + n_{\mathbb{B}})$	0.64

b. $p(b) = 0.58$

	$\mathbb{A}$	$\mathbb{C}$	$\mathbb{D}$		
w_k	2.00	0.75	0.75		
n_k	$n_{\mathbb{A}}$	$n_{\mathbb{C}}$	$n_{\mathbb{D}}$	$\mathscr{H}$	p
a. Candidate 1	-1			$-1(2.0 + n_{\mathbb{A}})$	0.42
b. Candidate 2		-1	-1	$-1(0.75 + n_{\mathbb{C}})$ $+ -1(0.75 + n_{\mathbb{D}})$	0.58

c. $p(b) = 0.50$

	$\mathbb{A}$	$\mathbb{E}$	$\mathbb{F}$	$\mathbb{G}$		
w_k	2.0	0.50	0.50	0.50		
n_k	$n_{\mathbb{A}}$	$n_{\mathbb{E}}$	$n_{\mathbb{F}}$	$n_{\mathbb{G}}$	$\mathscr{H}$	p
a. Candidate 1	-1				$-1(2 + n_{\mathbb{A}})$	0.50
b. Candidate 2		-1	-1	-1	$-1(0.5 + n_{\mathbb{E}})+$ $-1(0.5 + n_{\mathbb{F}})+$ $-1(0.5 + n_{\mathbb{G}})$	0.50

Another consequence of random terms in NHG is the sensitivity to violations of zero-weighted constraints. In Classical HG and MaxEnt, zero-weighted constraints are completely turned off, as (42) shows. In (c)NHG, by contrast, zero-weighted constraints affect candidate probability because their perturbed weights can be positive (Flemming, 2017, 2021; Hayes and Kaplan, 2025; Magri and Anttila, 2026).[3] This is shown by the doublet in (46), which mirrors the MaxEnt tableaux in (42). The violation of the zero-

[3]For $w_k = 0$, $\hat{w}_k > 0$ in one-half of evaluations; see Chapter 4 for the calculations.

weighted constraint $\mathbb{C}$ incurred by Candidate 4 in (46b) pushes its predicted probability well below 50%.

(46) Zero-weighted constraints affect probability in cNHG

a. Zero-weighted constraint $\mathbb{C}$ inactive

	$\mathbb{A}$	$\mathbb{B}$	$\mathbb{C}$		
w_k	1	1	0		
n_k	$n_{\mathbb{A}}$	$n_{\mathbb{B}}$	$n_{\mathbb{C}}$	$\mathscr{H}$	p
a. Candidate 1	-1			$-(1+n_{\mathbb{A}})$	0.50
b. Candidate 2		-1		$-(1+n_{\mathbb{B}})$	0.50

b. Zero-weighted constraint $\mathbb{C}$ active; affects probability

	$\mathbb{A}$	$\mathbb{B}$	$\mathbb{C}$		
w_k	1	1	0		
n_k	$n_{\mathbb{A}}$	$n_{\mathbb{B}}$	$n_{\mathbb{C}}$	$\mathscr{H}$	p
a. Candidate 3	-1			$-(1+n_{\mathbb{A}})$	0.61
b. Candidate 4		-1	-1	$-(1+n_{\mathbb{B}})$	0.39

Thus, in (c)NHG, every constraint in the grammar affects candidate probability. This property sets (c)NHG apart from other variants of HG, whose Harmony calculations involve no random noise component. In this regard, (c)NHG patterns with strict-ranking OT, where there is likewise no absolute switch-off: No constraint is ever turned off entirely, regardless of its ranking, and each constraint affects the output when all higher-ranked constraints are inactive (McCarthy and Prince, 1994; McCarthy, 2008, 26). Magri and Anttila (2026) argue that the absence of absolute switch-off in NHG is a desirable property (though without empirically substantiating the claim), whereas Flemming (2021) argues that NHG's sensitivity to candidates' violation profiles (as in 45) is falsified by variable schwa deletion/insertion in French. At any rate, NHG's sensitivity to candidates' violation profiles constitutes a testable prediction that could be

used to adjudicate between NHG and MaxEnt, and should therefore be tested against more copious data.

2.3 Stochastic Optimality Theory

StOT is the most widely used variant of OT for modeling free variation (foundational work includes Boersma and Hayes, 2001; see Ernestus and Baayen, 2003; Clark, 2004; Jäger and Rosenbach, 2006; Hayes and Londe, 2006; Zuraw, 2010, a.o. for applications, and Keller and Asudeh, 2002; Keller, 2006; Jäger and Rosenbach, 2006; Zuraw and Hayes, 2017; Smith and Pater, 2020; Kawahara, 2020 for criticism).[4] StOT retains much of Classical OT's apparatus, including ranked rather than weighted constraints and output selection based on the highest-ranking constraint(s) rather than all constraints. The most important departure of StOT from Classical OT is that constraint rankings are variable rather than fixed. Each constraint pair $\mathbb{A}$, $\mathbb{B}$ is assigned a ranking probability, $p(\mathbb{A} \gg \mathbb{B})$, which encodes the strictness of their ranking, i.e. whether one constraint nearly categorically dominates the other or their ranking fluctuates, and to what extent. The Hasse diagram in Figure 2.3a depicts a Classical OT-style fixed hierarchy in which $\mathbb{A} \gg \{\mathbb{B}, \mathbb{C}\} \gg \mathbb{D}$. Diagrammed in Figure 2.3b is a stochastic constraint hierarchy, where each pairwise ranking is assigned a certain probability.

In Classical OT (2.3a), $\mathbb{A}$ invariably dominates $\mathbb{B}$ following the fixed ranking $\mathbb{A} \gg \mathbb{B}$. The StOT grammar in Figure 2.3b assigns 64% probability to $\mathbb{A} \gg \mathbb{B}$, meaning that the weaker constraint $\mathbb{B}$ still outranks $\mathbb{A}$ 36% of the time. Further, $\mathbb{B}$ and $\mathbb{C}$ occupy the same ranking stratum in both grammars, but that means different things in deterministic and StOT. In deterministic OT, $\mathbb{B}$ and $\mathbb{C}$ are unranked, meaning that the OT learner

[4]Other prominent approaches to variation within OT (not considered in this dissertation) include partially ordered constraints (Anttila, 1997) and multiple grammars theory (Anttila, 2002, 2006, 2012).

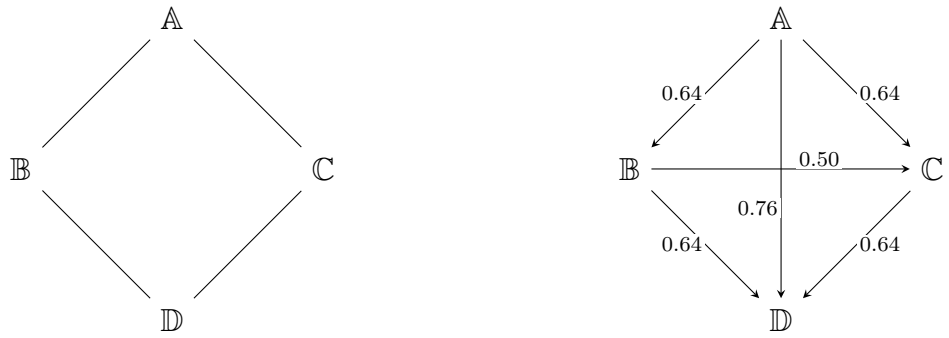


Figure 2.3: Ranking hierarchies in (**a**) deterministic OT and (**b**) StOT. In the StOT diagram, arrow direction encodes pairwise dominance relations, e.g., $\mathbb{A} \xrightarrow{0.64} \mathbb{B}$: $p(\mathbb{A} \gg \mathbb{B}) = 0.64$.

could not posit any crucial ranking between the two constraints. In StOT, $\mathbb{B} \gg \mathbb{C}$ has the probability of 50%, i.e. both possible rankings are equally likely.

Ranking probabilities are derived from constraints' ranking values. In StOT, each constraint k is assigned ranking value r_k as an approximation of its strength in the grammar. Ranking values are learned from variable input data using the Gradual Learning Algorithm (GLA; Boersma, 1997, 1998).[5] The grammar is made stochastic by perturbing each base ranking value r_k with independent noise n_k, a random normal variable with the mean $\mu = 0$ and standard deviation σ, as shown in (47). Throughout, unless stated otherwise, $\sigma = 2$ for StOT, which is the default OT-Soft setting (Hayes et al., 2013; Zuraw and Hayes, 2017).

(47) Perturbed ranking value

$$\hat{r}_k = r_k + n_k$$

At each evaluation, perturbed ranking values are converted into classical OT-style fixed rankings. Due to random noise perturbation, constraint rankings vary across

[5]See Pater (2008); Magri (2012); Magri and Storme (2020) for convergence properties of the GLA for OT.

evaluations, which potentially leads to different rankings at different evaluation times. Different rankings may in turn produce variable output across evaluations. Constraint $\mathbb{A}$ outranks $\mathbb{B}$ if $\mathbb{A}$'s perturbed ranking value is higher than $\mathbb{B}$'s at a given evaluation time:

(48) $\mathbb{A} \gg \mathbb{B}$ whenever $\hat{r}_{\mathbb{A}} > \hat{r}_{\mathbb{B}}$

The perturbed ranking value of $\mathbb{A}$ is higher than $\mathbb{B}$'s whenever the difference of noise terms applied to the two constraints, $n_{\mathbb{A}} - n_{\mathbb{B}}$, is greater than the negative of the difference of their fixed terms:

(49) $\mathbb{A} \gg \mathbb{B}$ whenever $n_{\mathbb{A}} - n_{\mathbb{B}} > -(r_{\mathbb{A}} - r_{\mathbb{B}})$

The pairwise ranking probability, $p(\mathbb{A} \gg \mathbb{B})$, is effectively the probability that (49) holds. Given that the individual constraints' noise terms are independent, the variance of the noise difference variable $n_{\mathbb{A}} - n_{\mathbb{B}}$ is the sum of variances of $n_{\mathbb{A}}$ and $n_{\mathbb{B}}$, $2\sigma^2$. The probability that $\mathbb{A}$ outranks $\mathbb{B}$ is determined by the normal cumulative distribution function, Φ (R; `pnorm()`), of their base ranking difference, $r_{\mathbb{A}} - r_{\mathbb{B}}$, divided by the standard deviation of the noise difference variable, $n_{\mathbb{A}} - n_{\mathbb{B}}$. The standard deviation of this random variable equals the square root of its variance: $\sqrt{2\sigma^2} = \sqrt{2}\sigma$.

(50) Pairwise ranking probability ($\mathbb{A}$, $\mathbb{B}$)

$$p(\mathbb{A} \gg \mathbb{B}) = \Phi\left(\frac{r_{\mathbb{A}} - r_{\mathbb{B}}}{\sqrt{2}\sigma}\right)$$

A possible ranking vector that produces the pairwise ranking probabilities in Figure 2.3b is $r_{\mathbb{A}} = 102$, $r_{\mathbb{B}} = 101$, $r_{\mathbb{C}} = 101$, $r_{\mathbb{D}} = 100$.[6]

As formalized in (49), the stronger constraint $\mathbb{A}$ in (51) dominates the antagonistic constraint $\mathbb{B}$ unless noise perturbation reverses the base ranking. By (50), for $r_{\mathbb{A}} - r_{\mathbb{B}} = 1$ and $\sigma = 2$, $\mathbb{A}$ outranks $\mathbb{B}$ in about 64% of evaluations.

[6]The absolute ranking values are not important; here they are kept close to OT-Soft's initial value of 100. What matters are the pairwise ranking differences between constraints.

(51) Predicted candidate probabilities for $r_{\mathbb{A}} - r_{\mathbb{B}} = 1$, $\sigma = 2$

	$\mathbb{A}$	$\mathbb{B}$	
r_k	102	101	
n_k	$n_{\mathbb{A}}$	$n_{\mathbb{B}}$	p
a. Candidate 1		$*$	0.64
b. Candidate 2	$*$		0.36

A point worth highlighting is that (50) is the closed-form expression for determining ranking probabilities from ranking values. The OT-Soft implementation of StOT evaluation uses the Monte Carlo procedure, whereby base ranking values are perturbed by noise drawn a finite number of times (usually 1,000,000) from a Gaussian distribution (Zuraw and Hayes, 2017, 511). In practice, analytic and Monte Carlo methods produce virtually indistinguishable results (see Appendix B).

2.4 Model selection

I introduced eight constraints in Chapter 1. These constitute the baseline model of poetic grammar. Ideally, a grammar model should include all and only the constraints that actively contribute to meter (that is, affect poetic word order), which raises two questions: Can we explain the patterns of word-order variation in BCMS folk meter with fewer constraints than currently assumed without significant information loss? And does the baseline set contain all constraints needed to capture these patterns, or is recourse needed to additional constraints?

The model selection procedure implemented here is rooted in the MINIMUM DESCRIPTION LENGTH principle (Rissanen, 1978; Grünwald, 2007), which dictates that added model complexity is justified if and only if the addition of a parameter (here, constraint) sufficiently improves model fit. The goal is to find the most parsimonious

model (in terms of the number of constraints) that minimizes information loss. As an approximation to the minimum description length principle, I use the Bayesian Information Criterion (BIC; Schwarz, 1978), which trades off model fit against complexity (52).

(52) $\text{BIC} = k\ \ln(n) - 2\ \ln(\hat{L})$

where:

k = number of constraints

$\ln(n)$ = log-transformed total number of observations

$\ln(\hat{L})$ = model's log-likelihood

The log-likelihood of a model measures how well the probabilities predicted by the model match the observed data (Hastie et al., 2009, 120). The values are nonpositive. Higher log-likelihood indicates better fit. The formula in (53) applies to the two-class (i.e. binomial) case. This is relevant for the present modeling setup because the dependent variable, word-order choice, is binary: Poets choose between adherence to the canonical order and deviation from it.

(53) $$\ln(\hat{L}) = \sum_{i=1}^{N} \Big[y_i \ln(\hat{p}_i) + (n_i - y_i) \ln(1 - \hat{p}_i) \Big]$$

where:

N = number of conditions (tableaux)

y_i = observed token count of a candidate (i.e. word order)

n_i = observed by-condition token count (canonical order + inversion)

$\hat{p}_i$ = candidate probability predicted by the model

Other criteria used for model selection in the MaxEnt literature include likelihood ratio test (LRT; Hayes et al., 2012; Breiss and Hayes, 2020) and the second-order Akaike Information Criterion (AIC_c; Shih, 2017a; Henriksson, 2022). BIC is a more conservative inclusion criterion for constraints than AIC_c, in that it applies a stronger

penalty for model complexity (Burnham and Anderson, 2004). Lower BIC indicates a better model. The rule of thumb is that the BIC change that results from dropping a single constraint, ΔBIC, should be higher than 2 to justify retaining that constraint. $\Delta\text{BIC} > 10$ constitutes very strong evidence for retention (Raftery, 1995).

Constraint selection is conducted in MaxEnt. The MaxEnt implementation in the `maxent.ot` R package facilitates model selection via BIC. I use the MaxEnt-selected constraint set for all stochastic grammars: MaxEnt, cNHG, and StOT.

2.4.1 Are all baseline constraints justified?

I compare the baseline, superset model ($k = 8$) with subset models that each omit a single constraint ($k = 7$). These are nested models in that each subset model is a proper subset of the baseline, differing only in the absence of one constraint. This procedure tests whether dropping a given constraint worsens model fit, isolating each constraint's individual contribution. A constraint is retained when dropping it results in a sufficiently higher BIC: $\Delta\text{BIC} = \text{BIC}_{\text{subset}} - \text{BIC}_{\text{superset}} > 2$. The results are shown in Figure 2.4.

The BIC comparison finds very strong evidence (in the sense of Raftery, 1995) for seven out of eight baseline constraints: Dropping any of these leads to an immense BIC increase, indicative of much poorer model fit. The only exception is VerbFirst, whose omission yields the lowest BIC of all models compared ($\Delta\text{BIC}_{\text{no VerbFirst} - \text{full model}} = -9.55$). Although the superset model has a slightly higher log-likelihood than the more parsimonious model without VerbFirst (Δlog-likelihood = 0.04), the gain is too small to justify the added complexity.

The outcome of MaxEnt model selection accords with the regression analysis in §1.4: All phonological constraints, along with the linearization constraints on adjective–noun and noun–possessive order, are robustly supported. I therefore retain all baseline

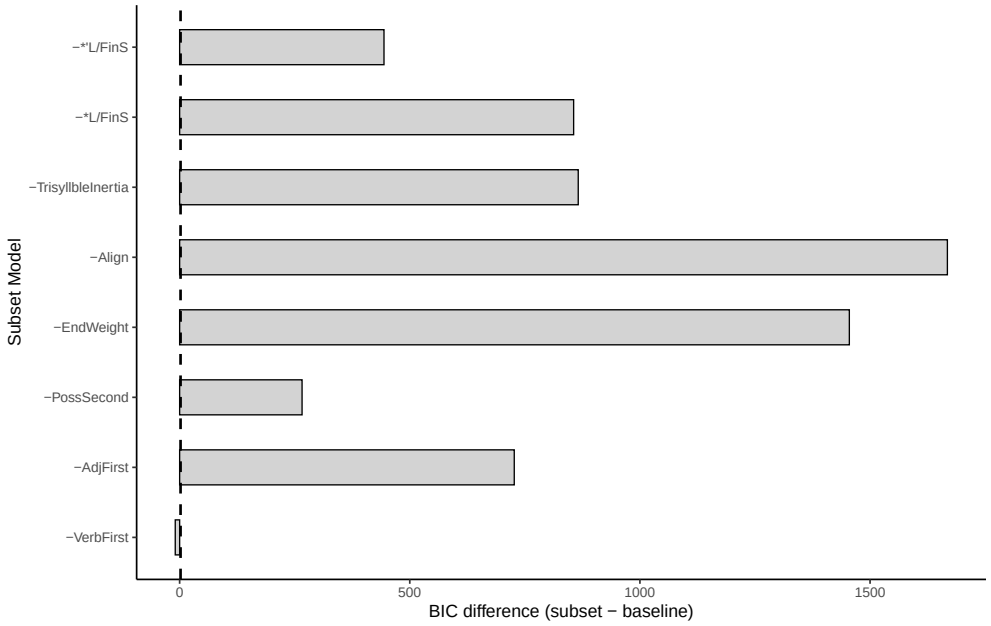


Figure 2.4: Model comparison (BIC change; baseline model BIC = 9,686). The dashed vertical line represents the inclusion threshold. A constraint is supported when BIC difference > threshold. VERBFIRST is the only constraint that remains below the inclusion threshold.

constraints except VERBFIRST. I refer to the resulting model as the *updated baseline model*.

The exclusion of VERBFIRST does not necessarily imply that BCMS folk poets have no preferred verb-object order. The more appropriate interpretation is that the corpus data do not provide sufficient evidence for including a dedicated linearization constraint that regulates verb-object order in meter.

2.4.2 Local conjunction and ban-only-the-worst-of-the-worst

Having excluded VERBFIRST from the constraint set, I now turn to the question of whether additional constraints may be needed in the grammar. Since constraints interact cumulatively, a reasonable starting point is to test the utility of interaction terms, known in constraint-based phonology as LOCAL CONJUNCTIONS (LC; Smolensky, 1993,

2006).[7] For a constraint pair $\mathbb{A}$, $\mathbb{B}$, their LC $\mathbb{A}$ & $\mathbb{B}$ assigns a violation whenever $\mathbb{A}$ and $\mathbb{B}$ are simultaneously violated in a designated local domain (here, verse line). MaxEnt grammars produce a wide range of cumulativity effects (linear and non-linear) with no recourse to LC (Breiss and Albright, 2022). The role of LC in MaxEnt, and HG more generally, is therefore different from strict-ranking OT, where LCs are used to accommodate all instances of ganging-up cumulativity (Smolensky, 2006; Pater, 2009b). In HG, LCs amplify the severity of coincident constraint violation, being particularly helpful in cases of extreme superlinearity, where the effect of simultaneously violating multiple constraints is much more severe than the combination of their independent effects (Green and Davis, 2014; Shih, 2017a; Milenković, 2026).[8]

Hayes (2021, 38) notes that liberal use of LC in MaxEnt undermines the "possibility of making strict predictions about quantitative signatures." Just as the simplex constraints used here are statistically corroborated and have precedent in the literature, so must their LCs be. One plausible conjunction is End-Weight & *Light/Final Strong, defined in (54).

(54) End-Weight & *Light/FinalStrong

Assign a violation for every verse line that violates both End-Weight and *Light/FinalStrong.

Verse lines may violate either End-Weight or *Light/FinalStrong, but the coincident violations of these independent constraints are robustly avoided. That these two independent metrical factors are interrelated has been known for more than a century. For example, Maretić (1907, 59) reports that folk poets disprefer placing light-initial disyllables (like *polje* ['pɔ.ʎɛ] 'field') in line-final position. Short words

[7]Interaction term is a term used in the context of regression analysis, whereas LC is used in OT and HG. Given the conceptual equivalence of the two terms, I use them interchangeably throughout.

[8]See Chapter 3 for a more detailed overview of the role of LC in OT and HG.

with a light penult violate *Light/FinalStrong and, when preceded by a longer word, End-Weight as well.[9]

The avoidance of doubly marked verse lines is a typical ban-only-the-worst-of-the-worst pattern (BOWOW; Smolensky, 2006; Legendre et al., 2006; Kager, 1999, 400; McCarthy, 2002, 18), whereby marked phonological structures are permitted in isolation but cannot co-occur. Similar patterns exist outside phonology, where this phenomenon is known as the 3/4 signature (Bobaljik and Wurmbrand, 2012). The key finding is that the effect of simultaneously violating End-Weight and *Light/FinalStrong is so strong that base MaxEnt cannot capture it adequately by means of its built-in additivity of constraint violations.

Given the independent violation rates of End-Weight or *Light/FinalStrong in the corpus, their coincident violations are severely underrepresented, as diagrammed in Figure 2.5.[10] About 21.5% of lines violate End-Weight (20.2% that violate only End-Weight, plus the 1.3% that also violate *Light/FinalStrong), and 45.4% of lines violate *Light/FinalStrong (44.1% alone, plus the same 1.3% that also violate End-Weight). Only 1.3% of lines simultaneously violate both constraints. Assuming independence, the chance rate of coincident violation is $0.215 \times 0.454 \approx 9.8\%$. The interaction is therefore superlinear (Legendre et al., 2006; Smith and Pater, 2020; Breiss and Albright, 2022), and rather extremely so: Violating End-Weight and

[9]This avoidance is partly attributable to *StressedLight/FinalStrong, since initial syllables of disyllabic words are invariably stressed in BCMS due to nonfinality (Daničić, 1925; Nikolić, 1970; Lehiste and Ivić, 1986; Inkelas and Zec, 1988), such that line-final light-initial disyllables supply a stressed light to the final strong position. As we shall see momentarily, however, the conjunction of End-Weight and *Light/FinalStrong is still needed to capture the robustness of the avoidance effect.

[10]The violation-count method is not completely nuanced, as it fails to distinguish between cases where constraint violations are incurred but avoidable, and those where violations are unavoidable (MaxEnt/logistic models are sensitive to this crucial distinction). However, Figure 2.5 is a convenient illustration of the extreme underrepresentation of the verse lines that violate both End-Weight and *Light/FinalStrong.

*Light/FinalStrong has a much more detrimental effect on line frequency than the combination of their independent contributions.

	End-W satisfied	End-W violated
*L/FinS satisfied	5,290 (34.4%)	3,112 (20.2%)
*L/FinS violated	6,792 (44.1%)	**203 (1.3%)**

Figure 2.5: BOWOW diagram: Independent violations of End-Weight and *Light/FinalStrong are tolerable and common, but their coincident violations are extremely rare.

I test the utility of End-Weight & *Light/FinalStrong using the selection procedure outlined in §2.4.1. I adopt the standard inclusion threshold: The LC model's BIC must be lower than the BIC of the updated baseline model by at least 2 units ($\Delta\text{BIC} = \text{BIC}_{\text{LC}} - \text{BIC}_{\text{updated baseline}} < -2$). The superset model has a substantially lower BIC ($\Delta\text{BIC}_{\text{LC - updated baseline}} = -171$), which strongly supports the inclusion of the conjoined constraint. The likelihood ratio test likewise indicates that the conjoined constraint significantly improves model fit: $\chi^2(1) = 181$, $p = 0.000$. Inclusion is further supported by the high weights assigned to the conjunction in the MaxEnt and cNHG models (see §2.7.1).

In sum, BCMS folk meter displays a robust BOWOW pattern. Verse lines whose final words are syllabically short (e.g., disyllables) and have a light penult are strongly avoided because they violate two important metrical constraints: End-Weight and *Light/FinalStrong. The grammar with simplex constraints only cannot adequately capture the magnitude of this interaction effect. This BOWOW pattern necessitates an interaction term even in MaxEnt, which otherwise models a wide range of nonlinear cumulative interactions (Breiss and Albright, 2022). This finding adds to

the growing evidence for LCs in MaxEnt and HG more generally (Hayes et al., 2012; Green and Davis, 2014; Shih, 2016, 2017a; Milenković, 2026).

2.5 Modeling

In this section, I describe the simulation setup for the MaxEnt (§2.5.1), cNHG (§2.5.2), and StOT models (§2.5.3).

2.5.1 MaxEnt

The MaxEnt model is fitted using the `maxent.ot` R package (Mayer et al., 2024). Weight estimation is based on Maximum Likelihood (ML; Myung, 2003; Jurafsky and Martin, 2026). The `optimize_weights` function uses the L-BFGS-B algorithm (Byrd et al., 1995; Zhu et al., 1997) to find weights that maximize the likelihood of the observed data given the grammar model.

Unless stated otherwise, all MaxEnt models in this dissertation use L2 regularization with weak Gaussian priors ($\mu = 0$, $\sigma = 100$) on all constraint weights. The regularization term biases the optimization towards zero weights, but the high standard deviation allows significant departures from zero when the data warrant it. Regularization prevents overfitting (Goldwater and Johnson, 2003; Jurafsky and Martin, 2026), though in this specific case, the effect of setting priors is negligible, since the weights of the unregularized model differ from the current model's weights at the sixth decimal place.[11]

Once weights are learned, the package's `predict_probabilities()` function calculates candidate probabilities for the input data. MaxEnt uses the closed-form ex-

[11]Other purposes of regularization terms in MaxEnt modeling include simulating learner biases (Wilson, 2006; Hayes and White, 2015) and capturing the baseline prose effects of metrical constraints (Henriksson, 2022).

pression for probability mapping outlined in §2.2.1.

2.5.2 cNHG

I use two methods for weight search for cNHG. The standard method includes learning simulations using the GLA, which is guaranteed to converge for HG (Boersma and Pater (2016) provide a convergence proof for HG-GLA). Following Flemming (2021), the other method for cNHG weight search uses ML as the optimization criterion, implemented using the Nelder-Mead algorithm (Nelder and Mead, 1965). The latter method is similar in spirit to MaxEnt's weight search, although the optimization procedures differ in practice (L-BFGS-B in MaxEnt vs. Nelder-Mead in ML-cNHG; L2 regularization in MaxEnt vs. no regularization in ML-cNHG).

Given that MaxEnt and GLA-NHG use different methods for finding weights, any comparison of model predictions is confounded by the contribution of the learning algorithms. Using ML as a weight search method in cNHG removes the contribution of the learning algorithm, which makes it possible to more directly compare the effects of grammar architecture (Flemming, 2021, 21*f.*). I still report the GLA-cNHG results, to get a sense of how different the GLA and ML implementations are and what the GLA contributes to the MaxEnt-NHG differences.

The differences between the GLA-cNHG and ML-cNHG pipelines concern only weight search. Once base weights are arrived at, the evaluation, i.e. probability mapping mechanism is identical for GLA-cNHG and ML-cNHG (as described in §2.2): Base weights are perturbed by random noise and negative weights are truncated to zero, candidate probabilities are estimated over 1,000,000 evaluation trials, and random selection is employed when two candidates are tied.

For the GLA method, cNHG simulations are conducted in OT-Soft (Hayes et al., 2013) using the default settings of the learner parameters. The learner passes through

the training data 999 times, using the GLA to find base weights. All initial weights are set to zero. Plasticity, which determines how much the grammar adjusts to new evidence during learning, starts at 2.00 and decreases to 0.01 over the course of learning. The learned grammar is evaluated over 1,000,000 trials. At each evaluation, the base weights are perturbed by independent noise drawn from a censored Gaussian distribution ($\mu = 0$, $\sigma = 1$). Noise is added to base weights, not to individual cells in a tableau (as in Goldrick and Daland, 2009). Constraint weights are perturbed by random noise before multiplication by violation tallies, although this choice makes no practical difference in the present setup, given that no constraint is ever violated more than once (see Hayes, 2017, 2021 for the differences between pre- and post-multiplication noise).

The GLA-NHG pipeline introduces randomness at two stages: (i) random sampling of training data, and (ii) random noise perturbation of learned base weights at the evaluation phase. Thus, as implemented in OT-Soft, repeated GLA-cNHG runs converge on different grammars for the same input data, which may produce different probability distributions.

Existing studies (e.g., Zuraw and Hayes, 2017; Smith and Pater, 2020) usually report results from individual GLA runs in OT-Soft or Praat (Boersma and Weenink, 2022). Given the doubly random nature of the GLA-cNHG pipeline, individual GLA-cNHG runs may differ considerably in their estimates of constraint weights fit to the data. To get a more robust sense of how stable parameter estimates and generated probability distributions are, I conduct 100 GLA-cNHG runs in OT-Soft (each involving 999 training and 1,000,000 evaluation trials, as outlined) and use the best-fitting cNHG model for between-framework comparison. Although the "best" model is not guaranteed to be the theoretically optimal model, it represents the best available approximation given GLA-cNHG's non-determinism. Following Flemming (2021), I use residual deviance

(McCullagh and Nelder, 1989; Agresti, 2013) to identify the best-fitting cNHG model.[12] Residual deviance measures the model's departure from the saturated model, i.e. the hypothetical "perfect" model in which every fitted candidate probability corresponds to its observed probability (55). Lower deviance indicates better fit.

(55) Residual deviance

$$\text{Deviance} = -2\sum_{i} N_i[\log(\hat{p}_i) - \log(p_i)]$$

where:

N_i = observed count of candidate i

$\hat{p}_i$ = model's fitted probability of candidate i

p_i = observed probability of candidate i (and saturated model's fitted probability)

Figure 2.6 shows residual deviance for all 100 GLA-cNHG models. The residual deviance of the best-fitting model is 1,571. The distribution clusters towards lower deviances.

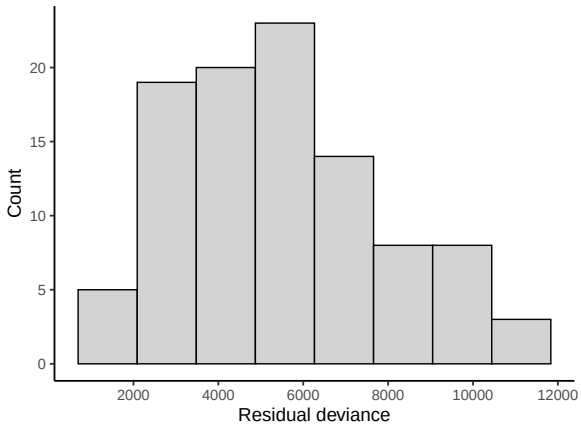


Figure 2.6: Histogram of residual deviances of 100 GLA-cNHG models.

The second method for cNHG weight search is based on Flemming's (2021, 22) ML

[12]Using error metrics such as MAE in lieu of deviance yields the same results.

implementation. Like MaxEnt, ML-cNHG searches for constraint weights that maximize the likelihood of observed data, i.e. minimize deviance. cNHG has no closed-form expression for candidate probabilities, so deviance must be estimated via simulation for every weight vector considered during the search. For a given weight vector, candidate probabilities are estimated from 1,000,000 simulated evaluation trials (which, given the size of the dataset, makes the procedure extremely time-consuming).[13] The fitted probabilities at each iteration are used to calculate deviance. The Nelder-Mead algorithm, implemented in R's `optim()` function, uses deviance to propose the weight vector to be considered at the next iteration. The procedure continues until it converges on the weight vector that minimizes deviance, but no more than 1,000 iterations may be conducted. The starting weights for the cNHG optimization are weights from uncensored NHG found by the L-BFGS-B algorithm, also implemented in the `optim()` function in R.[14] The starting weights for the uncensored NHG optimization are set to 1 for all constraints.

Given that the Nelder-Mead optimization does not impose a lower bound on weights, *base* weights can be negative. Whereas base cNHG weights from the Nelder-Mead optimization are not left-capped at zero, perturbed weights cannot go below zero (Magri, 2015; Boersma and Pater, 2016; Hayes, 2017; Hayes and Kaplan, 2025). Clipping ensures that cNHG eschews the well-known adverse effects of negative weights in HG (Keller, 2000; Potts et al., 2010). A negative base weight in cNHG with clipping simply means that the resulting perturbed weight will be zero (by clipping) in more than half of evaluation trials. For example, for a base weight of -1, the perturbed weight will be positive in only 16% of evaluations (and zero in the other 84%).

[13]Lowering the number of evaluation trials per weight vector makes the algorithm converge faster (I verified this with 10,000 trials), but the resulting weights provide a slightly inferior fit to the setup with 1,000,000 trials.

[14]Random seed is set for replicability before cNHG optimization begins.

The present approach departs from Flemming's original implementation in how ties between candidates are resolved during evaluation. Flemming's `cNHG()` function, which estimates candidate probabilities for a set of candidate violations and constraint weights given a cNHG grammar, uses the R function `which.max()` to identify the winning candidate in each evaluation trial. With multiple maxima, `which.max()` returns the position of the first maximum value in a vector. Consider:

```
1 > which.max(c(7,7,8,5)) # returns the position of the maximum value
2 [1] 3
3 > which.max(c(7,8,8,5)) # in case of multiple maxima, returns the
      position of the first maximum
4 [1] 2
```

When two or more candidates have identical Harmony scores, the winner is always whichever one happens to come first in the tableau. The `which.max()` selection method does not correspond to any known tie resolution strategy in NHG (Boersma and Pater, 2016; Hayes and Kaplan, 2025; Kaplan, 2025b). Given the crucial relevance of tie resolution for harmonic bounding (see Chapter 4), I replaced Flemming's `which.max()` with a function that distributes probability equally among all tied candidates, mimicking random selection (Boersma and Pater, 2016).

I then use the ML weights to estimate the fitted probabilities of the final model over 1,000,000 evaluation trials (analogous to the OT-Soft evaluation procedure). The deviance of the final ML-cNHG model is 870, down from the best GLA-cNHG model's 1,571, though still worse than MaxEnt's 611.[15]

[15]This comparison is just meant to be orientational. One should approach cross-framework deviance comparisons with utmost caution, for reasons discussed in §2.6.

2.5.3 StOT

For StOT, the procedure is identical to the GLA-cNHG setup. I use OT-Soft to conduct OT-GLA learning simulations. All learner parameters are set to their default values. All constraints have an initial ranking value of 100. Plasticity decreases from 2.00 to 0.01 over the course of learning. The learner passes through the training data 999 times. The simulations employ Magri (2012)'s update rule, which was designed to resolve the convergence issues of Boersma (1997, 1998)'s OT-GLA identified by Pater (2008).[16] The learned grammar is evaluated over 1,000,000 cycles. The standard deviation of random Gaussian noise is set to 2.

Again, given that individual OT-GLA runs on the same data produce different ranking values, I conduct 100 independent OT-GLA runs in OT-Soft with the same input dataset and select the best-fitting model based on residual deviance (55). Figure 2.7 provides a histogram of residual deviances of all 100 StOT models. The residual deviance of the best-fitting model is 21,202. (See §2.6 on why StOT's deviance is much higher than MaxEnt's and NHG's.)

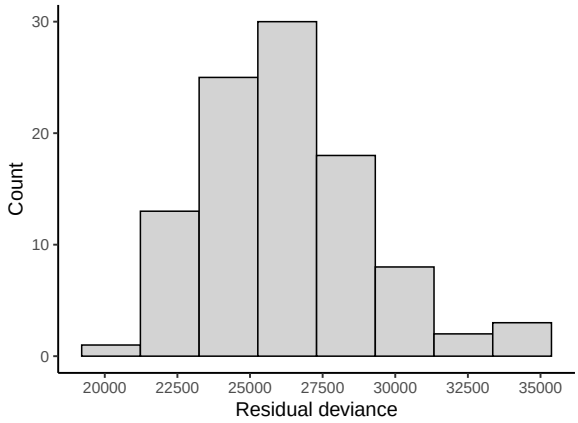


Figure 2.7: Residual deviances for 100 StOT models

[16] Although see Magri and Storme (2020) for some remaining limitations of Magri (2012)'s update rule.

2.6 Goodness of fit and model comparison

Before proceeding to the discussion of model performance, I briefly introduce the goodness-of-fit metrics used throughout this dissertation. I employ two types of goodness-of-fit metrics. The first includes error-based metrics, which quantify the discrepancy between the observed probabilities and the probabilities predicted by a grammar. I report the mean absolute error (MAE) and the root-mean-square error (RMSE). Both metrics are weighted by the number of observations per tableau, which reduces the contribution of large prediction errors in conditions with sparse data. The weighted MAE (56) measures the average absolute deviation between observed and predicted probabilities, whereas the weighted RMSE (57) squares prediction errors before averaging, thereby being more sensitive to large deviations. Both measures are non-negative. Smaller error indicates better fit.

(56) Weighted MAE

$$MAE = \frac{\sum_{i=1}^{n} w_i |\hat{p}_i - p_i|}{\sum_{i=1}^{n} w_i}$$

where:

n = total number of tableaux,

w_i = total number of observations per tableau,

$\hat{p}_i$ = fitted probability of the selected candidate in tableau i,

p_i = observed probability of the selected candidate in tableau i

(57) Weighted RMSE

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} w_i (\hat{p}_i - p_i)^2}{\sum_{i=1}^{n} w_i}}$$

The error metrics are computed using $n - 1$ free parameters per condition. In each

condition, the predicted probabilities of the candidates are complementary and sum to one, such that prediction errors are also complementary. For example, if a grammar overestimates the probability of inversion by 5%, it underestimates the probability of the canonical order by 5%. To avoid counting the same prediction error twice, I compute the error metrics using only the second candidate in each tableau.

I also report the squared Pearson correlation between observed and predicted probabilities (r^2). Like MAE and RMSE, the correlation is weighted by the sample size per tableau. r^2 ranges from 0 (no linear relationship) to 1 (perfect correspondence). Higher r^2 indicates closer correspondence between the observed and predicted probabilities.

The second type of goodness-of-fit metrics includes likelihood-based metrics: the residual deviance (§2.5.2) and BIC (§2.4). Likelihood-based measures are used only for within-framework model comparison (e.g., for selecting the best OT-GLA/HG-GLA grammar among multiple optimization runs, or for comparing alternative MaxEnt models). I do not use likelihood-based metrics for between-framework comparisons (StOT vs. cNHG vs. MaxEnt), mainly because stochastic grammars use different procedures to generate probabilities. Frameworks that employ analytic solutions (like MaxEnt) can never produce a probability of zero, because their probability functions are strictly positive. Frameworks that estimate probability over a finite number of evaluation trials (cNHG, StOT) can produce zero probabilities, since some candidates may simply never win during the simulation. This difference has direct ramifications for likelihood-based evaluation. If a model assigns a probability of zero to an observed candidate, its log-likelihood is negative infinity (or, in practice, it incurs an extremely large penalty). Likelihood-based metrics are therefore extremely sensitive to zero probabilities. For this reason, probabilistic modeling employs smoothing techniques that avoid assigning zero probability to attested events (Jurafsky and Martin, 2026, 51). Given that simulation-based grammars may produce zero probabilities whereas MaxEnt cannot,

likelihood-based metrics inherently favor MaxEnt in cross-framework comparisons.

StOT is disproportionately affected by absolute zeros, which explains the large deviance gap between StOT and cNHG/MaxEnt (§§2.5.2–2.5.3). StOT invariably assigns zero probability to harmonically bounded candidates (Keller and Asudeh, 2002; Boersma, 2004), many of which are attested in the corpus, so that these observations incur extremely large log-likelihood penalties. For example, the phonologically unmotivated non-canonical order of [ˈguːja ˈʃuːta] 'snake angry' is attested in 85 out of 418 relevant verse lines (20%); see §4.1.1. Being harmonically bounded, this candidate always loses out to the canonical order in StOT. This datum alone incurs −2,349 log-likelihood, half of the entire MaxEnt model's log-likelihood (−4,714), and it is just one of several such conditions in the data. This fact explains the extremely large deviances of the StOT models shown in Figure 2.7. MaxEnt never assigns zero probability to any candidate, and NHG, as implemented here,[17] need not assign zero probability to simply harmonically bounded candidates (Hayes and Kaplan, 2025; Chapter 4 of this dissertation), avoiding much of StOT's penalty.

I therefore report weighted error metrics and r^2 to provide a general picture of model fit across frameworks. Another purpose of the error metrics is to evaluate the viability of the proposed constraint set. If all models provide a reasonably good fit to the corpus data, the proposed constraint set provides a good approximation of the poetic grammar.

I do not use raw quantitative fit as a knockout argument in the theoretical discussion. The main purpose of the discussion is not to compare StOT and HG models based on their ability to fit particular data, but to model classes of phenomena that are ubiquitous in variable linguistic data. Using BCMS folk meter as a case study, the discussion draws on two phenomena with far-reaching theoretical consequences:

[17]Censored NHG with random selection as the tie resolution strategy.

ganging-up cumulativity (Chapter 3) and simple harmonic bounding (Chapter 4). For ganging-up interactions, I compare the range of cumulative interactions (linear and non-linear) predicted by competing stochastic frameworks with the gang effects attested in BCMS folk meter. Further, I show that MaxEnt and cNHG make different testable predictions in the domain of simple harmonic bounding and evaluate these predictions against BCMS metrical data, which provide an ideal test case.

2.7 Results

2.7.1 Weights and ranking values

This section summarizes the learned constraint weights (or ranking values for StOT). Table 2.1 reports the learned constraint weights for the three stochastic HG models: MaxEnt, GLA-cNHG (the best-fitting grammar out of 100 independent runs), and ML-cNHG.

Constraint	Weight (MaxEnt)	Weight (GLA-cNHG; best fit)	Weight (ML-cNHG)
3σInertia	2.29	1.98	1.19
Align	2.24	2.15	2.00
End-Weight	2.04	2.05	1.55
*Light/FinalStrong	1.81	1.79	1.47
End-Weight & *Light/FinalStrong	1.53	2.51	2.56
*StressedLight/FinalStrong	0.99	0.77	−0.15
PossessiveSecond	1.52	0.58	1.06
AdjectiveFirst	0.97	0.71	0.57

Table 2.1: Constraint weights for three stochastic HG models.

Recall that the MaxEnt and ML-cNHG weights are the ML estimates obtained algorithmically, whereas the reported GLA-cNHG weights correspond to the best-fitting grammar out of 100 independent GLA runs rather than the theoretical optimum.

*StressedLight/FinalStrong receives a negative base weight in the ML-cNHG model, but this simply means that its perturbed weight is zero in more than half of evaluation trials, since perturbed weights are left-capped at zero (§2.5.2).

The different procedures for weight search and probability mapping employed by MaxEnt and cNHG render the absolute weights not amenable to direct cross-framework comparison. The weights are reported here to provide a general picture of constraint contributions in the fitted grammars. The discussion in Chapters 3–4 instead focuses on the substantive predictions the grammars generate.

Independent GLA-cNHG runs converge on different grammars, of which the best-fitting one is reported in Table 2.1. It is also useful to get a sense of how stable parameter estimates are across the GLA learning runs. Figure 2.8 shows the distribution of base weights across 100 independent GLA-cNHG runs in OT-Soft. Although the absolute weights differ from run to run, the relative contributions of the constraints are largely stable.

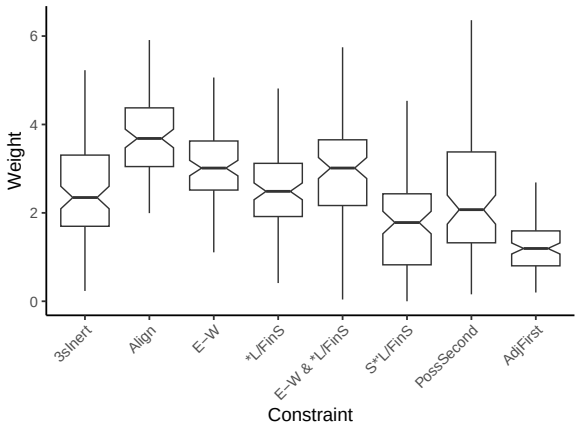


Figure 2.8: Base weights obtained from GLA-cNHG simulations in OT-Soft.

For StOT, Table 2.2 reports the ranking values of the best-fitting grammar (lowest deviance) obtained from 100 independent OT-GLA runs in OT-Soft. The base ranking values translate into the pairwise ranking probabilities shown in the Hasse diagram in

Figure 2.9.

Constraint	Ranking value
3σInertia	97.69
End-Weight & *Light/FinalStrong	97.47
PossessiveSecond	94.43
*StressedLight/FinalStrong	93.84
Align	93.60
End-Weight	89.88
*Light/FinalStrong	89.12
AdjectiveFirst	87.67

Table 2.2: Ranking values in the best-fitting StOT model.

As with GLA-cNHG, individual runs of the OT-GLA pipeline converge on different grammars. I therefore also examine the stability of the learned parameter estimates across runs. Figure 2.10 shows the distribution of ranking values across 100 independent OT-GLA runs.

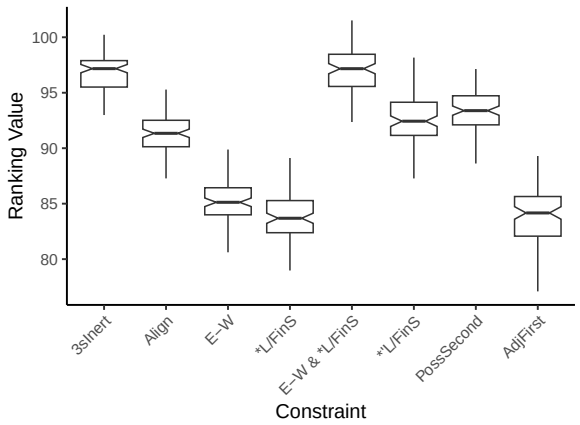


Figure 2.10: Ranking values across 100 independent OT-GLA runs in OT-Soft.

2.7.2 Model fit

Table 2.3 reports the weighted goodness-of-fit metrics introduced in §2.6: MAE, RMSE, and r^2. Figure 2.11 complements Table 2.3 by plotting the observed and predicted

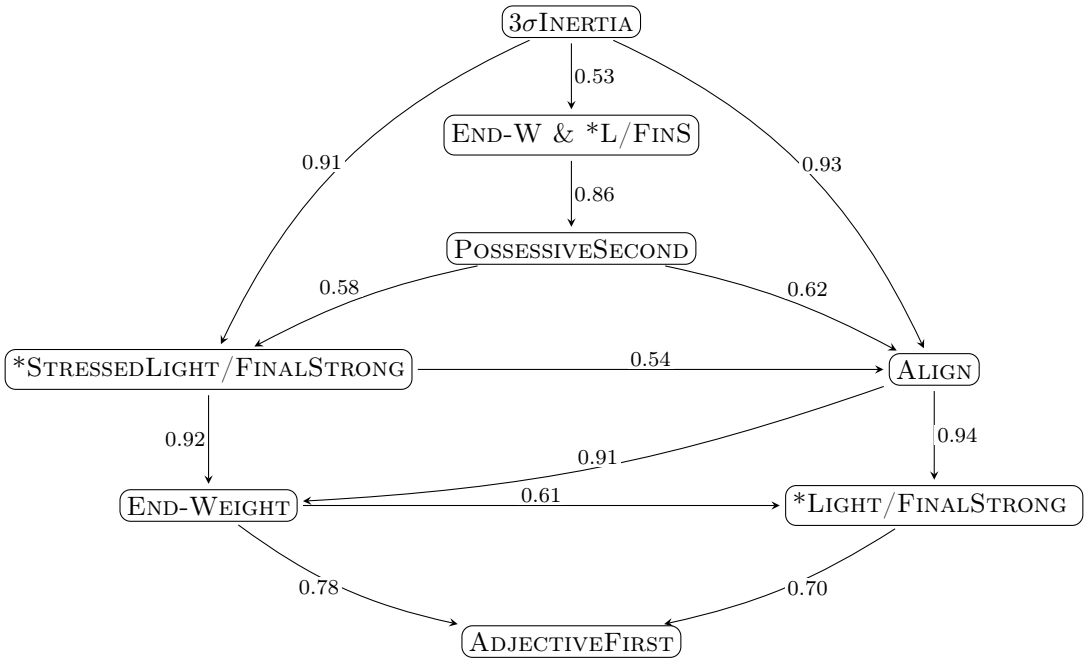


Figure 2.9: Hasse diagram depicting the constrait hierarchy with pairwise ranking probabilities for the StOT grammar.

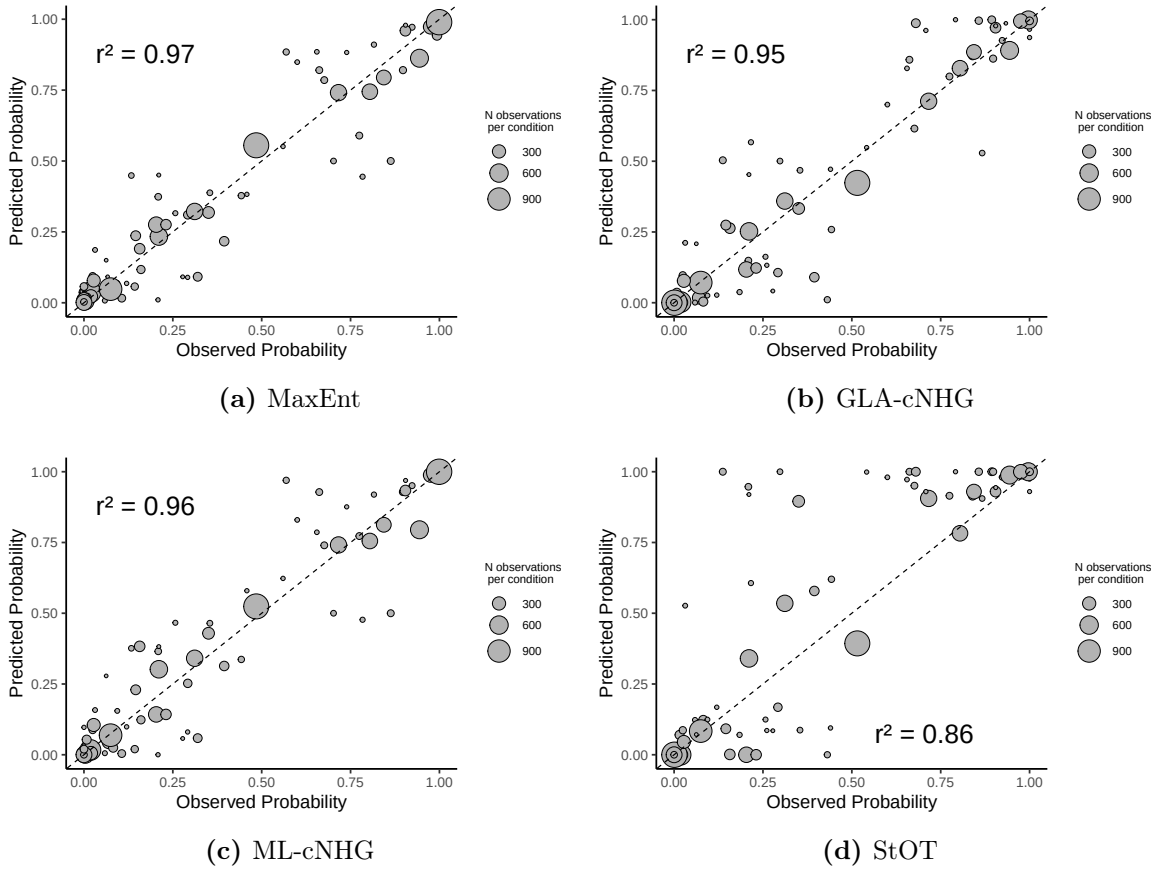
candidate probabilities for each fitted grammar.

	MaxEnt	GLA-cNHG	ML-cNHG	StOT
MAE	0.03	0.05	0.04	0.09
RMSE	0.06	0.09	0.07	0.16
r_w^2	0.97	0.95	0.96	0.86

Table 2.3: Goodness-of-fit metrics for all four fitted grammars.

All four models provide a reasonably good fit to the corpus data, suggesting that the constraint set used here is a sensible choice. There are several framework-level differences that merit closer examination.

The StOT model performs conspicuously worse than its HG counterparts (e.g., MAE = 0.09 vs. 0.03–0.05 for the stochastic HG models; $r_w^2 = 0.86$ vs. 0.95–0.97). As I argue in Chapters 3–4, StOT's poorer fit falls out from two major limitations of its probability mapping. StOT consistently assigns too weak a penalty to coincident



(a) MaxEnt **(b)** GLA-cNHG

(c) ML-cNHG **(d)** StOT

Figure 2.11: Scatterplots of observed (horizontal axis) vs. predicted probabilities (vertical axis). Point thickness is proportional to tableau-level sample size. Only one candidate per tableau is plotted, to avoid redundant symmetrical points.

constraint violations and invariably assigns zero probability to harmonically bounded word orders, some of which are abundantly attested in meter.

Of the stochastic HG models, MaxEnt provides the best overall fit, although the difference between MaxEnt and the cNHG models is very subtle, judging from the error metrics alone. As discussed in Chapter 4, the main source of divergent behavior between stochastic HGs is their handling of harmonic bounding.

Finally, using ML weights for cNHG isolates the effect of the MaxEnt vs. cNHG probability mapping functions, independent of the learning algorithm. Replacing GLA learning with ML optimization narrows the gap between cNHG and MaxEnt (deviance

= 1,571 in GLA-cNHG, 870 in ML-cNHG, and 611 in MaxEnt; for the same result, see Flemming's 2021 ML solution versus Smith and Pater's 2020 GLA solution for French schwa data).

The focus of Chapter 3 is the comparison between StOT and HGs. I only use GLA-cNHG because it is more similar to StOT, in that they both use the GLA to learn the stochastic grammar from input data and employ similar mechanisms to introduce stochasticity. In Chapter 4, the focus shifts to the MaxEnt-cNHG comparison, where isolating the contribution of their evaluation componets becomes essential. To control for the contribution of the learning algorithm, I also consider the ML-cNHG model.

Chapter 3

Ganging-up cumulativity

3.1 Introduction

Linguistic factors interact CUMULATIVELY. In typical cases of GANGING-UP CUMULATIVITY, the co-occurrence of two or more marked structures erodes the acceptability or production frequency of linguistic forms (Jäger and Rosenbach, 2006; Pater, 2009b; Kenstowicz, 2009; Green and Farris-Trimble, 2010; Albright, 2012; McPherson, 2016; Shih, 2017a; Ryan, 2017a; Smith and Pater, 2020; Breiss, 2020; Breiss and Albright, 2022; Milenković, 2026; Flemming and Magri, 2026). For example, schwa deletion in French is less likely when it simultaneously gives rise to a triconsonantal cluster and stress clash, as in *la terre se vend* [la ˈtɛʁ s(œ) ˈvɑ̃] 'the land sells,' than when it produces stress clash only, as in *le vín se vénd* [lœ vɛ̃ s(œ) vɑ̃] 'the wine sells,' or a triconsonantal cluster only, as in *la térre se vend bíén* [la ˈtɛʁ s(œ) vɑ̃ ˈbjɛ̃] 'the land sells well' (Smith and Pater, 2020).

Absolute grammaticality thresholds can also be defined by multiple interacting factors. In BCMS, two independent prosodic dimensions shape the inventory of permissible feet: foot size (mora count) and the tonal profile of head moras. The language

tolerates degenerate (i.e. monomoraic) feet, as in /brát/ → [ˈ(brát)] 'brother.NOM.SG', as well as feet whose head moras are toneless, as in /ʊuːk/ → [ˈ(ʊuːk)] 'wolf.NOM.SG.' However, no foot can simultaneously be degenerate and have a toneless head mora: /lɛd/ → [ˈ(lɛːd)], not *[ˈ(lɛd)] 'ice.NOM.SG' (Zec, 1999; Milenković, 2024, 2026).[1]

In COUNTING CUMULATIVITY, multiple occurrences of the same marked configuration have an increasingly detrimental effect on probability or acceptability (Jäger and Rosenbach, 2006; Jäger, 2007; Sorace and Keller, 2005; Keller, 2006; Kawahara, 2020; Breiss, 2020; Kawahara and Kumagai, 2021; Kawahara, 2021; Kim, 2022; Cabrera, 2025).[2] In Breiss's (2020) experiments, for instance, two violations of nasal harmony are consistently judged worse by participants than single violations or no violations.

Cumulativity is one of the most extensively debated phenomena in constraint-based phonology, with major consequences for the OT-HG debate (Pater, 2009a,b, 2016). This chapter focuses on ganging-up cumulativity, since no constraint that modulates word order in BCMS folk meter can be violated more than once per candidate. OT and HG make conflicting predictions for ganging-up interactions. In HG, candidate well-formedness is evaluated in a holistic fashion, such that all constraint violations contribute to candidates' Harmonies. HG therefore predicts cumulative effects of constraint violations, as in (58), where violating two lower-weighted constraints $\mathbb{B}$ and $\mathbb{C}$ produces a more severe aggregate penalty than violating higher-weighted $\mathbb{A}$ alone.

(58) Ganging-up cumulativity in HG

[1]Kumaran (2024) argues that gang effects are restricted to variable data, casting doubt on the purported cases of categorical ganging-up cumulativity discussed by Farris-Trimble (2008). Flemming and Magri (2026) independently arrive at the same conclusion. Aside from BCMS foot minimality, documented cases of categorical gang effects include ATR harmony in Kinande (Kenstowicz, 2009), Awa tonal patterns (McPherson, 2016), Sanskrit retroflex harmony (Ryan, 2017a), Yakut rounding harmony (Walker, 2017), Assamese ATR harmony (Tebay and Zimmermann, 2020), and Tamil geminate licensing (Ryan, 2026).

[2]But see Kawahara and Kumagai (2023) on the validity of some of Kim's (2022) data.

		$\mathbb{A}$	$\mathbb{B}$	$\mathbb{C}$	
		3	2	2	$\mathscr{H}$
a. ☞	Candidate 1	-1			-3
b.	Candidate 2		-1	-1	-4

Strict-ranking OT employs "fast and frugal" assessment of candidate well-formedness (Kawahara, 2021): The winner is selected based on candidates' performance on the top-ranking constraint(s). Lower-ranked constraints cannot affect the outcome, even when they overwhelmingly favor a candidate that fares poorly on the highest-ranking constraint. As a consequence, OT predicts no cumulative constraint interactions. In (59), the winner is the candidate favored by the highest-ranking constraint $\mathbb{A}$, despite faring worse on two lower-ranked constraints.

(59) No ganging-up cumulativity in OT

		$\mathbb{A}$	$\mathbb{B}$	$\mathbb{C}$
a.	Candidate 1	$*!$		
b. ☞	Candidate 2		$*$	$*$

To model gang effects, OT employs local conjunction (LC; Smolensky, 1993, 2006) of the cumulatively interacting constraints (60).

(60) Ganging-up cumulativity in OT with LC

		$\mathbb{B}$ & $\mathbb{C}$	$\mathbb{A}$	$\mathbb{B}$	$\mathbb{C}$
a.	Candidate 1		$*$		
b. ☞	Candidate 2	$*!$		$*$	$*$

OT with LC (OT-LC) is prone to overgeneration, since it produces bizarre interdependencies between otherwise unrelated phonological processes (Kirchner, 1996; Itô and Mester, 1998; McCarthy, 2003a; Pater, 2009b, 2016; Hayes and White, 2015). HG's

ability to model gang effects without recourse to nondefault machinery is widely regarded as its major advantage (Farris-Trimble, 2008; Jesney and Tessier, 2008; Pater, 2009a,b, 2016; Potts et al., 2010; Irvine and Dredze, 2017, and related work), since it provides a more parsimonious account of ganging-up interactions while eschewing the problematic predictions of OT-LC.[3]

When HG and OT-LC are compared as theories of ganging-up cumulativity, the focus is usually on whether HG's built-in additivity of constraint violations can reproduce the full range of effects of LC in strict-ranking OT grammars. Early work in the OT-HG debate generally assumed that HG entirely dispenses with the mechanism of LC because it can capture the full range of gang effects predicted by OT-LC grammars (Pater, 2009a,b, 2016; Farris-Trimble, 2008; Potts et al., 2010). However, the equivalence between OT-LC and HG holds only for a subset of ganging-up interactions (Legendre et al., 2006, 350*ff.*; Mai and Bakovic, 2020). Subsequent work has shown that, although HG can model many cases of ganging-up cumulativity without nondefault machinery (Smith and Pater, 2020; Breiss and Albright, 2022), it must still resort to LCs to model extreme cases of superlinearity (Albright, 2009; Hayes et al., 2012; Green and Davis, 2014; Shih, 2016, 2017a; Smith, 2022; Milenković, 2026; Chapter 2).

The question of whether HG obviates the need for LC is orthogonal to the validity of OT-LC as a theory of cumulativity, this chapter's main concern. The relevant question is therefore not whether HG supplants OT-LC, but rather whether OT-LC reproduces the full range of cumulativity effects generated by plain HG. The latter question has received less attention in the literature. This dearth of attention reflects the widespread view that OT-LC, as the more complex theory of ganging-up cumulativity, should

[3]An early argument against strict-ranking OT on grounds of cumulativity was made by Guy (1997), who compares OT with the Variable Rule framework of Labov (1969); Cedergren and Sankoff (1974).

subsume the empirical coverage of HG, the more parsimonious of the two approaches.[4] Thus, the most commonly raised objection against OT-LC concerns its overgeneration (Pater, 2009a,b, 2016; Potts et al., 2010).

This complexity-based reasoning is somewhat misplaced. The two approaches differ not only in complexity but also in the grammar architectures they employ (OT's strict ranking + fast and frugal evaluation vs. HG's numeric weights + holistic evaluation). Thus, despite being the more complex theory, OT-LC need not be as expressive as HG, given their architectural differences. Indeed, cases have been reported where OT-LC fails to model gang effects that plain HG readily captures (Walker, 2017; Ryan, 2026). Undergeneration of this kind constitutes a stronger argument against OT-LC as a theory of ganging-up cumulativity. The problem is not that OT-LC models gang effects at the expense of undesirable predictions elsewhere, but rather that, despite its added complexity, OT-LC fails to reproduce the full range of ganging-up interactions generated by plain HG.

Using phonologically conditioned syntactic variation in BCMS folk meter as a case study, this chapter extends the debate between HG and OT-LC to their stochastic implementations. The motivation for examining StOT-LC is twofold. Unlike Classical OT, StOT derives some patterns of ganging-up cumulativity (Jäger and Rosenbach, 2006; Hayes and Londe, 2006; Maslova, 2007; Smith and Pater, 2020). Moreover, the role of LC in StOT has not, to my knowledge, been explored.

I show that OT-LC's inherent limitations, identified in categorical data (Farris-Trimble, 2008; Pater, 2009b; Potts et al., 2010; Walker, 2017; Ryan, 2026), not only persist but are exacerbated in variable data. Without nondefault machinery, StOT cannot accommodate the full range of gang effects in the data nearly as successfully

[4]Complexity is understood here in information-theoretic terms, where the number of parameters in the grammar is the measure of complexity.

as MaxEnt and NHG. Augmenting StOT with LCs does not close the gap. Despite added complexity, StOT-LC shares the undergeneration problem of plain StOT, albeit in a somewhat milder form. This result lends further support to the undergeneration argument of Walker (2017); Ryan (2026). At the same time, StOT-LC still produces bizarre interdependencies between apparently unrelated aspects of poetic word order. In other words, the StOT grammar augmented with LCs simultaneously undergenerates and overgenerates. Given the preponderance of gang effects in linguistic data (Pater, 2009b; Breiss, 2020; Kawahara, 2021), StOT's inability to accommodate such interactions, even with additional machinery, calls into question its viability as a model of variation.

The chapter is structured as follows. In §3.2, I provide background on constraint cumulativity in StOT and stochastic variants of HG. In §3.3, I compare the baseline models. In §3.4, I test whether augmenting the baseline StOT model with conjunctions of cumulatively interacting constraints improves its handling of ganging-up cumulativity. §3.5 concludes.

3.2 Ganging-up cumulativity in stochastic grammars

This section reviews the treatment of ganging-up cumulativity in StOT (§3.2.1) and stochastic HGs (§3.2.2).

3.2.1 StOT

Unlike classical OT, StOT derives a limited form of ganging-up cumulativity (Jäger and Rosenbach, 2006; Hayes and Londe, 2006; Zuraw and Hayes, 2017; Smith and Pater, 2020; Flemming and Magri, 2026). Cumulativity of constraint violations manifests itself in the fact that multiple cooperating constraints generally have greater chances

of defeating a rival constraint when violated jointly than individually.

As an illustration, consider the minimal grammar in (61). Two cooperating constraints, $\mathbb{B}$ and $\mathbb{C}$, gang up against another constraint, $\mathbb{A}$. Tableaux a–b isolate the independent contributions of $\mathbb{B}$ and $\mathbb{C}$ relative to $\mathbb{A}$: $\mathbb{A}$ clashes with only $\mathbb{B}$ (61a) or only $\mathbb{C}$ (61b). In tableau c, $\mathbb{A}$ takes up $\mathbb{B}$ and $\mathbb{C}$ together: Compliance with $\mathbb{A}$ results in the coincident violation of $\mathbb{B}$ and $\mathbb{C}$, and vice versa.

(61)

		$\mathbb{A}$	$\mathbb{B}$	$\mathbb{C}$
a.	Candidate 1		*	
	Candidate 2	*		
b.	Candidate 3			*
	Candidate 4	*		
c.	Candidate 5		*	*
	Candidate 6	*		

At each evaluation, perturbed ranking values map onto classical OT-style strict rankings. The grammar is deterministic, outputting a single winner that fares best on the highest-ranking active constraint(s). Accordingly, when two or more constraints gang up in StOT, their preferred candidate wins whenever at least one of them dominates the opposing constraint. For the constraint triplet in (61), there are $3! = 6$ possible total rankings. Of these, four have $\mathbb{B}$ and/or $\mathbb{C}$ on top of $\mathbb{A}$ (62). These rankings select the violator of $\mathbb{A}$, Candidate 6, as optimal, avoiding the simultaneous violations of $\mathbb{B}$ and $\mathbb{C}$.

(62) Rankings that favor Candidate 6 in (61c)

a. $\mathbb{B} \gg \mathbb{A} \gg \mathbb{C}$

b. $\mathbb{B} \gg \mathbb{C} \gg \mathbb{A}$

c. $\mathbb{C} \gg \mathbb{A} \gg \mathbb{B}$

d. $\mathbb{C} \gg \mathbb{B} \gg \mathbb{A}$

In the remaining two ranking permutations (63), $\mathbb{A}$ dominates both ganging constraints. As a result, violating $\mathbb{B}$ and $\mathbb{C}$ together becomes less costly than violating $\mathbb{A}$ alone.

(63) Rankings that favor Candidate 5 in (61c)

a. $\mathbb{A} \gg \mathbb{B} \gg \mathbb{C}$

b. $\mathbb{A} \gg \mathbb{C} \gg \mathbb{B}$

Rankings (63) constitute the only scenario in which the doubly violating Candidate 5 is selected as optimal. The probability that the doubly violating candidate surfaces equals the probability that $\mathbb{A}$ dominates $\mathbb{B}$ and $\mathbb{C}$ at the same time. This happens whenever the pairwise noise differences between $\mathbb{A}$ and its rival constraints are both greater than the negative of the corresponding base ranking differences, as formalized in (64).

(64) For Candidate 5 in (61c) to win, both of the following must hold:

a. $n_{\mathbb{A}} - n_{\mathbb{B}} > -(r_{\mathbb{A}} - r_{\mathbb{B}})$;

b. $n_{\mathbb{A}} - n_{\mathbb{C}} > -(r_{\mathbb{A}} - r_{\mathbb{C}})$.

Given that the two noise difference variables $n_{\mathbb{A}} - n_{\mathbb{B}}$ and $n_{\mathbb{A}} - n_{\mathbb{C}}$ share noise term $n_{\mathbb{A}}$, they are not independent. Their covariance is $Cov = \sigma^2$, and their correlation is $Corr = \frac{1}{2}$ (Evans and Rosenthal, 2004). The probability of $\mathbb{A} \gg \mathbb{B} \wedge \mathbb{A} \gg \mathbb{C}$ is obtained using the standardized bivariate normal cumulative distribution function, Φ_2 (Genz and Bretz, 2009),[5] whose arguments are the threshold ranking differences from

[5]Implemented in R, using the `pmvnorm()` function of the `mvtnorm` package (Genz et al., 2020).

(61) divided by the standard deviations of the corresponding noise difference variables ($\sqrt{2}\sigma$ in both cases), and the correlation between the noise difference variables, $Corr$. See Appendix B for a detailed derivation.

(65) Probability of the doubly violating candidate

$$P(\mathbb{A} \gg \mathbb{B} \wedge \mathbb{A} \gg \mathbb{C}) = \Phi_2\left(\frac{r_{\mathbb{A}} - r_{\mathbb{B}}}{\sqrt{2}\sigma}, \frac{r_{\mathbb{A}} - r_{\mathbb{C}}}{\sqrt{2}\sigma}; Corr\right)$$

To put these general remarks on a more concrete footing, consider (66), where $\mathbb{A}$ individually outranks both $\mathbb{B}$ and $\mathbb{C}$, with $r_{\mathbb{A}} - r_{\mathbb{B}} = 1$ and $r_{\mathbb{A}} - r_{\mathbb{C}} = 2$.[6] $\mathbb{A}$'s base ranking advantage over the rival constraints maps, by (50), onto 64% and 76% probabilities of $\mathbb{A}$-compliant candidates in (66a–b) with the standard deviation of random noise $\sigma = 2$ for all constraints (OT-Soft default for StOT). In tableau c, where $\mathbb{B}$ and $\mathbb{C}$ gang up against $\mathbb{A}$, four out of six possible rankings favor the violator of $\mathbb{A}$ (62), and the remaining two (63) favor the doubly violating candidate. By (65), the probability of $\mathbb{A}$ outranking both ganging constraints is 55%. That is, despite the two-thirds of possible rankings favoring the singly violating Candidate 6, their aggregate probability given the base ranking differences in (66) is only 45%.

(66) Ganging-up cumulativity in StOT ($\sigma = 2$)

		$\mathbb{A}$	$\mathbb{B}$	$\mathbb{C}$	
	r_k	102	101	100	p
a.	Candidate 1		*		0.64
	Candidate 2	*			0.36
b.	Candidate 3			*	0.76
	Candidate 4	*			0.24
c.	Candidate 5		*	*	0.55
	Candidate 6	*			0.45

[6]This toy grammar is a slightly modified example from Smith and Pater (2020, 18–20), who use inherently unranked constraints (all $r_x - r_y = 0$) to make the same theoretical point.

The grammar in (66) produces ganging-up cumulativity in that the cooperating constraints give each other some degree of probability boost. Violations of $\mathbb{B}$ and $\mathbb{C}$ add up in (66c), yielding a more severe penalty (i.e. lower probability) for the doubly violating Candidate 5 than for the singly violating Candidates 1 and 3 in (66a–b). In other words, the probability of $\mathbb{B}$ and $\mathbb{C}$ jointly defeating $\mathbb{A}$ (45%) is greater than both of their individual probabilities of doing so (36% and 24%). In this respect, StOT departs from Classical OT, whose strict domination rules out any cumulative effect of multiple constraint violations.

The range of cumulative patterns produced by StOT is highly restricted, however. Cooperating constraints invariably receive a relatively small probability boost in multiply violating environments. The observation goes back to Maslova (2007), who notes that the combined effect of multiple constraints in StOT is always only slightly stronger than the independent effect of the strongest of them. This property of StOT cumulativity has since been documented in a series of case studies (e.g. Hayes and Londe, 2006; Smith and Pater, 2020; Flemming and Magri, 2026). The present study is no exception; see §§3.3–3.4.

In particular, StOT grammars tend to assign higher-than-chance probabilities to multiply violating candidates. In the toy grammar in (66), the independent violations of $\mathbb{B}$ and $\mathbb{C}$ are observed in 64% and 76% of the cases, respectively (66a–b). Assuming independence, we expect $0.64 \times 0.76 = 49\%$ coincident violations of $\mathbb{B}$ and $\mathbb{C}$. The model predicts 55% (66c). This is SUBLINEARITY: The grammar predicts the coincident violation of multiple cooperating constraints to be more frequent than chance, meaning that the severity of their coincident violation is less severe than expected (Smith and Pater, 2020; Breiss and Albright, 2022).[7]

[7]Note that the method used here to determine the degree of (non-)linearity of cumulative interactions is formally equivalent to Breiss and Albright's (2022) procedure. Smith and Pater (2020) use a different method (see *ibid.*, pp. 20–22).

The sublinearity identified in the StOT grammar in (66) is not an idiosyncratic property of that particular grammar. I demonstrate this by assessing, over a wide range of ranking values, the space of possible ganging-up interactions generated by StOT. The Con in this minimal example involves three constraints, $\mathbb{A}$, $\mathbb{B}$, and $\mathbb{C}$. As in (66), there are three tableaux: one in which $\mathbb{A}$ clashes with $\mathbb{B}$, one in which $\mathbb{A}$ clashes with $\mathbb{C}$ (the independent violation conditions), and one in which $\mathbb{A}$ opposes $\mathbb{B}$ and $\mathbb{C}$ simultaneously (the ganging-up condition). Pairwise ranking differences between $\mathbb{A}$ and each ganging constraint ($\mathbb{B}/\mathbb{C}$) are varied over the interval $[-10, 10]$ in steps of 0.1. (For negative values, $\mathbb{A}$ is inherently weaker than the rival constraint; for positive values, $\mathbb{A}$ is inherently stronger than the rival constraint; zero means the two constraints are ranked equally.) This results in 40,401 unique ranking vectors (i.e. stochastic grammars).

For each stochastic grammar, the independent violation probability of a ganging constraint is the probability of that constraint ranking below $\mathbb{A}$ when the two are pitted against each other. Assuming independence, the expected probability of coincident violation is given in (67):

(67) $p(\mathbb{A} \gg \mathbb{B} \wedge \mathbb{A} \gg \mathbb{C})_{\text{expected}} = p(\mathbb{A} \gg \mathbb{B}) \times p(\mathbb{A} \gg \mathbb{C})$

The model's predicted probability of coincident violation and the expected probability of coincident violation under independence are transformed to the logit scale. The logit of the expected probability is subtracted from the logit of the predicted probability to obtain the logit-difference. Unlike the raw predicted/expected ratio, this measure treats deviations from independence symmetrically across the probability space. For example, the predicted/expected ratio for 2% predicted and 3% expected probability of coincident violation (0.67) is much lower than for, say, 96% predicted and 97% expected (0.99). The predicted/expected ratio thus falsely indicates different magnitudes of superlinearity (see the general discussion of scale-dependence in interaction

measures in VanderWeele and Knol, 2014).

Figure 3.1 plots the logit-difference between predicted and expected probabilities of coincident violation as a function of the pairwise ranking differences between $\mathbb{A}$ and each of the ganging constraints $\mathbb{B}$ and $\mathbb{C}$. The relationship between expected and predicted probabilities diagnoses the type of cumulativity generated by the grammar. If the predicted and expected probabilities coincide or closely match each other (logit-difference ≈ 0; green-shaded areas in Figure 3.1), the interaction is (APPROXIMATELY) LINEAR. If the predicted probability of coincident violation is higher than the combined probabilities of independent violations (logit-difference > 0; red/orange-shaded areas in Figure 3.1), the interaction is SUBLINEAR: The grammar penalizes coincident violations of $\mathbb{B}$ and $\mathbb{C}$ less strongly than chance. If the predicted probability is lower than expected (logit-difference < 0; blue-shaded areas), the interaction is SUPERLINEAR: Coincident violation of two constraints gives rise to a stronger penalty than the combination of their independent effects (see Smith and Pater, 2020; Breiss and Albright, 2022).

Notably absent from Figure 3.1 are superlinear interactions (blue-shaded areas). StOT grammars produce only sublinear and approximately linear cumulativity (confirming Maslova, 2007; Smith and Pater, 2020). Linearity arises trivially in grammars in which $\mathbb{A}$ nearly categorically outranks both ganging constraints. Given that the independent violation probabilities of the ganging constraints both approach 1, and the predicted probability of their coincident violation also approaches 1, they trivially converge.[8]

In sum, this section demonstrates that while StOT predicts some ganging-up interactions, these are always sublinear (or trivially linear). This is further substantiated in §3.3.

[8]With per-constraint noise $\sigma = 2$, a ranking difference of 5 results in a pairwise ranking probability of roughly 96% (accounting for the combined noise of both constraints).

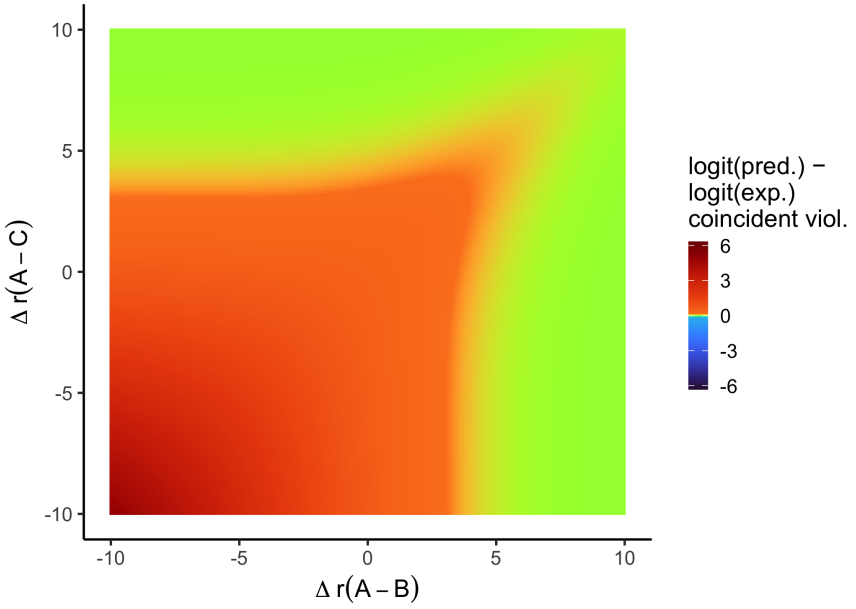


Figure 3.1: Ganging-up cumulativity patterns predicted by StOT ($\sigma = 2$). Tile color reflects the logit-difference between predicted and expected probabilities of coincident violation (logit(predicted) − logit(expected).) The grammars produce only sublinear (red/orange areas in the heatmap) and approximately linear interactions (green areas). $\Delta r(x, y) = r_x - r_y$.

3.2.2 Stochastic HGs

Stochastic HGs predict a wider range of gang effects than StOT, producing both linear and different types of non-linear interactions. For MaxEnt, this property has been extensively documented in the literature (see Breiss and Albright, 2022; Flemming and Magri, 2026, a.o.). The predictions of NHG in this domain are largely unexplored in previous work. This section shows that MaxEnt and (censored) NHG both produce diverse arrays of ganging-up interactions, though the two exhibit some curious differences that most likely stem from the divergent probability mappings they employ.

Given that in HG, all violations incurred by a candidate count towards its Harmony, constraint violations add up by default, each contributing to candidate Harmony unless zero-weighted (Pater, 2009a,b). (In NHG, even zero-weighted constraints affect candidate probability (Flemming, 2021; Hayes and Kaplan, 2025).)

MaxEnt's probability mapping is straightforward. In simple cases of ganging-up cumulativity like (66), the probability of coincident violation of $\mathbb{B}$ and $\mathbb{C}$, that is, of $\mathbb{A}$ prevailing against the combination of the interacting constraints, is determined directly from the weight difference between the opposing constraints, $w_{\mathbb{A}} - (w_{\mathbb{B}} + w_{\mathbb{C}})$.

$$(68) \quad p(\text{coincident violation}\,\mathbb{B},\mathbb{C}) = \frac{1}{1 + e^{-(w_{\mathbb{A}} - (w_{\mathbb{B}} + w_{\mathbb{C}}))}}$$

Note that HG's holistic evaluation obviates the need for assessing each ganging constraint individually against the opposing constraint $\mathbb{A}$ in a pairwise fashion, as was the case in StOT. Thus, the probability mapping functions for stochastic HGs directly compare the combined weight of $\mathbb{B}$ and $\mathbb{C}$ against the weight of $\mathbb{A}$.

In NHG, $\mathbb{A}$ prevails against the combination of $\mathbb{B}$ and $\mathbb{C}$ whenever its perturbed weight $\hat{w}_{\mathbb{A}}$ exceeds the sum of the perturbed weights of the ganging constraints, $\hat{w}_{\mathbb{B}} + \hat{w}_{\mathbb{C}}$:[9]

$$(69) \quad p(\text{coincident violation}\,\mathbb{B},\mathbb{C}) = p(\hat{w}_{\mathbb{A}} > \hat{w}_{\mathbb{B}} + \hat{w}_{\mathbb{C}})$$

Given the base weights of this constraint triplet, $w_{\mathbb{A}}$, $w_{\mathbb{B}}$, and $w_{\mathbb{C}}$, (69) holds whenever the noise difference variable $n_{\mathbb{A}} - (n_{\mathbb{B}} + n_{\mathbb{C}})$ does not go below the negative base weight difference, $-(w_{\mathbb{A}} - (w_{\mathbb{B}} + w_{\mathbb{C}}))$:[10]

$$(70) \quad p(\text{coincident violation}\,\mathbb{B},\mathbb{C}) = p(n_{\mathbb{A}} - (n_{\mathbb{B}} + n_{\mathbb{C}}) > -(w_{\mathbb{A}} - (w_{\mathbb{B}} + w_{\mathbb{C}})))$$

The noise difference variable follows a normal distribution because all constraint-specific noise terms, $n_{\mathbb{A}}$, $n_{\mathbb{B}}$, and $n_{\mathbb{C}}$, are normally distributed. Because $n_{\mathbb{A}}$, $n_{\mathbb{B}}$, and $n_{\mathbb{C}}$ are independent, the variance of the noise difference variable is the sum of the

[9] As a reminder, for NHG, I distinguish constraints' base weights w_k from their perturbed weights $\hat{w}_k$.

[10] When $n_{\mathbb{A}} - (n_{\mathbb{B}} + n_{\mathbb{C}}) = -(w_{\mathbb{A}} - (w_{\mathbb{B}} + w_{\mathbb{C}}))$, the two competitors have the same Harmony score. I assume that in such cases, the winner is picked at random (Boersma and Pater, 2016). See Hayes and Kaplan (2025); Kaplan (2025b) for an alternative.

variances of the constraint-specific noise terms (each σ^2): $3\sigma^2$. Its standard deviation is $\sqrt{3\sigma^2} = \sqrt{3}\sigma$. The probability that (70) holds, and that, accordingly, the output simultaneously violates $\mathbb{B}$ and $\mathbb{C}$, is determined by the normal cumulative distribution function of the base weight difference between $\mathbb{A}$ and the combined weight of $\mathbb{B}$ and $\mathbb{C}$, divided by the standard deviation of the noise difference variable:

$$(71) \quad p(\text{coincident violation } \mathbb{B}, \mathbb{C}) = \Phi\left(\frac{w_{\mathbb{A}} - (w_{\mathbb{B}} + w_{\mathbb{C}})}{\sqrt{3}\sigma}\right)$$

The probability formula in (71) holds for uncensored NHG, in which no lower bound is imposed on perturbed constraint weights. For censored NHG, the main concern of this dissertation, candidate probabilities must be obtained from the Monte Carlo procedure (Flemming, 2021, 22). The cNHG simulations reported in this section were conducted in R using a custom function for cNHG probability mapping.[11]

Examples (72–74) illustrate the range of possible ganging-up interactions in stochastic HGs. I use MaxEnt, the more extensively studied stochastic variant of HG, as an example. Depending on the weighting conditions, the cumulative contribution of $\mathbb{B}$ and $\mathbb{C}$ may be approximately linear (72), superlinear (73), or sublinear (74).

Consider (72) first.

(72) Linearity in MaxEnt

[11] $\sigma = 1$; the default number of evaluations per grammar is 1,000,000. The function uses random selection as the tie-resolution strategy.

		$\mathbb{A}$ 2.0	$\mathbb{B}$ 1.4	$\mathbb{C}$ 0.3	$\mathscr{H}$	$e^{\mathscr{H}}$	p
a.	Candidate 1		-1		-1.4	0.25	0.65
	Candidate 2	-1			-2.0	0.14	0.35
b.	Candidate 3			-1	-0.3	0.74	0.85
	Candidate 4	-1			-2.0	0.14	0.15
c.	Candidate 5		-1	-1	-1.7	0.18	0.57
	Candidate 6	-1			-2.0	0.14	0.43

The grammar predicts independent violation rates of 65% for $\mathbb{B}$ and 85% for $\mathbb{C}$, and a joint violation rate of 57%. The predicted coincident violation rate approaches the expected rate ($0.65 \times 0.85 = 55\%$, assuming independence), indicative of approximate linearity.

In (73), $\mathbb{A}$ outweighs $\mathbb{B}$ and $\mathbb{C}$ individually, but the combined weight of $\mathbb{B}$ and $\mathbb{C}$ is now greater than $\mathbb{A}$'s. These weighting conditions produce superlinearity:

(73) Superlinearity in MaxEnt

		$\mathbb{A}$ 3.0	$\mathbb{B}$ 2.0	$\mathbb{C}$ 2.0	$\mathscr{H}$	$e^{\mathscr{H}}$	p
a.	Candidate 1		-1		-2.0	0.14	0.73
	Candidate 2	-1			-3.0	0.05	0.27
b.	Candidate 3			-1	-2.0	0.14	0.73
	Candidate 4	-1			-3.0	0.05	0.27
c.	Candidate 5		-1	-1	-4.0	0.02	0.27
	Candidate 6	-1			-3.0	0.05	0.73

The probability of simultaneously violating $\mathbb{B}$ and $\mathbb{C}$ is considerably lower than chance. Given the independent violation rates of $\mathbb{B}$ and $\mathbb{C}$, we expect $0.73^2 = 53\%$

coincident violation. The predicted probability of the doubly violating Candidate 5 is 27%.

The MaxEnt grammar in (74) displays sublinearity. Both ganging constraints individually outweigh 𝔸. Given that both ganging constraints are individually strong (a–b), their joint effect is attenuated.

(74) Sublinearity in MaxEnt

		𝔸	𝔹	ℂ			
		0.1	0.5	0.5	$\mathscr{H}$	$e^{\mathscr{H}}$	p
a.	Candidate 1		−1		−0.5	0.61	0.40
	Candidate 2	−1			−0.1	0.90	0.60
b.	Candidate 3			−1	−0.5	0.61	0.40
	Candidate 4	−1			−0.1	0.90	0.60
c.	Candidate 5		−1	−1	−1.0	0.37	0.29
	Candidate 6	−1			−0.1	0.90	0.71

The doubly violating candidate is assigned higher-than-chance probability, which is reminiscent of StOT cumulativity (Figure 3.1). Given that both ganging constraints in (74) individually outweigh 𝔸, their joint violations are expected to be comparably rare: $0.40^2 = 16\%$. The grammar, however, predicts a much higher probability for the doubly violating candidate: 29%.

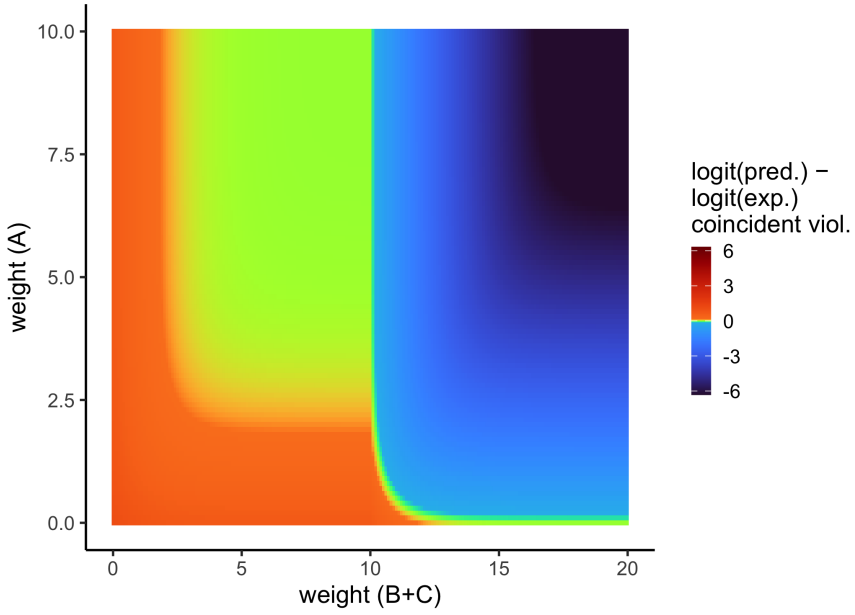
The key takeaway is that stochastic HGs are less restrictive than StOT with respect to ganging-up cumulativity. Figure 3.2 further substantiates this conclusion, considering cNHG alongside MaxEnt. The setup is similar to the StOT setup in §3.2.1. Logit-differences between predicted and expected probabilities of coincident violation are estimated over a large space of weighting conditions for the minimal grammar in (66). Constraint weights range from 0 to 10 in steps of 0.1, resulting in over one million unique weight vectors. For MaxEnt, an analytic solution is available for probability

estimation (68). cNHG probabilities are estimated through simulation. cNHG simulations employ 10,000 evaluation trials per grammar instead of the standard 1,000,000. This was done to accommodate the vast number of stochastic grammars under consideration (the simulation was still excruciatingly slow). The empty areas in the cNHG heatmap reflect probabilities of exactly zero, which render the logit-difference undefined ($\text{logit}(0) = -\infty$). Absolute zeros are possible in stochastic grammars that utilize Monte Carlo probability estimation (cNHG and the OT-Soft implementations of StOT and NHG). By contrast, analytic solutions prevent absolute zeros, since their probability functions are strictly positive (like MaxEnt, uncensored NHG as implemented in this section, and StOT as implemented in §3.2.1). In practice, I found analytic and Monte Carlo procedures to produce virtually indistinguishable probability distributions for the same weight vectors (see Appendix B).

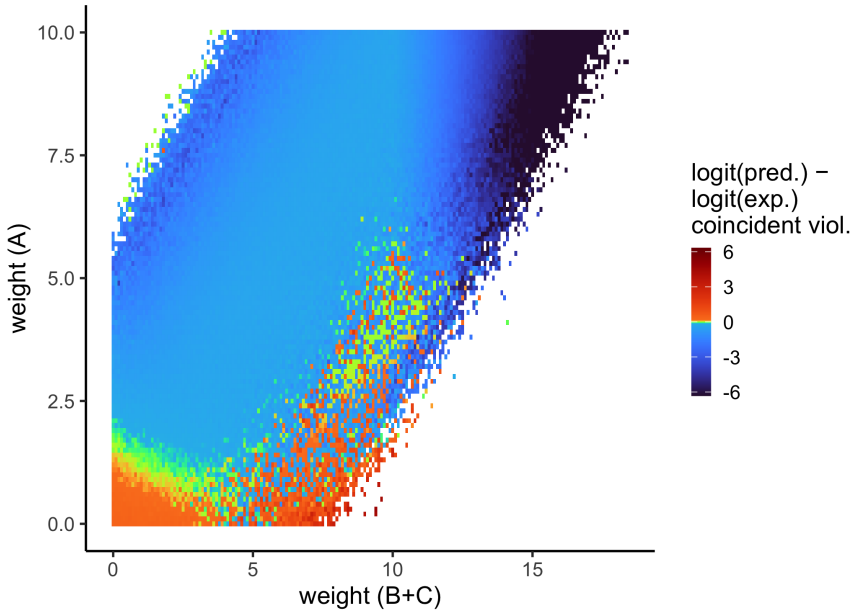
Like MaxEnt, cNHG produces both linear and various types of non-linear cumulativity. However, the two stochastic variants of HG often generate different types of cumulativity under identical weighting conditions. One difference evident from Figure 3.2 is that cNHG generates more widespread superlinearity and more restricted sublinearity than MaxEnt, with more erratic transitions between the sublinear and superlinear regions. The divergent behavior is hardly surprising given the different mechanisms by which MaxEnt and cNHG map weights onto candidate probabilities.

As an illustration, MaxEnt and cNHG predict different types of cumulative interactions under the weighting conditions in (75).

(75) MaxEnt and cNHG predict different cumulativity types for the same weighting conditions



(a) MaxEnt



(b) cNHG ($\sigma = 1$; 10,000 evaluations per model)

Figure 3.2: Cumulativity types predicted by stochastic HGs. Tile color indicates the logit-difference between predicted and expected probabilities of coincident violation.

		$\mathbb{A}$	$\mathbb{B}$	$\mathbb{C}$		
		1.2	0.3	0.3	p(cNHG)	p(MaxEnt)
a.	Candidate 1		-1		0.73	0.71
	Candidate 2	-1			0.27	0.29
b.	Candidate 3			-1	0.73	0.71
	Candidate 4	-1			0.27	0.29
c.	Candidate 5		-1	-1	0.55	0.65
	Candidate 6	-1			0.45	0.35

NHG predicts independent violation probabilities of 73% for both $\mathbb{B}$ and $\mathbb{C}$. The predicted probability of their coincident violation, 55%, is close to the expected rate of $0.73^2 = 54\%$, indicative of approximate linearity. By contrast, MaxEnt produces a sublinear pattern under the same weighting conditions. The grammar's predicted probability of joint violation, 65%, is higher than the expected rate of $0.71^2 = 50\%$, that is, the simultaneous violation of $\mathbb{B}$ and $\mathbb{C}$ is penalized less strongly than expected.

To my knowledge, ganging-up cumulativity has not previously been recognized as a source of divergent predictions between stochastic HGs, since the differences between the two frameworks in this domain remain largely unexplored. For example, Flemming and Magri's (2026) survey of gang effects compares StOT and MaxEnt HG but leaves NHG aside. Gang effects thus appear to offer a potentially fruitful avenue for probing the differences between stochastic HGs. For the time being, it suffices to note that stochastic HGs predict a more diverse range of gang effects than StOT.

This section brings to the fore two important properties of stochastic HG cumulativity. First, stochastic HGs predict pervasive superlinear patterns, which StOT cannot generate under any circumstances. Although both stochastic variants of HG produce some sublinear interactions, the scope of sublinearity is considerably more limited than in StOT, where sublinearity is the norm (Maslova, 2007; Smith and Pater,

2020). Moreover, the sublinear patterns produced by stochastic HGs do not appear to be as extreme as those predicted by StOT, as evidenced by the absence of dark red areas in Figure 3.2. Sublinear cumulativity has been documented in phonological data (Pizzo, 2015; Pater, 2023), but, contrary to what StOT predicts, it is by no means the norm (Smith and Pater, 2020; Breiss and Albright, 2022).

Second, the degree of (non-)linearity in stochastic HGs depends on the weighting conditions, that is, on how strongly the cumulatively interacting factors are enforced in the grammar (Breiss and Albright's (2022) experiments bear this out for MaxEnt). The new observation of the present study is that MaxEnt and cNHG can predict different types of cumulative interaction under identical weighting conditions and violation profiles, which leads to new testable predictions for stochastic HGs. Future work should explore these differences in greater detail.

3.3 Results

Keeping in mind the general properties of StOT and HG cumulativity laid out in §3.2, I now turn to ganging-up cumulativity in BCMS folk meter. Stochastic HGs substantially outperform StOT, which predicts only a limited form of ganging-up cumulativity. This highly restricted cumulativity is grossly insufficient to accommodate the widespread ganging-up interactions between the metrical constraints, particularly in cases where a phonological and syntactic constraint jointly oppose another phonological constraint.

Given the ubiquity of ganging-up interactions in the data, StOT's suppression of interaction effects has consequences well beyond cases of ganging-up cumulativity: Because cumulativity is suppressed, the grammar allocates the entirety of cumulative effects to individual constraints, whose independent contributions are compensatorily

inflated. That is, the model's failure to properly account for interaction effects contaminates the estimates of independent effects.

Examples (76–77) illustrate these points. In (76a), *Light/FinalStrong and AdjectiveFirst gang up against higher-ranking End-Weight, whereas in (76b), End-Weight and AdjectiveFirst jointly oppose *Light/FinalStrong. The canonical order of (76a), ['bi.jɛ.li.jɛm 'graː.dɔm] 'white city.ins' has a disyllabic noun preceded by a quadrisyllabic adjective in line-final position, incurring an End-Weight penalty. Inversion does away with the violation of End-Weight by placing the syllabically longer modifier line-finally, but it does so by violating both AdjectiveFirst and *Light/FinalStrong, since the modifier's penult is light.

In (76b), the canonical order ['mlaː.da u.'dɔ.ʋi.tsa] 'young widow.nom' is End-Weight-compliant, but has a light syllable in the final strong position. Inversion ([u.'dɔ.ʋi.tsa 'mlaː.da]) provides the penultimate position of the line with a heavy syllable, satisfying *Light/FinalStrong, but in turn moves a disyllable over a quadrisyllable, offending both AdjectiveFirst and End-Weight.

(76) a. *bijelijem gradom* ['bi.jɛ.li.jɛm 'graː.dɔm] 'white city.ins'

	End-W	*L/FinS	Adj1st	p(MaxEnt)	p(cNHG)	p(StOT)	Observed
w(MaxEnt)	2.04	1.81	0.97				
w(cNHG)	2.05	1.79	0.71				
r(StOT)	89.88	89.12	87.67				
a. Adj N 'bi.jɛ.li.jɛm 'graː.dɔm	−1			0.68	0.64	0.46	0.69
b. N Adj 'graː.dɔm 'bi.jɛ.li.jɛm		−1	−1	0.32	0.36	0.54	0.31

b. *mlada udovica* ['mlaː.da u.'dɔ.ʋi.tsa] 'young widow.nom'

	End-W	*L/FinS	Adj1st	p(MaxEnt)	p(cNHG)	p(StOT)	Observed
w(MaxEnt)	2.04	1.81	0.97				
w(cNHG)	2.05	1.79	0.71				
r(StOT)	89.88	89.12	87.67				
a. Adj N ˈmlaː.da u.ˈdɔ.ʋi.tsa		−1		0.77	0.75	0.66	0.79
b. N Adj u.ˈdɔ.ʋi.tsa ˈmlaː.da	−1		−1	0.23	0.25	0.34	0.21

In both conditions, the doubly violating noncanonical order is dispreferred to the canonical order in the corpus. Stochastic HGs capture the qualitative generalization, with solid quantitative fit. By contrast, StOT erroneously predicts the noncanonical order in (76a) to be slightly more likely than the violator of End-Weight (54% fitted inversion rate vs. 31% observed).

The cumulative contribution of two individually weaker constraints, *Light/Final Strong and AdjectiveFirst, is reduced. Given the ranking values of the constraint triplet in (76), End-Weight has pairwise ranking probabilities of 61% and 78% against *Light/FinalStrong and AdjectiveFirst, respectively. Given the predicted violation rates of *Light/FinalStrong and AdjectiveFirst when up against End-Weight, the chance rate of their coincident violation is $0.61 \times 0.78 = 48\%$, assuming independence. The interaction is weakly sublinear, since the fitted rate of coincident violation is slightly higher than the chance rate (54%), meaning that the cumulative effect of the ganging constraints is weaker than expected.

In (76b), StOT correctly predicts the preferred word order because it happens to be favored by the highest-ranking active constraint, End-Weight, though the grammar fits the data more poorly than its HG counterparts.

A similar pattern obtains in (77). End-Weight and AdjectiveFirst jointly oppose Align in (77a), while in (77b), Align and AdjectiveFirst gang up against End-Weight. Line-final trisyllables are penalized by Align because the right edge

of the preceding prosodic word is misaligned (§1.3.1.3). In (77a), the head noun [sla.ˈʊuː.ja] 'nightingale.GEN' is trisyllabic, meaning that ALIGN functions as an inversion trigger. Moving a disyllabic adjective over a trisyllabic noun incurs penalties from ADJECTIVEFIRST and END-WEIGHT. The canonical order is attested in almost two-thirds of lines in the corpus.

The opposite is true in (77b). The adjectival modifier is trisyllabic, while the head noun is disyllabic. ALIGN and END-WEIGHT therefore switch roles: ALIGN is an inversion inhibitor, whereas END-WEIGHT is an inversion trigger. The preference for the canonical order is even stronger than in (77a).

(77) a. *i mlada slavuja* [i=ˈmlaːda sla.ˈʊuː.ja] 'young nightingale.GEN'

	ALIGN	END-W	ADJ1ST	p(MaxEnt)	p(cNHG)	p(StOT)	Observed
w(MaxEnt)	2.24	2.04	0.97				
w(cNHG)	2.15	2.05	0.71				
r(StOT)	93.60	89.88	87.67				
a. Adj N i=ˈmlaːda sla.ˈʊuː.ja	−1			0.68	0.67	0.10	0.65
b. N Adj i=sla.ˈʊuː.ja ˈmlaːda		−1	−1	0.32	0.33	0.90	0.35

b. *od suvoga zlata* [ɔd=ˈsuː.vɔː.ga ˈzlaː.ta] 'of pure (lit. dry) gold.GEN'

	ALIGN	END-W	ADJ1ST	p(MaxEnt)	p(cNHG)	p(StOT)	Observed
w(MaxEnt)	2.24	2.04	0.97				
w(cNHG)	2.15	2.05	0.71				
r(StOT)	93.60	89.88	87.67				
a. Adj N ɔd=ˈsuː.vɔː.ga ˈzlaː.ta		−1		0.76	0.73	0.91	0.85
b. N Adj ɔd=ˈzlaː.ta ˈsuː.vɔː.ga	−1		−1	0.24	0.27	0.09	0.15

As in (76), adjectives preferably occur prenominally in both conditions, because inversion incurs multiple penalties. Stochastic HGs again capture the general pattern comparably well (the quantitative fit to (77b) is not great, however). StOT makes

the same kind of flawed prediction as in (76): For both mirror-image conditions, the candidate that satisfies the top-ranking constraint ALIGN is predicted to be preferred to its rival. Note also that the ranking difference between ALIGN and lower-ranking constraints is immense, which ensures that ALIGN-preferred candidates receive high probabilities in both tableaux.

This prediction is accurate for (77b), where ALIGN is one of the ganging constraints. However, StOT reverses a robust empirical trend in (77a), where doubly violating inversion is dispreferred to the canonical order, but the model predicts a very high inversion rate (90% fitted vs. 35% observed).

In sum, examples (76–77) show a robust pattern of suppression of interaction effects in StOT. Across the board, the grammar predicts word orders favored by the highest-ranking active constraint to be more likely than their rivals, suppressing, in some cases rather dramatically, the effects of coincident weaker penalties.

3.3.1 Other consequences of StOT's sublinearity

Since gang effects are widespread in meter, failing to properly account for them leads to a series of flawed predictions in other corners of the grammar. The StOT grammar cannot distribute explanatory power to multiple interacting constraints. Thus, most of the power is concentrated in a single constraint. ALIGN is a case in point.

The distortion of ALIGN's independent effect is evident from (78). In both conditions, the ALIGN-compliant disyllable-final orders are strongly preferred to their trisyllable-final alternatives, which are favored by END-WEIGHT. In both cases, ALIGN gangs up with *LIGHT/FINALSTRONG. Crucially, ADJECTIVEFIRST sides with ALIGN and *LIGHT/FINALSTRONG in (78a), but with END-WEIGHT in (78b). The effect of ADJECTIVEFIRST is tangible: END-WEIGHT almost categorically gives way to the three opposing constraints in (78a), but in (78b), ADJECTIVEFIRST and END-WEIGHT

together push back against the opposing constraints. As a result, the canonical order is observed in more than a quarter of verse lines in the corpus.

(78) a. *u careve dvore* [u='tsa.rɛ.ʊɛ 'dʊɔː.rɛ] 'in tsar's palace.ACC.PL'

	ALIGN	END-W	*L/FINS	ADJ1ST	*p*(MaxEnt)	*p*(cNHG)	*p*(StOT)	Observed
w(MaxEnt)	2.24	2.04	1.81	0.97				
w(cNHG)	2.15	2.05	1.79	0.71				
r(StOT)	93.60	89.88	89.12	87.67				
a. Adj N u='tsa.rɛ.ʊɛ 'dʊɔː.rɛ		−1			0.95	0.93	0.92	0.93
b. N Adj u='dʊɔː.rɛ 'tsa.rɛ.ʊɛ	−1		−1	−1	0.05	0.07	0.08	0.07

b. *u crnu zemljicu* [u='tsr̩ː.nu 'zɛm.ʎi.tsu] 'in black ground.ACC'

	ALIGN	END-W	*L/FINS	ADJ1ST	*p*(MaxEnt)	*p*(cNHG)	*p*(StOT)	Observed
w(MaxEnt)	2.24	2.04	1.81	0.97				
w(cNHG)	2.15	2.05	1.79	0.71				
r(StOT)	93.60	89.88	89.12	87.67				
a. Adj N u='tsr̩ː.nu 'zɛm.ʎi.tsu	−1		−1		0.26	0.29	0.09	0.28
b. N Adj u='zɛm.ʎi.tsu 'tsr̩ː.nu		−1		−1	0.74	0.71	0.91	0.72

In (78a), END-WEIGHT is the only inversion trigger. ALIGN, *LIGHT/FINAL STRONG, and ADJECTIVEFIRST gang up to prevent inversion, which takes place in only 7% of the lines. All models predict the pattern fairly well.

A better diagnostic of StOT's distorted estimate of ALIGN's contribution is (78b). The canonical order [u='tsr̩ː.nu 'zɛm.ʎi.tsu] 'in black ground.ACC' violates ALIGN and *LIGHT/FINALSTRONG: The former because of a misaligned word edge (u=$_{s}$ 'tsr̩ː-$_{w}$ $\boxed{\text{nu}_{s}}$) and the latter because the line's penultimate position is assigned a light syllable. Inversion ([u='zɛm.ʎi.tsu 'tsr̩ː.nu]) is the preferred option because it does away with both markedness penalties incurred by the canonical order. However, this is done at the expense of END-WEIGHT and ADJECTIVEFIRST. The coincident violation of

Align and *Light/FinalStrong is more costly, but the order favored by End-Weight and AdjectiveFirst still wins out in over a quarter of verse lines. StOT inflates the effect of Align, predicting, as a consequence, a much higher inversion rate than observed (91% fitted vs. 72% observed).

In the actual data, the strong preference for the Align-compliant orders in (78) is not solely attributable to Align, since this constraint gangs up with another phonological constraint, *Light/FinalStrong. The StOT model nonetheless appears to attribute almost all of this effect to Align, which renders this constraint overly powerful.

The poor fit in (78b) thus follows directly from the StOT grammar's inability to distribute explanatory power across multiple constraints when Align gangs up with other constraints. Because cumulative effects are suppressed, Align ends up absorbing much of the joint contribution, effectively doing the work of multiple cooperating constraints.

3.3.2 Constraints in stringency relations

Another manifestation of StOT's suppression of interaction effects is its tendency to inflate the effects of more specific constraints in subset-superset pairs. Constraints in stringency relations invariably gang up because the violation of the more specific constraint is always accompanied with the violation of the more general one. Being unable to distribute ample power to both ganging constraints, StOT overestimates the effect of the more specific constraints across the board.

Consider 3σInertia. Trisyllabic adjectives resist moving over trisyllabic head nouns to comply with this constraint. However, 3σInertia is not the only constraint responsible for this robust avoidance. Every violation of 3σInertia in NP environments is by definition also a violation of AdjectiveFirst. The two constraints jointly

oppose inversion, meaning that at least some of this effect is attributable to ADJECTIVE FIRST.

Unsurprisingly, StOT assigns nearly all the power to the stronger of the two interacting constraints — 3σINERTIA. This inflates the overall contribution of 3σINERTIA well beyond its actual power, resulting in near-floor inversion probabilities in virtually all conditions involving two trisyllables. This is at odds with the data in (79), where inversion takes place at non-negligible rates.

(79) a. *sitnoga bisera* ['sit.nɔː.ga 'bi.sɛ.ra] 'small pearl.GEN'

	3σINERT	*'L/FINS	*L/FINS	ADJ1ST	p(MaxEnt)	p(cNHG)	p(StOT)	Observed
w(MaxEnt)	2.29	0.99	1.81	0.97				
w(cNHG)	1.98	0.77	1.79	0.71				
r(StOT)	97.69	93.83	89.12	87.67				
a. Adj N 'sit.nɔː.ga 'bi.sɛ.ra			−1		0.81	0.74	1.00	0.84
b. N Adj 'bi.sɛ.ra 'sit.nɔː.ga	−1			−1	0.19	0.26	0.00	0.16

b. *sivoga sokola* ['siː.vɔː.ga sɔ.'kɔ.la] 'gray falcon.GEN'

	3σINERT	*'L/FINS	*L/FINS	ADJ1ST	p(MaxEnt)	p(cNHG)	p(StOT)	Observed
w(MaxEnt)	2.29	0.99	1.81	0.97				
w(cNHG)	1.98	0.77	1.79	0.71				
r(StOT)	97.69	93.83	89.12	87.67				
a. Adj N 'siː.vɔː.ga sɔ.'kɔ.la		−1	−1		0.61	0.53	0.91	0.65
b. N Adj sɔ.'kɔ.la 'siː.vɔː.ga	−1			−1	0.39	0.47	0.09	0.35

In both tableaux in (79), the adjectival modifier has a heavy penult, while the head noun's penult is light. *LIGHT/FINALSTRONG thus functions as an inversion trigger in both cases. The only difference between the two conditions lies in the head noun's stress position: antepenultimate in (79a) and penultimate in (79b). Inversion is more likely in (79b) because the canonical order violates both weight-mapping constraints,

placing a stressed light in the final strong position. StOT derives some additive effect of *StressedLight/FinalStrong in (79b), but frequency matching is far from satisfactory. The model predicts near-floor inversion because 3σInertia is too strong. Stochastic HGs capture the patterns considerably better, with MaxEnt achieving a better fit than cNHG.

3.3.3 Local conclusion

The results presented thus far accord with the picture emerging from previous work on gang effects in StOT (Maslova, 2007; Hayes and Londe, 2006; Smith and Pater, 2020; Flemming and Magri, 2026). StOT produces a highly restricted form of ganging-up cumulativity, whereby the contribution of two or more interacting constraints is only slightly larger than the individual contribution of the strongest of the interacting constraints (Maslova, 2007). Along these lines, the StOT model of poetic word order in BCMS consistently underestimates the severity of coincident constraint violations. As a result, multiply violating word orders are consistently assigned higher probabilities than observed.

Failure to adequately model constraint cumulativity leads to a series of adverse effects. Unable to attribute power to all interacting constraints, the StOT grammar compensates by inflating the independent effects of higher-ranking constraints. The inflated ranking values of the higher-ranked constraints make ranking permutations across evaluations highly unlikely. This results in strong preferences for word orders that satisfy the highest-ranking active constraint in the tableau, irrespective of how they fare on lower-ranked constraints. By contrast, stochastic HGs predict robust cumulativity of constraint violations. In what follows, I examine whether adding conjoined constraints can help StOT derive gang effects more successfully.

3.4 Constraint conjunction in StOT

The baseline StOT grammar fails to capture two robust ganging-up interactions in BCMS folk meter: between *Light/FinalStrong and AdjectiveFirst in (76a), and between End-Weight and AdjectiveFirst in (77a). In this section, I test whether augmenting the baseline grammar with the LCs of these interacting constraints (80) enables StOT to accommodate their cumulative effects.

(80) a. End-Weight & AdjectiveFirst Assign a violation for each line that:

1. has a line-final prosodic word that is preceded (within the second hemistich) by a syllabically longer word; and
2. contains an inverted (adjective-final) NP.

b. *Light/FinalStrong & AdjectiveFirst

Assign a violation for each line that:

1. contains a light syllable in the final strong position (position 9); and
2. contains an inverted (adjective-final) NP.

The rationale behind complementing the baseline grammar with LCs is that LCs additionally penalize multiply violating candidates, further reducing their probabilities. This mechanism has been shown to work in HG (Classical and MaxEnt) in cases of extreme superlinear cumulativity (Shih, 2017a; Milenković, 2026). Whether LCs have a similar effect in StOT has not, to my knowledge, been tested. I demonstrate here that, unlike in HG, the additional violations that LCs assign can result in only a marginal decrease in probability for multiply violating candidates.

In §3.4.1, I review previous work on OT-LC as a theory of ganging-up cumulativity. In §3.4.2, I present and evaluate a StOT-LC model of poetic word order.

3.4.1 OT-LC as a theory of cumulativity

As a prelude to the discussion of StOT-LC, I provide a brief literature review of deterministic OT-LC. Several arguments have been advanced in favor of HG over OT-LC. The most straightforward argument is parsimony: HG accommodates ganging-up cumulativity with fewer degrees of freedom than OT-LC (Jäger and Rosenbach, 2006, 965; Farris-Trimble, 2008, 112, 248; Blumenfeld, 2015, 82).

Another oft-cited argument for HG concerns OT-LC's lack of restrictiveness, that is, its propensity to overgenerate. OT-LC captures gang effects, but predicts pathological interactions that are not reproduced by corresponding HG grammars with simplex constraints.[12]

The overgeneration argument has two facets. On the typological side, OT-LC is argued to predict unattested, pathological, or otherwise unlikely grammars (Pater, 2009a, 218; Pater, 2009b, 1014; Pater, 2016; Potts et al., 2010, 103; Farris-Trimble, 2008, 166). This view builds on the broader observation that LCs are prone to overgeneration (Itô and Mester, 1998; Kirchner, 1996; Padgett, 2002; McCarthy, 2003a; Pater, 2009a; Hayes and White, 2015). Studies that evaluate the broader predictions of OT-LC (e.g., Moreton and Smolensky, 2002; Mai and Bakovic, 2020; Milenković, 2026) suggest that, with restrictions on which constraint types may be conjoined (à la Fukazawa and Miglio, 1998), OT-LC's overgeneration problem can be substantially curtailed. On this view, conjoining only functionally related constraints may add closer to the right amount of power to OT, deriving gang effects while limiting (if not eliminating) pathological mappings, though whether a fully general, theory-independent

[12]On the flip side, Tebay and Zimmermann (2020, 41) argue that OT-LC produces simpler ganging-up interactions than HG. This is because, in HG, all constraint violations contribute to Harmony, whereas interactions between independent constraints are inhibited in strict-ranking OT. OT-LC predicts simple two-way interactions between the conjoined constraints, but no interactions involving other constraints.

criterion for 'functional relatedness' can be stated remains an open question.

On the language-internal side, LCs may be involved in ranking paradoxes: A conjunction may need to be ranked differently for different tableaux in a single OT grammar. A conjunction may therefore help the OT grammar capture gang effects, while the ranking required for that analysis conflicts with rankings independently required by other, unrelated processes. I suspect that such cases are more common than the literature suggests. One reason may be that gang effects are often analyzed in isolation, without evaluating their interaction with other processes in the grammar. A more holistic approach could potentially identify more pervasive ranking inconsistencies involving LCs.

In contrast to the repeatedly raised overgeneration concerns, undergeneration is a relatively underexplored limitation of OT-LC. OT-LC is typically regarded as undesirably powerful (Pater, 2009a,b; Potts et al., 2010), but in some cases, OT-LC accounts of gang effects are not powerful enough. A few case studies suggest that OT-LC cannot derive certain gang effects that are readily captured by plain HG (Walker, 2017; Ryan, 2026). In what follows, I document the stochastic analogue of this limitation. Despite its greater complexity, StOT-LC fails to derive some gang effects that plain HG captures straightforwardly by means of its built-in additivity of constraint violations. Undergeneration constitutes the strongest argument for HG over OT-LC, as it shows that LCs do not salvage strict-ranking OT in the face of gang effects.

Apart from the criticism of OT-LC as a theory of ganging-up cumulativity, the mechanism of LC has drawn several further objections. First, to date, no learning theory has been devised for LC (Pater, 2009a, 223–224; Pater, 2009b, 1014). It thus remains unclear under what conditions learners posit such interaction terms in the first place. In addition, the presence of LCs in the grammar is argued to undermine one of the central postulates of OT — strict ranking — in that one constraint may

appear in several positions in the hierarchy, as part of a conjunction and as a simplex constraint (Kager, 1999, 400; McCarthy, 2002, 18). Relatedly, the availability of freely positing LCs during phonological acquisition leads to the credit problem (Dresher, 1999), making it difficult for OT learners to choose between inducing a conjunction and reranking simplex constraints to match the target grammar (Idsardi, 2000, 348; Jesney, 2005, 38).

Co-relevance, or functional relatedness, of the conjoined constraints has likewise featured prominently in the literature. LC brings together independent, often disparate, linguistic dimensions (Pater, 2009a, 222; see also Hayes, 2021, 38 on the phonetic justification of plausible conjunctions). In HG, by contrast, independent factors gang up by default, which obviates the need to posit any formal relatedness between them through LC. Breiss and Albright (2022, 24) provide a compelling example of an interaction between two independent dimensions, vowel backness and consonant nasality, which show "no formal, typological, or phonetic connection." Some of the ganging-up interactions documented in this dissertation fall into this category, as they involve unrelated factors, namely language-specific linearization requirements and prosodic factors such as end-weight.

The proper definition of locality domains for LC constitutes another frequently raised objection to LC as a mechanism. This issue connects naturally to overgeneration, since the broader the domain of an LC, the more prone it is to producing bizarre interdependencies between unrelated configurations. To limit the power of LC, Łubowicz (2005) proposes general restrictions on locality domains. However, Pater (2009a, 221) shows that OT-LC's overgeneration problems persist even when such locality restrictions are in place. Relatedly, Walker (2017) reports that the undergeneration of the OT-LC account of Yakut vowel harmony stems primarily from a locality inconsistency: The domain of LC must be broad enough for the conjoined constraint to take effect,

eliminating a dangerous contender, but with a broad local domain, the intended winner loses to a different contender. Importantly, however, Farris-Trimble (2008, 162) shows that some cumulative interactions arise over large domains, which falls out naturally from HG's additivity of constraint violations.

The foregoing review is confined to deterministic OT-LC. One might expect StOT to mitigate some of OT's inherent shortcomings, given that StOT, unlike Classical OT, derives a limited form of ganging-up cumulativity without LCs. Crucially, the gang effects produced by plain StOT are too weak to match the actual magnitude of the interaction effects in the data. LCs hold promise in this regard, since they amplify the severity of coincident constraint violations, inducing a greater aggregate penalty for doubly violating candidates. In what follows, however, I show that the known shortcomings of deterministic OT-LC are, if anything, exacerbated in its stochastic counterpart.

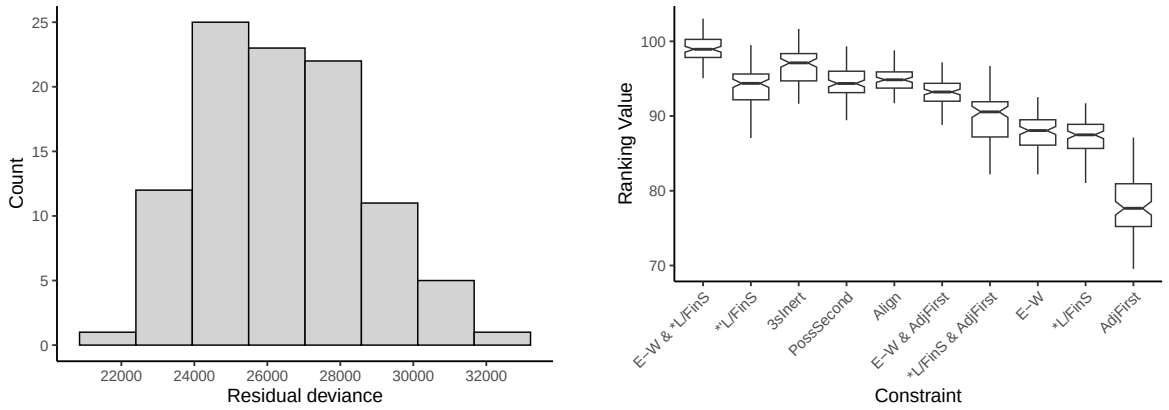
3.4.2 StOT-LC

This section focuses on the StOT-LC model of poetic word order in BCMS oral folk tradition. In §3.4.2.1, I lay out the OT-LC simulation setup and results. In §3.4.2.2, I analyze StOT-LC's handling of the ganging-up interactions that are problematic for the baseline model. In §3.4.2.3, I present some detrimental effects of the added LCs in other corners of the grammar.

3.4.2.1 Procedure and results

The OT-GLA procedure for the grammar enriched with the LCs is the same as in Chapter 2: I conduct 100 OT-GLA runs in OT-Soft and choose the best-fitting model (with the lowest deviance) for framework assessment. Figure (a) in 3.3 plots the deviances of all 100 StOT-LC models. The deviance of the best-fitting grammar is 21,966. Figure

(b) shows the distribution of ranking values across 100 runs.



(a) **(b)**

Figure 3.3: (**a**) Histogram of residual deviances across 100 StOT-LC models (100 independent GLA runs in OT-Soft); (**b**) Ranking values for the best-fitting LC-augmented model across these 100 runs.

Table 3.1 shows the base ranking values in the best-fitting StOT-LC grammar, along with the ranking strata for each constraint in the baseline and superset grammars.

	ranking value stoch. OT-LC (best fit)	ranking stratum stoch. OT-LC (best fit)	ranking stratum stoch. OT (best fit)
End-Weight & *Light/FinalStrong	98.26	#1	#2
*StressedLight/FinalStrong	95.85	#2	#4
3σInertia	94.56	#3	#1
PossessiveSecond	94.12	#4	#3
Align	93.78	#5	#5
End-Weight & AdjectiveFirst	91.89	#6	N/A
*Light/FinalStrong & AdjectiveFirst	90.55	#7	N/A
End-Weight	88.71	#8	#6
*Light/FinalStrong	88.01	#9	#7
AdjectiveFirst	83.26	#10	#8

Table 3.1: Base ranking values in the best-fitting LC-augmented StOT grammar, with ranking strata for the baseline and LC-augmented models.

The baseline and superset grammars differ in the relative order of the top four constraints. At the bottom of the hierarchy, the relative order of End-Weight, *Light/FinalStrong, and AdjectiveFirst is stable. The most striking change

from the baseline to the superset model is that the lowest-ranking constraint, AdjectiveFirst, is considerably weaker relative to the immediately superordinate constraints End-Weight and *Light/FinalStrong in the LC-augmented model. For example, in the baseline model, the base ranking difference (AdjectiveFirst minus End-Weight) is -2.27, which maps onto a 21% probability that AdjectiveFirst $\gg$ End-Weight. The gap between the two constraints is considerably wider in the superset model: The base ranking difference is -5.45, such that AdjectiveFirst almost categorically gives way to End-Weight, with p(AdjectiveFirst $\gg$ End-Weight) of only 3%.

AdjectiveFirst is attenuated in the superset model because of the presence of two conjunctions involving this linearization constraint, which absorb some of its contribution. Given that in StOT, more specific constraints tend to absorb contributions of their general counterparts (as shown in §3.3.2), the two conjunctions involving AdjectiveFirst end up chipping away at some of AdjectiveFirst's baseline effect size. The attenuation of AdjectiveFirst's contribution has detrimental effects that I discuss in greater detail in §3.4.2.3.

Turning to the added conjoined constraints: End-Weight & AdjectiveFirst and *Light/FinalStrong & AdjectiveFirst outrank all of their conjuncts, but rank below all other constraints in the grammar (the exception is *Light/Final Strong & AdjectiveFirst, which outranks one other constraint, End-Weight). Table 3.2 shows the goodness-of-fit metrics for the baseline and superset StOT models.

Considering raw fit alone (i.e. without a complexity penalty introduced by e.g. BIC), the superset grammar fits the data worse than the baseline, since it has a lower log-likelihood and higher deviance. It is somewhat surprising that the superset StOT-LC grammar underperforms the baseline grammar (StOT; no conjunctions) despite two added parameters. More complex models are expected to at least match the

	updated baseline	LC-augmented
k	8	10
MAE	0.09	0.09
RMSE	0.16	0.16
r_w^2	0.86	0.86
log-likelihood	−15,009	−15,391
deviance	21,202	21,966

Table 3.2: Goodness-of-fit comparison between the updated baseline and LC-augmented StOT grammars (as a reminder, k is the number of constraints in the grammar). The simpler model fits the data better than the model with two additional conjunctions.

performance of the baseline, which happens when the added parameters are completely useless. This holds in MaxEnt (and Classical HG), where an unhelpful constraint can be assigned zero weight. In the superset StOT model, the added conjunctions are not only useless, in that they do not help much with the ganging-up interactions, but also outright detrimental, since they hurt the grammar's handling of some aspects of poetic word order (82). However, unlike HG, strict-ranking OT has no means to completely switch off a constraint: Irrespective of how low it ranks, a constraint can never be completely turned off in an OT grammar (McCarthy and Prince, 1994; McCarthy, 2008, 26). Specifically, the conjunctions add to the severity of the violations incurred by some of the attested candidates, and the StOT grammar has no way to neutralize that harm.

3.4.2.2 Ganging-up cumulativity in StOT-LC

This section addresses the role of conjoined constraints in modeling gang effects in StOT. I consider ganging-up interactions within two constraint triplets that were shown in §3.3 to pose a major problem for StOT. The first triplet, illustrated in (81–82), includes End-Weight, *Light/FinalStrong, and AdjectiveFirst. In (81), *Light/FinalStrong and AdjectiveFirst jointly oppose higher-ranked End-Weight,

whereas in (82), End-Weight and AdjectiveFirst gang up against *Light/Final Strong. The baseline model captures only the latter interaction, where the preferred winner is favored by the highest-ranked constraint in the triplet, End-Weight. However, the baseline model fails to accommodate the cumulative effect of the two lower-ranked constraints in (81), erroneously assigning greater probability to the End-Weight-compliant, doubly violating inversion candidate. This defect of the baseline grammar is the rationale for introducing the LC of *Light/FinalStrong and AdjectiveFirst, which additionally penalizes the doubly violating inversion candidate in (81):

(81) *bijelijem gradom* ['bi.jɛ.li.jɛm 'graː.dɔm] 'white city.ins'

	End-W & Adj1st	*L/FinS & Adj1st	End-W	*L/FinS	Adj1st	p(StOT-LC)	p(StOT; baseline)	Observed
r(StOT-LC)	91.89	90.55	88.71	88.01	83.26			
a. Adj N 'bi.jɛ.li.jɛm 'graː.dɔm			−1			0.79	0.46	0.69
b. N Adj 'graː.dɔm 'bi.jɛ.li.jɛm		−1		−1	−1	0.21	0.54	0.31

With the added conjoined constraint, the noncanonical order violates three constraints: *Light/FinalStrong, AdjectiveFirst, and their conjunction. *Light/FinalStrong & AdjectiveFirst is stronger than End-Weight, the constraint that the conjuncts gang up against. The base ranking difference is $r_{*\text{L/FinS \& Adj1st}} - r_{\text{End-W}} = 1.84$. This translates into the pairwise ranking probability $p(*\text{L/FinS \&Adj1st} \gg \text{End-W}) = \Phi(\frac{1.84}{2\sqrt{2}}) = 0.74$.

The preferred candidate in the baseline model is the noncanonical order, given that End-Weight is individually stronger than both ganging constraints, and their cumulative effect is attenuated, in line with the general properties of StOT cumulativity (Maslova, 2007; Smith and Pater, 2020; §3.2). The added LC remedies this shortcom-

ing of the baseline model: The preferred winner in (81) is the canonical order. The conjoined constraint is the main contributor to this cumulative effect. The probability that *Light/FinalStrong, AdjectiveFirst, or their conjunction outranks End-Weight (79%) is only slightly higher than the 74% probability contributed by the conjunction alone.

Example (82) instantiates a different ganging-up configuration within the same constraint triplet: AdjectiveFirst sides with End-Weight rather than *Light/FinalStrong. *Light/FinalStrong & AdjectiveFirst is inactive in this condition, but the other added LC, End-Weight & AdjectiveFirst, militates against inversion.

(82) *mlada udovica* [ˈmlaːda u.ˈdɔ.ʋi.ʦa] 'young widow.NOM'

	End-W & Adj1st	*L/FinS & Adj1st	End-W	*L/FinS	Adj1st	p(StOT-LC)	p(StOT; baseline)	Observed
r(StOT-LC)	91.89	90.55	88.71	88.01	83.26			
a. Adj N ˈmlaːda u.ˈdɔ.ʋi.ʦa				−1		0.93	0.66	0.79
b. N Adj u.ˈdɔ.ʋi.ʦa ˈmlaːda	−1		−1		−1	0.07	0.34	0.21

Although inversion is dispreferred, it takes place in 21% of verse lines. The model augmented with the LCs conspicuously underestimates the inversion rate, predicting only 7% inversion. The main reason is that the noncanonical order simultaneously violates End-Weight and AdjectiveFirst, incurring a costly penalty from their conjunction. Because End-Weight & AdjectiveFirst is considerably stronger than *Light/FinalStrong, the model's predicted 7% inversion rate is well below the observed 21%.

The second constraint triplet considered here includes Align, End-Weight, and AdjectiveFirst. In (83), two lower-ranking constraints, End-Weight and Ad-

jectiveFirst, gang up against Align, while in (84), Align and AdjectiveFirst jointly counter End-Weight. In both cases, the doubly violating noncanonical order is dispreferred to the canonical order, as indicated by 35% inversion in (83) and only 15% in (84). In both conditions, the baseline model assigns greater probability to the candidates favored by Align, the highest-ranked of the three constraints.

Consider (83) first. The cumulative contribution of End-Weight and AdjectiveFirst in this condition is grossly attenuated in the baseline model, which assigns only 10% probability to the preferred winner, the Align-violating canonical order (vs. 65% observed). The conjunction of End-Weight and AdjectiveFirst could therefore mitigate this severe prediction error, by adding to the aggregate penalty incurred by the Align-compliant inversion candidate:

(83) *i mlada slavuja* 'and young nightingale.acc' [i=ˈmlaː.da sla.ˈʋuː.ja]

	Align	End-W & Adj1st	End-W	Adj1st	*p*(StOT-LC)	*p*(StOT; baseline)	Observed
r(StOT-LC)	93.78	91.89	88.71	83.26			
a. Adj N i=ˈmlaː.da sla.ˈʋuː.ja	−1				0.26	0.10	0.65
b. N Adj sla.ˈʋuː.ja i=ˈmlaː.da		−1	−1	−1	0.74	0.90	0.35

However, even the grammar with the LC of the ganging constraints fails to capture the qualitative generalization in (83). The predicted preferred winner is the Align-compliant inversion candidate, with 74% fitted probability. Unlike (81), where the conjunction outranks the constraint its conjuncts gang up against, End-Weight & AdjectiveFirst ranks below Align here, with only a 25% pairwise ranking probability. Consequently, the superset grammar, much like the baseline grammar, grossly underestimates the severity of simultaneously violating End-Weight and AdjectiveFirst, despite the presence of their conjunction. The added conjoined constraint

thus only slightly reduces the prediction error of the baseline grammar but does not bring about any substantive improvement.

Thus, the StOT grammar still undergenerates: It fails to capture the gang effect of End-Weight and AdjectiveFirst against Align, even with the added LC. This is a particularly strong argument against OT-LC as a theory of cumulativity (to my knowledge, one that has not previously been made in the stochastic setting), since the more parsimonious HG account straightforwardly captures the ganging-up interaction that OT-LC cannot model despite its added complexity. Recall from (77a) that both NHG and MaxEnt capture the qualitative generalization in this condition, with excellent quantitative fit as well.

The opposite ganging-up pattern obtains in (84): Align and AdjectiveFirst jointly favor the canonical order, whereas End-Weight functions as an inversion trigger. This interaction is unproblematic for StOT because the preferred winner (the canonical order) is favored by Align, the strongest constraint in the triplet. However, both the baseline and superset StOT grammars underestimate the effect of inversion-inducing End-Weight, predicting less inversion than observed. The suppression is somewhat exacerbated in the OT-LC model, since the ranking gap between Align and End-Weight is widened compared to the baseline.

(84) *od suvoga zlata* 'of pure gold.GEN' [ɔd='suː.vɔ.ga 'zlaː.ta]

	Align	End-W & Adj1st	End-W	Adj1st	p(StOT-LC)	p(StOT; baseline)	Observed
r(StOT-LC)	93.78	91.89	88.71	83.26			
a. Adj N ɔd='suː.vɔ.ga 'zlaː.ta			−1		0.96	0.91	0.85
b. N Adj ɔd='zlaː.ta 'suː.vɔ.ga	−1			−1	0.04	0.09	0.15

In sum, the picture that emerges is not favorable for StOT: The added conjunc-

tions only partly improve its handling of ganging-up cumulativity, failing to salvage the model overall. Whereas *Light/FinalStrong & AdjectiveFirst helps the grammar capture the ganging-up interaction in (81), the other conjunction, End-Weight & AdjectiveFirst, fails to impose a sufficiently severe penalty on the coincident violation of its conjuncts in (83). The latter instantiates the stochastic analogue of the undergeneration of OT-LC identified by Walker (2017) and Ryan (2026) in categorical data. This result strongly favors stochastic HGs (MaxEnt and NHG) over StOT. StOT fails to model ganging-up interactions that are readily amenable to simple HG analyses, even when equipped with nondefault machinery (LCs) designed specifically to accommodate gang effects in OT.

3.4.2.3 Further consequences of LC in StOT grammar

The undergeneration identified in the previous section is not the only issue of the StOT-LC model. The role of conjoined constraints is not confined to the ganging-up interactions; they often bear unexpectedly on unrelated patterns elsewhere in the grammar, resulting in ranking paradoxes (Pater, 2009b).

The present model is no exception. For instance, End-Weight & AdjectiveFirst gives rise to a ranking contradiction in the grammar. In (83), this conjunction is overridden by Align, which leads to a rather large prediction error; the data appear to warrant the opposite of the current model's ranking: End-Weight & AdjectiveFirst ≫ Align. Why does then the conjunction rank below Align?

The answer lies in (85). The canonical order [u=ˈtsr̩ː.nu ˈzɛm.ʎi.tsu] 'into the dark earth' violates Align and *Light/FinalStrong, while its noncanonical alternative violates End-Weight and AdjectiveFirst. In the corpus, inversion is robustly preferred, with 72% of verse lines displaying the adjective-final order.

(85) *u crnu zemljicu* [u=ˈtsr̩ː.nu ˈzɛm.ʎi.tsu] 'in black ground.acc'

	Align	End-W & Adj1st	End-W	*L/Fin$S$	Adj1st	p(StOT-LC)	p(StOT; baseline)	Observed
r(StOT-LC)	93.78	91.89	88.71	88.01	83.26			
a. Adj N u='tsɻː.nu 'zɛm.ʎi.tsu	−1			−1		0.26	0.09	0.28
b. N Adj u='zɛm.ʎi.tsu 'tsɻː.nu		−1	−1		−1	0.74	0.91	0.72

The intended preferred winner, [u='zɛm.ʎi.tsu 'tsɻː.nu] 'in the earth dark,' incurs a penalty from End-Weight & AdjectiveFirst. The superset StOT grammar fits the data almost perfectly (74% predicted inversion rate vs. 72% observed), in fact, much better than the baseline grammar. This is because the conjunction ranks below Align, the higher-ranked of the two inversion triggers in (85). Were this conjunction to outrank Align, the canonical order would prevail against the intended preferred winner.

Thus, Align ≫ End-Weight & AdjectiveFirst is the right ranking for (85), whereas (83) entails the opposite ranking: End-Weight & AdjectiveFirst ≫ Align. The current OT-LC model ranks Align over the conjunction, fitting (85) exceptionally well, but failing in (83). This grammar-internal contradiction is the StOT analogue of Classical OT's ranking paradoxes. Because the two patterns require mutually exclusive rankings, no StOT grammar that involves End-Weight & AdjectiveFirst will be able to capture both of them. This further reinforces the conclusion that the LC of End-Weight and AdjectiveFirst is not an adequate solution for their cumulative interaction in the grammar.

Another detrimental effect of the added LCs, albeit a more indirect one, is that the gap between AdjectiveFirst's ranking value and the higher-ranked constraints widens considerably relative to the baseline model. This happens because the two conjunctions involving AdjectiveFirst do much of the work of the simplex constraint,

which is demoted as a result. This attenuation effect is consistent with StOT's general suppression of the more specific of two constraints in a stringency relation. LCs are by definition in a stringency relation with their conjuncts, since the violation of an LC entails the violation of its conjuncts.

Given that AdjectiveFirst is weakened in the superset model, its independent ability to counteract inversion-inducing phonological constraints is markedly reduced. This is shown in (86). The only inversion trigger is End-Weight. Inversion is strongly preferred in the corpus, but the canonical order is still observed in about one-fifth of the lines, which shows that AdjectiveFirst does not categorically give way to higher-ranking End-Weight. The baseline model captures the data well. By contrast, the superset model predicts near-ceiling inversion, because of the increased ranking gap between End-Weight and AdjectiveFirst.

(86) *bijeloga grada* ['bi.jɛ.lɔː.ga 'graː.da] 'white city.GEN'

		End-W	Adj1st	p(StOT-LC)	p(StOT; baseline)	Observed
	r(StOT-LC)	88.71	83.26			
a. Adj N	'bi.jɛ.lɔː.ga 'graː.da	−1		0.03	0.22	0.20
b. N Adj	'graː.da 'bi.jɛ.lɔː.ga		−1	0.97	0.78	0.80

Summing up, this section has identified two major issues in StOT-LC. First, even with the added LCs of cumulatively interacting constraints, StOT cannot accommodate the full range of gang effects produced by stochastic HGs, as shown in (83). Second, the insufficient improvement that comes with the LCs carries a major cost. The LCs introduce the stochastic analogue of a ranking paradox (83; 85) and absorb some of the effects of their conjuncts, diluting their contribution to the grammar. This

is particularly true of AdjectiveFirst, whose ability to counteract higher-ranked phonological constraints on its own is effectively nullified in the superset model (86). Thus, not only is the gain from the added LCs insufficient, but the LCs are also detrimental to the model's handling of unrelated patterns. This results in inferior overall fit relative to the baseline model (Table 3.2).

3.5 Conclusion

Putting the pieces together, this chapter examines ganging-up interactions between constraints that modulate word order in BCMS folk meter and how different stochastic models of poetic word order, namely StOT, cNHG, and MaxEnt, accommodate this phenomenon. Across the board, coincident violations of multiple phonological constraints, as well as coincident violations of phonological and linearization constraints, erode word-order probability in the corpus. Stochastic variants of HG (cNHG and MaxEnt) do considerably well on ganging-up cumulativity, predicting widespread interactions between independent metrical constraints. StOT is an outlier: Whereas coincident violations of multiple constraints add up to yield stronger penalties than individual constraint violations, the cumulative effects are severely attenuated. Because the severity of simultaneously violating multiple constraints is weakened, multiply violating candidates are generally assigned much higher probabilities than their corpus rates. As a result, the baseline StOT grammar of poetic word order fails to capture cases where multiple weaker constraints gang up against a stronger constraint, consistently predicting word orders favored by the highest-ranking active constraint to be preferred winners. In other words, StOT produces a limited range of ganging-up interactions between constraint violations, but nowhere near the actually observed interaction effects.

StOT's failure to accommodate the ganging-up interactions in the data has adverse effects in other corners of the grammar. Because constraint cumulativity is effectively suppressed, the grammar compensates by attributing greater power to individual constraints. This trade-off yields distorted estimates of independent constraint contributions, in that the effects of high-ranked constraints are inflated well beyond their actual contributions. This leads to virtually fixed rankings, which can rarely be altered by random noise perturbation. The issue does not carry over to the stochastic variants of HG, whose estimates of constraint effects are more commensurate with their actual contributions given that interaction effects are properly accounted for.

StOT's suppression of gang effects is not an isolated property of the BCMS case study reported here, as suggested by previous work (Maslova, 2007; Hayes and Londe, 2006; Smith and Pater, 2020). I conduct a large-scale typological analysis, which assesses the types of ganging-up interactions that emerge over a vast space of StOT and HG grammars. The results indicate that StOT cumulativity is severely limited compared to stochastic HGs. StOT grammars generally predict sublinear cumulativity, whereby the effect of simultaneously violating two or more constraints is weaker than chance. In some cases, StOT predicts approximately linear cumulativity, but linearity arises trivially when the ganging constraints virtually categorically yield to the opposing constraints, so their interaction effect cannot be attenuated any further. By contrast, both MaxEnt and cNHG produce a diverse range of cumulativity effects, including approximately linear, sublinear, and superlinear cumulativity. To the extent that attested cumulative interactions in phonological data display superlinearity and (non-trivial) linearity (Hayes et al., 2012; Green and Davis, 2014; Shih, 2017a; Smith and Pater, 2020; Breiss and Hayes, 2020; Breiss and Albright, 2022; Milenković, 2026), StOT undergenerates, and rather severely so.

Given that plain StOT predicts highly limited ganging-up interactions, I test whether

augmenting the model with the LCs of cumulatively interacting constraints helps close the gap between the predicted and observed probabilities of multiply violating word orders. LCs impose additional penalties on doubly violating candidates, which should in principle reduce their probabilities. However, LCs do not salvage StOT. The added complexity of OT-LC does not translate into ample explanatory power, since the grammar still fails to capture the full range of gang effects observed in the corpus data and predicted by plain HG (MaxEnt and cNHG). The LCs thus fail at the very task they are designed to do: capturing ganging-up cumulativity. Stochastic HGs therefore not only offer a more parsimonious account of gang effects than StOT-LC, but also constitute a more powerful theory of ganging-up interactions in variable linguistic data. This study thus identifies the stochastic analogue of the undergeneration of OT-LC previously reported by Walker (2017); Ryan (2026) in categorical data.

The added LCs are not just insufficiently effective; they are also quite detrimental to the grammar's overall fit to the data. In particular, the LCs introduce ranking inconsistencies, whereby some aspects of poetic word order require an LC to outrank a simplex constraint while others require the opposite ranking, forcing the learner to choose between contradictory rankings, failing to accommodate both. Moreover, the conjunctions end up absorbing much of the power of their conjuncts, making the conjuncts too weak to counteract higher-ranked constraints on their own. These undesirable effects of the LCs, paired with the insufficient improvement in the ganging-up conditions, lead to a considerably worse overall fit in the OT-LC grammar than in the baseline grammar, as shown by a log-likelihood of $-15{,}009$ for the baseline model ($k = 8$) versus $-15{,}391$ for the OT-LC model ($k = 10$). That the more complex model fits the data worse than the more parsimonious model confirms that the overall effect of adding LCs in the StOT grammar is rather detrimental.

Ganging-up cumulativity has figured prominently in the debate between HG and

OT-LC (Pater, 2009b; Potts et al., 2010; Farris-Trimble, 2008), but the argument has so far been limited to the categorical implementations of these competing frameworks. The chapter extends the debate to the stochastic terrain. Variable data further expose the limitations of OT-LC: LCs do not help StOT capture all gang effects derived by plain stochastic HGs. In addition, other limitations of OT-LC identified in categorical data, such as ranking inconsistencies, carry over to its stochastic implementation. The phenomenon of ganging-up cumulativity is used here primarily as a testing ground for StOT versus stochastic HGs. In the next chapter, I use another phenomenon with major theoretical bearing, namely harmonic bounding, to probe the differences between the stochastic variants of HG.

On a final note, this chapter provides evidence against OT-LC (deterministic or stochastic) as a general theory of ganging-up cumulativity, lending further support to HG as a theory of phonology. The results do not argue against the necessity of LC as a mechanism in Con. Conjoined constraints have been argued to be necessary in HG under certain conditions, although the status of these analyses and the availability of alternatives are matters of ongoing research (Albright, 2009; Hayes et al., 2012; Green and Davis, 2014; Shih, 2016, 2017a; Milenković, 2026; Chapter 2 of this dissertation). The argument for HG is therefore not that it dispenses with LC (as sometimes argued; Farris-Trimble, 2008; Pater, 2009a,b; Potts et al., 2010), but rather that it adequately constrains the usage of this device, invoking it only for a limited subset of patterns involving extreme superlinearity.

Chapter 4

Harmonic bounding: Unmotivated inversions in meter

4.1 Introduction

In constraint-based theories, two or more competitors are in a harmonic bounding relation if one of them, the bounded, cannot emerge as optimal under any ranking or weighting condition (Samek-Lodovici, 1992; Prince and Smolensky, 2004; Samek-Lodovici and Prince, 1999, 2002; Prince, 2002; Pater, 2016). Harmonic bounding comes in two forms: SIMPLE and COLLECTIVE. Simple harmonic bounding (87a) involves pairwise relations in which the bounded's violations are a proper superset of the bounder's, that is, the bounder outperforms the bounded on at least one constraint, and no constraint favors the bounded over the bounder (Prince and Smolensky, 2004). A collective harmonic bounding relation holds between the bounded and multiple bounders (87b). The bounded violates all discriminator constraints (i.e. constraints differentiating the bounders from the bounded), but incurs fewer penalties to individual discriminator constraints than the bounder that incurs a penalty on that constraint (Samek-Lodovici

and Prince, 1999, 2002). The bounded is therefore intermediate between the bounders on all discriminator constraints: It fares better than one bounder but worse than the other, being invariably suboptimal.

(87) Harmonic bounding

a. Simple harmonic bounding

	$\mathbb{C}_1$	$\mathbb{C}_2$	$\mathbb{C}_n$
a. bounder		*	
b. bounded		*	*!

b. Collective harmonic bounding

	$\mathbb{C}_1$	$\mathbb{C}_2$
a. bounder 1	**	
b. bounded	*	*
c. bounder 2		**

The focus of this chapter is simple harmonic bounding (87a); see Jesney, 2007; O'Hara, 2022; Kaplan, 2021, 2025a; Hayes and Schuh, 2019; Hayes, 2022; Hayes and Kaplan, 2025 for simple harmonic bounding in the context of variation. The phenomenon holds much promise as a diagnostic for distinguishing between competing stochastic frameworks. Constraint-based frameworks differ profoundly in their handling of simple harmonic bounding, as diagrammed in Figure 4.1. The most restrictive class of frameworks enforces ABSOLUTE harmonic bounding, categorically ruling out the bounded. A weaker form is RELATIVE harmonic bounding, whereby the bounded is not invariably assigned zero probability, but it can never receive a higher probability than the bounder. The least restrictive pattern is NO HARMONIC BOUNDING: The bounded can be assigned a higher probability than the bounder.

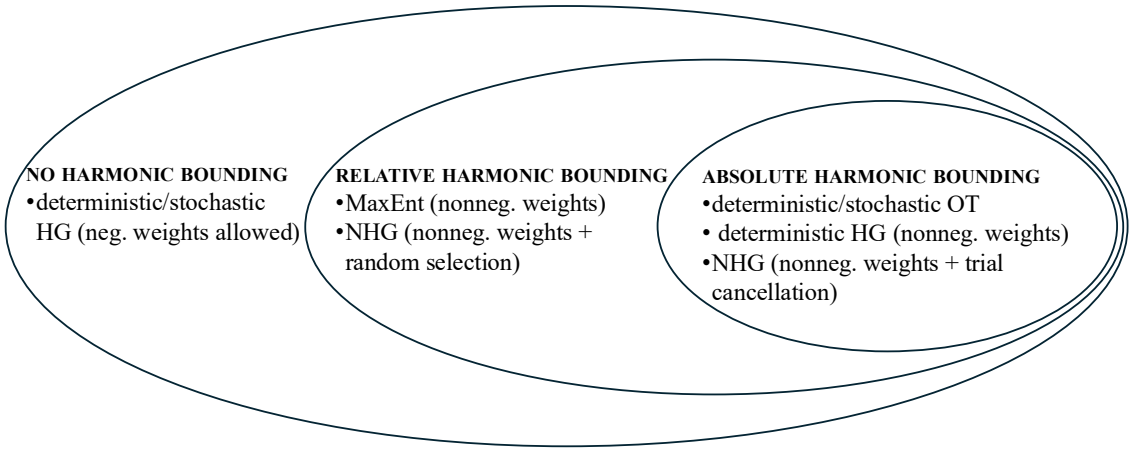


Figure 4.1: Venn diagram of the types of simple harmonic bounding predicted by different versions of OT and HG. Subset relations encode greater restrictiveness.

Strict-ranking OT (both deterministic and stochastic) upholds absolute harmonic bounding: Under all logically possible constraint rankings, the bounded loses out to the bounder. Deterministic HG maintains absolute harmonic bounding provided negative weights are prohibited (Keller, 2000; Potts et al., 2010).

Stochastic variants of HG produce a diverse range of harmonic bounding effects. Insofar as negative weights are prohibited, all stochastic HGs uphold at least relative harmonic bounding. In MaxEnt, all candidates, including simply harmonically bounded ones, are assigned nonzero probabilities (Jesney, 2007; Hayes, 2017) and may, depending on the magnitude of Harmony difference, be assigned relatively high probabilities, as in (88).

(88) Relative harmonic bounding in MaxEnt

	$\mathbb{A}$ 1	$\mathscr{H}$	p
a. bounder		0.00	0.73
b. bounded	−1	−1.00	0.27

cNHG produces variable output by perturbing constraint weights with independent Gaussian noise at each evaluation. After noise perturbation, the resulting weights are fed into a deterministic weighted-constraint grammar, which selects the best-performing candidate as the winner. By definition, simply harmonically bounded forms can never outperform their bounders. The best-case scenario for the bounded is to be tied with the bounder after noise perturbation. The type of harmonic bounding enforced in cNHG depends on how ties between candidates are resolved. One way of breaking ties is by randomly selecting one of the tied candidates as the winner (Boersma and Pater, 2016). When tied with the bounder post-noise perturbation, Random tie resolution is the only mechanism by which the bounded can be selected as winner in cNHG.

In (89a), given the base weight of $\mathbb{A}$, noise perturbation cancels out the bounder's base advantage over the bounded in 16% of evaluations (the precise calculations are provided in §4.2). Because ties are resolved at random, each contender wins in approximately half of the evaluations resulting in a tie (assuming there are two tied candidates in the tableau), which explains the 8% probability assigned to the bounded, hereafter, p(bounded). Alternatively, Hayes and Kaplan (2025) and Kaplan (2025b) propose that evaluation trials resulting in a tie are canceled. Trial cancellation deprives the bounded of its only chance of ever being selected as winner. cNHG-cum-trial-cancellation thus produces absolute harmonic bounding, as in (89b), matching the predictions of strict-ranking OT.

(89) Harmonic bounding in cNHG

a. Relative harmonic bounding with random tie resolution

	$\mathbb{A}$ 1	$\mathscr{H}$	p
a. bounder		$0(1 + n_{\mathbb{A}})$	0.92
b. bounded	-1	$-1(1 + n_{\mathbb{A}})$	0.08

b. Absolute harmonic bounding with trial cancellation

	$\mathbb{A}$ 1	$\mathscr{H}$	p
a. bounder		$0(1 + n_{\mathbb{A}})$	1.00
b. bounded	-1	$-1(1 + n_{\mathbb{A}})$	0.00

This chapter compares stochastic HGs that produce relative harmonic bounding: MaxEnt and cNHG with random tie resolution. Whether upholding absolute harmonic bounding is desirable is a matter of ongoing debate in phonological theory. Maintaining absolute harmonic bounding is customarily viewed as instrumental for typological restrictiveness and tractability in constraint-based models (Prince, 2002; Riggle, 2004; Anttila and Magri, 2017; Mai and Bakovic, 2020; Kaplan, 2021). The inability of frameworks like MaxEnt to enforce absolute harmonic bounding has been argued to cause overgeneration (Anttila and Magri, 2017). By contrast, a series of studies demonstrate that simply harmonically bounded candidates are attested in corpus and experimental data (Hayes and Wilson, 2008; Hayes and Moore-Cantwell, 2011; Hayes and Schuh, 2019; Kumagai, 2023; Siah et al., 2025), treating the ability to assign nonzero probability to simply harmonically bounded forms as a virtue rather than a liability. This study subscribes to the latter view.

Importantly, simple harmonic bounding reveals new falsifiable predictions of MaxEnt and NHG, advancing the debate between stochastic constraint-based frameworks. Whereas ample work has shown that stochastic HGs are to be preferred to StOT (Zuraw and Hayes, 2017; Smith and Pater, 2020; Hayes, 2022, among others; see also

Chapter 3), work probing the differences between stochastic variants of HG has been sparse (see Flemming, 2021; de la Fuente et al., 2025). However, just looking at the literature, there are different, and sometimes even contradictory, explanations as to why this is so. For example, Alderete and Finley (2023, 590) argue that the two frameworks are simply way too similar for there to be any empirical tests to differentiate them. A different position is taken by Flemming (2021, 2), who argues that at least superficially, MaxEnt and Noisy Harmonic Grammar are quite distinct, and it is in fact these differences that have so far resisted clear empirical testing. I use simple harmonic bounding to identify new testable predictions of MaxEnt and cNHG and evaluate these predictions against South Slavic poetic corpus data.

4.1.1 Phonologically unmotivated inversions in BCMS folk meter

The BCMS folk meter is rife with instances of simply harmonically bounded word orders. Syntactic inversion is harmonically bounded whenever no phonological constraint justifies deviation from the canonical order. The key data are given in (90–91): The modifier and the head noun are both disyllabic, their penults are both heavy (90) or both light (91), and stress falls on the initial syllables in all relevant items.

(90) Phonologically unmotivated syntactic inversion (1)

a. ˈɔ.na ˈʊri.snuː ‖ ka.kɔ ˈʎuː.ta ˈguː.ja
3SG.F.NOM scream.AOR.3SG like angry.NOM.SG.F snake.NOM.SG
'She screamed like an angry snake.' [III, 57:93]

b. ˈpi.snuː ˈdɛː.tɛ ‖ ˈkaː.nɔ ˈguː.ja ˈʎuː.ta
squeak.AOR.3SG child like snake.NOM.SG angry.NOM.SG.F
'The child squeaked like an angry snake.' [III, 7:212]

(91) Phonologically unmotivated syntactic inversion (2)

a. ˈɔd.maːx ˈpɔ.znaː ‖ ˈsʊɔː.ga ˈmi.la
right.away recognize.AOR.3SG self's beloved.ACC.SG.M
ˈbra.ta
brother.ACC.SG
'She immediately recognized her beloved brother.' [II, 31:155]

b. ˈjaː nɛ ˈxɔ.ʨu ‖ ˈtʊɔː.ga ˈbra.ta
1SG.NOM NEG want.PRS.1SG your brother.ACC.SG
ˈmi.la
beloved.ACC.SG.M
'I do not want your beloved brother.' [III, 54:205]

None of the phonological constraints introduced in Chapter 1 compel the adjectives in [ˈguː.ja ˈʎuː.ta] 'snake angry' (90b) and [ˈbra.ta ˈmi.la] 'brother beloved' (91b) to occur postnominally, since weight, stress, syllable count, and boundary alignment are all moot. This is a textbook case of simple harmonic bounding: The noncanonical order violates AdjectiveFirst for no apparent reason, while the canonical order is violation-free. Nevertheless, inversion takes place in about one-fifth of lines in (92a). In (92b), the canonical and noncanonical orders share the violations of *Light/FinalStrong and *StressedLight/FinalStrong, while inversion gratuitously violates AdjectiveFirst. (The gratuitous penalties incurred by the bounded are boxed.)

(92) Harmonic bounding

a. *ljuta guja* [ˈʎuː.ta ˈguː.ja] 'angry snake'

	End-W	Align	*L/FinS	*L/FinS	Adj1st	p
a. Adj N ˈʎuː.ta ˈguː.ja						0.80
b. N Adj ˈguː.ja ˈʎuː.ta					$\boxed{-1}$	0.20

b. *mila brata* [ˈmi.la ˈbra.ta] 'beloved brother.GEN'

	End-W	Align	*L/Fin S	*'L/Fin S	Adj1st	p
a. Adj N 'mi.la 'bra.ta			−1	−1		0.77
b. N Adj 'bra.ta 'mi.la			−1	−1	$\boxed{-1}$	0.23

Discussion of simple harmonic bounding is inherently confounded by the possibility that the apparently harmonically bounded candidate is favored by a constraint not included in the current set. This, I argue, is not the case here. The adjective and noun in (92) are identical along all prosodic parameters known to affect word order. Moreover, since all examples come from non-rhymed verse, rhyme cannot be responsible for inversion. Other stylistic factors are also controlled for.

Unexpected inversion cannot be attributed to sampling noise either. In both crucial conditions in (92), the sample size is large enough that the observed inversion rates are unlikely to reflect sampling noise: 418 verse lines in (92a) and 182 in (92b). (95% Wilson confidence intervals exclude zero in both conditions: [0.17, 0.24] in (92a) and [0.18, 0.30] in (92b).) Given the same single discriminator constraint distinguishing the two candidates in (92a–92b), inversion rates are nearly identical in the two conditions: 20% (85 out of 418) in (92a) and 23% (42 out of 182) in (92b); Fisher's exact test OR = 1.17, $p = 0.45$.

It is worth noting that the aggregate rate of unexpected inversion (21.2% for both conditions in (92) combined) is not evenly reflected in all bigrams. Pooling across (92a–92b), eleven adjective–noun bigrams out of 140 attested types in these two prosodic conditions (7.9%) jointly account for 47% of all tokens. Six high-frequency bigrams (e.g. *star, majka* 'old mother', $n = 81$; *vjeran, sluga* 'faithful servant', $n = 43$) show categorically no inversion. Five bigrams show the opposite skew. The most frequent of these is *sinj ,more* 'gray sea' ($n = 28$, 78.6% inverted), which alone contributes 17.3%

of all inverted tokens. Several of these items have the character of fixed epithets, conventionalized pairings characteristic of this poetic tradition. If so, their categorical behavior reflects lexical listing rather than the productive metrical grammar. Crucially, when all eleven frequent idiosyncratic bigrams are excluded, the overall pattern does not disappear: The remaining 129 bigrams still invert at 23.9% (76 of 318 tokens), which is nearly indistinguishable from the original (pre-exclusion) 21.2%. The overall inversion rates in the two key conditions are therefore not reducible to a handful of idiosyncratic items. Unexpected inversion holds as a general property of the adjective–noun construction, even when formulaic exceptions are excluded.

In stochastic HGs, this unmotivated inversion is explained by the inertia of AdjectiveFirst. This constraint ensures that the harmonically bounded word order is dispreferred to the canonical order, but its weight is not high enough to push the bounder to the ceiling.

To preview the results, both MaxEnt and NHG-cum-random-selection can match the observed p(bounded) in (92). However, the concessions that cNHG must make to do so give rise to weighting contradictions that compromise the grammar's handling of other, unrelated aspects of poetic word order. This undesirable result is a direct consequence of NHG's handling of simple harmonic bounding.

Across the board, cNHG produces stronger relative harmonic bounding effects than MaxEnt, assigning lower p(bounded) for all Harmony differences and violation profiles. For example, cNHG imposes a lower ceiling value on p(bounded), predicting a maximum of 25% p(bounded), compared to MaxEnt's 50%. NHG's aggressive harmonic bounding severely suppresses the contribution of constraints that otherwise play an active role in the grammar. Being more permissive, MaxEnt does not suffer from this issue. This result adds to Flemming (2021); Hayes (2022)'s evidence for MaxEnt over cNHG.

The chapter proceeds as follows. §4.2 identifies new testable predictions of MaxEnt and cNHG in the domain of simple harmonic bounding, setting the stage for an empirical evaluation of the two frameworks. §4.3 compares the models' predictions in this domain, bringing to light a trade-off in cNHG between fitting harmonic bounding and capturing the independent strength of linearization pressure in meter. §4.4 addresses the overgeneration concerns surrounding MaxEnt's (and, to a lesser extent, cNHG's) permissiveness vis-à-vis simple harmonic bounding. §4.5 concludes.

4.2 Simple harmonic bounding in stochastic HGs

The MaxEnt–NHG differences in probability mapping extend to their handling of simple harmonic bounding. For pairwise harmonic bounding relations, p(bounded) in MaxEnt is derived from $\Delta\mathscr{H}$, via (93).

(93) p(bounded) in MaxEnt

$$p(\text{bounded}) = \frac{1}{1 + e^{-\Delta\mathscr{H}}}$$

where $\Delta\mathscr{H} = \mathscr{H}_{\text{bounded}} - \mathscr{H}_{\text{bounder}}$ (always nonpositive)

Crucially, the determination of p(bounded) uniformly follows (93) irrespective of the candidates' violation profiles.

The estimation of p(bounded) is more complex in cNHG, given the way NHG introduces stochasticity. Take the simplest possible case of simple harmonic bounding. In (89a), repeated here as (94), a single discriminator constraint $\mathbb{A}$ favors the bounder over the bounded.

(94)

	$\mathbb{A}$		
w_k	1		
n_k	$n_{\mathbb{A}}$	$\mathscr{H}$	p
a. bounder		$0(1 + n_{\mathbb{A}})$	0.92
b. bounded	-1	$-1(1 + n_{\mathbb{A}})$	0.08

Since weights cannot be perturbed below zero, the bounded's Harmony score will always be lower than or equal to the bounder's. The best the bounded can do is be tied with the bounder. This happens whenever the perturbed weight $\hat{w}_{\mathbb{A}}$ is nonpositive, that is, when the noise term $n_{\mathbb{A}}$ is less than or equal to $\mathbb{A}$'s negative base weight, $-w_{\mathbb{A}}$, effectively canceling out the bounder's starting advantage. The probability of $\hat{w}_{\mathbb{A}}$ being nonpositive corresponds to the normal cumulative distribution function of $\mathbb{A}$'s negative base weight, $-w_{\mathbb{A}}$, divided by the standard deviation of random Gaussian noise, σ (here, $\sigma = 1$), as in (95).

(95) Probability of $\hat{w}_{\mathbb{A}} = 0$

$$p(\text{nonpositive}) = \Phi\left(\frac{-w_{\mathbb{A}}}{\sigma}\right)$$

When a single discriminator constraint favors the bounder, as in (94), p(nonpositive) equals p(tie) (for cases involving multiple discriminator constraints, see below).

Varieties of NHG employ two tie resolution strategies: RANDOM SELECTION and TRIAL CANCELLATION. On the former approach (Boersma and Pater, 2016), ties are resolved by randomly selecting one of the equally harmonic candidates as winner. This version of cNHG produces relative harmonic bounding (Hayes and Kaplan, 2025): When the bounded ties with the bounder, it wins approximately half of the time (96).

(96) $$p(\text{bounded}) \approx \frac{1}{2}p(\text{tie})$$

Substituting (95) into (96):

$$(97) \quad p(\text{bounded}) \approx \frac{1}{2}\Phi\left(\frac{-w_{\mathbb{A}}}{\sigma}\right)$$

For $w_{\mathbb{A}} = 1$ in (94), (95) returns 16% p(tie). (97) derives the 8% p(bounded) reported in (94).

An alternative tie resolution strategy is TRIAL CANCELLATION (Kaplan, 2025b; Hayes and Kaplan, 2025). Evaluation trials that result in multiple winners are suspended and simply harmonically bounded candidates can therefore never win. Accordingly, cNHG with trial cancellation produces absolute harmonic bounding.

The purpose of this chapter is not to determine which tie resolution strategy is more appropriate for NHG (see Kaplan, 2025b for discussion). However, to the extent that a theory should enforce relative rather than absolute harmonic bounding (the position I take here), one should adopt a version of cNHG that produces relative harmonic bounding. I show that MaxEnt and cNHG with random selection produce different forms of relative harmonic bounding and argue that cNHG is overly restrictive. The argument extends a fortiori to cNHG with trial cancellation, the more restrictive of the two cNHG variants.

Consequently, MaxEnt and cNHG differ in two respects in their determination of p(bounded): (i) cNHG always assigns lower p(bounded) than MaxEnt (§4.2.1), and (ii) cNHG, unlike MaxEnt, is crucially sensitive to the source of the Harmony difference between bounded and bounder (§4.2.2).

4.2.1 cNHG bounds more aggressively

cNHG[1] pushes the bounded more aggressively towards zero than MaxEnt: For any given violation profile and resulting $\Delta\mathscr{H}$, cNHG assigns a lower p(bounded) than MaxEnt under identical weighting conditions. This is a direct consequence of NHG's

[1]Henceforth, cNHG refers to cNHG with random selection, unless stated otherwise.

deterministic output selection, due to which the bounded can prevail only when it is tied with the bounder. cNHG's restrictiveness is evident from the stark contrast between (88) and (94): The bounded violates a single constraint $\mathbb{A}$ ($w = 1$), which maps to 8% p(bounded) in cNHG vs. MaxEnt's 27%. This is further illustrated by Figure 4.2, which plots p(bounded) as a function of $\mathbb{A}$'s (base) weight. Note that the cNHG curve holds only for the single-discriminator-constraint case; the MaxEnt curve generalizes to any violation profile yielding the same $\Delta\mathscr{H}$.

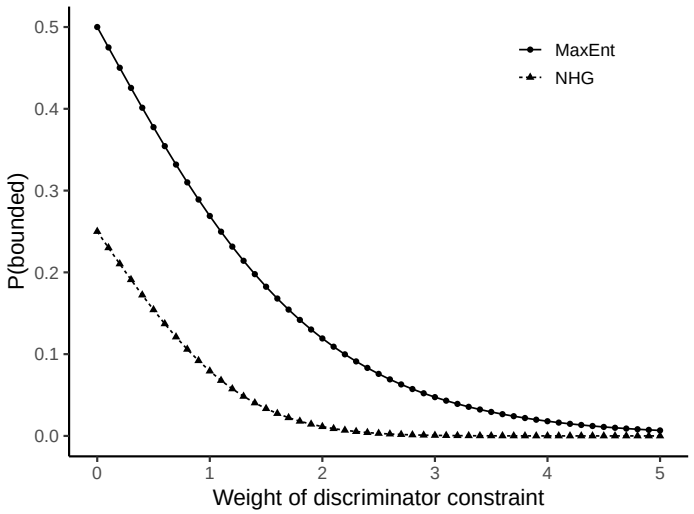


Figure 4.2: Probability of the bounded (vertical axis) by the weight of the discriminator constraint (horizontal axis) in MaxEnt vs. cNHG-cum-random-selection (point shape).

MaxEnt and cNHG impose different ceiling values on p(bounded). In MaxEnt, the upper bound is 50%, whereas in cNHG, p(bounded) can never exceed 25%. p(bounded) reaches 50% in MaxEnt when $\Delta\mathscr{H} = 0$, i.e. when the bounder and bounded are equally harmonic. In cNHG, p(bounded) reaches the 25% ceiling when the bounded violates a single zero-weighted discriminator constraint. Violations of zero-weighted constraints have no contribution in MaxEnt. In cNHG, even zero-weighted constraints affect probability because their perturbed weights are positive in half of evaluations

(Flemming, 2021; Hayes and Kaplan, 2025).[2]

This is illustrated in (98). When $w_{\mathbb{A}} = 0$, $\hat{w}_{\mathbb{A}}$ is positive in half of evaluations, in which the bounder wins. In the other half of evaluations, the candidates are tied. Because winners are selected at random, each candidate wins in approximately half of the evaluations resulting in a tie, yielding $p(\text{bounded}) \approx 25\%$.

(98) Upper bound on p(bounded)

	$\mathbb{A}$	p	p
	0	cNHG	MaxEnt
a. bounder		0.75	0.50
b. bounded	-1	0.25	0.50

(98) is as good as things can get for the bounded in cNHG. Violating a higher-weighted constraint, or even violating multiple zero-weighted constraints results in a lower p(bounded).

Harmonically bounded candidates in cNHG can be assigned higher probabilities by increasing the standard deviation of the Gaussian noise added to the base weight(s) of the discriminator constraint(s). A higher σ widens the spread of perturbed weights around their base values across evaluations. This increases the likelihood of the discriminator constraint's perturbed weight being nonpositive. Higher noise therefore translates into a higher p(tie), which in turn yields a higher p(bounded). In (94), the bounded violates only one discriminator constraint $\mathbb{A}$ ($w_{\mathbb{A}} = 1$), receiving 8% probability with $\sigma = 1$. All other things being equal, increasing σ yields a higher p(bounded), as (99) shows.

(99) Harmonic bounding in cNHG with $\sigma > 1$

[2] $p(\text{positive}) = \Phi(0) = 0.50$.

	$\mathbb{A}$			
w_k	1			
n_k	$n_{\mathbb{A}}$	$\mathscr{H}$	$p(\sigma = 2)$	$p(\sigma = 5)$
a. bounder		$0(1 + n_{\mathbb{A}})$	0.85	0.79
b. bounded	-1	$-1(1 + n_{\mathbb{A}})$	0.15	0.21

Importantly, the 25% ceiling on p(bounded) holds in cNHG irrespective of how noisy weight perturbation becomes: Simply harmonically bounded candidates may not, under any circumstances, receive a probability higher than 25% in cNHG. This is shown in Figure 4.3.

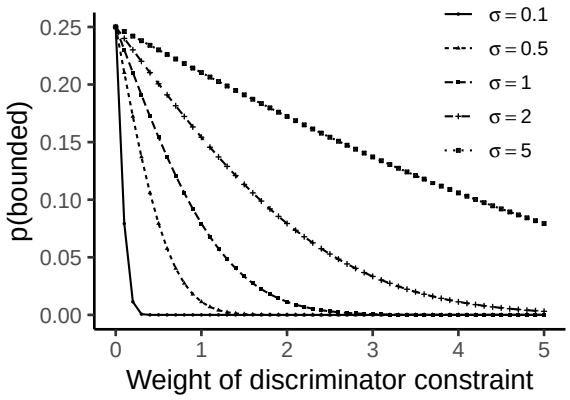


Figure 4.3: p(bounded) in cNHG as a function of the weight of the discriminator constraint, for different standard deviations of random noise (point and line shape). Compared to the standard setting used throughout ($\sigma = 1$), the range of p(bounded) increases at $\sigma > 1$ and decreases at $\sigma < 1$.

4.2.2 cNHG is sensitive to source of Harmony difference

The second major difference between stochastic HGs lies in cNHG's SENSITIVITY TO THE BOUNDED'S VIOLATION PROFILE (Flemming, 2021). As (100) shows, violating two or more zero-weighted constraints will have a different effect on probability from

violating a single zero-weighted constraint in cNHG, whereas in MaxEnt, the effect will be the same irrespective of the number of penalties.

(100) Effect of violating multiple zero-weighted constraints

	$\mathbb{A}$ 0	$\mathbb{B}$ 0	p cNHG	p MaxEnt
a. bounder			0.875	0.500
b. bounded	-1	-1	0.125	0.500

In cNHG, the bounder and the bounded are tied when both $\mathbb{A}$'s and $\mathbb{B}$'s perturbed weights are nonpositive. The probability that each of the individual $\hat{w}$s is nonpositive is determined by (95), yielding 50% of nonpositive $\hat{w}$ for both $\mathbb{A}$ and $\mathbb{B}$. p(tie) is the likelihood that these two independent events occur simultaneously, which equals their intersection, i.e. the product of individual p(nonpositive), as formalized in (101). For two zero-weighted constraints, both of whose p(nonpositive) is 0.50, p(tie) is $0.50 \times 0.50 = 0.25$. Random tie resolution accounts for the p(bounded) of 12.5% reported in (100).

(101) p(tie) for N discriminator constraints

$$p(\text{tie}) = \prod_{i=1}^{N} \Phi\left(\frac{-w_i}{\sigma}\right)$$

Another example is (102). In NHG, violating two constraints in (102b) has a more detrimental effect on p(bounded) than violating a single constraint in (102a), despite the same aggregate penalty incurred by the bounded. MaxEnt assigns the same p(bounded) in both cases. The tableaux in (102) illustrate both properties of cNHG identified above: its sensitivity to the source of $\Delta\mathscr{H}$, and its aggressive harmonic bounding.

(102) $x \neq 2 \times \frac{1}{2}x$ in NHG

a.

	$\mathbb{A}$ 0.50	p cNHG	p MaxEnt
a. bounder		0.85	0.62
b. bounded	-1	0.15	0.38

b.

	$\mathbb{B}$ 0.25	$\mathbb{C}$ 0.25	p cNHG	p MaxEnt
a. bounder			0.92	0.62
b. bounded	-1	-1	0.08	0.38

cNHG's sensitivity to the origin of $\Delta\mathscr{H}$ leads to some borderline bizarre predictions for simple harmonic bounding, like the one in (103).

(103)

		$\mathbb{A}$ 1.0	$\mathbb{B}$ 0.1	$\mathbb{C}$ 0.1	$\mathbb{D}$ 0.1	p(MaxEnt)	p(cNHG)
a.	Candidate 1	-1				0.33	0.43
	Candidate 2		-1	-1	-1	0.67	0.57
b.	Candidate 3	-1				0.27	0.08
	Candidate 4					0.74	0.92
c.	Candidate 5		-1	-1	-1	0.43	0.05
	Candidate 6					0.57	0.95

In (103a), three constraints, $\mathbb{B}$, $\mathbb{C}$, and $\mathbb{D}$, gang up against higher-weighted $\mathbb{A}$. $\mathbb{A}$ defeats $\mathbb{B}$, $\mathbb{C}$, and $\mathbb{D}$ combined: Both MaxEnt and cNHG assign lower probabilities to the violator of $\mathbb{A}$ than its rival that simultaneously violates $\mathbb{B}$, $\mathbb{C}$, and $\mathbb{D}$.

In MaxEnt, the severity of the two violation profiles carries over to harmonic bounding. The gratuitous violation of $\mathbb{A}$ in (103b) yields a more severe penalty for the

bounded, i.e. lower p(bounded), than the coincident violation of $\mathbb{B}$, $\mathbb{C}$, and $\mathbb{D}$ in (103c). In other words, $\mathbb{A}$ is stronger and bounds more aggressively than $\mathbb{B}$, $\mathbb{C}$, and $\mathbb{D}$ combined.

Such transitivity does not hold in cNHG. The simultaneous gratuitous violations of the three weaker constraints in (103c) yield a slightly lower p(bounded) (5%) than a single higher-weighted penalty in (103b) (8%). The bounder and bounded are tied in (103) whenever all of $\mathbb{B}$, $\mathbb{C}$, and $\mathbb{D}$'s weights are perturbed to zero at the same time, which, by (101), corresponds to $(\Phi(-0.1))^3 = 0.10$, which explains the p(bounded) $\approx$ 5% obtained in the simulation. Thus, $\mathbb{A}$ prevails against $\mathbb{B}$, $\mathbb{C}$, and $\mathbb{D}$ but bounds less aggressively than the combination of these lower-weighted constraints.

4.2.3 Local conclusion

Summing up, MaxEnt and cNHG produce relative harmonic bounding, but exhibit two major differences in this domain. First, cNHG restricts p(bounded) more aggressively than MaxEnt, assigning lower probabilities to simply harmonically bounded candidates across the board. In cNHG, p(bounded) is always lower than p(bounder), with a ceiling of 25% (104a). In MaxEnt, p(bounded) is lower than or equal to p(bounder) (104b).

(104) a. cNHG: $p(\text{bounder}) > p(\text{bounded}), \quad p(\text{bounded}) \leq 0.25$

b. MaxEnt: $p(\text{bounder}) \geq p(\text{bounded}), \quad p(\text{bounded}) \leq 0.50$

Further, the determination of p(bounded) is sensitive only to Harmony difference in MaxEnt, whereas cNHG shows sensitivity to the source of Harmony difference, i.e. which discriminator constraints the bounded violates, and how many. Therefore, simply harmonically bounded candidates that violate multiple constraints generally receive much lower probabilities in cNHG than in MaxEnt. This is because every constraint violated only by the bounded must simultaneously be perturbed to zero (100). The likelihood of all discriminator constraints simultaneously being zero-weighted becomes

increasingly unlikely as the number of discriminator constraints increases. Even when this happens, the bounded wins only half of the time assuming random selection.

Both properties yield testable predictions for stochastic HGs. This study uses data from South Slavic folk meter to probe the first prediction (aggressiveness). The results suggest that cNHG bounds overly aggressively. For sensitivity to violation profiles in cases not involving simple harmonic bounding, see Flemming (2021).

4.3 Results

This section zooms in on the stochastic HG models' predictions regarding harmonic bounding. StOT's predictions in this domain are largely uninteresting, since the framework enforces absolute harmonic bounding. All simply harmonically bounded word orders in the dataset are therefore categorically ruled out, irrespective of the aggregate severity of their violations. This property sets StOT apart from the stochastic variants of HG, both of which assign nonzero probabilities to some harmonically bounded syntactic inversions.

In §4.3.1, I identify a trade-off effect in cNHG (both GLA-cNHG and ML-cNHG). To produce a high p(bounded) in (92), the linearization constraint AdjectiveFirst receives a minuscule weight in cNHG. The near-zero weight of AdjectiveFirst ensures a high proportion of evaluations in which the bounder and bounded are tied, which in turn yields a higher p(bounded) assuming random selection. However, AdjectiveFirst's role elsewhere in meter warrants a higher weight, which creates a weighting contradiction.

In §4.3.2, I demonstrate that this trade-off is not an isolated property of any particular cNHG model, given that all 100 GLA-cNHG models, along with their ML-cNHG counterpart, exhibit it in some form. Though ML-cNHG alleviates some issues of its

GLA counterparts, it still exhibits a milder form of the trade-off effect. Models that predict high p(bounded) tend to underestimate AdjectiveFirst's overall contribution and predict higher inversion rates than observed across the board. By contrast, cNHG models that accurately capture AdjectiveFirst's independent role in meter predict much lower p(bounded) in (92) than observed. The trade-off is thus directly attributable to a weighting contradiction that stems from cNHG's aggressive harmonic bounding (§4.2).

4.3.1 Trade-off

Simply harmonically bounded word orders are pervasive in the candidate set. The noncanonical order is harmonically bounded whenever no phonological constraint favors it over the canonical order. The canonical order itself is never harmonically bounded, being inherently preferred to inversion by AdjectiveFirst.

The frequency of harmonically bounded candidates correlates with the severity of their aggregate penalties. It is therefore useful to distinguish inversions dispreferred by both syntax and phonology from those dispreferred by syntax alone. In the former case, inversions are unattested or only marginally attested, whereas in the latter, inversions are attested in meter, and at considerable rates. An analogy from segmental phonology can help further pin down the distinction between the two types of harmonically bounded word orders in BCMS folk meter. Some candidates are gratuitously unfaithful to the input but are not inherently more marked than their faithful competitors. An example would be spontaneous assibilation, as in /tam/ → [sam]. There are also forms that violate a faithfulness constraint for no reason and are, in addition, more marked than the faithful candidate, like coda voicing, as in /pat/ → [pad]. BCMS folk meter abundantly attests word orders of the former kind (gratuitous violation of the linearization constraint, no added markedness), but almost never displays harmonically

bounded word orders of the latter kind.

All models do reasonably well on harmonic bounding, predicting near-zero p(bounded) in cases where multiple constraints favor the bounder, and sufficiently high p(bounded) in cases where only AdjectiveFirst favors the bounder.

Turning first to harmonic bounding with multiple discriminator constraints, consider (105). In both tableaux, the canonical order incurs no penalties, whereas its inversion counterpart violates AdjectiveFirst along with multiple phonological constraints. Folk poets choose the violation-free canonical order nearly categorically, as predicted by both MaxEnt and cNHG.

(105) Near-zero p(bounded) with multiple discriminator constraints[3]

a. *mila majka* [ˈmi.la ˈmaːj.ka] 'beloved mother'

	End-W	*L/FinS	*ˈL/FinS	Adj1st	p(MaxEnt)	p(ML-cNHG)	p(GLA-cNHG)	Observed
w(MaxEnt)	2.04	1.81	0.99	0.97				
w(ML-cNHG)	1.55	1.47	−0.15	0.57				
w(GLA-cNHG)	2.05	1.79	0.77	0.71				
a. Adj N ˈmi.la ˈmaːj.ka					0.98	0.99	1.00	0.98
b. N Adj ˈmaːj.ka ˈmi.la		−1	−1	−1	0.02	0.01	0.00	0.02

b. *mila gospodara* [ˈmi.la gɔ.spɔ.ˈdaːra] 'beloved master.gen'

[3]Recall from Chapter 2 that, despite *StressedLight/FinalStrong's base weight being negative in ML-cNHG, its perturbed weight is never negative due to clipping. The negative base weight merely indicates that the perturbed weight will be nonpositive (and thus zero by fiat) in more than 50% of evaluations (for $w_k = -0.15$, p(nonpositive) = 56%.) The negative base weight results from the fact that the Nelder-Mead optimization used for ML-cNHG is not bounded to nonnegative weights, unlike the L-BFGS-B optimization used for MaxEnt.

		End-W	*L/Fin S	*L/Fin S	Adj1st	p(MaxEnt)	p(ML-cNHG)	p(GLA-cNHG)	Observed
	w(MaxEnt)	2.04	1.81	0.99	0.97				
	w(ML-cNHG)	1.55	1.47	−0.15	0.57				
	w(GLA-cNHG)	2.05	1.79	0.77	0.71				
a. Adj N	ˈmi.la gɔ.spɔ.ˈdaː.ra					1.00	1.00	1.00	0.99
b. N Adj	gɔ.spɔ.ˈdaː.ra ˈmi.la	−1	−1	−1	−1	0.00	0.00	0.00	0.01

Of greater theoretical interest are the data in (106): Inversion and the canonical order fare equally on all phonological constraints. The only constraint distinguishing between the two is AdjectiveFirst.

(106) Higher p(bounded) with single discriminator constraint

a. *ljuta guja* [ˈʎuː.ta ˈguː.ja] ‘angry snake’

		*L/Fin S	*L/Fin S	Adj1st	p(MaxEnt)	p(ML-cNHG)	p(GLA-cNHG)	Observed
	w(MaxEnt)	1.81	0.99	0.97				
	w(ML-cNHG)	1.47	−0.15	0.57				
	w(GLA-cNHG)	1.79	0.77	0.71				
a. Adj N	ˈʎuː.ta ˈguː.ja				0.72	0.86	0.88	0.80
b. N Adj	ˈguː.ja ˈʎuː.ta			−1	0.28	0.14	0.12	0.20

b. *mila brata* [ˈmi.la ˈbra.ta] ‘beloved brother.gen’

		*L/FinS	*L/FinS	Adj1st	p(MaxEnt)	p(ML-cNHG)	p(GLA-cNHG)	Observed
	w(MaxEnt)	1.81	0.99	0.97				
	w(ML-cNHG)	1.47	−0.15	0.57				
	w(GLA-cNHG)	1.79	0.77	0.71				
a. Adj N	ˈmi.la ˈbra.ta	−1	−1		0.72	0.86	0.88	0.77
b. N Adj	ˈbra.ta ˈmi.la	−1	−1	−1	0.28	0.14	0.12	0.23

The fitted probabilities of the harmonically bounded noncanonical order are relatively high in all three stochastic HG models, though the quantitative fit is not perfect in any of them. The directionality of error is different, since MaxEnt predicts a higher p(bounded) than observed (28% fitted vs. 20%/23% observed), while the cNHG models underestimate p(bounded) in both tableaux (14%/12% fitted vs. 20%/23% observed).

None of the reported stochastic HG models have the weights of AdjectiveFirst that produce perfect frequency matching in the harmonic bounding conditions (106). In the MaxEnt model, AdjectiveFirst's weight (0.97) is lower than necessary to suppress the harmonically bounded inversion candidates in (106) to their observed rates. The MaxEnt weight that predicts p(bounded) = 20% in (106a) can be found via the logit function (107), the inverse of the logistic cumulative distribution function. The logit function of a given probability (here, p(bounded) = 20%) returns the Harmony difference between the two competitors (here, $\mathscr{H}_{\text{bounded}} - \mathscr{H}_{\text{bounder}}$; always nonpositive), i.e. the negative of the target weight of AdjectiveFirst.

(107) Mapping p(bounded) to $\Delta\mathscr{H}$(bounded − bounder) in MaxEnt

$$\Delta\mathscr{H}_{\text{bounded}-\text{bounder}} = \ln\left(\frac{p(\text{bounded})}{1-p(\text{bounded})}\right)$$

For p(bounded) = 20% in (106a), (107) returns a $\Delta\mathscr{H}$ of −1.39. The weight of AdjectiveFirst that matches the observed inversion rate in (106a) is therefore 1.39

(vs. the ML weight of 0.97). Importantly, this mismatch has no detrimental effects elsewhere in the grammar: As I show below, MaxEnt's weight for AdjectiveFirst is consistent with this constraint's independent contribution in (109–111), unlike in cNHG.

By contrast, the weights assigned to AdjectiveFirst by the cNHG models are higher than the target weight that produces $p(\text{bounded}) = 20\%$. In cNHG with random selection, $p(\text{bounded}) = 20\%$ arises when the bounder and the bounded are tied in approximately 40% of evaluations. The weight of AdjectiveFirst that gives rise to 40% $p(\text{tie})$ can be found via the probit function (108), the inverse of the normal cumulative distribution function.[4] For $p(\text{tie}) = 0.40$, (108) returns the negative target weight of the discriminator constraint AdjectiveFirst, -0.25, meaning that the target weight that produces 20% $p(\text{bounded})$ in cNHG is 0.25.

(108) Mapping $p(\text{tie})$ to the weight of the discriminator constraint in cNHG

$$-w(\text{discriminator constraint}) = \Phi^{-1}(p(\text{tie}))$$

Crucially, however, AdjectiveFirst must not only produce reasonable frequency matching in cases like (106) but also be consistent with the general strength of the linearization pressure in the poetic grammar. As shown in §4.2, cNHG reaches a ceiling value of 25% $p(\text{bounded})$ when a single zero-weighted constraint distinguishes the bounder from the bounded. The observed $p(\text{bounded})$ in (106) (20% and 23%) are very close to cNHG's ceiling $p(\text{bounded})$, implying that AdjectiveFirst must receive near-zero weight, specifically, around 0.25, to match the attested word-order rates in (106).

AdjectiveFirst receives relatively low weights in both cNHG models reported here (0.57 and 0.71), but still significantly higher than the 0.25 weight that would be

[4] `qnorm()` in base R.

needed for perfect frequency matching in (106). As a result, even these comparatively low weights fail to capture the observed 20%/23% p(bounded). Meanwhile, the low weights assigned to AdjectiveFirst by the cNHG models cannot be fully reconciled with its role elsewhere in the grammar: The constraint makes a tangible impact on word-order probabilities by counteracting inversion-inducing phonological constraints. The assigned cNHG weights are thus, in a way, doubly problematic, in that they are not low enough to predict as high a p(bounded) as observed, and not high enough to capture AdjectiveFirst's counteracting effect elsewhere in the grammar.

In (109–110), AdjectiveFirst pushes back against inversion triggers by ganging up with other phonological constraints. (The data were discussed more extensively in Chapter 3 in the context of cumulativity.) Consider (109). The quadrisyllabic adjective ['bi.jɛ.li.jɛm] 'white.ins.sg.m' is longer than the disyllabic head noun ['graː.dɔm] 'city.ins.sg.' End-Weight thus functions as an inversion trigger. Inversion satisfies End-Weight at the expense of two lower-weighted constraints: AdjectiveFirst and *Light/FinalStrong. Folk poets prefer the canonical order to its doubly-violating inversion counterpart.

(109) AdjectiveFirst gangs up with *Light/FinalStrong:

bijelijem gradom ['bi.jɛ.li.jɛm 'graː.dɔm] 'white city.ins'

	End-W	*L/Fin S	Adj1st	p(MaxEnt)	p(ML-cNHG)	p(GLA-cNHG)	Observed
w(MaxEnt)	2.04	1.81	0.97				
w(ML-cNHG)	1.55	1.47	0.57				
w(GLA-cNHG)	2.05	1.79	0.71				
a. Adj N 'bi.jɛ.li.jɛm 'graː.dɔm	−1			0.68	0.66	0.64	0.69
b. N Adj 'graː.dɔm 'bi.jɛ.li.jɛm		−1	−1	0.32	0.34	0.36	0.31

All models capture the data with considerable quantitative precision. The cNHG

models slightly overestimate inversion rates due to AdjectiveFirst's low weight, but the error is for the most part benign.

An even more dire effect of assigning a low weight to AdjectiveFirst is provided in (110), the mirror image of (109). Inversion satisfies *Light/FinalStrong but in turn violates both AdjectiveFirst and End-Weight. The canonical order is strongly preferred to its doubly-violating *Light/FinalStrong-compliant alternative.

(110) AdjectiveFirst gangs up with *End-Weight (1):

mlada udovica [ˈmlaː.da u.ˈdɔ.ʋi.tsa] 'young widow'

	End-W	*L/FinS	Adj1st	*p*(MaxEnt)	*p*(ML-cNHG)	*p*(GLA-cNHG)	Observed
w(MaxEnt)	2.04	1.81	0.97				
w(ML-cNHG)	1.55	1.47	0.57				
w(GLA-cNHG)	2.05	1.79	0.71				
a. Adj N ˈmlaː.da u.ˈdɔ.ʋi.tsa		−1		0.77	0.70	0.75	0.79
b. N Adj u.ˈdɔ.ʋi.tsa ˈmlaː.da	−1		−1	0.23	0.30	0.25	0.21

All three models accurately capture the qualitative generalization, but ML-cNHG assigns a higher probability than observed to the doubly-violating noncanonical order, with a higher error than in (109). Due to AdjectiveFirst's low weight in ML-cNHG, the combined effect of the ganging constraints End-Weight and AdjectiveFirst is underestimated.

The ML-cNHG model displays a similar behavior in other cases of ganging-up interactions involving AdjectiveFirst. In (111), the linearization constraint and End-Weight jointly oppose higher-weighted Align. Moving a disyllabic modifier to line-final position avoids the word-foot misalignment of the canonical order, but at the expense of both AdjectiveFirst and End-Weight. The noncanonical order is

therefore dispreferred to the violator of ALIGN.

(111) ADJECTIVEFIRST gangs up with *END-WEIGHT (2):

mlada slavuja [i='mlaː.da sla.'vuː.ja] 'and young nightingale'

	ALIGN	END-W	ADJ1ST	p(MaxEnt)	p(ML-cNHG)	p(GLA-cNHG)	Observed
w(MaxEnt)	2.24	2.04	0.97				
w(ML-cNHG)	2.00	1.55	0.57				
w(GLA-cNHG)	2.15	2.05	0.71				
a. Adj N i='mlaː.da sla.'vuː.ja	−1			0.68	0.57	0.67	0.65
b. N Adj i=sla.'vuː.ja 'mlaː.da		−1	−1	0.32	0.43	0.33	0.35

ML-cNHG again predicts a higher inversion rate than observed, because ADJECTIVEFIRST's weight is low. As in (110), the magnitude of ML-cNHG's error exceeds the error in (109). This again shows the cost of suppressing ADJECTIVEFIRST's contribution, particularly when this constraint is needed to counteract higher-weighted inversion-inducing constraints.

ADJECTIVEFIRST's weight in ML-cNHG is lower than in the best-fitting GLA-cNHG model. This lower weight leads to slightly higher p(bounded) in (106), though still not nearly as high as in MaxEnt. This partial improvement comes at a cost: A weight low enough to approach the observed p(bounded) is too low to capture ADJECTIVEFIRST's independent role elsewhere in the grammar, causing ML-cNHG to overpredict inversion across the board, most severely in (110) and (111). ML-cNHG thus narrows cNHG's trade-off, but the issue does not vanish. In the next section, I show that this trade-off is not an isolated property of any particular cNHG model: All cNHG models exhibit the trade-off in some form, regardless of how the weights are obtained.

4.3.2 Weighting contradiction in cNHG

The takeaway of the previous section is that cNHG grammars of poetic word order exhibit a trade-off: Fitting a high p(bounded) requires a minuscule weight for the discriminator constraint AdjectiveFirst, but AdjectiveFirst's inversion-inhibiting role in meter requires a considerably higher weight. A question arises of whether this trade-off can be avoided under some weighting conditions in cNHG. The answer, I contend, is no. The first indicator is the ML-cNHG model, which finds the optimal weights for cNHG. The ML weights do not rescue cNHG from this trade-off, as shown in §4.3.1.

In this section, I demonstrate that the trade-off identified in §4.3.1 is unavoidable. Identical trade-off effects emerge in all 100 GLA-cNHG models fitted for this study, as a direct consequence of the way cNHG translates Harmony into p(bounded). No cNHG model, including the one with ML weights, can simultaneously accommodate AdjectiveFirst's near-zero contribution in (106) and its tangible role in (109–111), since the two require conflicting weights for AdjectiveFirst.

For all 100 GLA-cNHG models, I analyze the relationship between the fitted p(bounded) in (106a) and inversion probabilities in other corners of the grammar. Figure 4.4 plots each model's p(bounded) in (106a) against its fitted inversion probability in (110).[5] As shown by a beta regression analysis (Smithson and Verkuilen, 2006; implemented in R using the `betareg` package; Cribari-Neto and Zeileis, 2010), the two dimensions are positively correlated: $\beta = 5.32$, $p < 0.001$, pseudo-$R^2 = 0.41$. Models that produce a high enough p(bounded) tend to predict undesirably high inversion rates in (110).

This trade-off stems from a weighting contradiction. Being close to cNHG's ceiling of 25% (Figure 4.2), a p(bounded) of 20% entails a near-zero weight for Adjective-

[5]Recall that in (110), AdjectiveFirst and End-Weight gang up against *Light/FinalStrong.

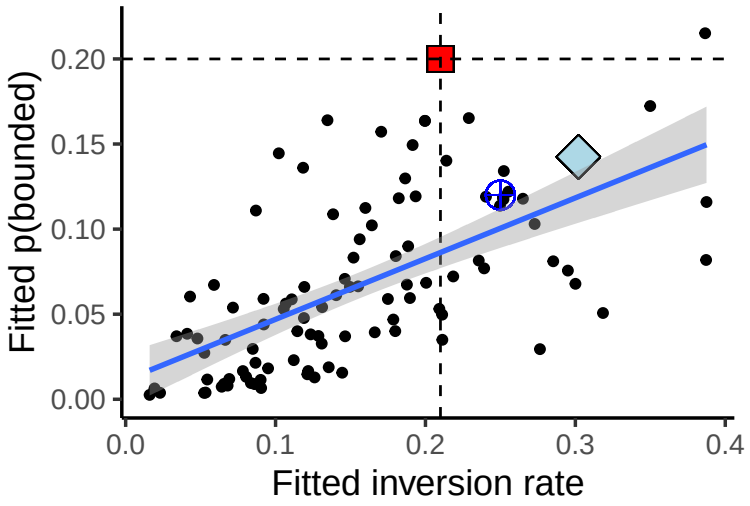


Figure 4.4: Correlation between predicted p(bounded) in (106a) (vertical axis) and predicted inversion probability in (110) (horizontal axis). Each black point represents one of the 100 GLA-cNHG runs. The red square marks the observed, target rates. The dark blue circle singles out the best-fitting GLA-cNHG model. The light blue diamond represents the fitted probabilities of the ML-cNHG model.

First. However, a near-zero weight limits AdjectiveFirst's ability to push back against inversion-inducing phonological constraints elsewhere in the grammar. Models that fit (106a) well therefore tend to overestimate inversion rates in (110) and similar cases. Conversely, increasing AdjectiveFirst's weight keeps the inversion rate in (110) close to the observed 21%, but pushes p(bounded) well below the observed rate.

Some GLA-cNHG models fit both (106a) and (110) conspicuously better than the best-fitting GLA-cNHG and ML-cNHG models reported in §4.3.1. A case in point is the model in (112).[6] This model predicts only a slightly lower p(bounded) than observed, as (112a) shows, and closely approximates the inversion rate in (112b). However, this solid performance comes at a cost elsewhere. The model predicts a low inversion rate in (112b) because one of the inversion-inhibiting constraints, End-Weight, is strong. In (112c), where End-Weight is an inversion trigger, the model erroneously predicts

[6]Though the overall fit is not great: residual deviance = 3,524, MAE = 0.07, RMSE = 0.11.

near-ceiling inversion, because AdjectiveFirst is too weak to fend off End-Weight.

(112) a. *ljuta guja* [ˈʎuː.ta ˈguː.ja] ‘angry snake’

	End-W	*L/FinS	AdjFirst	Pred	Obs
	2.93	2.39	0.44		
a. Adj-N ˈʎuː.ta ˈguː.ja				0.83	0.80
b. N-Adj ˈguː.ja ˈʎuː.ta			−1	0.17	0.20

b. *mlada udovica* [ˈmlaː.da u.ˈdɔ.ʋi.tsa] ‘young widow’

	End-W	*L/FinS	AdjFirst	Pred	Obs
	2.93	2.39	0.44		
a. Adj-N ˈmlaː.da u.ˈdɔ.ʋi.tsa		−1		0.77	0.79
b. N-Adj u.ˈdɔ.ʋi.tsa ˈmlaː.da	−1		−1	0.23	0.21

c. *bijeloga grada* [ˈbi.jɛ.lɔː.ga ˈgraː.da] ‘white city.Gen’

	End-W	*L/FinS	AdjFirst	Pred	Obs
	2.93	2.39	0.44		
a. Adj-N ˈbi.jɛ.lɔː.ga ˈgraː.da	−1			0.04	0.20
b. N-Adj ˈgraː.da ˈbi.jɛ.lɔː.ga			−1	0.96	0.80

Example (112c), where AdjectiveFirst clashes with End-Weight, is important because it reveals another trade-off effect in cNHG, depicted in Figure 4.5. The figure shows the correlation between fitted p(bounded) and fitted inversion probability in (112c). The two are positively correlated, though the effect is borderline: $\beta = 1.26$, $p = 0.05$, pseudo-$R^2 = 0.05$. Models that predict a high p(bounded) tend to overestimate inversion in (112c), whereas models that approach the observed inversion rate in (112c) tend to predict a lower p(bounded). This is a direct consequence of the weighting contradiction: AdjectiveFirst’s low weight yields a high p(bounded) but pushes inversion in (112c) towards ceiling.

Strikingly, besides the general trade-off, many fitted GLA-cNHG models fare poorly along both dimensions. The dense clustering in the bottom-right region of Figure 4.5 shows that a number of models simultaneously overpredict inversion in (112c) and underpredict p(bounded) in (106a). The source of this doubly erroneous pattern is that AdjectiveFirst's weight in these models is relatively high (hence near-floor p(bounded)), but still too low relative to End-Weight to prevent near-ceiling inversion in (112c).

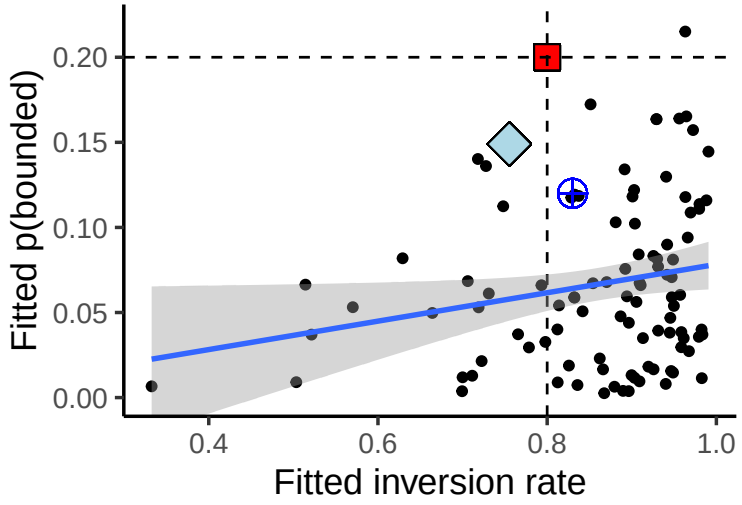


Figure 4.5: Predicted p(bounded) in (106a) against predicted inversion probability in (112c). Markers as in Figure 4.4.

In both cases, the weighting contradiction is a direct consequence of cNHG's aggressive harmonic bounding. Accommodating the high p(bounded) in (106) requires a minuscule weight for AdjectiveFirst, which is at odds with AdjectiveFirst's independently established contribution to poetic word order. The trade-off is therefore not an isolated prediction of the models reported in §4.3.1: It obtains, in some form, in all cNHG models, including the ML-cNHG model. The findings thus point to a substantive limitation of cNHG. The issue is not simply a matter of quantitative fit to a specific dataset; the weighting contradiction stems directly from the mechanism by

which cNHG assigns probability to harmonically bounded forms (§4.2).

4.4 Harmonic bounding and overgeneration

The upshot of §4.3 is that cNHG, as implemented here, bounds overly aggressively. Both cNHG and MaxEnt can assign nonzero p(bounded), but under the same weighting conditions and violation vectors, MaxEnt grammars consistently assign greater probabilities to harmonically bounded candidates. BCMS folk meter, in which some harmonically bounded word orders are attested at relatively high rates, supports MaxEnt's predictions in this domain.

On the flip side, MaxEnt's ability to assign relatively high probabilities to harmonically bounded candidates is often viewed as a serious, if not insurmountable, issue. In Classical OT, harmonic bounding plays a critical role in maintaining typological universals and excluding unnatural patterns (Iosad, 2017; Beguš, 2018, among others). Failure to enforce absolute harmonic bounding reportedly leads to problematic typological predictions. Along these lines, Anttila and Magri (2017) observe that MaxEnt grammars fail to uphold some implicational universals.[7]

Before turning to the specifics of this argument, it is worth pausing to deconstruct the assumption on which it rests. Relative harmonic bounding, as established throughout this chapter, is a relation that holds *within* a single tableau: The probability of the bounded is constrained relative to its own bounders. Nothing about this relation entails, or even suggests, that a bounded candidate in one tableau should receive lower probability than an unbounded candidate in an unrelated tableau, with its own distinct violation profile. Anttila and Magri's (2017) implicational-universal argument

[7]Though the exact form of the relational universals utilized by Anttila and Magri (2017) is called into question by Magri and Durresi (2023).

tacitly assumes exactly this kind of cross-tableau comparison, i.e., that harmonically bounded candidates, as a class, should receive lower probabilities than unbounded candidates from other tableaux. This is an important assumption, and this dissertation gives independent reason to reject it. The discussion of (116) below shows that some harmonically bounded candidates may surface more frequently than non-bounded losing candidates from other tableaux. The harmonically bounded candidate in (116a) receives a higher probability (20%) than the unbounded candidate in the unrelated tableau (116b). Anttila and Magri's implicational universals discussed here involve this type of cross-tableau relation. The question, then, is not whether we should assume that bounded candidates are generated with less frequency than unbounded candidates, since the obvious answer is no, as (116) clearly shows. I now focus on the specific typological claims discussed by Anttila and Magri.

In probabilistic grammars, natural processes are often assumed to be universally more likely than their unnatural counterparts (Beguš, 2018, 192*ff.*), though whether this holds as a gradient statistical tendency is an open empirical question that, to my knowledge, has not been systematically tested across languages. Setting this aside for now, MaxEnt at minimum predicts grammars in which, e.g., unnatural onset deletion is more likely than onset insertion, its natural counterpart (113).

(113) Universal-subverting MaxEnt grammar: $p(/\text{CV}/ \rightarrow [\text{V}]) > p(/\text{V}/ \rightarrow [\text{CV}])$

a. $p(/\text{CV}/ \rightarrow [\text{V}]) = 27\%$

/CV/	Onset 0.5	NoCoda 0.5	Max 0.5	Dep 5.0	$\mathscr{H}$	$e^{\mathscr{H}}$	p
a. CV					0.0	1.00	0.73
b. V	−1		−1		−1.0	0.37	0.27
c. CVC		−1		−1	−5.5	0.00	0.00
d. VC	−1	−1	−1	−1	−6.5	0.00	0.00

b. $p(/\text{V}/ \rightarrow [\text{CV}]) = 1\%$

/V/	Onset 0.5	NoCoda 0.5	Max 0.5	Dep 5.0	$\mathscr{H}$	$e^{\mathscr{H}}$	p
a. CV				−1	−5.0	0.01	0.01
b. V	−1				−0.5	0.61	0.99
c. CVC		−1		−2	−10.5	0.00	0.00
d. VC	−1	−1		−1	−6.0	0.00	0.00

While the argument is customarily framed as pertaining exclusively to MaxEnt (Anttila and Magri, 2017; Magri, 2018; Magri and Anttila, 2026), it applies to any theory that fails to enforce absolute harmonic bounding, given that universal-subverting grammars fall out from the failure to categorically rule out harmonically bounded candidates. The most likely source of the view that MaxEnt is the only stochastic variant of HG susceptible to the universal-subverting pattern is a conviction, widespread in the literature predating Hayes and Kaplan (2025), that any version of NHG upholds absolute harmonic bounding provided clipping is enforced (Jesney, 2007; Hayes, 2017; Flemming, 2017; O'Hara, 2022).[8]

Indeed, cNHG with random selection produces a universal-subverting pattern under the same weighting conditions, as (114) shows. The issue is somewhat alleviated in cNHG, which, due to its aggressive harmonic bounding, predicts a smaller probability difference between unnatural /CV/ → [V] in (114a) and natural /V/ → [CV] in (114b) than its MaxEnt counterpart in (113). Nevertheless, the added restrictiveness of cNHG with random selection is not, by itself, enough to rule out the universal-subverting pattern. The only remedy is adopting trial cancellation as the tie resolution strategy (Hayes and Kaplan, 2025), which prevents the onset-deleting candidate from winning in any single evaluation.

[8]For example: "Noisy HG, like Standard HG, never allows simply harmonically-bounded candidates to win, assuming all weights are positive" (Jesney, 2007, 3); "Under a suitable interpretation, Classical NHG preserves the OT principle that harmonically-bounded candidates never win; i.e. they are assigned the probability zero. The essential auxiliary assumption is that weights, even after perturbation under noise, are always kept positive" (Hayes, 2017, 5).

(114) Universal-subverting cNHG grammar: $p(/\text{CV}/ \rightarrow [\text{V}]) > p(/\text{V}/ \rightarrow [\text{CV}])$

a. $p(/\text{CV}/ \rightarrow [\text{V}]) = 5\%$

/CV/	Onset	NoCoda	Max	Dep		
w_k	0.5	0.5	0.5	5.0		
n_k	n_1	n_2	n_3	n_4	$\mathscr{H}$	p
a. CV					0.0	0.95
b. V	−1		−1		$-(0.5 + n_1)-(0.5 + n_3)$	0.05
c. CVC		−1		−1	$-(0.5 + n_2)-(5.0 + n_4)$	0.00
d. VC	−1	−1	−1	−1	$-(0.5 + n_1)-(0.5 + n_2)$ $-(0.5 + n_3)-(5.0 + n_4)$	0.00

b. $p(/\text{V}/ \rightarrow [\text{CV}]) = 0\%$

/V/	Onset	NoCoda	Max	Dep		
w_k	0.5	0.5	0.5	5.0		
n_k	n_1	n_2	n_3	n_4	$\mathscr{H}$	p
a. CV				−1	$-(5 + n_4)$	0.00
b. V	−1				$-(0.5 + n_1)$	1.00
c. CVC		−1		−2	$-(0.5 + n_2)-2(5.0 + n_4)$	0.00
d. VC	−1	−1		−1	$-(0.5 + n_1)-(0.5 + n_2)$ $-(5.0 + n_4)$	0.00

Example (114) demonstrates that cNHG can produce the universal-subverting pattern with hand-assigned weights. A separate question is whether the cNHG learner, when presented with the data that display the universal-subverting pattern, converges on a grammar that reproduces the universal-subverting pattern. The answer is positive: cNHG-GLA routinely learns universal-subverting grammars. I conduct a cNHG simulation in OT-Soft using the tableaux in (113–114) and the input frequencies as shown in Table 4.1.

Input	Output	Description	Input frequency
/CV/	[CV]	faithful realization	8,000
/CV/	[V]	onset deletion	2,000
/CV/	[CVC]	coda insertion	0
/CV/	[VC]	onset deletion + coda insertion	0
/V/	[CV]	onset insertion	500
/V/	[V]	faithful realization	9,500
/V/	[CVC]	onset insertion + coda insertion	0
/V/	[VC]	coda insertion	0

Table 4.1: Input to the cNHG-GLA simulation in OT-Soft. The data display the universal-subverting pattern: Onset insertion has a lower application rate than unnatural onset deletion.

The base weights learned in the cNHG-GLA simulation, reported in the tableaux in (115), produce the universal-subverting grammar. The model predicts a higher application rate for unnatural onset deletion (8%) than for onset insertion (0%).

(115) cNHG-GLA can learn universal-subverting grammars

a. $p(/\text{CV}/ \rightarrow [\text{V}]) = 8\%$

/CV/	Onset	NoCoda	Max	Dep		
w_k	0.46	2.00	0.03	4.69		
n_k	n_1	n_2	n_3	n_4	$\mathscr{H}$	p
a. CV					0.0	0.92
b. V	−1		−1		$-(0.46 + n_1)-(2.00 + n_3)$	0.08
c. CVC		−1		−1	$-(2.00 + n_2)-(4.69 + n_4)$	0.00
d. VC	−1	−1	−1	−1	$-(0.46 + n_1)-(2.00 + n_2)$ $-(0.03 + n_3)-(4.69 + n_4)$	0.00

b. $p(/\text{V}/ \rightarrow [\text{CV}]) = 0\%$

/V/	Onset	NoCoda	Max	Dep		
w_k	0.46	2.00	0.03	4.69		
n_k	n_1	n_2	n_3	n_4	$\mathscr{H}$	p
a. CV				-1	$-(4.69 + n_4)$	0.00
b. V	-1				$-(0.46 + n_1)$	1.00
c. CVC		-1		-2	$-(2.00 + n_2) - 2(4.69 + n_4)$	0.00
d. VC	-1	-1		-1	$-(0.46 + n_1) - (2.00 + n_2) - (4.69 + n_4)$	0.00

Along similar lines, Magri and Anttila (2026, 13) adduce natural coda deletion (/CVC/ → [CV]) versus unnatural coda insertion (/CV/ → [CVC]) as another example of a probabilistic implicational universal that does not hold in MaxEnt, but, per the authors, does hold in NHG, because it holds in classical HG and "HG and NHG share the same universals" (*loc.cit*).[9] Again, the implicational universal does not hold in Boersma and Pater (2016)'s standard implementation with random selection, which, much like MaxEnt, predicts the universal-subverting pattern with $p(/\text{CV}/ \rightarrow [\text{CVC}]) > p(/\text{CVC}/ \rightarrow [\text{CV}])$. For instance, a grammar in which Max is sufficiently strong, while the other constraints in (114) are inert,[10] predicts more probability for coda insertion than coda deletion in both MaxEnt (27% coda insertion vs. 1% coda deletion) and cNHG (5% coda insertion vs. 0% coda deletion).

In both stochastic variants of HG considered here, the universal-subverting pattern stems from the failure to enforce absolute harmonic bounding. A harmonically bounded candidate with a small Harmony deficit in one tableau can receive a much greater probability than a non-harmonically bounded candidate incurring a more severe aggregate penalty in another tableau. In (113–114), the weight of Dep relative to all other constraints is immense, which ensures that Dep-violating candidates, even markedness-reducing ones like /V/ → [CV], receive vanishing probabilities. Meanwhile,

[9]It is unclear which variant of NHG the authors refer to.

[10](Base) weights: Onset = 0.5, NoCoda = 0.5, Max = 5.0, Dep = 0.5.

onset deletion is more likely because both constraints it violates, Onset and Max, have low weights.

In the MaxEnt and cNHG models of poetic word order in BCMS, this sensitivity to the magnitude of harmony deficit (not harmonic bounding per se) turns out to be a virtue: Both models capitalize on the fact that some harmonically bounded candidates may be more harmonic (and thus more probable) than non-harmonically bounded candidates in other conditions. The harmonically bounded non-canonical order of (106a), repeated here as (116a), has a relatively high frequency (20%), violating only the weakest constraint in the grammar, AdjectiveFirst. Recall that, because canonical orders are always favored by AdjectiveFirst, they are never harmonically bounded. However, some canonical orders are highly phonologically marked, as in (116b). Though not harmonically bounded, the canonical order violates three strong phonological constraints: End-Weight, *Light/FinalStrong, and *Stressed-Light/FinalStrong. Inversion takes place almost without exception: Only 2 out of the 574 NP bigrams in this condition resist inversion. All stochastic HG models correctly predict near-categorical inversion. The non-canonical order of (116a) is overall more harmonic than the canonical order of (116b), despite the former's being harmonically bounded.

(116) a. *ljuta guja* [ˈʎuː.ta ˈguː.ja] 'angry snake'

		*L/FinS	*L/FinS	Adj1st	*p*(MaxEnt)	*p*(ML-cNHG)	*p*(GLA-cNHG)	Observed
	w(MaxEnt)	1.81	0.99	0.97				
	w(ML-cNHG)	1.47	−0.15	0.57				
	w(GLA-cNHG)	1.79	0.77	0.71				
a. Adj N	ˈʎuː.ta ˈguː.ja				0.72	0.86	0.88	0.80
b. N Adj	ˈguː.ja ˈʎuː.ta			−1	0.28	0.14	0.12	0.20

b. *velikoga čuda* [ˈʋɛ.li.kɔː.ga ˈʧu.da] ‘great miracle.GEN’

	End-W	*L/Fin S	*ˈL/Fin S	Adj 1st	*p*(MaxEnt)	*p*(ML-cNHG)	*p*(GLA-cNHG)	Observed
w(MaxEnt)	2.04	1.81	0.99	0.97				
w(ML-cNHG)	1.55	1.47	−0.15	0.57				
w(GLA-cNHG)	2.05	1.79	0.77	0.71				
a. Adj N ˈʋɛ.li.kɔː.ga ˈʧu.da	−1	−1	−1		0.00	0.00	0.80	0.00
b. N Adj ˈʧu.da ˈʋɛ.li.kɔː.ga				−1	1.00	1.00	1.00	1.00

Failure to enforce absolute harmonic bounding is therefore a double-edged sword. Permissiveness is beneficial in cases like BCMS folk meter, where harmonically bounded candidates have probability > 0 (see also Hayes and Moore-Cantwell, 2011; Hayes and Schuh, 2019), but it leads to problematic typological predictions elsewhere (Anttila and Magri, 2017; Magri, 2018; Magri and Anttila, 2026). How, then, can these seemingly contradictory results be reconciled?

The answer may ultimately lie outside the grammar. O’Hara (2022)’s iterated learning simulations suggest that MaxEnt learners are biased against assigning probability to harmonically bounded forms. Grammars involving universal-subverting mappings take more learning effort and are more diachronically unstable, being susceptible to being replaced by grammars that assign no probability to harmonically bounded candidates.[11] This raises the possibility that the rarity of such patterns reflects learner bias rather than a grammar-internal restriction. I leave for future work whether solutions relying on learner biases (à la O’Hara, 2022) provide a viable alternative to the grammar-internal solutions proposed by Anttila and Magri (2017); Magri and Anttila (2026).

[11]The idea falls along the general lines of a growing body of work suggesting that some unlikely grammars are underattested because they are difficult to learn (Pater and Moreton, 2012; Staubs, 2014; Stanton, 2016; Hughto, 2020).

In sum, this section demonstrates that MaxEnt's universal-subverting pattern (Anttila and Magri, 2017, and subsequent work), which arises as a direct consequence of the failure to categorically exclude harmonically bounded candidates, likewise obtains, in a milder form, in cNHG with random selection. The ability to assign nonzero probabilities to harmonically bounded candidates is considered a virtue (Hayes and Moore-Cantwell, 2011; Hayes and Schuh, 2019; Kumagai, 2023; Siah et al., 2025; this chapter), but at the same time, it could be the source of overgeneration (Anttila and Magri, 2017; Magri and Anttila, 2026). A potential solution lies in the MaxEnt learner bias against assigning nonzero probability to harmonically bounded forms (O'Hara, 2022).

4.5 Conclusion

This chapter uses simple harmonic bounding as a test case for two stochastic variants of HG: MaxEnt and cNHG. MaxEnt was found to make superior predictions to cNHG.

cNHG's main weakness is not that it cannot fit the observed candidate rates in cases of simple harmonic bounding, but rather that doing so undermines the model's performance elsewhere in the grammar. The model fails to simultaneously accommodate harmonic bounding (106) and the counteracting effect of AdjectiveFirst against phonological constraints (109–111). Fitting (106) entails suppressing AdjectiveFirst's contribution, at the expense of the model's fit to (110–111). In MaxEnt, no such trade-off is observed: AdjectiveFirst can be weighted commensurately with its contribution to meter, while still producing a high p(bounded) in (106) (if anything, even higher than observed).

ML-cNHG fits the data considerably better than GLA-cNHG: residual deviance drops from 1,571 (best-fitting GLA-cNHG) to 870 (ML-cNHG), approaching MaxEnt's deviance of 611. However, the overall conclusion still holds. Once optimal weights

are found for cNHG (which the GLA is not designed to do), cNHG comes closer to MaxEnt's fit, with remaining differences more subtle (though not nonexistent). ML-cNHG still underpredicts p(bounded) in (106) (14% vs. 20%/23% observed), which is similar in magnitude, though opposite in direction, to MaxEnt's overprediction (28% vs. 20%/23% observed).

More revealing are fitted inversion rates outside harmonic bounding conditions. ML-cNHG tends to overpredict inversion across the board. The culprit is AdjectiveFirst's low weight in ML-cNHG, which derives a reasonably high p(bounded), but fails to fully accommodate this constraint's inversion-blocking role elsewhere in meter. Deriving a high p(bounded) in cNHG entails suppressing AdjectiveFirst's contribution to meter, which leads to a weighting contradiction.

The crux of the argument is therefore that MaxEnt predicts no trade-off between p(bounded) and independent inversion propensities in other parts of the grammar, capturing both patterns reasonably well. In cNHG, moving closer to target probabilities in one dimension pushes the predictions in the other dimension further away from the target values, as §4.3.2 shows. MaxEnt's advantage over cNHG persists regardless of the weight estimation method. The ML implementation of cNHG strikes the best possible balance between the conflicting demands on AdjectiveFirst's weight, but still exhibits the trade-off: The grammar predicts a somewhat lower p(bounded) than observed, while at the same time overpredicting inversion rates across the board. This indicates that the weighting contradiction is not an artifact of the weight estimation method, but is rooted in cNHG's probability mapping.

MaxEnt's failure to enforce absolute harmonic bounding is often viewed as a liability, leading to overgeneration (Anttila and Magri, 2017; Magri, 2018; Magri and Anttila, 2026). I demonstrate that the universal-subverting pattern identified by Anttila and Magri (2017) and subsequent work does not arise exclusively in MaxEnt, since cNHG

likewise produces it.

Chapter 5

Syntax–phonology interface in poetic meter

5.1 Introduction

In meter, phonological factors affect word order far more extensively than in ordinary language. However, linearization constraints do not uniformly give in to phonological markedness; they still have a tangible effect on word-order choice in meter. Following (Fitzgerald, 2007), the most straightforward way to capture this state of affairs is to assume that phonological and linearization constraints are interleaved in metrical grammar: Phonological constraints may override baseline linearization pressures, but not all phonological constraints dominate their syntactic counterparts. Phonological constraints on poetic word order operate as fine-grained statistical tendencies suggesting that the metrical grammar must be probabilistic. This dissertation therefore argues for the probabilistic interleaving model of metrical grammar. Chapters 2–4 focused on accommodating the probabilistic nature of grammatical generalizations in meter. This chapter turns to the other ingredient of the proposed model, the interleav-

ing of phonological and linearization constraints, and its bearing on the theories of the syntax–phonology interface.

The goal of the chapter is twofold. First, I aim to determine which existing theories of the syntax–phonology interface are compatible with the relatively free interaction of phonological factors and constraints on the linear order of constituents. The second goal is to identify which architectural differences set the grammar of poetic meter apart from the grammar of natural language, where phonology's modulation of constituent order is more stringently constrained.

The chapter proceeds as follows. In §5.2, I set out the parameters along which existing models of the syntax–phonology interface differ. In §§5.3–5.4, I introduce and ultimately refute two prominent theories of the phonology's relationship with syntax: the filtering theory (§5.3) and the PF movement theory with indirect reference (§5.4). Neither theoretical model, I demonstrate, permits the degree of phonological modulation of poetic word order attested in the South Slavic oral tradition. In §5.5, I explore interactionist theories, which hold that phonological information is directly accessible by the generative component and may directly affect syntactic operations. I show that the interleaving model is equally compatible with the view that linearization is a post-syntactic, PF phenomenon, such that the interleaving of phonological and linearization constraints in fact instantiates an interaction between two constraint types within the phonological grammar.

5.2 Syntax–phonology interface

Before evaluating competing theories against the data, I outline three major parameters along which models of the syntax–phonology interface differ, defined in (117).

(117) Parameters differentiating models of the syntax–phonology interface

a. Locus of linearization

   Is linear precedence between words assigned by syntax proper or post-syntactically, e.g., at its interface with phonology?

b. Directionality

   Is communication between syntax and phonology *unidirectional* (syntax feeds information into phonology, but cannot access phonological information), or *bidirectional* (the two modules feed information into each other)?

c. Sequentiality

   Do syntactic and phonological processes apply in a strict, ordered sequence, or in parallel?

Word-order variation attested in BCMS folk meter bears on questions (117b–117c) only insofar as linear order is considered part of narrow syntax (117a). If linearization is external to syntax proper, then the competition between phonological factors and linearization pressures of the kind obtaining in the BCMS oral folk tradition belongs to PF, with little to no bearing on the nature of the syntax–phonology interface in poetic meter.

In what follows, I review each of these parameters in greater detail.

5.2.1 Locus of linearization

Syntax is the generative component of the grammar: It builds hierarchical structure, determining the structural dependencies between sentence constituents. The question at issue here is whether the syntactic component, in addition to its structure-building role, also determines the linear order between words, or whether this task is left to post-syntactic Spell-Out. Views on this issue differ profoundly, as I show below, and the question bears directly on the present discussion. If syntax only builds hierarchical

structure and linearization is post-syntactic, variable linear order, although interesting for a plethora of other reasons (Chapters 1–4), has very little to offer the debate between theories of the syntax–phonology interface. The phenomenon is purely post-syntactic, falling outside the jurisdiction of syntax proper.

The view that linear order is not part of narrow syntax, being part of post-syntactic Spell-Out, is maintained, in some form, by a variety of theoretical approaches within the generative tradition (e.g. Hornstein et al., 2005, 218). Notably, Chomsky (1995, 334) argues that "ordering is part of the phonological component." Fox and Pesetsky (2005) view linearization as one aspect of Spell-Out, the mapping between syntax and phonology, rather than of syntax proper. Essentially the same position is taken by Agbayani and Golston (2010), who likewise hold that linear precedence relations between nodes are set at the syntax–phonology interface. The same view is standardly assumed in Distributed Morphology (Halle and Marantz, 1994): For example, Embick and Noyer (2007, 293) argue that "[l]inear ordering of nodes in this hierarchical structure is, however, plausibly a relation that is defined by operations on the PF branch." More recently, Kusmer (2019, 2020) devises a system in which the linear position of words is assigned post-syntactically, by the parallel OT grammar which also governs prosodification (see also Johnson, 2024). Similarly, Ryan's (2010) analysis of variable affix order argues for post-syntactic linearization constraints on morpheme order.

By contrast, Kayne (1994) derives linear precedence between nodes from the principles of narrow syntax (famously, via the Linear Correspondence Axiom). In the literature on the nature of the syntax–phonology interface, the view that linear order is determined by syntax is particularly widespread. Among phonologists, the most explicit claim along these lines is found in Shih and Zuraw (2018), who maintain that "word order and linearization principles [are] the purview of (morpho)syntax." This implies that, where phonology affects the linear order of constituents, as it does in Tagalog

NPs, Shih and Zuraw's case study, such phenomena bear directly on the nature of the syntax–phonology interface. Miller et al.'s (1997) approach likewise appears to fall into this category, although the connection is somewhat less direct: The authors argue that variable adjective–noun order in French does not constitute evidence against Zwicky and Pullum's (1986a) principle of phonology-free syntax, which holds that phonological information cannot affect operations in the syntactic component. The defense of phonology-free syntax in the face of variable adjective order in French seems to suggest that Miller et al. treat linear order as part of syntax proper.

The locus of linearization is a crucial parameter because if linearization is external to syntax, then the variable linear order of constituents has effectively no bearing on the nature of phonology's relationship with the syntactic component.

5.2.2 Directionality of the syntax–phonology interaction

Arguably the most extensively debated issue at the syntax–phonology interface is whether information flows unidirectionally between the two modules, from syntax to phonology, or bidirectionally, such that the generative component may directly access phonological information.

The widely accepted, NON-INTERACTIONIST APPROACH, also known as the FEED-FORWARD APPROACH, maintains that there is no direct communication between syntax and phonology, and that syntax is PHONOLOGY-FREE (Zwicky and Pullum, 1986a,b; Miller et al., 1997; Anttila, 2016). This accords with the Y-model of grammar (Chomsky, 1995), in which syntax is the generative core and phonology is an interpretive branch with no direct impact on the generative component. The feed-forward approach does not, however, rule out phonological effects on word order. Phonologically conditioned syntactic variation has received two principled explanations in this line of work. One is to consider word-order variation as external to grammar, being usage-

related, i.e. a matter of performance (in the sense of Chomsky, 1965). For instance, Miller et al. (1997, 74) maintain that phonological effects on variable adjective–noun order in French represent "a usage tendency that the grammar does not mandate." The authors seem to imply that such gradient effects are extraneous to grammar because they involve statistical tendencies rather than categorical rules, and "grammatical theory does not aim to predict statistical facts about usage" (*op. cit.*, 69). I follow Anttila (2016, 319*f.*) in rejecting this view. The other approach is to attribute phonologically conditioned word-order variation to post-syntactic filtering by phonology (Bošković, 2001; Anttila, 2016).

The INTERACTIONIST APPROACH holds that phonological factors can directly influence syntactic processes, that is, that syntax has access to phonological information and its output can be directly shaped by it (Shih and Zuraw, 2018; Bruening, 2019; Breiss and Hayes, 2020; see also Richards, 2016, who argues that syntactic derivation can be sensitive to prosodic information). Architectures involving bidirectional flow of information between syntax and phonology are commonly assumed for poetic meter (Fitzgerald, 2007).

Since non-interactionist models involve more restrictive architectures of grammar than their interactionist counterparts (Zwicky and Pullum, 1986a; Golston, 1995; Anttila, 2016), they are to be preferred on grounds of restrictiveness, provided they handle the data with equal success. As I show in what follows, two prominent non-interactionist models fail to account for the data, which at first glance lends support to the interactionist model as the only viable solution. However, the interactionist model is not the necessary consequence of the data. The proposal can be framed in such a way as to maintain the restrictive, non-interactionist architecture, leaving the generative component essentially untouched by phonology.

5.2.3 Sequentiality

Finally, existing models of the syntax–phonology interface differ as to whether the two modules are sequentially ordered or apply in parallel. The standard Y-model instantiates a DERIVATIONAL APPROACH, in which the syntactic component first generates structures that are then fed into the interpretive branches: phonology (PF) and semantics (LF). Models assuming PARALLEL ARCHITECTURES maintain that distinct modules, i.e. dimensions of linguistic structure (syntax, semantics, and phonology), are processed simultaneously rather than in a fixed sequence (Bresnan, 2000; Jackendoff, 2002; Breiss and Hayes, 2020).

At first glance, it is tempting to conflate the directionality of the syntax–phonology interaction with sequentiality, equating non-interactionist with derivational models, and interactionist with parallel models. The reasoning is that precedence of application of syntactic operations, or lack thereof, may determine whether phonological information is available to syntax at all. However, the two parameters are orthogonal: Not all non-interactionist models are couched in derivational theories, and not all interactionist models assume parallel architectures. Both interactionist and non-interactionist models of the syntax–phonology interface have derivational and parallel implementations.

An example of a non-interactionist approach that can be implemented in both derivational and parallel models of grammar is Anttila's (2016) variant of the filtering theory, discussed in greater detail in §5.3. The derivational version holds that syntactic structure-building operations precede phonology. Once fed into phonology, syntactic structures cannot be sent back to the syntactic component for further modification. The model thus assumes a feed-forward architecture with a unidirectional flow of information from syntax to phonology.

The same non-interactionist architecture can be maintained in a parallelist system

like Classical OT (Anttila, 2016, 128), in which syntactic and phonological constraints are part of a unified constraint grammar, and all syntactic alternatives are evaluated in parallel. Unidirectionality is ensured by ranking all syntactic constraints over all phonological constraints (à la Golston, 1995; Tranel, 1998; Rice and Svenonius, 1998). That way, phonological constraints invariably yield to higher-ranked syntactic constraints and can affect word order, or the choice of syntactic configuration, only when syntactic constraints are moot, that is, when syntactic alternatives are equally well-formed.

Interactionist models, which permit bidirectional flow of information between syntax and phonology, likewise come in both derivational and parallelist variants. Shih and Zuraw (2018) develop a derivational interactionist model in which syntax initially generates a structure with default order (since the authors hold that linear order is set by syntax proper), which is then fed into the phonological component. If the default order is phonologically ill-formed (the details of which remain unclear), phonology may send it back to syntax, which then generates the alternative order. This looping model is derivational in that syntactic and phonological operations apply in an ordered sequence, but the flow of information between the modules is bidirectional, in that phonology may affect the output of syntax.

Parallel interactionist models also exist. Under such models, syntactic and phonological constraints are interleaved, that is, freely rankable within a single grammar, such that some phonological constraints may dominate some syntactic constraints.

5.3 The filtering theory

The FILTERING THEORY, also known as the VARIATION + FILTERING THEORY, holds that phonology cannot feed information into the syntactic component, which puts it into the category of non-interactionist models of the syntax–phonology interface. Syn-

tax generates structures fully oblivious to phonological well-formedness requirements, and phonology functions as a post-syntactic filter over the structures admitted by the generative component. Phonology therefore has no direct impact on the output of syntactic operations, including word order if linear order is truly regulated by narrow syntax. It is only when syntax passes to phonology multiple structural alternatives that the phonological component can select the surface realization based on the phonological well-formedness of the otherwise equally syntactically well-formed variants. The idea of phonology as a post-syntactic filter has many proponents in the generative tradition (see Bošković, 2001, among others). In this section, I focus on a particular version of the filtering theory advanced by Anttila (2016), which has gained much attention among phonologists.

The highly restricted post-syntactic intervention by phonology assumed in the filtering theory predicts that phonology affects word order only when word-order alternatives are equally syntactically well-formed. This readily explains why phonological effects on sentence structure are cross-linguistically so pervasive in coordinate constructions and the choice of syntactic construction (such as the so-called genitive and dative alternations), but not in headed XPs. The latter have one default order generated by the syntax and it is only this default order that gets passed to phonology.

In coordinate constructions, there is no basic linear order, unlike, e.g., BCMS adjectival modifiers, which are mandatorily prenominal (§1.2.2). This is, of course, not to say that coordinate constructions are *hierarchically* flat. The hierarchical structure of coordinate constructions (on which see e.g., Munn, 1993) is orthogonal to the linear order of conjuncts and as such in no way affects the present discussion. What matters is that per syntax, the surface linear order of e.g., *black and white* is not inherently better (or worse) than its transposition *white and black*.

Another environment in which the filtering theory predicts (and rightly so) robust

phonological effects on sentence structure is the choice between syntactic constructions that convey the same real-world events. An oft-quoted example is the English dative alternation (Wasow, 1997b, 2002; Bresnan et al., 2007; Bresnan and Nikitina, 2009; Bresnan and Ford, 2010; Anttila et al., 2010; Grafmiller and Shih, 2011). Speakers express the same underlying idea with two postverbal argument configurations: the prepositional *to*-dative (V NP PP), as in (118a), and the double object construction (V NP NP), as in (118b).

(118) English dative alternation (Bresnan and Ford, 2010, 169, ex. 2)

a. Who gave that wonderful watch to you? [V NP PP]

b. Who gave you that wonderful watch? [V NP NP]

Importantly, it is because the syntactic alternatives in (118) are equally syntactically well-formed that phonological factors like word size and rhythm may affect the choice between the prepositional dative and double object constructions in English.

5.3.1 Derivational and parallel implementations

The filtering theory has both derivational and non-derivational variants. The feed-forward architecture assumed in Anttila (2016)'s version of the filtering theory is sketched in Figure 5.1. The communication between the modules is unidirectional, with no looping back from phonology to syntax. The example illustrates phonology's filtering effect on syntactic configurations: Syntax fails to select a single output, since the structural alternatives [AB] and [BA] are equally plausible, passing the choice to the phonological component. Phonology selects [BA] over [AB]. When syntax designates one structural alternative as more well-formed than the others, phonology has no means to help the syntactically inferior structures because it receives only the best-performing syntactic structure from the generative component; others are knocked out

already in the syntax.

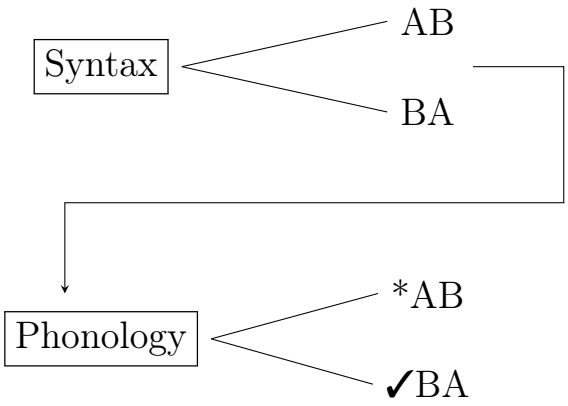


Figure 5.1: Phonology as a post-syntactic filter (adapted from Anttila, 2016, 118).

In a parallelist OT setting, the filtering effect of phonology on sentence structure is viewed as the emergence of the unmarked (TETU; McCarthy and Prince, 1994). Phonology is overridden by syntax and can affect sentence structure only when syntax itself does not discriminate between the structural variants. The precedence of syntax over phonology is ensured by universally ranking all syntactic constraints over all phonological constraints (Golston, 1995; Tranel, 1998; Rice and Svenonius, 1998; Anttila, 2016). The tableaux in (119) illustrate the inhibition of phonology by syntax under the syntax-over-phonology proviso (119a) and phonology's filtering effect when higher-ranked syntactic constraints are moot (119b).

(119) Phonological effects on sentence structure as TETU

a. Syntax overrides phonology

{AB, BA}	$\mathbb{S}$	$\mathbb{P}$
a. ☞ AB		***
b. BA	*!	

b. Phonology determines the winner when syntax is indifferent

{CD, DC}	$\mathbb{S}$	$\mathbb{P}$
a. CD		$*!$
b. ☞ DC		

In (119a), the winning candidate [AB] is more phonologically marked than its competitor [BA], but it fares better on the syntactic constraint. Syntax has a preferred candidate and no degree of phonological ill-formedness can put it out of the running, at least in strict-ranking OT. In (119b), the candidates fare equally well on the syntactic constraint. In the absence of the syntactically preferred order, the winner is determined by the lower-ranked phonological constraint.

One caveat is that the priority of syntax over phonology via $\mathbb{S} \gg \mathbb{P}$ can be strictly enforced only in deterministic OT. In a system that allows cumulativity effects, like HG, even if all syntactic constraints outweigh all phonological constraints, phonology can still overcome syntax by means of ganging-up or counting cumulativity (Jäger and Rosenbach, 2006). It thus follows that the parallelist implementation of the filtering theory crucially rests on OT-style strict ranking to enforce phonology-free syntax. This commitment to strict-ranking OT need not be a sensible choice for a plethora of other reasons, given OT's general limitations (Pater, 2009b; Zuraw and Hayes, 2017; Smith and Pater, 2020; Breiss, 2020; Hayes, 2022; Chapter 3 of this dissertation).

5.3.2 Application to poetic meter

The appeal of the filtering theory is that it allows phonology to modulate word order and other aspects of sentence structure, while preserving the restrictiveness of the non-interactionist architecture. Proponents of the filtering theory argue that interactionist models are overly permissive (see Golston, 1995), in that they fail to restrict phonological effects on word order to environments where they frequently occur across languages

(e.g., binomials, argument structure alternations) and predict more widespread phonological effects on word order in headed XPs, which are less sensitive to phonology than e.g., binomials.

Despite these restrictiveness concerns, the filtering theory is clearly too restrictive to provide a general model of the syntax–phonology interface in poetic meter. For one, the theory makes the falsifiable prediction that phonology can only modulate word order in the absence of a syntactic default, as in binomials (Golston, 1995; Anttila, 2016). BCMS folk meter, along with other metrical traditions, shows robust phonological effects on word order even in constructions that exhibit a default order. Adjectives are by default prenominal, as indicated by the regression analysis in §1.4, but may invert to the postnominal position in response to phonological constraints. The same is true of possessive pronouns, which are postnominal by default, but may precede the noun to satisfy phonological constraints. Phonological effects on poetic word order do not invariably instantiate TETU, which renders the filtering theory an inadequate model of poetic grammar.

Whether the filtering theory offers a plausible model of the syntax–phonology interface in natural language is a matter of ongoing debate, and certainly goes beyond the limits of the present discussion. However, it should be noted that cases have been reported in which phonological factors trigger deviations from the canonical order or otherwise affect syntactic operations in natural language (Harford and Demuth, 1999; Teeple, 2008; Shih and Zuraw, 2017, 2018). Strong evidence to this effect is offered by the variable adjective–noun order in Tagalog. The Tagalog pattern is akin to the word-order alternation documented in the BCMS folk meter: Adjectives are prenominal by default, but the canonical order may shift to noun–adjective in response to non-syntactic constraints, including phonological ones.

5.4 Post-syntactic linearization and PF movement

In the Y-model of grammar, movement operations can take place in the narrow syntax (overt movement), and on the interpretative branches: at LF (covert movement) and, importantly, at PF (PF movement). This section examines whether post-syntactic PF movement can provide a viable account of phonologically conditioned word-order variation in poetic meter. Similarly to the filtering theory, the PF movement account is inherently attractive because it preserves the principle of phonology-free syntax: Deviations from the canonical order admitted by poetic meter arise post-syntactically, by phonologically triggered movement at PF; hence phonology does not directly tamper with the generative component.

Various forms of PF movement have been invoked in the literature to accommodate a diverse range of phenomena (see Aoun and Benmamoun, 1998; Sauerland and Elbourne, 2002; Göbbel, 2007, 2013, 2020; Agbayani and Golston, 2010, 2016; Agbayani et al., 2015; Bennett et al., 2016; Edmiston and Potsdam, 2017; Potsdam, 2022, among others).[1] Potsdam (2022) distinguishes three types of PF movement operations: (i) SYNTACTIC PF movement, which targets syntactic constituents and affects the syntactic structure, being therefore subject to the movement constraints (Aoun and Benmamoun, 1998; Sauerland and Elbourne, 2002; Weir, 2014), (ii) PHONOLOGICAL PF movement, which displaces prosodic (not syntactic) constituents, introduces no additional syntactic structure, and shows no sensitivity to constraints on syntactic movement (Agbayani and Golston, 2010, 2016; Agbayani et al., 2015; Bennett et al., 2016; Clemens and Coon, 2018; Dolatian and Meadows, 2023), and (iii) MIXED PF movement, which has an impact on the syntactic structure, but is driven by purely phonological factors and is exempt from syntactic movement constraints (Göbbel, 2007,

[1]Though see e.g., Bošković (2001) for criticism of the mechanism of PF movement.

2013).

If PF movement is indeed the source of word-order variation in BCMS oral tradition, the pattern no doubt instantiates *phonological* PF movement: Deviations from the canonical order are driven by phonological well-formedness requirements, such as prosodic end-weight and prosodic boundary alignment. At the same time, inversions do not, as far as I can tell, derive additional syntactic structure (as would be the case under syntactic PF movement), and have no semantic import.

This overview focuses on the version of the PF movement approach developed by Agbayani and Golston (2010 and subsequent work), widely engaged with in the phonological literature. Much like Chomsky (1995); Fox and Pesetsky (2005), and related work (see §5.2.1), Agbayani and Golston maintain that the syntactic component only determines the hierarchical dominance relations between constituents, but not their linear order. Linear precedence between words is assigned on the branch from Spell-Out to PF, where the output of syntax is prosodified. Later, at PF, the linearized and prosodified structures may be subjected to further manipulation by PF movement, driven by phonological well-formedness requirements. That is, the default linear order initially generated at the interface between syntax and phonology may be inverted at PF by way of phonological optimization. The division of labor between the modules is diagrammed in Figure 5.2.

One major prediction of this brand of the PF movement approach is that phonological grammar, and PF movement as its part, never refers directly to syntactic labeling (Agbayani and Golston, 2010, 158). Domain reference by phonological processes (including PF movement) is strictly mediated by prosodic structure. In other words, syntactic categories such as NPs, VPs, and the like are subject to overt movement operations in the narrow syntax, whereas PF movement can only access *prosodic* constituents generated at the interface between syntax and phonology. The PF movement

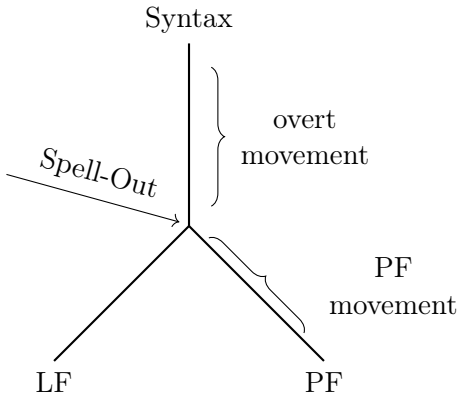


Figure 5.2: Y-model of grammar, under the post-syntactic-linearization-cum-PF-movement approach of Agbayani and Golston (2010).

approach thus maintains the strict INDIRECT REFERENCE HYPOTHESIS (Selkirk, 1981, 1986, 2011; Nespor and Vogel, 2007; Elfner, 2012; Bennett and Elfner, 2019), whereby phonology does not refer back to syntax, but rather to prosodic categories, onto which syntactic structures map.

5.4.1 Application to meter

Figure 5.3 shows how Agbayani and Golston's PF movement approach derives the non-canonical adjective-final order of BCMS [ˈgraː.da ˈbi.jɛ.lɔː.ga] 'city white.' First, the syntactic component generates the core NP structure via Merge, determining the hierarchical relation between the AdjP and the head noun, but no other aspect of the surface structure. The output of syntax is then passed to the post-syntactic Spell-Out at the interface between syntax and phonology. It is at this stage that the linear order of the inserted vocabulary items is set up. In accordance with the linearization rule for BCMS adjectival modifiers, the initially generated order is adjective–noun: [ˈbi.jɛ.lɔː.ga ˈgraː.da] 'white city,' which, we see, is not our attested surface order in this case. In addition, the structure is prosodified: The NP maps onto a prosodic phrase (φ), whose

daughters are prosodic words (ω): $\big[[\text{'bi.jɛ.lɔː.ga}]_\omega\ [\text{'graː.da}]_\omega\big]_\varphi$. Finally, the spelled-out, linearized, and fully prosodified structure is fed into PF, where its phonological well-formedness is assessed. In ordinary language, phonological constraints usually give way to linearization constraints, such that the language's linearization preferences are enforced at the expense of phonological markedness requirements (Golston, 1995; Anttila, 2016). However, meter heightens sensitivity to phonological factors, such that phonological markedness constraints may license deviation from the canonical order via PF movement. In our case, the input to PF ($\big[[\text{'bi.jɛ.lɔː.ga}]_\omega\ [\text{'graː.da}]_\omega\big]_\varphi$) is phonologically marked: It displays a quadrisyllabic prosodic word preceding a disyllable, contravening End-Weight. This prosodic constraint (optionally) triggers the PF movement of the longer, quadrisyllabic prosodic word over the disyllable, resulting in the adjective-final order: $\big[[\text{'graː.da}]_\omega\ [\text{'bi.jɛ.lɔː.ga}]_\omega\big]_\varphi$.

PF movement is not cost-free: Since the input to phonology comes with a predetermined order, any deviation from it constitutes an unfaithful mapping and incurs a faithfulness penalty. Agbayani and Golston (2010, 155, ex. 94) use the anti-movement Stay constraint from OT syntax (Grimshaw, 1997) to penalize displacement of prosodic constituents at PF. The constraint that opposes End-Weight in Figure 5.3 is Stay-φ, which assigns a violation for every φ node whose dependents shift their input order (120).

(120) Stay-φ (Agbayani and Golston, 2010)

No daughter of φ may move.

The inverted order of $\big[[\text{'graː.da}]_\omega\ [\text{'bi.jɛ.lɔː.ga}]_\omega\big]_\varphi$ 'city white' prevails against the canonical order because End-Weight dominates the antagonistic constraint Stay-φ. This is shown in the OT tableau in (121). I simplify the BCMS facts by setting aside, for the time being, the variation in the data.

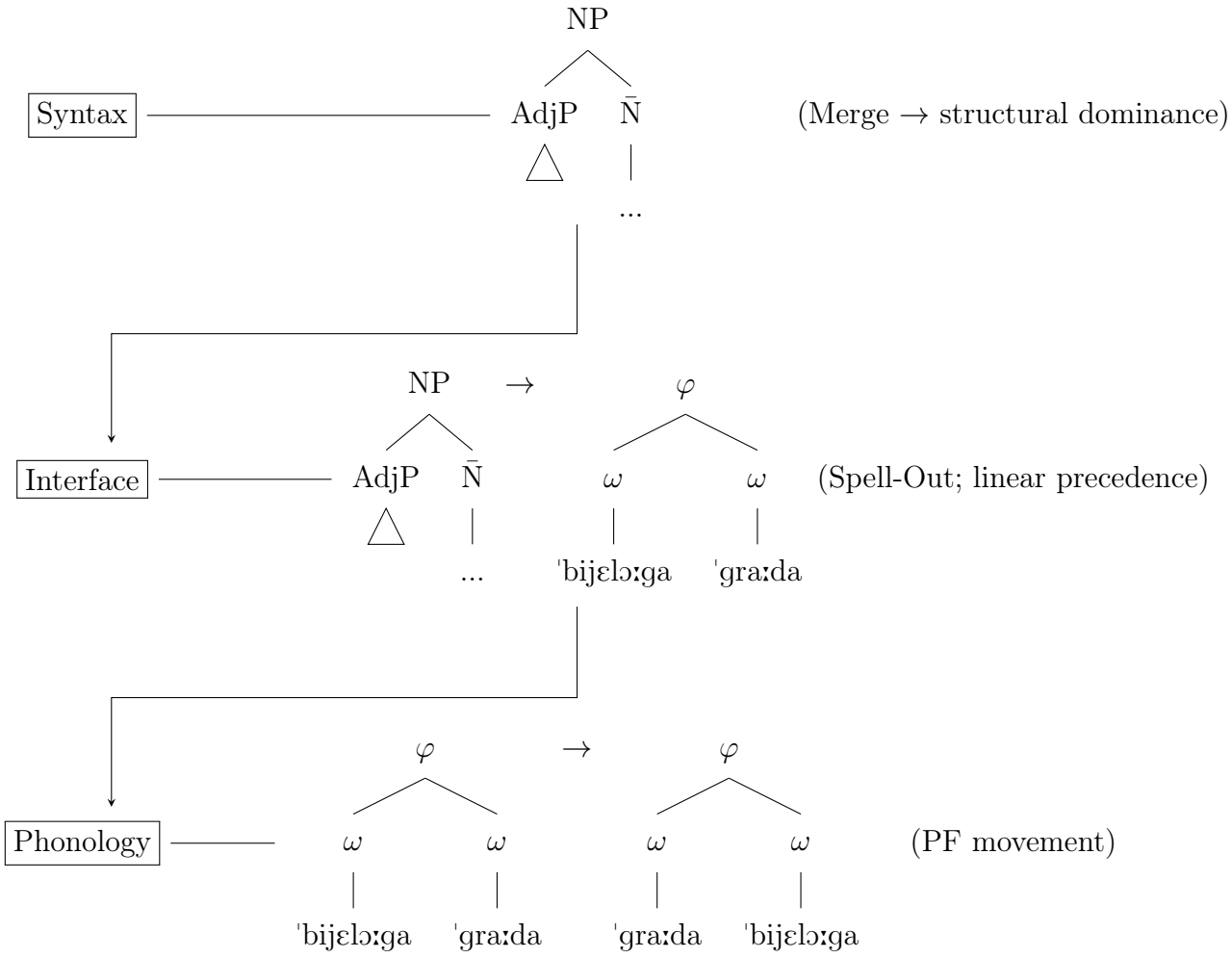


Figure 5.3: Derivation of the non-canonical postnominal adjective order in BCMS under the post-syntactic PF movement model (adapted from Agbayani and Golston, 2010, 155; ex. 94).

(121) Poetic meter: interleaving/free ranking

$\left[\left[\text{ˈbi.jɛ.lɔː.ga}\right]_{\omega}\ \left[\text{ˈgraː.da}\right]_{\omega}\right]_{\varphi}$	End-Weight	Stay-φ
a. $\left[\left[\text{ˈbi.jɛ.lɔː.ga}\right]_{\omega}\ \left[\text{ˈgraː.da}\right]_{\omega}\right]_{\varphi}$	$*!$	
b. ☞ $\left[\left[\text{ˈgraː.da}\right]_{\omega}\ \left[\text{ˈbi.jɛ.lɔː.ga}\right]_{\omega}\right]_{\varphi}$		$*$

The fixed prenominal order of adjectival modifiers in ordinary language is derived under the opposite ranking: Stay-$\varphi \gg$ End-Weight, as in (122). In general, ordinary language can be assumed to inhibit inversions that are readily available in poetic meter by prioritizing the anti-movement Stay constraints over phonological markedness constraints. The fixed ranking is reminiscent of the $\mathbb{S} \gg \mathbb{P}$ proviso of Golston (1995); Tranel (1998); Anttila (2016), with one important distinction: Since linearization constraints are part of PF, their fixed ranking over phonological markedness constraints instantiates a relationship between two constraint types within the same module (phonology), not a fixed ranking across modules.

(122) Ordinary language: fixed ranking ($\mathbb{S} \gg \mathbb{P}$)

$\left[\left[\text{ˈbi.jɛ.lɔː.ga}\right]_{\omega}\ \left[\text{ˈgraː.da}\right]_{\omega}\right]_{\varphi}$	Stay-φ	End-Weight
a. ☞ $\left[\left[\text{ˈbi.jɛ.lɔː.ga}\right]_{\omega}\ \left[\text{ˈgraː.da}\right]_{\omega}\right]_{\varphi}$		$*$
b. $\left[\left[\text{ˈgraː.da}\right]_{\omega}\ \left[\text{ˈbi.jɛ.lɔː.ga}\right]_{\omega}\right]_{\varphi}$	$*!$	

It thus follows that the derivation proceeds in essentially the same way in poetic grammar and ordinary language until PF, and that their different linearization preferences fall out exclusively from the PF differences, i.e. meter's interleaving of linearization and phonological markedness constraints, as opposed to ordinary language's strict priority given to linearization constraints.

5.4.2 Critical assessment

Given that the PF movement approach assumes indirect reference, it makes a falsifiable prediction about construction-specific inversion rates: Construction types can display different inversion propensities only if they are prosodified differently. This is because phonological constraints may drive movement of *prosodic* constituents, not syntactic movement, which takes place in the syntactic component under purely syntactic conditions. If, for instance, NPs and VPs are both prosodified as prosodic phrases, PF movement of prosodic words within a phrase should be penalized equally by Stay-φ in both construction types.

With that in mind, consider the difference between adjectives and possessive pronouns. Recall first that in meter, adjectives are prenominal by default, while possessives are postnominal. The default orders for the examples in (123) and (124) below are therefore *rođena brata* [ˈrɔ.ʤɛ.na ˈbra.ta] 'blood brother.GEN' and *družina moja* [ˈdru.ʒi.na ˈmɔ.ja] 'crew my,' respectively.

(123) *rođena brata* [ˈrɔ.ʤɛ.na ˈbra.ta] 'blood brother.GEN'

a.	Adj N:	i=ˈrɔ.ʤɛ.na ˈbra.ta	0.16
b.	N Adj:	i=ˈbra.ta ˈrɔ.ʤɛ.na	0.84

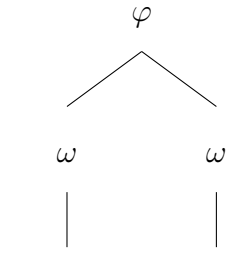
(124) *družina moja* [ˈdru.ʒi.na ˈmɔ.ja] 'my crew' (lit. 'crew my')

a.	N Poss:	i=ˈdru.ʒi.na ˈmɔ.ja	0.29
b.	Poss N:	i=ˈmɔ.ja ˈdru.ʒi.na	0.71

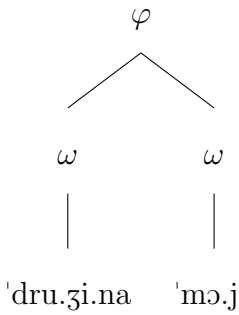
Both construction types map onto prosodic phrases. Thus, the two examples display identical prosodic structure, as (125) shows.

(125) Adjectives and possessives prosodified identically

a. Adj–Noun: *rođena brata* [ˈrɔ.ʤɛ.na ˈbra.ta] 'blood brother.GEN'



b. Noun–Poss *družina moja* [ˈdru.ʒi.na ˈmɔ.ja] ‘crew my’



Under meter’s default order, the final member in both cases is a disyllable with an initial stressed light, while the preceding element is a trisyllable whose penult is light and unstressed. The phonological markedness profiles of the default and non-default orders are identical. Multiple markedness constraints gang up to move the trisyllable over the disyllable (with the exception of ALIGN, which prefers the disyllable to remain line-final). With STAY-φ as the anti-movement constraint, the two construction types are predicted to display the same rates of movement because they both constitute prosodic phrase domains. Identical probability distributions across the two tableaux are produced regardless of the weight assigned to STAY-φ.

(126) STAY-φ predicts identical inversion rates for the adjective–noun and noun–possessive constructions

a. *rođena brata* [ˈrɔ.ʤɛ.na ˈbra.ta] ‘blood brother.GEN’

	Stay-φ 1.00	Align 2.24	End-W 2.04	*L/FinS 1.81	End-W & *L/FinS 1.53	*'L/FinS 0.99	H	Predicted	Observed
a. ˈrɔ.dʑɛ.na ˈbra.ta			−1	−1	−1	−1	−6.37	0.21	0.16
b. ˈbra.ta ˈrɔ.dʑɛ.na	−1	−1		−1			−5.05	0.79	0.84

b. *družina moja* [ˈdru.ʒi.na ˈmɔ.ja] 'crew my'

	Stay-φ 1.00	Align 2.24	End-W 2.04	*L/FinS 1.81	End-W & *L/FinS 1.53	*'L/FinS 0.99	H	Predicted	Observed
a. ˈdru.ʒi.na ˈmɔ.ja			−1	−1	−1	−1	−6.37	0.21	0.29
b. ˈmɔ.ja ˈdru.ʒi.na	−1	−1		−1			−5.05	0.79	0.71

Contrary to this prediction, the two constructions behave differently: The noun–possessive construction resists inversion more strongly than the adjective–noun construction (Fisher's exact test OR = 0.45, $p = 0.002$). If PF movement were responsible for word-order variation, the difference between adjectives and possessives in (126) could only be explained by assuming that they map to different prosodic structures in meter. However, the two constructions both map to prosodic phrases. The linearization constraint(s) must therefore directly reference syntactic constructions, as is the case with the linearization constraints proposed here: AdjectiveFirst and PossessiveSecond. The tableaux in (127) illustrate this point.

(127) a.

	AdjFirst	Align	End-W	*L/FinS	End-W & *L/FinS	*'L/FinS	*H*	Predicted	Observed
	0.97	2.24	2.04	1.81	1.53	0.99			
a. ˈrɔ.ʤɛ.na ˈbra.ta			−1	−1	−1	−1	−6.37	0.21	0.16
b. ˈbra.ta ˈrɔ.ʤɛ.na	−1	−1		−1			−5.02	0.79	0.84

b.

	PossSecond	Align	End-W	*L/FinS	End-W & *L/FinS	*'L/FinS	*H*	Predicted	Observed
	1.52	2.24	2.04	1.81	1.53	0.99			
a. ˈdru.ʒi.na ˈmɔ.ja			−1	−1	−1	−1	−6.37	0.31	0.29
b. ˈmɔ.ja ˈdru.ʒi.na	−1	−1		−1			−5.49	0.69	0.71

Given that each *syntactic* construction has its own linearization constraint, the grammar in (127) captures the observed construction-specific inversion rates despite their identical prosodification, unlike its Stay-φ counterpart in (126). The results suggest that phonology directly refers to syntactic structure in meter, that is, domain reference is not strictly mediated by prosodic structure. I conclude that the PF movement approach, as implemented in Agbayani and Golston (2010) and subsequent work, does not provide an adequate model of phonologically conditioned syntactic variation in meter.

5.5 Probabilistic interleaving model

A number of studies on the syntax–phonology interface in natural language have called for restrictive theoretical models, ones that prevent phonology from directly tampering with syntactic operations. In meter, however, more permissive models are widely recognized as unproblematic, even desirable. Indeed, proposals have been made that the $\mathbb{S} \gg \mathbb{P}$ proviso is suspended in poetic meter, such that phonological constraints may override syntactic ones (Rice, 1997; Golston, 1998; Golston and Riad, 2000; Fitzgerald, 2003, 2007; Kentner and Franz, 2019). Poets avail themselves of syntactic inversions that make verse lines more metrically well-formed (Youmans, 1983, 1989; Fitzgerald, 2007; Hayes et al., 2012). As Fitzgerald (*ibid.*) demonstrates, suspending $\mathbb{S} \gg \mathbb{P}$ does not entail that phonology invariably dominates syntax in meter, as occasionally assumed (e.g., Golston's 1998, 742 $\mathbb{P} \gg \mathbb{S}$ in meter). Were syntax to invariably give way to phonology, metered verse would license any deviation from ordinary-language syntax, as long as it is phonologically optimizing. There are, however, inversions that meter does not admit, suggesting that syntactic constraints can figure as a counterweight to their phonological counterparts. Fitzgerald therefore concludes that the most adequate model of poetic grammar is one in which syntactic and phonological constraints are interleaved, i.e. freely rankable (e.g., $\mathbb{P}_1 \gg \mathbb{S}_1 \gg \mathbb{P}_2 \gg \mathbb{S}_2$).

Given the limitations of the non-interactionist models of the syntax–phonology interface, I now turn to models that permit the phonological component of poetic grammar to directly access syntactic information. The most obvious choice is a parallel architecture in which syntactic and phonological information is evaluated simultaneously, allowing constraints from one domain to interact directly with those from the other. Such an architecture abandons both the derivational Y-model and the principle of phonology-free syntax (Zwicky and Pullum, 1986a). The stochastic OT/HG models

of poetic word order devised throughout Chapters 1–4, as well as other approaches assuming that phonological and syntactic constraints are interleaved in a single parallel constraint grammar (Fitzgerald, 2007), naturally lend themselves to this interpretation.

I argue, however, that this interactionist architecture is not the only framework capable of capturing the empirical generalizations identified in Chapters 1–4. An alternative is to treat linear order as lying outside the jurisdiction of narrow syntax, with linearization constraints being part of the phonological grammar. Under this approach, which I call the PROBABILISTIC INTERLEAVING MODEL WITH POST-SYNTACTIC LINEARIZATION, linear precedence between words is determined entirely at PF. Phonology therefore affects word order not because the two modules operate in parallel, but because linearization constraints are effectively part of the phonological grammar.

5.5.1 Interactionist variant

A line of work on the syntax–phonology interface assumes direct communication between syntax and phonology (Shih and Zuraw, 2018; Breiss and Hayes, 2020). Against the backdrop of Zwicky and Pullum's (1986a) principle of phonology-free syntax, interactionist approaches maintain that the generative component has direct access to phonological information, and that phonological well-formedness requirements may affect the output of syntax, including constituent order. A prominent example is Shih and Zuraw's (2018) analysis of variable adjective-noun order in Tagalog, where the linear position of adjectival modifiers on the surface is highly sensitive to phonological constraints, as shown in (2) in Chapter 1, repeated here as (128). Example (128) demonstrates the effect of nasal cluster avoidance: The canonical adjective–noun order of (128a) creates adjacent nasals in sandhi and is therefore dispreferred to the noncanonical adjective-final order of (128b).

(128) OCP-NASAL-driven inversion in Tagalog NPs

a. Canonical order (Adj–N)

itim na pelúka
black LINK wig

b. Inversion (N–Adj)

pelúka -ng itim
wig -LINK black
'black wig' [Shih and Zuraw, 2017: e325, ex. (6)]

Per Shih and Zuraw, variable word order in Tagalog NPs offers particularly strong evidence that narrow syntax has access to phonological information. The syntactic component must refer not only to the input properties of its terminals, but also to surface forms that have not been derived yet. Nasal cluster avoidance is a case in point. Whether a nasal cluster is brought about depends on the phonological shape of the linker morpheme, which is realized as [-ŋ] after vowels and as [na] after consonants. The possibility of a nasal cluster on the surface in turn affects the ordering of the adjective and the head noun. Syntax must therefore anticipate the surface realization of the linker before the morpheme is spelled out, which instantiates a cross-modular lookahead effect.

Interactionist models are more permissive than non-interactionist architectures, but it is a common misconception that they are effectively unconstrained, permitting arbitrary phonological effects on syntax. Interactionist architectures do not fall into the category of "anything goes" models. The pathological patterns that motivated feed-forward architectures (Zwicky and Pullum, 1986a), e.g., "head-move if labial-initial," do not arise from the *interaction* of independent phonological and syntactic constraints, as assumed here, but from constraints that conflate unrelated phonological and syntactic dimensions (e.g., place of articulation and head movement). As long as such hybrid

phonology–syntax constraints are excluded from CON, interactionist models retain a certain level of restrictiveness and do not freely generate the pathological patterns discussed by Zwicky and Pullum (1986a); see also Breiss and Hayes (2020, 363). This is the case in the interleaving model à la Fitzgerald (2007): Syntactic and phonological constraints are independent, being evaluated in a single parallel constraint grammar while still constituting separate constraint families. The modules are distinct, albeit interleaved, and this independence is what makes the model more restrictive than "anything goes."

Interactionist models along the lines of Shih and Zuraw (2018) fit naturally with theories that assume a parallel architecture of grammar, in which information from different modules is processed simultaneously (Bresnan, 2000; Jackendoff, 2002; Breiss and Hayes, 2020). The simultaneous evaluation of syntactic and phonological information provides a straightforward explanation for the feedback from phonology to syntax assumed in Shih and Zuraw's model.

Interactionist models of the syntax–phonology interface need not be parallel, however; derivational implementations are also possible. Indeed, Shih and Zuraw (2018) entertain a derivational version of their proposed model, sketched in Figure 5.4. The syntactic component first generates the default word order and passes it to the phonological component, where its well-formedness is evaluated. If the output is phonologically ill-formed, it is sent back to syntax, which generates an alternative, non-default order. The non-default order is then fed into the phonological component for evaluation. The derivation of [ˈgraːda ˈbi.jɛ.lɔː.ga] 'city white' shows how this derivational model works (setting aside, for simplicity, the fact that the phonologically inferior canonical order is attested in about one-fifth of the relevant corpus data). Syntax initially generates the canonical order, [ˈbi.jɛ.lɔː.ga ˈgraː.da] 'white city,' which phonology rejects because it violates END-WEIGHT. The derivation then returns to syntax, which

generates the alternative order [ˈgraː.da ˈbi.jɛ.lɔː.ga], which is subsequently accepted by phonology as more prosodically well-formed.

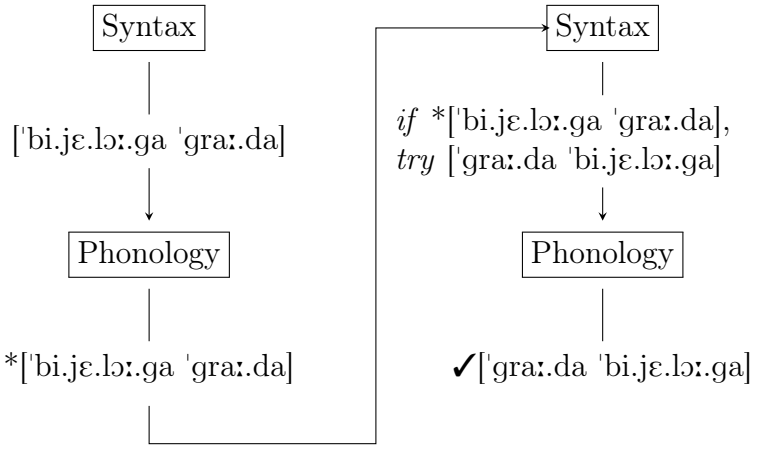


Figure 5.4: Interactionist model of the syntax–phonology interface (derivational variant; adapted from Shih and Zuraw, 2018, 3, ex. 1).

The upshot of the derivational model considered by Shih and Zuraw is that phonological evaluation is interleaved with syntactic derivation. Syntax first generates the preferred, default order and submits it to phonology for evaluation. If the output is phonologically ill-formed, syntax derives an alternative order and submits it back to phonology.

This looping architecture leaves much room for interpretation, especially in a stochastic system where phonological effects on word order are gradient. For one, it is unclear under what conditions phonology gets to reject the initial output of syntax and prompt an alternative derivation. Unless properly restricted, the architecture could in principle allow phonology to continue rejecting syntactic output until a phonologically unmarked structure is arrived at. The grammar must therefore specify some well-formedness threshold below which the phonological component may reject the output of syntax. But how exactly should such a threshold be defined? How phonologically marked does a structure need to be for phonology to send it back to the generative

component? Should this threshold be defined in terms of relative rather than absolute markedness, that is, how marked a structure is relative to its alternatives? But if the modules apply in an ordered sequence, how does phonology assess the default order's well-formedness relative to non-default orders that have yet to be generated by syntax?

A second challenge concerns probability mapping. In a derivational system, it is not immediately clear how probability distributions over word-order alternatives could be estimated as rigorously as in a parallel MaxEnt grammar. Although serial and multi-level variants of MaxEnt have been proposed (Staubs and Pater, 2016; Nazarov and Pater, 2017), it is unclear how such models would assign probabilities to competing outputs in a looping architecture such as that in Figure 5.4.

Neither of these concerns constitutes a prohibitive objection against derivational interactionist models. Future implementations must address the issues of well-formedness thresholds and probability mapping raised here. The parallelist alternative proposed here readily avoids both, obviating the need for additional mechanisms. I therefore argue that the parallelist implementation adopted here is superior to the derivational model outlined in Shih and Zuraw (2018), because it provides a simple, quantitatively grounded account of the gradient nature of phonological effects on word order and eschews some of the indeterminacy inherent to the existing derivational implementation.

What is important for the present discussion is that, unlike the non-interactionist approaches reviewed in §§5.3–5.4, interactionist models offer a plausible account of phonologically conditioned word-order variation in poetic meter. In particular, Fitzgerald's (2007) interleaving model, of which the probabilistic interleaving model proposed here is a derivative, is interactionist by design. Because phonological and syntactic constraints are evaluated in a single parallel OT grammar and are freely rankable, phonological markedness requirements may directly affect the output of syntactic operations. This architecture therefore abandons the principle of phonology-free syntax (Zwicky

and Pullum, 1986a) and predicts pervasive phonological effects on constituent order, which are unattainable under the filtering theory (Anttila, 2016) and the post-syntactic PF movement approach (Agbayani and Golston, 2010). The prediction is indeed borne out by BCMS folk meter and many other traditions (Youmans, 1983, 1989; Rice, 1997; Golston, 1998; Golston and Riad, 2000; Fitzgerald, 2003; Hayes et al., 2012).

The question, however, is whether an interactionist architecture with direct syntax–phonology communication (Shih and Zuraw, 2018; Breiss and Hayes, 2020) is the only way of accounting for phonologically conditioned word-order variation in poetic meter. In the next section, I argue that it is not. Whether variable word order has consequences for the nature of the syntax–phonology interface depends crucially on where linearization takes place. If linear order is established not in narrow syntax but during post-syntactic Spell-Out, phonological markedness constraints can modulate word order without abandoning the derivational architecture of the standard Y-model or the feed-forward organization of the grammar.

5.5.2 Probabilistic interleaving model with post-syntactic linearization

While the parallel interactionist model, unlike its non-interactionist counterparts discussed in §§5.3–5.4, is compatible with the variable word order in poetic meter, the question remains whether the data in fact require such a model, that is, whether bidirectional communication between syntax and phonology is the only way to explain phonological effects on word order in the BCMS oral tradition. The answer, I contend, is negative. As Shih and Zuraw (2018) point out, word-order variation bears on the nature of the syntax–phonology interface only insofar as linear order between words is taken to be part of narrow syntax. Building on Zwicky and Pullum (1986a), the

authors maintain that "word order and linearization principles [are] the purview of (morpho)syntax."

If, by contrast, linear order is established during post-syntactic Spell-Out, being essentially a PF phenomenon (Chomsky, 1995; Fox and Pesetsky, 2005; Embick and Noyer, 2007; Agbayani and Golston, 2010; Kusmer, 2020), word-order variation of the kind attested in Tagalog and BCMS folk meter likewise arises post-syntactically, at PF, and therefore has no direct implications for whether syntax has access to phonological information. On this view, syntax determines only hierarchical structure, while phonology establishes linear precedence between terminals. Phonology therefore does not affect syntactic computation but still determines word order during Spell-Out. The crucial point is thus not whether phonology affects word order, but whether word order itself belongs to syntax or post-syntactic Spell-Out.

The proposal is similar in spirit to Agbayani and Golston's (2010) approach, in that it assumes post-syntactic linearization. As discussed in §5.4, however, Agbayani and Golston's approach does not provide a viable account of word-order variation in poetic meter because it assumes indirect reference. Poetic word order calls for several important modifications. The most important departure, naturally, concerns indirect reference. In poetic meter, syntactic constructions that map onto the same prosodic structures have different inversion propensities (123–125). This suggests that PF cannot operate exclusively over prosodic representations, as the indirect reference hypothesis maintains. Assuming that linear order is established at PF, phonology must have direct access to syntactic categories through linearization constraints.

The version of the probabilistic interleaving model proposed here simplifies the architecture of Agbayani and Golston (2010), sketched in Figures 5.2–5.3. Recall that Agbayani and Golston assume that linearization takes place on the branch from Spell-Out to PF. The linearization procedure initially generates only the preferred, canonical

order, so the input to phonology is already linearized (see tableau 121). Deviation from the canonical order can be derived only by the PF movement of prosodic constituents. As argued in §5.4, this mechanism is insufficient to accommodate the phonologically conditioned word-order variation in poetic meter.

I assume that Spell-Out, linearization, and prosodic structure assignment are carried out all at once by the phonological grammar. This eliminates Agbayani and Golston's (2010) intermediate step (the syntax–phonology interface) and allows phonology to determine surface linear order directly through the interaction of phonological markedness constraints and linearization constraints. The input to phonology is therefore an unlinearized hierarchical structure generated by narrow syntax. Competing word orders constitute the candidate set from which the phonological grammar selects the optimal output. Consequently, PF movement is no longer required. The default order is enforced by linearization constraints, rather than being built into the input to phonology. Phonology in turn need not perform movement operations of any kind; it simply evaluates the candidates emitted by GEN. The tableau in (129) illustrates the phonological stage of the derivation, after syntax has generated the hierarchical structure and passed it to PF.

(129) Probabilistic interleaving model with post-syntactic Spell-Out: PF

Input: [NP [AdjP] [$\bar{N}$]]	END-W 2.04	ADJ1ST 0.97	$\mathscr{H}$	$e^{\mathscr{H}}$	p
a. Adj–N ˈbi.jɛ.lɔː.ga ˈgraː.da	−1		−2.04	0.13	0.26
b. N–Adj ˈgraː.da ˈbi.jɛ.lɔː.ga		−1	−0.97	0.38	0.74

In contrast to (121), there is no need for the anti-movement STAY constraints in (129): Deviation from the canonical order is not achieved via movement at PF. The noncanonical order violates the linearization constraint ADJECTIVEFIRST, which requires that adjectival modifiers be prenominal.

Table 5.1 summarizes the major differences between the proposed version of the probabilistic interleaving model which assumes post-syntactic Spell-Out and Agbayani and Golston's PF movement approach.

	PF movement approach	Probabilistic interleaving model
Role of syntax	Hierarchical structure building	Hierarchical structure building
Phonology's access to syntactic categories	Indirect, mediated by prosodic structure	Direct, through linearization constraints
Locus of linearization	Syntax–phonology interface	PF
Input to phonology	Linearized structure (default order)	Unlinearized structure
Source of word-order variation	PF movement	Competition between phonological markedness and linearization constraints

Table 5.1: Differences between Agbayani and Golston's (2010) PF movement approach and the proposed probabilistic interleaving model with post-syntactic Spell-Out.

In sum, the probabilistic interleaving approach does not require an interactionist architecture in which syntax directly accesses phonological information. The separation of syntax and phonology can be maintained, provided that linearization is treated as a post-syntactic PF process rather than part of narrow syntax. On this view, surface word order is derived from the interaction between phonological markedness constraints and linearization constraints in a probabilistic MaxEnt grammar. The main departure from previous post-syntactic accounts is the abandonment of indirect reference: Linearization constraints must have direct access to syntactic categories rather than their prosodic reflexes.

5.6 Chapter conclusion

In this dissertation, I devise a model of poetic word order in which phonological markedness constraints compete directly against constraints enforcing default word order, such as AdjectiveFirst and PossessiveSecond. This model submits itself readily to an interpretation in terms of parallel architecture, whereby phonological and syntactic information is processed simultaneously, allowing bidirectional flow of information between the two modules. It generally accords with Fitzgerald's (2007) model, which assumes an interleaving of phonological and syntactic constraints within a parallelist OT system; the most notable point of departure is that I use a probabilistic MaxEnt grammar in lieu of strict-ranking OT, while retaining the same interleaving structure for poetic grammar.

I considered two non-interactionist alternatives to this model, each implemented in both derivational and parallelist settings: the filtering theory and the post-syntactic-linearization-cum-PF-movement approach. I demonstrated that neither can be made to work for the patterned word-order variation found in the BCMS oral tradition.

The probabilistic interleaving model proposed in this dissertation is compatible with two interpretations. Under the first, the probabilistic interleaving model is an interactionist approach, which allows direct communication between phonology and syntax: Linearization constraints, which belong to narrow syntax, are freely ranked (or weighted) relative to phonological markedness constraints. Under the second, the probabilistic interleaving model is, in fact, a model of PF grammar, of which both phonological constraints and, importantly, linearization constraints are part. I leave for future work the question of whether linearization is part of syntax proper or of post-syntactic spell-out; the present data do not allow us to tease apart these two possibilities.

Chapter 6

Conclusion

This dissertation advances the idea of a METRICAL (or POETIC) GRAMMAR: An internalized body of knowledge, shared among the community of folk poets, that dictates, among other things, how they order words in verse lines. Although the surface order of constituents in the BCMS oral folk tradition is highly variable (for example, one and the same collocation can surface in different orders just a few lines apart), this variation is patterned. Folk meter's word order is largely rule-governed, subject to phonological and other grammatical constraints: Folk poets internalize the fine-grained statistical generalizations that they use to fit linguistic substance in verse lines during live performance. The word order of poetic language is less rigid than in ordinary language. Many word orders that are impossible in ordinary language become available in poetic meter, so long as they make verse lines *sound better*, that is, increase their phonological well-formedness. Variation arises because several such phonological pressures operate simultaneously within the metrical grammar and, in a given line, can conflict, and because baseline linearization pressures, though severely weakened compared to their ordinary-language effects, can still push back against phonology.

Generative metrics, including MaxEnt metrics as its widely adopted offshoot, has

customarily modeled line frequency as a function of metricality (Hayes and Moore-Cantwell, 2011; Hayes et al., 2012; Hayes and Schuh, 2019; Hayes and Minkova, 2023; McPherson and Ryan, 2018; Henriksson, 2022), treating word order as a static property of verse lines rather than a system of alternatives from which poets actively choose. Studies that examine poetic word order typically sample instances of syntactic inversions from a corpus and show that they are phonologically motivated (e.g., Youmans, 1983; Fitzgerald, 2007). This line of work thus provides an account of only one aspect of poetic word order, namely syntactic inversions, not full-fledged models of poetic word order. This dissertation directly models the competition between word-order alternatives in meter as a function of the phonological properties of the constituents. The most immediate contribution is to linguistic metrics and poetics, although poetic word order, I show, tells us a surprisingly great deal about the inner workings of language.

Meter has been shown to deviate from the linearization rules of ordinary language in two principled ways. Meter displays more flexible order in NPs and VPs than ordinary language: Adjectival modifiers, for example, are mandatorily prenominal in ordinary language, but may be either prenominal or postnominal in folk meter. Conversely, the order of conjuncts in coordinate constructions, relatively free in ordinary language, is virtually fixed in meter. To give just one example, BCMS allows both [ˈruː.kɛ i ˈnɔ.gɛ] 'arms and legs' and [ˈnɔ.gɛ i ˈruː.kɛ] 'legs and arms,' the former being preferred to the latter for semantic reasons (see §1.3.2.2 in Chapter 1), but only [ˈnɔ.gɛ i ˈruː.kɛ] 'legs and arms' is attested in meter. This is because the semantically inferior order ('legs and arms') is more phonologically well-formed in BCMS than its transposition (Maretić, 1901, 1907; Zec, 2008).

Both discrepancies were shown to follow from meter's heightened sensitivity to phonological factors. In NPs and VPs, phonological factors uniformly give way to linearization requirements in ordinary language but may prevail in poetic meter. In coor-

dinate constructions, there is no baseline, syntactically preferred word order. Phonological constraints on conjunct order (e.g. end-weight) compete with semantic factors in ordinary language (Cooper and Ross, 1975; Wright et al., 2005; Benor and Levy, 2006; Ryan, 2019a), whereas in meter, non-phonological factors are severely weakened, which effectively makes phonology the only predictor of conjunct order.

To accommodate meter's amplified sensitivity to phonological markedness requirements, along with the fundamentally probabilistic nature of phonology's conditioning of poetic word order, I have adopted a PROBABILISTIC INTERLEAVING MODEL of poetic grammar, whose main ingredients are the following:

1. The metrical grammar is stochastic rather than deterministic;

2. Phonological and linearization constraints are part of a single parallel constraint grammar (Fitzgerald, 2007), in which the two constraint types are freely ranked or weighted relative to each other, with no fixed dominance between them.

Postulate (1) is motivated in Chapters 3–4. Chapter 5 explores the consequences of postulate (2).

***

Although this dissertation has focused on poetic meter, it makes contributions to linguistics proper. Most immediately, it adds to our understanding of variation and probability in language, and their role in grammar. Drawing on the research tradition that holds our linguistic knowledge to be stochastic, Chapters 3–4 inform the study of variation and probability and constraint-based phonology by addressing the following questions:

1. The source of stochasticity: Is grammar inherently probabilistic (as in MaxEnt), generating probability distributions over linguistic events (i.e. outputs), or is

it essentially categorical, generating ones and zeros, but speakers change the parameters of their grammar every time they use their grammar to evaluate output forms (as in StOT and NHG)?

2. The mode of evaluation: Does constraint-based grammar employ strict ranking (as in StOT) or numeric weights and holistic evaluation (as in NHG and MaxEnt)?

I have advanced a two-pronged argument that isolates MaxEnt, the inherently probabilistic weighted-constraint grammar, as the most viable model of variation given the data:

1. As shown in Chapter 3, stochastic variants of HG (MaxEnt and NHG) provide superior accounts of ganging-up cumulativity to StOT. StOT fails to accommodate the full range of ganging-up interactions in variable data, even when augmented with constraint conjunction. This limitation effectively eliminates StOT from contention.

2. Chapter 4 eliminates NHG on the grounds of its excessively restrictive harmonic bounding. To assign high probabilities to harmonically bounded candidates, NHG must effectively nullify the contribution of otherwise active constraints in the grammar, which results in a weighting paradox. MaxEnt produces a more flexible form of harmonic bounding, readily avoiding any trade-off effects.

The first result is consistent with a growing consensus that stochastic variants of HG outperform StOT (Zuraw and Hayes, 2017; Smith and Pater, 2020; Hayes, 2022). Despite limited success in the domain of gang effects (Jäger and Rosenbach, 2006; Hayes and Londe, 2006; Jarosz, 2010; Pater, 2023), StOT produces highly restricted, invariably sublinear ganging-up cumulativity, consistently attenuating the effects of

coincident constraint violations (Maslova, 2007; Smith and Pater, 2020; Flemming and Magri, 2026). This study is no exception: The StOT grammar assigns undesirably high probabilities to doubly violating word orders across the board.

Other than contributing another case study to the growing body of work on StOT's limitations, this dissertation offers, to my knowledge, the first demonstration that StOT augmented with conjoined constraints cannot accommodate the full range of ganging-up interactions. LCs, I show, are doubly detrimental in StOT: They fail to capture all gang effects in the data, while further degrading the grammar's overall fit by introducing ranking paradoxes. Only two prior studies document undergeneration effects in OT-LC (Walker, 2017; Ryan, 2026), both using categorical data. The present study therefore provides the first demonstration of OT-LC's undergeneration in a stochastic setting.

This finding bears on the OT–HG debate, particularly on the difference between HG and OT-LC as theories of ganging-up cumulativity. The bulk of work comparing HG and OT-LC draws entirely on categorical data (Pater, 2009a,b, 2016; Farris-Trimble, 2008; Potts et al., 2010; Walker, 2017; Tebay and Zimmermann, 2020; Mai and Bakovic, 2020; Ryan, 2026). Chapter 3 has shown that the stochastic implementations of OT and OT-LC inherit the damning limitations of their deterministic counterparts. This ultimately suggests that, in the face of attested patterns of variation and cumulativity, OT-style ranking is not a tenable mode of constraint interaction, and that weighted-constraint grammars offer a more plausible alternative (e.g., Hayes, 2022, 476).

Zooming out, this dissertation's findings bear on the role of conjoined constraints in the grammar. One of the central questions of the OT–HG debate has been whether HG, thanks to its holistic evaluation and additivity of constraint violations, can replicate the effects of OT-LC, the more complex, and presumedly more powerful, theory. The answer is no: Despite earlier claims that HG dispenses with LC, HG has been shown to

necessitate this device (Green and Davis, 2014; Shih, 2017a; Milenković, 2026; Chapter 2 of this dissertation), albeit for a more limited subset of ganging-up interactions than OT. The opposite question may well be asked: Can OT-LC replicate all gang effects generated by HG? Somewhat surprisingly, the answer is again no. Despite added machinery, OT-LC fails to capture some gang effects produced by base HG (Walker, 2017; Ryan, 2026; Chapter 3 of this dissertation).

These results support a nuanced position on the mechanism of LC. OT-LC is a failed theory of cumulativity: Even when augmented with LC, OT is not guaranteed to replicate the predictions of base HG. However, this failure does not warrant the conclusion that LC has no place in grammar. LC is still needed in HG, the superior theory of cumulativity, including its stochastic implementations such as NHG and MaxEnt, to model cases of extreme superlinearity (*pace* Potts et al., 2010; Farris-Trimble, 2008; Pater, 2016). OT's failure thus follows from the fact that it relies on LC across the board, whereas HG desirably restricts the role of this device to a narrow subset of ganging-up interactions.

Having eliminated StOT in Chapter 3, Chapter 4 turns to the differences between the stochastic variants of HG, MaxEnt and NHG. The two frameworks add stochasticity to HG by fundamentally different means, and are therefore expected, *a priori*, to make distinct predictions. In practice, however, it has been difficult to identify testable predictions that meaningfully tease MaxEnt and NHG apart (Flemming and Magri, 2026; Alderete and Finley, 2023), let alone evaluate those predictions empirically. Several factors complicate the framework-level comparison. First, the MaxEnt and NHG models are typically fit to data using different learning algorithms. MaxEnt grammars are standardly fit by searching for the Maximum Likelihood (ML) constraint weights, whereas NHG weights are estimated using the Gradual Learning Algorithm (GLA), whose learning objective differs from ML optimization. Some of the surface

differences between fitted MaxEnt and NHG grammars are therefore attributable to the learning algorithm, which makes it difficult to isolate the predictions that follow directly from each framework's probability mapping function (Flemming, 2021). Second, NHG comes in several variants (Jesney, 2007; Boersma and Pater, 2016; Hayes, 2017; Hayes and Kaplan, 2025; Kaplan, 2021, 2025b), which make distinct empirical predictions. It is not always obvious which variant should be employed and compared against MaxEnt. Finally, even once falsifiable predictions that discriminate between MaxEnt and NHG have been identified, they are difficult to evaluate empirically, since ideal test cases are not easily found in naturalistic data.

Due to these major architectural differences, MaxEnt and GLA-NHG should not be compared exclusively based on model performance on particular data. The objective of MaxEnt's weight search is to find weights that maximize the likelihood of the observed data given the model of grammar. It is therefore unsurprising that MaxEnt models in general tend to fit specific empirical frequencies better than corresponding NHG models. A more sensible approach is to compare the frameworks in terms of their qualitative predictions, that is, the kinds of qualitative generalizations that they predict for variable data and gradient acceptability judgments.

Along these lines, Magri (2025) claims that "[t]he discussion of alternatives [to MaxEnt; A.M.] has been limited to Noisy Harmonic Grammar [...] and based mostly on circumstantial evidence about M[ax]E[nt]'s and NHG's relative ability to fit specific patterns of empirical frequencies given specific choices of candidates and constraints (Smith and Pater 2020; Flemming 2021; Breiss and Albright 2022)" (see also Magri and Anttila, 2026, 2 for a similar claim).

This characterization undersells the current state of the MaxEnt–NHG debate, however. The case for MaxEnt does not rest exclusively on quantitative fit to specific datasets, as Magri (2025) and Magri and Anttila (2026) present it. There are substan-

tive, architectural reasons why MaxEnt outperforms NHG on concrete case studies, which are rooted in how each framework determines candidate probability.

Flemming (2021) offers a particularly instructive case study. Given that in NHG, constraint weights are perturbed by independent, constraint-specific noise, the determination of probability is crucially sensitive to the source of Harmony difference between candidates, i.e. candidates' violation profiles, unlike MaxEnt, whose probability mapping is only sensitive to the Harmony difference, not its source (recall examples 41–45 in Chapter 2). In NHG, violating one constraint with the weight x does not have the same effect on probability as violating multiple constraints whose base weights sum to x, precisely because of constraint-specific noise. This is a general property of NHG's probability mapping, which falls out from its architecture and therefore emerges independently of any specific dataset. The French schwa data examined in Flemming's study offer a rare test case to empirically probe this prediction. The results directly support MaxEnt's lack of sensitivity to the source of the Harmony difference.

Chapter 4 of this dissertation provides further evidence for MaxEnt over NHG, extending the results of Flemming (2021). The study leverages the phenomenon of simple harmonic bounding to identify another testable prediction that distinguishes MaxEnt from NHG and uses the BCMS corpus data to probe this newly identified prediction.

Both MaxEnt and the version of NHG adopted here (censored NHG with random tie resolution) produce only relative harmonic bounding, whereby the bounded can receive nonzero probability but can never receive more probability than the bounder. In MaxEnt, harmonically bounded candidates receive probability through the framework's general probability mapping function. cNHG is more interesting. Given that candidates are evaluated by a deterministic HG grammar at each evaluation, the bounded can never outperform the bounder, assuming negative weights are prohibited (Keller,

2000; Potts et al., 2010). At best, the two candidates can be tied after the addition of noise, in which case the grammar selects the winner at random. It is due to deterministic evaluation combined with random selection that NHG limits the probability range of harmonically bounded candidates more aggressively than MaxEnt. Since the bounded can surface only by being tied with its bounder, and it receives only a share of the probability mass assigned to ties, cNHG consistently assigns lower probabilities to the bounded than corresponding MaxEnt grammars. For example, MaxEnt and cNHG predict different ceiling probabilities for simply harmonically bounded forms: 50% in MaxEnt vs. 25% in cNHG.

The different forms of relative harmonic bounding enforced by cNHG and MaxEnt constitute a novel testing ground for stochastic HGs. Variable word order in the BCMS oral tradition provides an ideal test case, given that some simply harmonically bounded candidates, that is, phonologically unmotivated syntactic inversions, are attested at high rates in the corpus. Chapter 4 therefore not only identifies a new testable prediction of MaxEnt and NHG, but also subjects the newly identified prediction to empirical evaluation using naturalistic corpus data.

The results favor MaxEnt over cNHG. Although cNHG can derive sufficiently high probabilities for harmonically bounded word orders, it can do so only by making concessions that compromise its handling of other patterns in the data. Specifically, in the harmonic bounding conditions, the linearization constraint must receive weights close to zero to match the observed frequencies of phonologically unmotivated syntactic inversions. This is inconsistent with this constraint's independent contribution elsewhere in the grammar. MaxEnt avoids this trade-off, being simultaneously able to produce desirably high probabilities of harmonically bounded word orders and assign to the linearization constraint the weight that is commensurate with its independent role in the grammar.

Thus, fitting harmonic bounding in cNHG entails poor fit elsewhere, and vice versa. This trade-off falls out from the fact that cNHG does not directly assign probabilities to candidates (like MaxEnt), but rather employs deterministic winner selection. The circumstances under which the bounded can win in a deterministic grammar are severely limited, which leads to undesirably low rates of harmonically bounded candidates across the board. This result, I contend, supports a view of stochastic grammar as intrinsically probabilistic, as in MaxEnt, rather than categorical and sampling-based, as in NHG.

The argument for MaxEnt advanced in Chapter 4 is therefore not based on quantitative fit. MaxEnt outperforms NHG not merely as a "pure data fitting device" (Jäger and Rosenbach, 2006, 966), but as a theory of stochastic grammar (see also Breiss and Hayes, 2020, 343–344; Hayes, 2022, 477 on this dual nature of MaxEnt). In fact, in the present case, the MaxEnt and corresponding NHG models are virtually indistinguishable in terms of their overall quantitative fit to the data (§2.7.2), despite the latter's inferior performance on harmonic bounding. It is the kinds of trade-off effects in the grammar that set MaxEnt apart from NHG.

Finally, the dissertation argues for a model of poetic grammar in which phonological and linearization constraints are part of a single parallel probabilistic constraint-based grammar. Chapter 5 addresses the question of what kind of grammar architecture can accommodate such a model. One obvious candidate is a parallel interactionist model, in which phonological and syntactic information is processed simultaneously, and phonological information can directly affect the output of syntactic operations (Bresnan, 2000; Jackendoff, 2002; Shih and Zuraw, 2018; Bruening, 2019; Breiss and Hayes, 2020). An interactionist model, I demonstrate, is only warranted assuming that surface linear order is established by narrow syntax. If linearization takes place post-syntactically, with syntax performing only primitive structure-building operations such

as Merge and Move, phonology's pervasive effects on poetic word order follow from the free ranking (or weighting) of phonological and linearization constraints at PF. This theory of poetic word order is at the same time flexible, in that it allows phonology to manipulate word order in phrase types with default linearization preferences in meter, and restrictive, in that it preserves the strict separation between the modules (on the assumption that linearization is external to narrow syntax).

A fruitful avenue for future research is to extend the approach and methodology used in this dissertation to the study of word-order variation in other metrical traditions, including both folk and literary traditions. Comparative work of this kind may help identify probabilistic universals of poetic word order and shed further light on the limits of how far poetic language can go to enforce phonological well-formedness.

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Appendix A

Full model predictions

The table below reports the predicted probabilities generated by four stochastic models:

- **StOT**: StOT (implemented in OT-Soft; $\sigma = 2$; 1,000,000 evaluation trials);
- **GLA-cNHG**: censored NHG (implemented in OT-Soft; $\sigma = 1$; 1,000,000 evaluation trials);
- **ML-cNHG**: censored NHG (implemented in R using a modified version of Flemming (2021)'s procedure; $\sigma = 1$; 1,000,000 evaluation trials);
- **MaxEnt**: MaxEnt HG (implemented in R using the `maxent.ot` package).

Input labels in the `.txt` and `.csv` files encode the prosodic and syntactic properties of each phrase. These labels serve as the inputs to the stochastic OT/HG models implemented in R and OT-Soft. (Binomial labels follow a slightly different convention, described below.)

Each label for NP and VP data consists of four components:

1. Relative constituent size (syllable count difference; values from -2 to 2):
    - NPs (adjective–noun and possessive–noun): modifier size − noun size;

- VPs: verb size − object size.

This value is calculated with respect to the default syntactic order, not the surface order. Thus, for NPs, it is always modifier size − noun size, and for VPs, verb size − object size, irrespective of whether a given token appears in the canonical or noncanonical order in the corpus.

2. Penultimate weight and stress:

   - `H` = heavy (stress irrelevant);
   - `L0` = light and unstressed;
   - `L1` = light and stressed.

   The first code always refers to the modifier (or verb), and the second to the head noun (or object), regardless of surface order.

3. Trisyllable marker (only when relative size = 0):

   - `3` = both constituents are trisyllabic (relevant for TrisyllableInertia);
   - otherwise omitted.

4. Syntactic category:

   - no suffix = adjective–noun construction;
   - `P` = possessive–noun construction;
   - `V` = VP.

For example, ['bi.jɛ.li.jɛm 'graː.dɔm] 'white city.INS' belongs to the `2HL0` condition:

- "`2`" (relative size): quadrisyllabic adjective followed by a disyllabic head noun; thus 4 − 2 = 2;

- "HL0" (penultimate weight): adjective's penult heavy (h); noun's penult light (l) and unstressed (0);
- " " (syntactic category): NP with an adjectival modifier (null).

For [ˈpiː.ʃɛː ˈkɲi.gu] '(s)he is writing the book,' the label is "0HL1V":

- "0" (relative size): disyllabic verb followed by a disyllabic object; thus $2 - 2 = 0$;
- "HL1" (penultimate weight): verb's penult heavy (h), object noun's penult light (l) and stressed (1);
- "V" (syntactic category): VP.

For [li.ˈjɛ.pu ˈʤɛ.vɔːj.ku] 'pretty girl.ACC,' the label is "0L1H3":

- "0" (relative size): trisyllabic adjective followed by a trisyllabic head noun; thus $3 - 3 = 0$;
- "L1H" (penultimate weight): adjective's penult light (l) and stressed (1), head noun's penult heavy (h);
- "3": two trisyllables.
- " " (syntactic category): adjective–noun.

Labels for binomial conditions begin with the prefix Bin. The remaining portion of the label encodes the relative size of the conjuncts and the weight and stress of their penults. For example, [ˈviː.nɔ i=ˈra.ki.ja] 'wine and rakija' is labeled Bin-1HL0:

- "Bin": binomial;

- "`-1`" (relative size): first conjunct disyllabic; second conjunct trisyllabic; thus $2 - 3 = -1$

- "`HL0`" (penultimate weight): first conjunct has a heavy penult (h); second conjunct has light penult (l), which is unstressed (0);

Table A.1: By-condition model predictions

Input	Output	AdjectiveFirst	PossessiveSecond	TrisyllableInertia	Align	End-Weight	*Light/FinalStrong	*StressedLight/FinalStrong	End-Weight & *Light/FinalStrong	Frequency	p(observed)	p(MaxEnt)	p(GLA-cNHG)	p(ML-cNHG)	p(StOT)
-1HH	Adj N				*					152	0.65	0.68	0.67	0.57	0.10
-1HH	N Adj	*				*				82	0.35	0.32	0.33	0.43	0.90
-1HHP	Adj N		*		*					12	0.40	0.15	0.30	0.17	0.02
-1HHP	N Adj					*				18	0.60	0.85	0.70	0.83	0.98
-1HHV	VO				*					11	0.44	0.45	0.47	0.38	0.09
-1HHV	OV					*				14	0.56	0.55	0.53	0.62	0.91
-1HL0	Adj N				*		*			132	0.28	0.26	0.29	0.26	0.09
-1HL0	N Adj	*				*				333	0.72	0.74	0.71	0.74	0.91
-1HL0P	Adj N		*		*		*			4	0.08	0.03	0.07	0.05	0.02
-1HL0P	N Adj					*				48	0.92	0.97	0.93	0.95	0.98
-1HL0V	VO				*		*			6	0.26	0.12	0.13	0.12	0.09
-1HL0V	OV					*				17	0.74	0.88	0.87	0.88	0.91
-1HL1	Adj N				*		*	*		10	0.34	0.11	0.17	0.21	0.03
-1HL1	N Adj	*				*				19	0.66	0.89	0.83	0.79	0.97

Input	Output	AdjectiveFirst	PossessiveSecond	TrisyllableInertia	Align	End-Weight	*Light/FinalStrong	*StressedLight/FinalStrong	End-Weight & *Light/FinalStrong	Frequency	p(observed)	p(MaxEnt)	p(GLA-cNHG)	p(ML-cNHG)	p(StOT)
-1HL1P	Adj N		*		*		*	*		0	0.00	0.01	0.03	0.04	0.01
-1HL1P	N Adj					*				22	1.00	0.99	0.97	0.96	0.99
-1L1H	Adj N				*					129	0.98	0.99	1.00	0.99	0.93
-1L1H	N Adj	*				*	*	*	*	2	0.02	0.01	0.00	0.01	0.07
-1L1HP	Adj N		*		*					22	0.88	0.93	0.97	0.90	0.83
-1L1HP	N Adj					*	*	*	*	3	0.12	0.07	0.03	0.10	0.17
-1L1HV	VO				*					22	1.00	0.98	0.99	0.98	0.93
-1L1HV	OV					*	*	*	*	0	0.00	0.02	0.01	0.02	0.07
-1L1L0	Adj N				*		*			255	0.93	0.96	0.98	0.96	0.93
-1L1L0	N Adj	*				*	*	*	*	19	0.07	0.04	0.02	0.04	0.07
-1L1L0P	Adj N		*		*		*			80	0.71	0.69	0.89	0.75	0.83
-1L1L0P	N Adj					*	*	*	*	33	0.29	0.31	0.11	0.25	0.17
-1L1L0V	VO				*		*			17	0.71	0.91	0.96	0.92	0.93
-1L1L0V	OV					*	*	*	*	7	0.29	0.09	0.04	0.08	0.07
-1L1L1	Adj N				*		*	*		13	0.72	0.91	0.96	0.94	0.92
-1L1L1	N Adj	*				*	*	*	*	5	0.28	0.09	0.04	0.06	0.08
-2HH	Adj N									216	0.99	0.95	1.00	0.99	1.00
-2HH	N Adj	*				*				3	0.01	0.05	0.00	0.01	0.00
-2HHP	Adj N		*							53	0.79	0.63	0.85	0.63	0.05
-2HHP	N Adj					*				14	0.21	0.37	0.15	0.37	0.95

Input	Output	AdjectiveFirst	PossessiveSecond	TrisyllableInertia	Align	End-Weight	*Light/FinalStrong	*StressedLight/FinalStrong	End-Weight & *Light/FinalStrong	Frequency	p(observed)	p(MaxEnt)	p(GLA-cNHG)	p(ML-cNHG)	p(StOT)
-2HL0	Adj N						*			427	0.79	0.77	0.75	0.70	0.66
-2HL0	N Adj	*				*				114	0.21	0.23	0.25	0.30	0.34
-2HL0P	Adj N		*				*			23	0.32	0.21	0.38	0.26	0.05
-2HL0P	N Adj					*				48	0.68	0.79	0.62	0.74	0.95
-2HL0V	VO						*			8	0.22	0.56	0.57	0.52	0.61
-2HL0V	OV					*				29	0.78	0.44	0.43	0.48	0.39
-2HL1	Adj N						*	*		15	0.79	0.55	0.55	0.62	0.08
-2HL1	N Adj	*				*				4	0.21	0.45	0.45	0.38	0.92
-2L1H	Adj N									108	0.99	1.00	1.00	1.00	1.00
-2L1H	N Adj	*				*	*	*	*	1	0.01	0.00	0.00	0.00	0.00
-2L1HP	Adj N		*							32	0.94	0.99	1.00	0.99	0.88
-2L1HP	N Adj					*	*	*	*	2	0.06	0.01	0.00	0.01	0.12
-2L1L0	Adj N						*			298	1.00	1.00	1.00	1.00	1.00
-2L1L0	N Adj	*				*	*	*	*	1	0.00	0.00	0.00	0.00	0.00
-2L1L0P	Adj N		*				*			123	0.92	0.95	1.00	0.97	0.88
-2L1L0P	N Adj					*	*	*	*	11	0.08	0.05	0.00	0.03	0.12
-2L1L0V	VO						*			19	0.79	0.99	1.00	1.00	1.00
-2L1L0V	OV					*	*	*	*	5	0.21	0.01	0.00	0.00	0.00
-2L1L1	Adj N						*	*		18	1.00	0.99	1.00	1.00	1.00
-2L1L1	N Adj	*				*	*	*	*	0	0.00	0.01	0.00	0.00	0.00

Input	Output	AdjectiveFirst	PossessiveSecond	TrisyllableInertia	Align	End-Weight	*Light/FinalStrong	*StressedLight/FinalStrong	End-Weight & *Light/FinalStrong	Frequency	p(observed)	p(MaxEnt)	p(GLA-cNHG)	p(ML-cNHG)	p(StOT)
0HH	Adj N									333	0.80	0.72	0.88	0.86	1.00
0HH	N Adj	*								85	0.20	0.28	0.12	0.14	0.00
0HH3	Adj N				*					154	1.00	0.96	1.00	0.98	1.00
0HH3	N Adj	*		*	*					0	0.00	0.04	0.00	0.02	0.00
0HH3V	VO				*					14	0.93	0.91	0.99	0.94	1.00
0HH3V	OV			*	*					1	0.07	0.09	0.01	0.06	0.00
0HHP	Adj N		*							24	0.34	0.18	0.14	0.07	0.00
0HHP	N Adj									47	0.66	0.82	0.86	0.93	1.00
0HHV	VO									14	0.30	0.50	0.50	0.50	1.00
0HHV	OV									33	0.70	0.50	0.50	0.50	0.00
0HL03	Adj N				*		*			156	0.84	0.81	0.74	0.62	1.00
0HL03	N Adj	*		*	*					29	0.16	0.19	0.26	0.38	0.00
0HL03V	VO				*		*			13	0.54	0.62	0.55	0.42	1.00
0HL03V	OV			*	*					11	0.46	0.38	0.45	0.58	0.00
0HL1	Adj N						*	*		32	0.06	0.14	0.11	0.21	0.01
0HL1	N Adj	*								538	0.94	0.86	0.89	0.79	0.99
0HL13	Adj N				*		*	*		31	0.65	0.61	0.53	0.54	0.91
0HL13	N Adj	*		*	*					17	0.35	0.39	0.47	0.46	0.09
0HL1P	Adj N		*				*	*		0	0.00	0.01	0.00	0.00	0.00
0HL1P	N Adj									58	1.00	0.99	1.00	1.00	1.00

Input	Output	AdjectiveFirst	PossessiveSecond	TrisyllableInertia	Align	End-Weight	*Light/FinalStrong	*StressedLight/FinalStrong	End-Weight & *Light/FinalStrong	Frequency	p(observed)	p(MaxEnt)	p(GLA-cNHG)	p(ML-cNHG)	p(StOT)
0HL1V	VO						*	*		1	0.01	0.06	0.00	0.02	0.00
0HL1V	OV									146	0.99	0.94	1.00	0.98	1.00
0L0H3	Adj N				*					552	1.00	0.99	1.00	1.00	1.00
0L0H3	N Adj	*		*	*		*			2	0.00	0.01	0.00	0.00	0.00
0L0H3P	Adj N		*		*					29	0.91	0.93	0.97	0.85	0.88
0L0H3P	N Adj			*	*		*			3	0.09	0.07	0.03	0.15	0.12
0L0H3V	VO				*					84	0.89	0.98	1.00	1.00	1.00
0L0H3V	OV			*	*		*			10	0.11	0.02	0.00	0.00	0.00
0L0L03	Adj N				*		*			755	0.98	0.96	1.00	0.98	1.00
0L0L03	N Adj	*		*	*		*			14	0.02	0.04	0.00	0.02	0.00
0L0L03P	Adj N		*		*		*			26	0.74	0.68	0.84	0.53	0.88
0L0L03P	N Adj			*	*		*			9	0.26	0.32	0.16	0.47	0.12
0L0L03V	VO				*		*			85	0.68	0.91	0.99	0.94	1.00
0L0L03V	OV			*	*		*			40	0.32	0.09	0.01	0.06	0.00
0L0L13	Adj N				*		*	*		80	0.98	0.91	0.90	0.91	0.91
0L0L13	N Adj	*		*	*		*			2	0.02	0.09	0.10	0.09	0.09
0L1H	Adj N									302	0.98	0.98	1.00	0.99	1.00
0L1H	N Adj	*					*	*		6	0.02	0.02	0.00	0.01	0.00
0L1H3	Adj N				*					72	1.00	1.00	1.00	1.00	1.00
0L1H3	N Adj	*		*	*		*	*		0	0.00	0.00	0.00	0.00	0.00

Input	Output	AdjectiveFirst	PossessiveSecond	TrisyllableInertia	Align	End-Weight	*Light/FinalStrong	*StressedLight/FinalStrong	End-Weight & *Light/FinalStrong	Frequency	p(observed)	p(MaxEnt)	p(GLA-cNHG)	p(ML-cNHG)	p(StOT)
0L1H3V	VO				*					18	1.00	0.99	1.00	1.00	1.00
0L1H3V	OV			*	*		*	*		0	0.00	0.01	0.00	0.00	0.00
0L1HP	Adj N		*							89	0.61	0.78	0.91	0.69	0.42
0L1HP	N Adj						*	*		58	0.39	0.22	0.09	0.31	0.58
0L1HV	VO									72	0.86	0.94	1.00	0.98	1.00
0L1HV	OV						*	*		12	0.14	0.06	0.00	0.02	0.00
0L1L03	Adj N				*		*			85	1.00	0.99	1.00	0.99	1.00
0L1L03	N Adj	*		*	*		*	*		0	0.00	0.01	0.00	0.01	0.00
0L1L03V	VO				*		*			12	1.00	0.96	1.00	0.97	1.00
0L1L03V	OV			*	*		*	*		0	0.00	0.04	0.00	0.03	0.00
0L1L1	Adj N						*	*		140	0.77	0.72	0.88	0.86	1.00
0L1L1	N Adj	*					*	*		42	0.23	0.28	0.12	0.14	0.00
0L1L1P	Adj N		*				*	*		8	0.10	0.18	0.14	0.07	0.00
0L1L1P	N Adj						*	*		70	0.90	0.82	0.86	0.93	1.00
0L1L1V	VO						*	*		10	0.14	0.50	0.50	0.50	1.00
0L1L1V	OV						*	*		63	0.86	0.50	0.50	0.50	0.00
1HH	Adj N					*				135	0.85	0.76	0.73	0.77	0.91
1HH	N Adj	*			*					23	0.15	0.24	0.27	0.23	0.09
1HHV	VO					*				39	0.87	0.55	0.53	0.62	0.91
1HHV	OV				*					6	0.13	0.45	0.47	0.38	0.09

Input	Output	AdjectiveFirst	PossessiveSecond	TrisyllableInertia	Align	End-Weight	*Light/FinalStrong	*StressedLight/FinalStrong	End-Weight & *Light/FinalStrong	Frequency	p(observed)	p(MaxEnt)	p(GLA-cNHG)	p(ML-cNHG)	p(StOT)
1HL1	Adj N					*	*	*	*	18	0.10	0.04	0.03	0.07	0.07
1HL1	N Adj	*			*					170	0.90	0.96	0.97	0.93	0.93
1L0H	Adj N					*				853	0.93	0.95	0.93	0.93	0.92
1L0H	N Adj	*			*		*			69	0.07	0.05	0.07	0.07	0.08
1L0HP	Adj N		*			*				34	0.56	0.62	0.74	0.66	0.38
1L0HP	N Adj				*		*			27	0.44	0.38	0.26	0.34	0.62
1L0HV	VO					*				89	0.84	0.88	0.87	0.88	0.91
1L0HV	OV				*		*			17	0.16	0.12	0.13	0.12	0.09
1L0L1	Adj N					*	*	*	*	59	0.16	0.21	0.11	0.19	0.07
1L0L1	N Adj	*			*		*			318	0.84	0.79	0.89	0.81	0.93
1L0L1P	Adj N		*			*	*	*	*	2	0.10	0.02	0.02	0.03	0.06
1L0L1P	N Adj				*		*			19	0.90	0.98	0.98	0.97	0.94
1L0L1V	VO					*	*	*	*	7	0.18	0.09	0.04	0.08	0.07
1L0L1V	OV				*		*			31	0.82	0.91	0.96	0.92	0.93
1L1H	Adj N					*				130	0.99	0.98	0.96	0.95	0.97
1L1H	N Adj	*			*		*	*		1	0.01	0.02	0.04	0.05	0.03
1L1HV	VO					*				24	1.00	0.95	0.94	0.90	0.97
1L1HV	OV				*		*	*		0	0.00	0.05	0.06	0.10	0.03
1L1L1	Adj N					*	*	*	*	16	0.23	0.41	0.20	0.23	0.08
1L1L1	N Adj	*			*		*	*		55	0.77	0.59	0.80	0.77	0.92

Input	Output	AdjectiveFirst	PossessiveSecond	TrisyllableInertia	Align	End-Weight	*Light/FinalStrong	*StressedLight/FinalStrong	End-Weight & *Light/FinalStrong	Frequency	p(observed)	p(MaxEnt)	p(GLA-cNHG)	p(ML-cNHG)	p(StOT)
2HH	Adj N					*				83	0.20	0.26	0.17	0.24	0.22
2HH	N Adj	*								341	0.80	0.74	0.83	0.76	0.78
2HHV	VO					*				25	0.43	0.12	0.01	0.03	0.00
2HHV	OV									33	0.57	0.88	0.99	0.97	1.00
2HL1	Adj N					*	*	*	*	2	0.00	0.00	0.00	0.00	0.00
2HL1	N Adj	*								572	1.00	1.00	1.00	1.00	1.00
2HL1V	VO					*	*	*	*	0	0.00	0.00	0.00	0.00	0.00
2HL1V	OV									85	1.00	1.00	1.00	1.00	1.00
2L0H	Adj N					*				325	0.69	0.68	0.64	0.66	0.47
2L0H	N Adj	*					*			147	0.31	0.32	0.36	0.34	0.53
2L0HV	VO					*				586	0.52	0.44	0.42	0.48	0.39
2L0HV	OV						*			551	0.48	0.56	0.58	0.52	0.61
2L0L1	Adj N					*	*	*	*	9	0.02	0.03	0.00	0.01	0.00
2L0L1	N Adj	*					*			354	0.98	0.97	1.00	0.99	1.00
2L0L1V	VO					*	*	*	*	1	0.00	0.01	0.00	0.00	0.00
2L0L1V	OV						*			1168	1.00	0.99	1.00	1.00	1.00
2L1H	Adj N					*				15	0.94	0.85	0.79	0.72	0.93
2L1H	N Adj	*					*	*		1	0.06	0.15	0.21	0.28	0.07
Bin-1HH	HX i XHX									273	1.00	0.99	1.00	1.00	1.00
Bin-1HH	XHX i HX				*	*				0	0.00	0.01	0.00	0.00	0.00

Input	Output	AdjectiveFirst	PossessiveSecond	TrisyllableInertia	Align	End-Weight	*Light/FinalStrong	*StressedLight/FinalStrong	End-Weight & *Light/FinalStrong	Frequency	p(observed)	p(MaxEnt)	p(GLA-cNHG)	p(ML-cNHG)	p(StOT)
Bin-1HL0	HX i XL0X						*			284	0.97	0.92	0.92	0.89	0.95
Bin-1HL0	XL0X i HX				*	*				8	0.03	0.08	0.08	0.11	0.05
Bin-1HL1	HX i XL1X						*	*		31	0.97	0.81	0.79	0.84	0.47
Bin-1HL1	XL1X i HX				*	*				1	0.03	0.19	0.21	0.16	0.53
Bin-1L1H	L1X i XHX									214	1.00	1.00	1.00	1.00	1.00
Bin-1L1H	XHX i L1X				*	*	*	*	*	0	0.00	0.00	0.00	0.00	0.00
Bin-1L1L0	L1X i XL0X						*			440	1.00	1.00	1.00	1.00	1.00
Bin-1L1L0	XL0X i L1X				*	*	*	*	*	0	0.00	0.00	0.00	0.00	0.00
Bin-1L1L1	L1X i XL1X						*	*		66	1.00	1.00	1.00	1.00	1.00
Bin-1L1L1	XL1X i L1X				*	*	*	*	*	0	0.00	0.00	0.00	0.00	0.00
Bin0L1H	i L1X i HX				*					94	1.00	0.94	1.00	0.98	1.00
Bin0L1H	i HX i L1X				*		*	*		0	0.00	0.06	0.00	0.02	0.00

Appendix B

Ganging-up cumulativity in stochastic OT

StOT is known to produce only sublinear cumulativity, whereby the joint contribution of two or more interacting constraints is only slightly stronger than the independent contribution of the stronger of those interacting constraints (Hayes and Londe, 2006; Smith and Pater, 2020; Flemming and Magri, 2026, Chapter 3). This property of StOT was, to my knowledge, first observed by E. Maslova in 2004 (published as Maslova, 2007). Sublinearity is observed in natural-language data (Pizzo, 2015; Pater, 2023), but it is by no means the norm (Smith and Pater, 2020; Breiss and Albright, 2022).

In this Appendix, I analyze the simplest case of ganging-up cumulativity, deriving the probability of the coincident violation of two cooperating constraints $\mathbb{B}$ and $\mathbb{C}$ when pitted against a single constraint $\mathbb{A}$. I provide a detailed derivation of the probability that the output simultaneously violates both ganging constraints, $\mathbb{B}$ and $\mathbb{C}$.

$\mathbb{A}$ outranks two constraints $\mathbb{B}$ and $\mathbb{C}$ simultaneously whenever $\mathbb{A}$'s perturbed ranking value, $\hat{r}_{\mathbb{A}}$, is higher than the perturbed ranking value of each ganging constraint:

$$\mathbb{A} \gg \mathbb{B} \wedge \mathbb{A} \gg \mathbb{C}$$

$$\text{iff} \tag{B.1}$$

$$\hat{r}_{\mathbb{A}} > \hat{r}_{\mathbb{B}} \wedge \hat{r}_{\mathbb{A}} > \hat{r}_{\mathbb{C}}$$

The perturbed ranking value for each constraint k has two components:

- the fixed term r_k (a.k.a. base ranking value)
- the random term n_k (a.k.a. random noise, drawn from a normal distribution with $\mu = 0$; independent for each constraint)

Thus:

$$\hat{r}_k = r_k + n_k \tag{B.2}$$

Constraint $\mathbb{A}$ outranks its rival constraint $\mathbb{B}$ whenever the noise difference $n_{\mathbb{A}} - n_{\mathbb{B}}$ is greater than the negative of the difference of their fixed terms:

$$\mathbb{A} \gg \mathbb{B}$$

$$\text{iff} \tag{B.3}$$

$$n_{\mathbb{A}} - n_{\mathbb{B}} > -(r_{\mathbb{A}} - r_{\mathbb{B}})$$

For example, in the sample tableau in (51), where $r_{\mathbb{A}} = 101$ and $r_{\mathbb{B}} = 100$, $\mathbb{A} \gg \mathbb{B}$ whenever $n_{\mathbb{A}} - n_{\mathbb{B}} > -1$.

Thus, for $\mathbb{A} \gg \mathbb{B}$ and $\mathbb{A} \gg \mathbb{C}$ to both hold at the same time, both of the following must hold:

$$n_{\mathbb{A}} - n_{\mathbb{B}} > -(r_{\mathbb{A}} - r_{\mathbb{B}}) \tag{B.4a}$$

$$n_{\mathbb{A}} - n_{\mathbb{C}} > -(r_{\mathbb{A}} - r_{\mathbb{C}}) \tag{B.4b}$$

We therefore need to determine the probability that both (B.4a) and (B.4b) hold simultaneously. This requires computing the joint distribution of the two noise difference variables $n_{\mathbb{A}} - n_{\mathbb{B}}$ and $n_{\mathbb{A}} - n_{\mathbb{C}}$. Because each noise term is normally distributed, the noise difference variables are also normally distributed. Both difference variables have the shared component $n_{\mathbb{A}}$, so they are not independent.

The noise difference variables $n_{\mathbb{A}} - n_{\mathbb{B}}$ and $n_{\mathbb{A}} - n_{\mathbb{C}}$ follow a bivariate normal distribution (Evans and Rosenthal, 2004) whose covariance is:

$$Cov(n_{\mathbb{A}}-n_{\mathbb{B}}, n_{\mathbb{A}}-n_{\mathbb{C}}) = Cov(n_{\mathbb{A}}, n_{\mathbb{A}})-Cov(n_{\mathbb{A}}, n_{\mathbb{C}})-Cov(n_{\mathbb{B}}, n_{\mathbb{A}})+Cov(n_{\mathbb{B}}, n_{\mathbb{C}}) \tag{B.5}$$

Note:

- The covariance of a variable with itself is its variance. Each noise term n_k has the variance of σ^2, hence $Cov(n_{\mathbb{A}}, n_{\mathbb{A}}) = \sigma^2$
- The covariance of two independent variables is 0, hence $Cov(n_{\mathbb{A}}, n_{\mathbb{C}})$, $Cov(n_{\mathbb{B}}, n_{\mathbb{A}})$, and $Cov(n_{\mathbb{B}}, n_{\mathbb{C}})$ all equal zero

Thus:

$$Cov(n_{\mathbb{A}} - n_{\mathbb{B}}, n_{\mathbb{A}} - n_{\mathbb{C}}) = \sigma^2 - 0 - 0 + 0 = \sigma^2 \tag{B.6}$$

Next, we need to construct the covariance matrix for $n_{\mathbb{A}} - n_{\mathbb{B}}$ and $n_{\mathbb{A}} - n_{\mathbb{C}}$. The covariance matrix of the two noise difference variables $n_{\mathbb{A}} - n_{\mathbb{B}}$ and $n_{\mathbb{A}} - n_{\mathbb{C}}$, Σ, is:

$$\Sigma = \begin{pmatrix} Var(n_{\mathbb{A}} - n_{\mathbb{B}}) & Cov(n_{\mathbb{A}} - n_{\mathbb{B}}, n_{\mathbb{A}} - n_{\mathbb{C}}) \\ Cov(n_{\mathbb{A}} - n_{\mathbb{C}}, n_{\mathbb{A}} - n_{\mathbb{B}}) & Var(n_{\mathbb{A}} - n_{\mathbb{C}}) \end{pmatrix}$$

Recall: $Cov(n_{\mathbb{A}} - n_{\mathbb{B}}, n_{\mathbb{A}} - n_{\mathbb{C}}) = \sigma^2$.

Because noise terms are independent, the variance of each pairwise noise difference variable is the sum of variances of the individual noise terms:

$$Var(n_{\mathbb{A}} - n_{\mathbb{B}}) = Var(n_{\mathbb{A}}) + Var(n_{\mathbb{B}}) = 2\sigma^2 \tag{B.7}$$

Thus, the covariance matrix Σ for the two difference variables is:

$$\Sigma = \begin{pmatrix} 2\sigma^2 & \sigma^2 \\ \sigma^2 & 2\sigma^2 \end{pmatrix}$$

The probability that $\mathbb{A}$ outranks both $\mathbb{B}$ and $\mathbb{C}$ is the probability that each noise difference variable exceeds its negative threshold $-(r_{\mathbb{A}} - r_{\mathbb{B}})$ and $-(r_{\mathbb{A}} - r_{\mathbb{C}})$, respectively. The negative thresholds specify the minimum noise differences required for $\mathbb{A}$ to outrank each ganging constraint. Since the noise difference variables are symmetric around zero (recall that for each noise term, $\mu = 0$), this is equivalent to evaluating the relevant bivariate normal cumulative distribution function, Φ_2 (Genz and Bretz, 2009), whose arguments are: (i) the pairwise ranking differences $r_{\mathbb{A}} - r_{\mathbb{B}}$ and $r_{\mathbb{A}} - r_{\mathbb{C}}$, and (ii) the covariance matrix Σ, which captures the dependence between the two noise difference variables.

$$P(\mathbb{A} \gg \mathbb{B} \wedge \mathbb{A} \gg \mathbb{C}) = \Phi_2\left((r_{\mathbb{A}} - r_{\mathbb{B}}), (r_{\mathbb{A}} - r_{\mathbb{C}}); \Sigma\right) \tag{B.8}$$

The bivariate normal cumulative distribution function, Φ_2, is computed in R using the `pmvnorm()` function from the `mvtnorm` package (Genz et al., 2020).

* * *

In Chapter 3, I use the standardized bivariate normal cumulative distribution function to compute the probability of $\mathbb{A} \gg \mathbb{B} \wedge \mathbb{A} \gg \mathbb{C}$. Each noise difference variable is divided by its standard deviation.

Since constraint-level noise terms are independent, the variance of the pairwise noise difference variables $n_{\mathbb{A}} - n_{\mathbb{B}}$ and $n_{\mathbb{A}} - n_{\mathbb{C}}$ equals the sum of the variances of the individual noise terms, $2\sigma^2$ (B.7). The standard deviation of each noise difference variable is therefore:[1]

$$\sigma_D = \sqrt{2\sigma^2} = \sqrt{2}\sigma \tag{B.9}$$

Standardization is obtained by dividing both sides of (B.4a–B.4b) by σ_D. For example,

$$n_{\mathbb{A}} - n_{\mathbb{B}} > -(r_{\mathbb{A}} - r_{\mathbb{B}}) \tag{B.10}$$

becomes

$$\frac{n_{\mathbb{A}} - n_{\mathbb{B}}}{\sqrt{2}\sigma} > -\frac{r_{\mathbb{A}} - r_{\mathbb{B}}}{\sqrt{2}\sigma}. \tag{B.11}$$

Standardization replaces the covariance matrix with the correlation matrix:

$$\begin{pmatrix} 1 & Corr \\ Corr & 1 \end{pmatrix}$$

The correlation between two random variables X and Y is:

[1]The notation differentiates the standard deviation of a constraint-level noise term, σ, from the standard deviation of a pairwise noise difference variable, σ_D.

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sigma(X)\sigma(Y)} \tag{B.12}$$

See Evans and Rosenthal (2004, 156).

So, in our case:

$$Corr(n_{\mathbb{A}} - n_{\mathbb{B}}, n_{\mathbb{A}} - n_{\mathbb{C}}) = \frac{Cov(n_{\mathbb{A}} - n_{\mathbb{B}}, n_{\mathbb{A}} - n_{\mathbb{C}})}{\sigma_D(n_{\mathbb{A}} - n_{\mathbb{B}})\sigma_D(n_{\mathbb{A}} - n_{\mathbb{C}})} \tag{B.13}$$

Because $Cov(n_{\mathbb{A}} - n_{\mathbb{B}}, n_{\mathbb{A}} - n_{\mathbb{C}}) = \sigma^2$ (see B.6) and each $\sigma_D = \sqrt{2}\sigma$ (B.9),

$$Corr(n_{\mathbb{A}} - n_{\mathbb{B}}, n_{\mathbb{A}} - n_{\mathbb{C}}) = \frac{\sigma^2}{(\sqrt{2}\sigma)^2} = \frac{1}{2}. \tag{B.14}$$

The correlation matrix fed into `pmvnorm()` is therefore:

$$\begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}$$

The arguments of the standardized bivariate normal cumulative distribution function are therefore:

- standardized pairwise ranking difference between $\mathbb{A}$ and $\mathbb{B}$: $\frac{r_{\mathbb{A}} - r_{\mathbb{B}}}{\sqrt{2}\sigma}$,
- standardized pairwise ranking difference between $\mathbb{A}$ and $\mathbb{C}$: $\frac{r_{\mathbb{A}} - r_{\mathbb{C}}}{\sqrt{2}\sigma}$,
- correlation between the two difference variables, $Corr(n_{\mathbb{A}} - n_{\mathbb{B}}, n_{\mathbb{A}} - n_{\mathbb{C}}) = \frac{1}{2}$

Accordingly,

$$P(\mathbb{A} \gg \mathbb{B} \wedge \mathbb{A} \gg \mathbb{C}) = \Phi_2\left(\frac{r_{\mathbb{A}} - r_{\mathbb{B}}}{\sqrt{2}\sigma}, \frac{r_{\mathbb{A}} - r_{\mathbb{C}}}{\sqrt{2}\sigma}; Corr(n_{\mathbb{A}} - n_{\mathbb{B}}, n_{\mathbb{A}} - n_{\mathbb{C}})\right), \qquad Corr = \frac{1}{2}. \tag{B.15}$$

The standardized variant is equivalent to its unstandardized counterpart in (B.8).

(B.15) is implemented in R, using the `mvtnorm` package:

```
# two-vs-one in Stochastic OT: the probability that one constraint
    outranks two constraints simultaneously
# pairwise ranking differences must be fed into the function
# default settings: st dev of random noise = 2 (OT-Soft default); corr =
    0.5 (see Appendix B)
p_doubly_viol_sot <- function(delta_ab, delta_ac, sd = 2, corr = 0.5) {
  z_ab <- delta_ab / (sqrt(2) * sd) # pairwise ranking difference (A - B)
     divided by the st dev of random noise difference (A - B)
  z_ac <- delta_ac / (sqrt(2) * sd)

  Sigma <- matrix(c(1, corr,
                    corr, 1), 2, 2) # correlation matrix

  as.numeric(mvtnorm::pmvnorm(lower = c(-z_ab, -z_ac),
                              upper = c(Inf, Inf),
                              sigma = Sigma)) # returns the probability
    of coincident violation of B and C
}
```

***

The foregoing shows an analytic (i.e. closed form) method for computing $p(\mathbb{A} \gg \mathbb{B} \wedge \mathbb{A} \gg \mathbb{C})$ in StOT. The OT-GLA implementation in OT-Soft (Hayes et al., 2013) uses a Monte Carlo (i.e. simulation-based) method, in which base ranking values are perturbed by taking N random draws from a Gaussian distribution with $\mu = 0$ and $\sigma = 2$ (N = 1,000,000; Zuraw and Hayes, 2017). The Monte Carlo method can also be simulated in R:

```
# alternative: Monte Carlo solution (like OT-Soft)
p_doubly_viol_sot_monte_carlo <- function(r_a, r_b, r_c, sd = 2, n =
    1000000) {
  rhat_a <- rnorm(n, mean = r_a, sd = sd)
  rhat_b <- rnorm(n, mean = r_b, sd = sd)
  rhat_c <- rnorm(n, mean = r_c, sd = sd)

  mean((rhat_a > rhat_b) & (rhat_a > rhat_c))
}
```

The rationale for using the analytic method for the typological survey in §3.2.1 is that its Monte Carlo alternative à la OT-Soft would take a very long time to run given the number of stochastic grammars (ranking vectors) to go over. For example, the Monte Carlo implementation of cNHG (see Figure 3.2b), for which there is no analytic solution (Flemming, 2021, 22), took more than an hour to run despite reducing the number of random draws to 10,000 per grammar.

In practice, the difference between the analytic and Monte Carlo estimates of $p(\mathbb{A} \gg \mathbb{B} \wedge \mathbb{A} \gg \mathbb{C})$ is minuscule. Consider:

```
> # compares the analytic and Monte Carlo solutions
> r_a <- 102
> r_b <- 101
> r_c <- 100
> # analytic solution
> p_doubly_viol_sot(r_a - r_b, r_a - r_c)
[1] 0.5487437
> # Monte Carlo solution
> set.seed(123) # sets random seed to ensure the output of p_doubly_viol_
```

```
      sot_monte_carlo() is the same at different runs
10 > p_doubly_viol_sot_monte_carlo(r_a, r_b, r_c)
11 [1] 0.548432
```

(This setup reproduces tableau (66) in Chapter 3.)